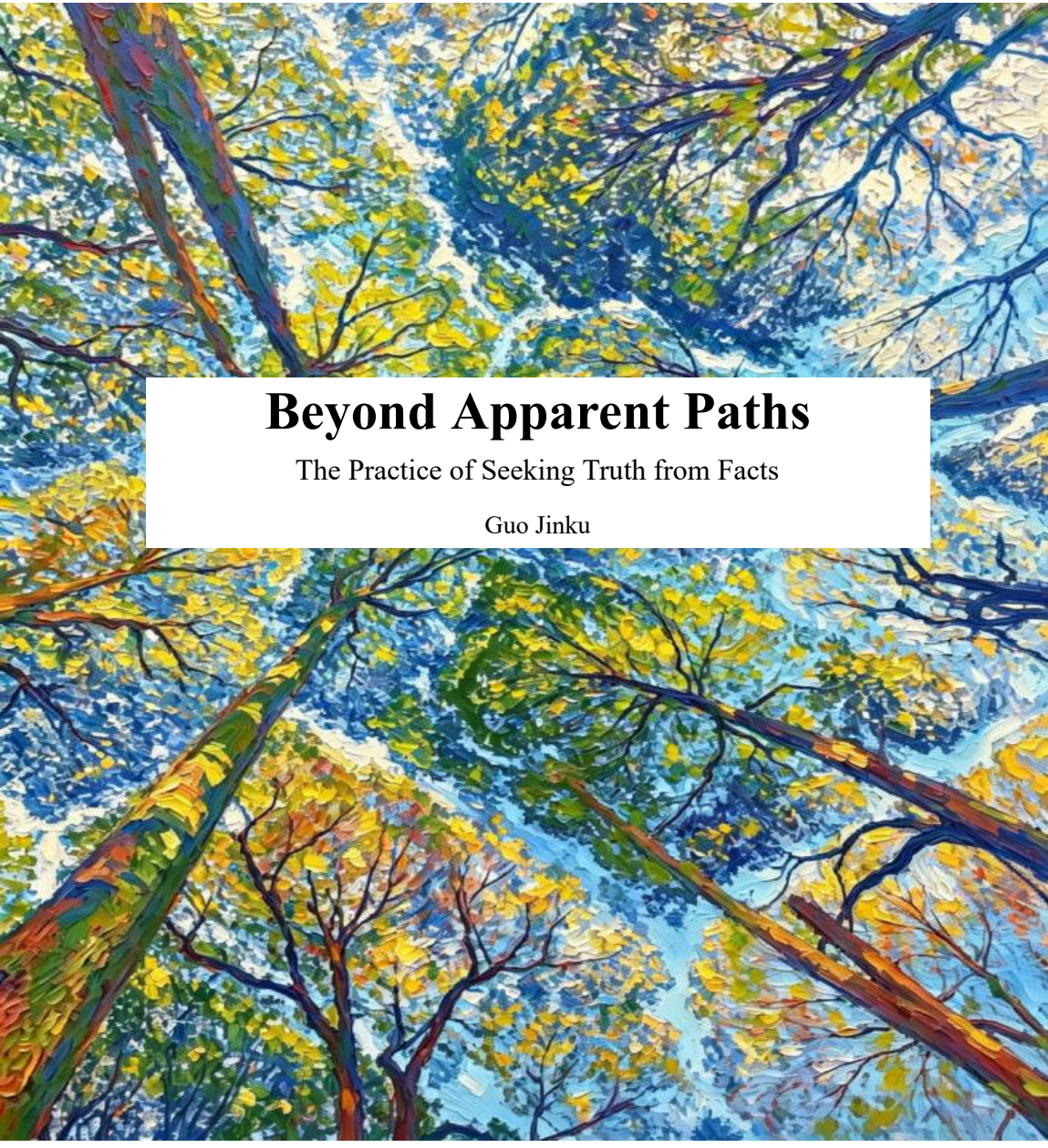


Beyond Apparent Paths

The Practice of Seeking Truth from Facts

Guo Jinku

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TO THE READER

Beyond Apparent Paths is a work of interdisciplinary philosophical nonfiction. It is written for readers drawn to complexity science, systems thinking, life and consciousness, Eastern thought, and the practical question of how human beings understand and act in a layered world. It brings physics and life, mind and society, ancient thought and ordinary experience into a single conversation, seeking a way of seeing that can pass through them all.

It is neither a conventional popular-science survey nor an academic monograph confined to one discipline. Its purpose is to explore how disorder opens possibility, how order takes form across levels, how new laws emerge, and how these patterns bear on knowledge, action, and human life. The rigor it pursues is fidelity to the question, coherence across levels, and the willingness to follow an idea beyond the field in which it first appeared.

The book is meant to be used as much as read: as a companion in observing, judging, acting, and understanding oneself. Some pages ask the reader to slow down, hold an unfamiliar connection in mind, and return to it when experience has supplied what words alone cannot. If these pages make one familiar thing appear differently, or allow two things long kept apart to speak to one another, the reading will have done its work.

Guo Jinku

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Chapter 1

The Finger Pointing to the Moon

I think the next century will be the century of complexity.

Stephen Hawking

However freely we revel in all the world has to offer when we are young, as we grow older we gradually long to break free from the grip of desire and pursue insight.

Each of us is constantly building an understanding of the outside world through observation and experience. Apart from the inner self embedded in body and brain, everything else is external to us. For this reason, every person is born alone, and the soul always walks by itself. When one person's way of understanding the world resonates with another's, we experience it as love. Family affection, friendship, romantic love, and the many other forms of love all ease the loneliness of the self.

At every moment we are living in the present, while also correcting the biases in how we understand the world. Yet bias is always there. It entangles us with the past, leaves us bewildered in the present, and fills us with excessive expectations for the future. From this arises a kind of suffering beyond bodily pain.

To reach a higher understanding is to resonate with more people, to let more people feel that resonance, to receive love and to give it. It is also to see more clearly how the world changes across its many levels. Do not cling to the past. Do not cloud the present. Do not rush ahead to meet the future. Face desire directly and lessen suffering. By extending cognition across more levels, we awaken deeper inner drive, pursue richer experiences of life, and pursue a life of relatively greater value.

It is not only scientists and philosophers who explore the truth of the world. Every ordinary person, in conscious and unconscious moments alike, is also experiencing the world and searching for answers. The knowledge in the universe is as vast as the sea. Even existence itself, together with all its changes, is knowledge. If there were truly an all-knowing and all-powerful being, perhaps it could only be the universe itself. Facing inexhaustible knowledge with a finite life, we should work all the more to seek a method for understanding the world. Only by finding the innate knowing within ourselves can we better illuminate the objective world.

I was born in a village in northern China, grew up in a small city, and studied in a large one. I spent many years teaching and conducting research at a university. After twenty years of military service, I returned to civilian life and started a business.

During my studies and research, I worked in control science and engineering, information and signal processing, and related fields. Knowledge of control, information, and systems not only proved

invaluable in my studies and work; more profoundly, it prompted me to ask how we come to know the world.

As early as the 1940s and 1950s, Norbert Wiener, Claude Shannon, and John von Neumann began to explore how to make machines think. Their work laid the foundations of cybernetics, information theory, systems theory, and computer science, and also formed the intellectual origins of artificial intelligence research.

Wiener described cybernetics as the study of the common laws of control and communication in nature, society, and thought. Shannon's information theory and von Neumann's computer architecture made universal information representation and general-purpose computation possible. The German philosopher Georg Klaus, in *Cybernetics and Epistemology* and *Cybernetics in Philosophical Perspective*, pointed out that the ideas and methodology revealed by cybernetics are very close to philosophy. Ludwig von Bertalanffy, the father of systems theory, also regarded systems theory as a new natural philosophy that would replace reductionism and mechanism.

Cybernetics, information theory, and systems theory therefore address broad and general questions in engineering. Their objects of study are not limited to the structure or change of any one kind of thing. Rather, they abstract from concrete objects to general things and the general laws of the real world. This professional experience gradually accustomed me to understanding things through systems thinking, and in everyday life, to sensing and understanding the universal laws of structure and change.

The earliest form of systems thinking I encountered held this view. Everything can be abstracted as a system. Complex systems

can be broken into combinations of simpler ones. Analyze the simple, and you arrive at the complex.

Matter is built from molecules and atoms. Atoms are built from neutrons, protons, and electrons. Living things are built from cells, and cells from organic macromolecules. Human society is built from organizations, and organizations from individuals. The complex is always made of the simple. So a universal method for understanding the world suggests itself. Take the simple things apart. Figure out their laws. Use those laws to grasp the complex. This way of thinking, with its roots in ancient Greece, fits the natural, intuitive way human beings make sense of things. It is what systems-analytic thinking embodies. In the language of science, it goes by the name reductionism.

Turn the logic around, and you can design and create complex systems by putting simple ones together. Take a car. It has a structural system, a power system, a lubrication system, an electronic information system, and many more. Each of these subsystems is itself an assembly of simpler parts. Put all the parts together in an orderly way, following a blueprint, and something new happens. The whole now has properties and functions that none of the individual parts had on its own. A new system, the automobile, comes into being.

Cybernetics, information theory, and systems theory played an exceptionally important part in advancing scientific thought and engineering, and profoundly accelerated industrialization and the information age. After the Second World War, rockets, satellites, computers, and communication systems developed rapidly, while automobiles, radar, television, and other technologies matured and

entered everyday life.

Without doubt, the closer a method of understanding the world comes to the essence of the world itself, the clearer and more accurate our understanding becomes. Matter leads to molecules, atoms, and elementary particles. Life leads to cells and proteins. A car leads to subsystems and parts. A country leads to companies and individuals. Things of every kind do present themselves as structures at many levels. Systems analysis gives us just such a way to understand the world through many levels.

Before the modern era, human understanding mostly stayed at the everyday and visible scale. The motion of the stars, the forces acting on objects, and mechanical movement were the main questions. To use systems analysis well is like holding a microscope with adjustable magnification. It lets us gain new understanding across a richer range of scales. Turn the magnification up, and we move toward the microscopic, exploring the more basic composition of things. Turn it down, and we gain a wider view, studying the stars and the cosmos from something close to the vantage point of a god.

With this cognitive microscope, humanity extended its reach into the evolution of the universe, the drift of tectonic plates, the life sciences, neuroscience, atomic and chemical theory, relativity, quantum mechanics. Phenomena and laws at vastly different scales came into view.

And as new knowledge poured in from all these scales, the limits of systems-analytic thinking, the limits of reductionism, began to show.

Immanuel Kant offered a pointed critique. He argued that reduc-

ing the understanding of a whole to the understanding of its parts is an isolated, static way of thinking. It erases the connections among things and ignores the fact that they change. Kant insisted that human understanding and scientific inquiry must go beyond analytic reduction. They need a dialectical, synthetic method as well.

Systems analysis focuses mainly on things and their component elements. It assumes that the relations among those elements are simple and mechanical. When we analyze forces between macroscopic objects and their relative motion, such relations can indeed be treated as mechanical and simple. Even in complex systems such as cars, rockets, and satellites, their functions depend on subsystems, yet the interfaces and relations among those subsystems remain simple at their core.

In the reductionist view, nature appears as an orderly world composed of mechanical systems: celestial bodies are linked by gravity and travel in fixed orbits, seemingly unchanged through the ages. Given such simple, mechanical relations, the motion of every object can be calculated, and the properties of a complex thing can be deduced from those of its constituent units. Every physical event is merely another link in the universe's chain of cause and effect. Such a mechanically ordered cosmos therefore necessarily requires a god to provide the 'first impulse' that drives its continuing change.

However, as scientists observed the world at more scales, they gradually came to see that complex systems such as living organisms and neural networks are different. In some cases, the cells in different systems do not differ greatly as constituent elements. Yet because their relations of combination differ enormously, higher systems can display entirely different functions and properties. We

cannot grasp an organ as a whole merely from the function of a single cell, nor can we grasp a living body merely from its organs. For many complex systems, analyzing their subsystems is far from enough. The relations among subsystems and the laws of their change are just as important, and sometimes even more important.

In 1937, drawing on his reflections in biology, the Austrian scientist Ludwig von Bertalanffy proposed general systems theory, usually shortened to systems theory. He argued that the relations among a system's elements could not be ignored and that complex systems must therefore be studied through analysis and synthesis together. His ideas attracted wide attention. Theories of dissipative structures and related approaches followed, helping general systems theory develop into a broader view of self-organization in systems.

Compared with the isolated, static way in which systems analysis grasps things, the idea of self-organization in systems offers a fuller way of understanding them. It holds that all things, or systems, can be seen as relatively disordered lower-level units that form ordered structures at many levels through random fluctuations, spontaneously moving from disorder to order or from order to disorder. Ordered arrangements of lower-level systems make up more complex systems at higher levels, while the relations among those lower-level systems remain indispensable parts of the higher-level whole.

Everything in the world is a layered arrangement of elementary particles. Move closer to the microscopic, and randomness dominates. Pull back toward the macroscopic, and determinacy takes over, because the randomness underneath is constrained layer by

layer. The world never stops changing, and the Prime Mover driving all of it is not God. It is the intrinsic randomness of the elementary particles at the very bottom. All of reality is the product of self-organized evolution.

You cannot reach this view by staring at a single scale. Only by bringing together what science has found across many scales does the picture grow clear. Systems self-organizing thought is the new cognitive microscope produced by this synthesis. It gives us a deeper sense of how levels of structure relate to one another, how disorder turns into order and back again, and how randomness and determinacy belong to the same process. From microscopic particles to the macroscopic cosmos, it opens a new way of understanding the self-organizing evolution of the objective material world. This process also forces humanity's methods of understanding to keep renewing themselves.

Bertalanffy once declared that the ideas of self-organizing systems and dialectical synthesis marked a genuine, necessary, and major development in our worldview. Ilya Prigogine, the Belgian physical chemist, and Ervin László, the Hungarian systems philosopher, went further, arguing that this mode of thought deserved to be called systems philosophy. The Chinese scientist Qian Xuesen[1] also held it in high esteem. He saw it as closely aligned with evolutionary thought and with dialectical materialism, especially their shared emphasis on universal connection and change. As he summarized it, "Systems thinking is a dialectical tool for analysis and

[1]Qian Xuesen (1911–2009), also romanized as Tsien Hsue-shen, was an aerospace engineer and a founding figure in China's rocket and space programs. He also developed "engineering cybernetics," linking control theory with large-scale systems engineering.

synthesis: it finds philosophical expression in dialectical materialism, quantitative expression in operations research and the systems sciences, and rich practical content in systems engineering."

In 1977, Prigogine received the Nobel Prize in Chemistry for his theory of dissipative structures. Synergetics, catastrophe theory, hypercycle theory, fractal theory, and chaos theory followed in succession. Their appearance showed that researchers were not content to treat self-organization in systems merely as a method of understanding; they hoped to develop from it a set of universal foundational theories capable of supporting a mature discipline of systems science.

The central aim of dissipative structure theory and related approaches was to uncover the underlying logic by which all things take shape: the universal laws through which interacting simple systems form more complex systems at higher levels. Although these theories successfully explained problems and phenomena in certain fields, their claim to universality remained contested. P. W. Anderson, the renowned condensed-matter physicist who received the 1977 Nobel Prize in Physics for his work on disordered and magnetic materials, disagreed with Prigogine on this point, arguing that dissipative structure theory described only certain special cases in chemistry.

Anderson was a towering figure in condensed-matter physics, and he also made major contributions to complex systems research and several other fields. He helped create what became the mecca of the discipline, the Santa Fe Institute, which studies complex systems across condensed-matter physics, biology, economics, neuroscience, computer science, and beyond. In 1972 he published a clas-

sic paper, “More Is Different.” It laid out a paradigm for complex systems that diverged sharply from Prigogine’s, pushed back hard against the reductionism that had long saturated the scientific community, and made a deep and lasting impact.

Anderson’s argument was straightforward: different fields, and even different scales within the same field, obey different laws of formation. No single universal law tells us how simple systems interact to produce complex higher-level ones. You can know everything about a water molecule and still not infer how quickly a glass of water will evaporate at room temperature. For that, you need thermodynamics: a macroscopic law.

Everything in nature is, of course, built from elementary particles. But the behavior of a system made of many particles cannot always be read off the properties of those particles alone. Nature keeps producing new kinds of complexity at new scales, and at each scale it generates its own characteristic laws. This is why we reach so often for the word emergence. It names the self-organization process by which vast numbers of elementary units form something at a higher level. And we use it because we have not found, or cannot find, a universal, unified law capable of describing every form of emergence.

If such a universal law truly existed, could complex systems not once again be reduced to and analyzed through simple ones? To insist on finding a universal law of system formation is, in effect, to fall back into the habits of reductionism. What systems thinking and the systems perspective offer us is fundamentally a methodology for understanding the world, not a fixed theory or law valid in every setting.

Anderson took the point further. We cannot rely on the foundational sciences alone to explain everything. Even if someone someday discovers a grand unified theory of physics, that will not be a final theory of everything. We will still need classical mechanics and chemistry and biology and neuroscience and medicine and political economy and all the rest, at all their different scales, to make sense of the staggering variety of things in the world and the laws that govern them at their own levels.

As Anderson observed, the laws of physics at the macroscopic scale are no less fundamental than those at the scale of elementary particles. Different things may differ because their constituent units differ, or because the relations among those units differ. The same thing at different scales differs not only in its constituent units, but also in the relations that organize them. Ultimately, all these differences appear as different forms of knowledge and law. We can even say that everything that exists in the universe, and every way in which it changes, is knowledge and law.

To expect a single theory to apply at every scale is tantamount to expecting an omniscient and omnipotent god capable of deducing the entire past and future. If such a universal law existed, the world would be rigid and dull. Instead, all things change under the impetus of random elementary particles at their lowest level, and every change manifests a law at a particular scale. Across every scale, the universe continually gives rise to an inexhaustible variety of complexity. It is this self-organizing evolution grounded in randomness that has produced the vivid and vital world we know.

Systems thinking has itself continued to absorb what science has learned about the world across scales, from the microscopic to

the macroscopic, and has kept developing in the process. Complex adaptive systems, emergence, and self-organized criticality remain active areas of research. John Holland of the Santa Fe Institute proposed the theory of complex adaptive systems, often described as the third generation of systems thinking. He argued that the driving force of a system's evolution arises within the system itself and that interactions among microscopic elements generate complex macroscopic phenomena. He also emphasized that research should not stop at describing things, but should reveal how they are constituted and how they evolve.

In 2021, the Nobel Prize in Physics was awarded to three scientists working on complex systems: Syukuro Manabe, Klaus Hasselmann, and Giorgio Parisi. The prize recognized their pioneering contributions to our understanding of Earth's climate and of disorder and random fluctuations in complex physical systems. In 2026, the Nobel Prize in Chemistry was awarded to Henri B. Kagan and Kenso Soai for the discovery of nonlinear effects and autocatalysis in asymmetric organic synthesis. Their work revealed, at the molecular level, mechanisms of self-organization and emergence in complex systems, further demonstrating the reach of systems thinking within chemistry. Following the 1977 prizes awarded to Prigogine in chemistry and Anderson in physics, these later awards show the scientific community's continuing recognition of systems research.

As Murray Gell-Mann, founder of the Santa Fe Institute and a Nobel laureate in physics, observed, "We had to set ourselves a really big task. And that was to tackle the great, emerging syntheses in science." From its beginnings, systems thinking sought to identify the fundamental laws shared across disciplines and to form

a broadly applicable scientific methodology for understanding the world, rather than remain confined to a single field. This is why it has had such a far-reaching influence throughout science.

It has been widely applied in astrophysics, condensed-matter physics, the life sciences, molecular biology, neurobiology, computer networks, meteorology, economics, society, and military affairs, where it has been used to study the origin of the universe, nanomaterials, artificial life, intelligence and consciousness, complex networks, climate change, urban evolution, human origins, organizational evolution, management optimization, and combat simulation.

As early as the 1970s and 1980s, Bertalanffy, Prigogine, László, and others were publishing popular works such as *A Systems View of Man*, *From Being to Becoming*, *Order Out of Chaos*, and *The Systems View of the World*. These books boldly carried systems thinking into philosophy, society, and other fields. Today, expressions shaped by this tradition—systems thinking, multidimensional thinking, underlying logic, top-level cognition, resisting entropy, managing dissipation, and organizational entropy—have entered economics, corporate management, and everyday speech. Books on systems thinking continue to appear in abundance.

Some specialists with particularly exacting standards remain cautious about extensions of this kind, regarding them as carrying a risk of misusing scientific concepts. Yet, as Anderson emphasized, different things, or the same thing at different scales, have laws of their own. The specific meanings and applications of these concepts must therefore differ across disciplines. From the standpoint of rigorous academic and technical research, they cannot simply be

transferred wholesale; but when they are used to describe a method of cognition, borrowing them in this way is not without reason.

My professional background gave me a cognitive framework grounded in systems thinking, one that made me habitually view complex things and phenomena through the lens of ordered structures at many levels. Later, the novels of Dou Dou,[2] *Betrayal*, *The Distant Savior*, and *The Red Dust beneath Heaven's Canopy*, pushed me beyond the limits of my own field and led me into deeper reflection.

I recommended these books to friends many times, and each reader came away with something different. Some saw a business story straining for profundity, or a love story filled with compassion for human suffering; some found a transcendence of human nature and morality; some read that there is no savior in this world and that we can rely only on ourselves; and others sensed a strong culture grounded in respect for seeking truth from facts.[3]

What stayed with me most was a student's spontaneous reaction after reading the books: "I feel as though a window has opened in my world." I often asked myself the same question: what was it in these works that had touched me so deeply, yet so quietly?

[2]Dou Dou is the pen name of a contemporary Chinese novelist known for philosophically charged stories of business, society, and individual awakening. The three titles named here are *Beipan*, *Yaoyuan de Jiushizhu*, and *Tianmu Hongchen*; published English editions are not widely available, so the English titles used here are descriptive translations.

[3]The phrase *shishi qiushi* literally means "seek truth from facts." Originating in classical Chinese scholarship, it became a central modern Chinese maxim of inquiry and practice.

The Distant Savior does not merely say that there is no savior. More importantly, it stresses independence of mind and independent thought, and urges us in everything we do to seek truth from facts rather than follow blindly.

Systems thinking tells us that everything is an ordered structure at many levels, forever changing, driven from below by randomness. Each level of a thing has its own characteristic trends of change. If you want to follow objective laws and truly seek truth from facts, you have to know the thing deeply. Only when you have grasped its trends of change across many levels, from surface to essence, can you correctly choose your direction and your path.

Hardly anyone thinks they are not seeking truth from facts. The real question runs deeper. Which level of the thing are you looking at when you claim to see the facts? Which level of truth are you seeking? If you blindly follow the experience and dogma of some particular level, you are, at its root, still leaning on someone else's thinking. You are still waiting for a path to appear before you walk. But every level of a thing, along with its external environment, is constantly changing. Fixed experience and fixed knowledge are not your savior. The way forward lies in using the right cognitive method, pulling together the information available in the present, and thinking across many levels, independently and critically. This is what it means to see a path and not follow it.

Elon Musk's advocacy of first principles is a typical example. It likewise urges us not to rely too readily on analogies with similar things around us or allow established experience to constrain us. Instead, we should look through appearances to underlying reality, analyze the problem level by level, and, when necessary, return to

foundational scientific laws to identify the level that truly matters. Only then can we hope to find the right solution.

The principle is not difficult to understand. The real difficulty is how to seek truth from facts in practice. The key lies in improving our cognitive ability and learning how to translate that understanding into action.

From early neural-network algorithms to today's much-discussed large models, these advances have continued the dream of Wiener, Shannon, von Neumann, and others: to make machines think. The central technical path of artificial intelligence has almost always followed a biomimetic principle, constructing artificial neural networks after the brain's neuronal networks and training them on labeled examples so that the machine's cognitive ability improves step by step.

An examination of how artificial intelligence works shows that, through repeated rounds of training, the structural parameters of an artificial neural network are continually adjusted and optimized until it becomes a cognitive system capable of rapidly recognizing and classifying unfamiliar inputs. Further study also shows that the resulting network structure and the evolution of its parameters do not have a clear physical interpretation.

This shatters the old paradigm of how knowledge is stored. An artificial neural network does not store sample knowledge directly. It learns from samples by changing its own structure and parameters. Concrete sample knowledge is transformed into abstract cognitive capacity. This is why we say the network has acquired intelligence. Neurobiological research tells us that the human brain is a complex biological neural network of nearly a hundred billion neu-

rons. Its cognitive mechanisms are built upon the dynamic reconfiguration of the topological network of synaptic connections. The brain's biological neural network, likewise, does not store knowledge directly. It changes the structure and strength of its synaptic connection network, and in doing so forms a cognitive model unique to each individual.

The human brain contains nearly one hundred billion neurons, each with thousands or even tens of thousands of synapses, while the behavior of every synapse reflects both connection strength and the influence of multiple neurotransmitters. The combinatorial complexity of this arrangement far exceeds that of today's most advanced artificial neural networks. If we want to improve our own cognitive ability, we too must keep training the brain with examples. Observing the outside world is, in essence, a way of training the brain, and the things of the world, with their ordered structures at many levels, are naturally labeled examples.

From childhood onward, when our parents first teach us to understand objects, to read, to speak, through our later years of formal education and wide reading and into working life, we are constantly exposed to a vast body of knowledge, laws, and experience verified by scientific experiment and social practice. These are all high-quality samples for training the brain's cognitive capacity.

By contrast, some forums, online communities, and short-video platforms use intelligent algorithms to pour fragmented information into us, mixing truth with falsehood and reality with fabrication. Poor-quality, disordered information that remains at the level of crude emotional resonance and continually caters to the lowest common denominator does not improve cognitive ability. It can in-

stead lead to cognitive disorientation and even anxiety, depression, and mania.

There is no question about what makes a genuinely high-quality sample. It must reflect the true nature of the outside world. The higher the quality of the sample, the more intelligent the cognitive model the brain produces and the more efficient the training process. We should take full advantage of the convenience of the information age and work hard to acquire systematic, high-quality samples, training our brains with purpose and method to raise our cognitive capacity.

Knowledge from the natural and social sciences, along with the laws and experience that people have accumulated, should not merely be memorized. We should expect these high-quality samples to genuinely change the structure and parameters of the brain's neural network, so that at the physiological level, knowledge, laws, and experience are transformed into the brain's own cognitive ability.

Systematic knowledge samples are especially valuable. They push the brain to establish internal connections across different dimensions and levels of knowledge, making it easier to grasp the common laws that run through them. Deep thinking built on top of such knowledge can, in particular, accelerate the formation of a personal cognitive model.

Every level of every thing, and the external environment around it, is always changing. Knowledge, laws, and experience are no more than our understanding of things at a certain level and at a certain moment. Only by continuously transforming them into our own ability can we respond rapidly to inputs we have never seen

before, and in doing so, cope with the unceasing change of things and circumstances.

This is why we must keep training our brain's neural network. Only then can we move, step by step, toward true independence of thought. Only then can we possibly become our own savior. This is not merely a shift in outlook; it is a lifelong process of self-learning that never stops.

Absorbing knowledge, laws, and experience is not enough; the work of building a personal cognitive model must be joined to practice. Learning to drive makes the point. Driving knowledge has many layers: how the car works, the rules of the road, the essentials of technique. But the coordination of brain, eyes, hands, and feet, along with the subtle control of emotion and muscle, can be acquired only through long hours behind the wheel, in real traffic, practicing again and again.

Extensive practice resembles the Confucian maxim "to learn and practice what one has learned." For an experienced driver, the actions of driving gradually become subconscious and no longer require deliberate thought. Repeated practice changes the strength and structure of synaptic connections through the brain's plasticity, as though driving ability had been carved into the brain as a physiological acquisition. Unlike general theoretical knowledge of driving, the coordination of brain, eyes, hands, and feet, together with the fine control of emotion and muscle, can be acquired only through personal practice. It becomes deeply coupled to the individual's physiology and develops into a distinctive capacity.

It is only through practice that we gain the chance to truly encounter and continuously observe the outside world, to grasp the

trends of change at its many levels, from surface to essence, and on that basis to correctly choose our direction and path and to influence how things change. The process of practice also resembles negative feedback control in automatic control theory. The gap between what we observe in the outside world and what we expect pushes us to continually correct our own cognition. Gradually, cognition converges, and in this way we realize the aim of understanding the world and changing it.

Seeking truth from facts, then, is necessarily a process of practice. When you make the effort, in everything you do, to analyze things and their changes from a multi-level perspective, when you insist on thinking for yourself instead of following blindly, the sustained discipline of this will itself bring about physiological changes in the neural networks of your brain. You will genuinely acquire the capacity to seek truth from facts. And when this capacity has been carved deeply into your mind, you will reach the point of knowing without deliberation. You will go beyond apparent paths and seek truth from facts without having to think about it. Perhaps this is what is meant by self-cultivation and seeking within. And perhaps this is what Wang Yangming[4] was pointing toward when he spoke of innate knowledge being complete in itself and of the unity of knowing and acting.

Dou Dou's books opened a window onto the world for me as well. They set me on a path of purposeful, systematic reading and deep reflection, and gave me the courage to step beyond my aca-

[4]Wang Yangming (1472–1529) was a Ming statesman and Neo-Confucian philosopher. His teachings of *liangzhi*, often rendered "innate knowing," and the unity of knowing and acting locate moral understanding in lived practice rather than in detached theory.

demic field onto the road of entrepreneurial practice.

When it comes to understanding things and the natural world, there is no authority more trustworthy than the natural sciences. Investigating things to extend knowledge means studying classical mechanics, electromagnetism, relativity, quantum mechanics, elementary particles, the origin of the universe, the riddle of life, and the emergence of consciousness. Reading widely across disciplines allows us to absorb the latest scientific understanding of the structure of things at every level. This is to obtain the fish. The Dao follows nature. The way science understands the world, especially the major breakthroughs of physics over the past century or more, also keeps renewing our methods of cognition. Therefore, beyond learning scientific knowledge, we should pay closer attention to the evolution of ideas and methodology in the natural sciences' effort to know the world. This is to seek the way to fish.

In the sixteenth century, Copernicus published *On the Revolutions of the Heavenly Spheres*, generally taken to mark the beginning of modern science. Galileo and Kepler carried his project forward, and through the contributions of Descartes, Hooke, Leibniz, Huygens, and others, Newton finally achieved a great synthesis in classical mechanics, laying the foundation for human understanding of material motion. In the mechanical, orderly world depicted by Newton, space and time were absolute, and the motion of every object in space and time seemed calculable in advance.

Extending the atomism proposed by Democritus and the ancient Greeks, Boyle, Lavoisier, and Dalton gradually established a scien-

tific atomic theory, which by the early nineteenth century had laid the basis for understanding the composition of matter. In the objective world described by scientific atomism, atoms were the building blocks of matter. All substances were simply atoms stacked together. This view carried a conviction. Once you had a clear picture of the basic units, you could understand everything.

Newtonian mechanics and scientific atomism together embodied reductionism, a central methodology through which modern science came to understand the world. Bacon's *Novum Organum* and Descartes' *Discourse on Method* both set out reductionist reasoning systematically. In the *Discourse*, for example, Descartes formulated several principles of rational analysis in scientific inquiry: proceed gradually from the simple to the complex; carry out a complete analysis that leaves nothing out; and divide difficult problems into as many simple and necessary parts as possible. These principles exemplify what has been called the analytic-reductive principle that marks the Western intellectual tradition.

To study the real world through analytic reduction is to isolate a thing from its complex relations and build a simplified model of a particular level, thereby revealing the laws of that level quickly and clearly. Reductionism was indeed highly effective in the scientific study of material existence and macroscopic motion through atomism and classical mechanics. Descartes' program in *Principles of Philosophy* is often summed up in the words, "Give me extension and motion, and I will construct the universe." The French mathematician Pierre-Simon Laplace went further, imagining an omniscient "Laplace's demon" that, if it knew the exact position and momentum of every atom in the universe, could calculate the

state of the universe at any moment in the past or future. This conception was almost the clearest and most thorough expression of reductionism.

From the late eighteenth century through the nineteenth, the Industrial Revolution drew science and technology into close partnership. Classical mechanics, thermodynamics, optics, chemistry, electricity, and magnetism all made striking advances, yet the mainstream of physics remained dominated by reductionism and mechanism. Still, as human observation pushed into more and more scales and absorbed the findings of astronomy, biology, and the life sciences, more and more scientists became inclined to see the world as a process of natural evolution. They no longer believed so firmly that the stars were eternal and unchanging, that species were fixed, that one form of motion could not transform into another.

In astronomy, Kant proposed the nebular hypothesis, arguing that the universe needed no god to deliver a "first impulse," but instead evolved spontaneously from chaos toward order in a process of continual motion and development. He wrote: "I assume that the matter of the entire universe was initially in a state of universal dispersion, producing complete chaos. Under the given law of attraction, I saw matter take shape; I also saw how repulsive force altered the motion of bodies. Without recourse to any fiction, and simply by following the established laws of motion, one can foresee an entire ordered system arising naturally."

In physics, Mayer, Joule, Helmholtz, Grove, Thomson, and others arrived at the law of conservation of energy from different directions, revealing both the inseparability of energy from material motion and its conservation. Things are forever changing: the mo-

tion of matter is eternal and spontaneous rather than created, and different forms of motion can be transformed into one another. This reveals a unity within the diversity of material motion.

In chemistry, Mendeleev's periodic table revealed the inner connections and commonalities among the elements, showing that at the level of chemical elements, the composition of matter is unified. Wöhler's synthesis of urea shattered the vitalist doctrine of organic compounds and demonstrated that no natural boundary separates the organic from the inorganic world.

In biology, Schleiden and Schwann used cell theory to reveal the basic structure of living things, showing that the difference between animals and plants is relative and that different forms of life possess an underlying unity. Darwin's theory of evolution went further: species are internally connected, and complex organic life was not divinely created but developed from the inorganic world. From inorganic matter to organic macromolecules, from single-celled to multicellular organisms, and from lower to higher forms of life, evolution has been a continuous movement from the simple to the complex and from the lower to the higher.

Most people's ideas about material motion and basic composition come from pre-twentieth-century physics, and for this reason they unconsciously accept a mechanical, reductionist worldview. And yet the world we perceive every day is one of color and sound and scent, full of life, everywhere displaying harmony and diversity. So our hearts also readily accept the self-organizing, evolutionary, emergentist worldview described by astronomy, biology, and the life sciences. The natural sciences, then, keep handing us new knowledge, and at the same time they present us with two dif-

ferent underlying logics of cognition.

The twentieth century brought even more brilliant achievements. Science and technology not only drove social development forward with great force. They also changed how human beings understood the world at a deeper level, overturning many old convictions in the process.

In 1905, Einstein formulated the special theory of relativity, doing away with the ether hypothesis and, more importantly, revealing the inseparability of matter and motion, the unity of space and time, and the connection between energy and mass, $E = mc^2$. In 1915, he went further with the general theory of relativity, once again transforming our view of space and time and revealing a still deeper inseparability. Matter tells spacetime how to curve. Spacetime tells matter how to move.

In 1900, Planck introduced the quantum concept while studying blackbody radiation. Einstein followed with the light quantum hypothesis. Bohr applied quantization to atomic structure. De Broglie proposed the matter wave hypothesis, laying the groundwork for unifying wave and particle descriptions. By the 1920s and 1930s, Heisenberg, Schrödinger, Dirac, Pauli, and others had together built and refined quantum mechanics and quantum field theory.

Relativity and quantum mechanics are the two revolutionary achievements of modern physics. Relativity, at the macroscopic level, overturned Newton's absolute space and time, binding spacetime tightly to matter and its motion. Quantum mechanics, at the microscopic level, describes the structure, motion, and laws of change in the microworld, and it does so in a way that places probability

and randomness at the very foundation of things, thereby overturning the mechanical determinism of early modern science.

At the macroscopic scale, between 1922 and 1924 Alexander Friedmann found new solutions to Einstein's equations, showing that within relativity the universe could not remain static. In 1927 Georges Lemaître proposed that the large-scale universe might have begun as a primeval atom and expanded into its present form; in 1948 George Gamow developed the Big Bang model. During the 1920s, Vesto Slipher and Edwin Hubble observed galactic redshifts and established the relation between recession velocity and distance; in the 1960s, Arno Penzias and Robert Wilson detected the cosmic microwave background. These observations closely matched the Big Bang theory and compelled scientists to accept that the universe may indeed have had a beginning, allowing the evolutionary view of the cosmos to prevail. Around the same time, plate tectonics, proposed in 1968 and supported by multiple lines of evidence, showed that Earth's geology, too, is in continual evolution.

At the microscopic scale, a series of physical experiments with X-rays, radioactivity, electrons, radium, and more pushed human understanding of matter deeper than the atom. The phenomenon of elemental transmutation made people realize that the atom, taken to be the brick of matter, was not only divisible but itself mutable. The Standard Model of quantum field theory went further, predicting that atoms could be broken down into still smaller elementary particles. From the discovery of the positron in 1932 to that of the Higgs boson in 2012, these particles were confirmed one after another.

In the Standard Model, elementary particles are understood, in essence, as quanta of fields, ripples in fields. They, too, are caught in an endless round of coming into being and passing away. At this point, even the new bricks of matter were no longer unchanging entities. This was yet another blow to reductionist thinking. The experimental physicists Alain Aspect, John Clauser, and Anton Zeilinger, through a series of precise experiments for which they received the 2022 Nobel Prize in Physics, confirmed that quantum entanglement violates Bell's inequality, thereby refuting the hypothesis of local realism. The nonlocality of quantum entanglement was verified, and the uncertainty of microscopic particles was shown to be an intrinsic property, not a disturbance introduced by measurement.

At the biological and life scale, Watson and Crick proposed the double helix model of DNA in 1953, clearly revealing the structure of the DNA molecule and the mechanism of heredity. It stands as one of the greatest biological discoveries of the twentieth century. Later, biologists gradually worked out the operating mechanisms of DNA, RNA, ATP, and other molecules, providing new clues and evidence for how life spontaneously generates order from disorder and how the order of life is transmitted and evolves.

On the question of soul and consciousness, ever since Santiago Ramón y Cajal, the father of neuroscience, established the neuron doctrine, and especially through the rapid advances of neurobiology in the second half of the twentieth century, scientists have grown increasingly confident. The core actors in the brain are neurons, the connections between them, the synapses, and the various neurotransmitters that operate within neural networks. There is no soul or consciousness entity existing independently of matter.

These discoveries brought new insight into perception, learning, memory, emotion, and action. They also strengthened the view that the soul and consciousness, long regarded as mysterious and unfathomable, are in essence emergent results of the complex connections and activity of vast numbers of neurons.

All things are in perpetual change: celestial bodies continually evolve, living beings are products of evolution, life has its own self-organizing process of growth, and consciousness emerges from networks of neurons. If the study of matter's basic composition and motion answers "Who am I?", then the study of the direction in which the universe and all things evolve asks, "Where did I come from?" and "Where am I going?"

As early as 1865, Clausius formulated the Second Law of Thermodynamics in terms of entropy. This is the famous principle of entropy increase. The Second Law reveals the directional character of energy transfer, and the principle of entropy increase goes further, revealing the irreversibility of natural processes. An isolated system always moves toward greater disorder, toward higher entropy. In the thermodynamic picture, the macroscopic motion of a system made of many particles is irreversible. As the Fourier heat equation makes plain, time-reversal symmetry does not hold.

In 1872, Boltzmann proposed his H-theorem, studying the evolution of many-particle systems from a microscopic standpoint and offering a microscopic statistical explanation for macroscopic irreversibility. It showed that the macroscopic evolution of a thermodynamic system made of many particles is likewise irreversible.

Extended to the cosmic scale, the entropy increase principle led Kelvin and Clausius to propose the heat death of the universe. The

universe as a whole, they argued, is also evolving toward a state of maximum entropy, eventually reaching a disordered condition of heat death. The idea caused an uproar when it was first proposed and set off a protracted debate across science and philosophy, a debate that remains unsettled even today. Black hole entropy and the holographic principle have since revised the picture somewhat, offering an explanation that moves from the despair of heat death toward an eternity of evolution.

Whether or not the universe will ultimately arrive at the limiting case of complete disorder, the trend of entropy increase certainly points, here and now, toward a direction of non-existence, of meaninglessness. Does this mean that all the ordered existence in the present universe, along with the entire development of human civilization, is meaningless?

Exactly the opposite. Meaning can only be defined against meaninglessness. Against the cosmic backdrop of entropy increase, moving toward non-existence and meaninglessness, everything that resists this entropic tendency, everything that resists being annihilated and voided of meaning, is thereby endowed with the meaning of its existence.

Darwin's account of evolution, depicting life progressing from the lower to the higher, from the simple to the complex, and the larger picture of an ever more civilized human society, both point in this direction. The meaning and direction of life and civilization lie in fighting entropy, in creating and sustaining order.

In the 1930s, Bertalanffy, the founder of General System Theory, advanced the insight that living organisms are open systems, thereby explaining the phenomenon of life without recourse to mys-

terious vitalism. Schrödinger, in his famous book What Is Life?, sounded the cry that life feeds on negative entropy.

By the 1960s and 1970s, a series of systems theories, led by Prigogine's dissipative structure theory, elaborated in detail how open systems, living organisms foremost among them, draw negative entropy from their environment, thereby enabling themselves to resist entropy increase and move toward greater order.

Taken as a whole, the natural sciences from the twentieth century to the present have made major advances at both extremes, the very large and the very small. At the macroscopic end, our picture of the material world now extends across the observable universe, roughly 93 billion light-years in diameter, or about 10^{26} meters, and back through 13.8 billion years of cosmic history, about 10^{17} seconds. At the microscopic end, it reaches below 10^{-18} meters, to the scale of elementary particles such as quarks and leptons, and to instants on the order of 10^{-23} seconds.

As science advances toward both the very large and the very small, we seek deterministic laws within the randomness found at every scale, and specialized disciplines multiply. Scientists have also gained the opportunity to connect knowledge across scales and form a synthetic understanding. What we now observe is a universe in continuous evolution at the largest scale, the smallest scale, and every scale between: a self-organizing, ever-evolving cosmos. Scientific inquiry therefore requires both analysis, or reduction, and synthesis, or a holistic and emergent perspective. In practice, the two underlying cognitive logics of reductionism and emergentism are gradually drawing together.

As Engels summed it up in *Dialectics of Nature*, "all rigidity

was dissolved, all fixity dissipated, all particularity that had been regarded as eternal became transient, the whole of nature shown as moving in eternal flux and cyclical course." Nature "has its existence in eternal coming into being and passing away."

The science of Descartes' time, of Newton's time, of Einstein's time, and of the quantum era each explored and explained nature at different scales, and the cognitive methods on which they relied and which they embodied differed accordingly. Nature can only show us how it works through successive approximations. Science can only reveal the truth of the world at different scales through successive approximations. The observational evidence we have at any given moment is always imperfect and incomplete, and so even the most advanced theories are often only approximations, not necessarily a final explanation of nature. This point was made by the 2019 Nobel laureate in physics, James Peebles.

We may get the feeling that scientific laws are always being replaced. But this does not justify the conclusion that science is unreliable. On the contrary, it is the correct way for humanity to understand the world, layer by layer.

For most of the problems ordinary people see and feel in daily life and work, modern science already has quite complete scientific models at the level of knowledge. To try to explain natural phenomena solely through life experience or the metaphysical cognition of religion and philosophy, to try to challenge the edifice of modern science, is like an ant trying to shake a great tree.

Theories such as relativity and quantum mechanics extend far beyond the reality of everyday experience. The laws at the scales they study cannot be guessed from ordinary intuition; to under-

stand them, we must adjust our habitual ways of thinking. As Frank Wilczek, the 2004 Nobel laureate in physics, observed, "If you want to be a serious student of reality or of mind you really should know quantum mechanics. To me, a philosopher who doesn't know quantum mechanics is like a swimmer with his or her hands tied behind their back."

Scientists, perhaps, are always inclined to stay busy working on new theories, and they can become exceedingly cautious, even timid, when it comes to drawing methodological conclusions above the level of disciplinary knowledge. But this does not justify the conclusion that science is not ultimate or not fundamental enough. On the contrary, the refusal to jump to universal conclusions before adequate verification is precisely why science is worthy of our trust.

In recent decades, physicists have been pursuing deep research on symmetry and symmetry breaking. This line of work is moving beyond the boundaries of specific subdisciplines and achieving a profound dialectical synthesis at the level of cognitive methodology. It beautifully bridges the traditional divides between reductionism and emergentism, determinacy and randomness, order and evolution, painting for us a unified picture for understanding the world. Nature is what it is, and it is forever becoming what it is not.

Science has shown us that all things exist as ordered structures at many levels, giving us an opportunity to reconcile relation and substance in our understanding of existence, and matter and consciousness in our understanding of reality. Disorder at the lowest level drives change in these layered structures: the lower the level,

the greater the randomness; the higher the level, the greater the determinacy. This allows us to bring macroscopic determinacy and microscopic randomness into a unified view. Different things, and different levels of the same thing, have different laws, which helps us reconcile the simple with the complex and the abstract with the concrete. At every level, entropy-increasing tendencies contend with tendencies that resist entropy increase. Entropy increase creates randomness, while randomness provides the first impetus for order. In this way, we can understand both the unity of opposing tendencies and the unity of being and non-being.

In the end, if we do not want to be the turkey on the farm, then verified scientific knowledge is the best labeled sample for training the brain's cognitive model. Deep thinking built on top of scientific knowledge at different scales is the shortest path to accelerating the formation of a cognitive framework.

Chinese traditional culture has always emphasized that the Dao follows what is naturally so. It teaches us to sense the laws of change in all things by watching the natural world, and it holds that human conduct and social activity should emulate and follow nature.

The I Ching, or Book of Changes, is revered as the first among the classics.[5] Through abstract elements such as the yin and yang

[5]The *I Ching*, or *Yijing*, is an ancient Chinese divination classic that became a major philosophical text. The *Xici*, commonly called the "Appended Statements," is one of its canonical commentaries.

lines, it reveals the laws of change running through nature, human life, and society. Thus, as the Xici commentary states, "In the Changes there is the Supreme Polarity, which gives rise to the Two Modes. The Two Modes give rise to the Four Images, and the Four Images give rise to the Eight Trigrams." From the undifferentiated chaos of the Supreme Polarity, Two Modes, Four Images, and Eight Trigrams unfold, describing a process in which all things spontaneously arise from disorder toward order. The trigrams are built from yin and yang lines, which are interrelated and mutually constraining, symbolizing the fundamental natural phenomena of heaven, earth, thunder, wind, water, fire, mountain, and marsh. The sixty-four hexagrams form a complete system of their own, a multi-level, multi-element structural model for describing the laws by which complex things evolve.

"The alternation of yin and yang is called the Dao"; "hardness and softness push against each other and give rise to change"; "the way of Heaven endures without ceasing." In the Eight Trigrams, changes in the lines at the lowest level provide the original impetus for all change. Through their interrelations, those changes drive change at every level, within and without, describing the cycle of the seasons and the ceaseless generation of all things. Yin and yang are treated as the two most fundamental aspects of the universe, opposed yet unified, through which the change and balance of all things can be understood. The later doctrines of yin-yang and the Five Phases, and of the Supreme Ultimate and the Eight Trigrams, all inherit this central insight: trust in the natural principles of the world and in unceasing change and generation.

Dependent origination is the fundamental doctrine realized

by Shakyamuni Buddha. Dependent Origination and Emptiness means that things arise and perish through the gathering and dispersing of causes and conditions. They have no independent self-nature. All things are provisional existences born of the confluence of causes. Their nature is Emptiness. Yet without the emptiness of self-nature, nothing could come into being. This is true emptiness giving rise to wondrous existence. As the saying goes, when conditions meet, things are born. When conditions scatter, they perish. Dependent origination likewise recognizes the universal interdependence of things and stresses that all things are caught in perpetual change and impermanence. From this it follows that human understanding of nature and the cosmos must be limited and must belong to the present moment.

The Dao expounded by Laozi and Zhuangzi[6] stands as the most fundamental category of ancient Chinese philosophy. The Daodejing says, "Human beings follow the Earth. The Earth follows Heaven. Heaven follows the Dao. The Dao follows what is naturally so." The Dao is the law governing the movement and change of Heaven, Earth, human beings, and the ten thousand things. The Dao gives birth to all things, and all things in turn give birth to the Dao. The Dao is the totality of all phenomena and principles in the natural world, the law of interconnection and interaction running through all things.

"The Dao gives birth to one; one gives birth to two; two gives birth to three; three gives birth to the ten thousand things." "The

[6]Dao literally means a way or path, but in the *Daodejing* and *Zhuangzi* it also names the generative course or ordering process of the world. Laozi and Zhuangzi are both traditional author names and titles of the texts associated with them.

Way of Heaven moves without accumulation, and so all things come into being." It is "independent and unchanging, moving in a cycle without exhaustion, and may be regarded as the mother of all under Heaven." The Dao inheres in the development and change of all things and governs their spontaneous generation and transformation. It is both the source from which all things arise and the universal law their evolution must follow.

Whether in Confucianism, Daoism, Buddhism, or Chan,[7] ancient Chinese philosophical thought more or less contains a deeply intuitive systems thinking and dialectical thinking, and at the metaphysical level it resonates profoundly with the underlying cognitive logic of the natural sciences and their evolutionary outlook.

The physicist Fritjof Capra was a great admirer of Chinese Dao culture. The covers and title pages of his books *The Tao of Physics* and *The Turning Point* both feature the taiji and bagua symbols.[8] In his writing, he observed that ancient Chinese culture possessed an insight of enormous importance, namely the recognition of motion as a fundamental aspect of the universe. As he wrote, "When we talk about the Taoist concept of change, it is important to realize that this change is not seen as occurring as a consequence of some force, but rather as a tendency which is innate in all things and situations." The universe is caught in never-ending movement and transformation, and this continuous cosmic process is what we call the Dao. This passage is quoted in Zeng Guoping's *The Self-*

[7] Chan is the Chinese Buddhist tradition that later developed in Japan as Zen; its name derives from the Sanskrit *dhyana*, meditation.

[8] The taiji symbol depicts the complementary movement of yin and yang; the bagua are the eight three-line trigrams of the *Book of Changes*. Together they are among the most recognizable visual schemata of traditional Chinese cosmology.

Organizing View of Nature.

In ancient Greece, Democritus and Aristotle lived at roughly the same time as Confucius, Shakyamuni, and Laozi. Democritus' atomism affirmed that systems are composed of elements, holding that all things evolved from atoms. Aristotle's reflections on the relationship between the whole and its parts were deeply admired by the modern systems thinker Bertalanffy, who wrote: "One formulation of this cosmic order was the Aristotelian world view with its holistic and teleological notions. Aristotle's statement, 'The whole is more than the sum of its parts,' is a definition of the basic system problem which is still valid."

German classical philosophy represents the culmination of over two thousand years of Western philosophical development, with Kant and Hegel at the turn of the nineteenth century as its most outstanding representatives. Kant was both the founder of German classical philosophy and a natural scientist. His critical epistemology valued analysis and synthesis in equal measure and stressed that knowledge should be understood as an ordered, layered, unified whole composed of elements. At the same time, his nebular hypothesis inaugurated a self-organizing, evolutionary view of the cosmos.

Hegel's dialectics mark the summit of German classical philosophy. At its core lies the idea of understanding the entire natural, historical, and spiritual world as a single process of continuous movement, change, transformation, and development, and of trying to uncover the inner connections within this process, as Engels noted in Anti-Dühring. What Hegel's philosophical system describes is a process in which the subject posits the object and then transcends

it, a developmental process moving from shallow to deep, from affirmation to negation and then to the negation of the negation.

Marx and Engels critically inherited the finest achievements of German classical philosophy and created dialectical materialism and the materialist dialectics of nature. Its basic propositions hold that things are universally interconnected: all things are, in essence, ordered structures at many levels, with lower-level structures interrelating to form higher-level ones. Things are always changing, as quantitative changes at the microscopic level accumulate into qualitative changes at the macroscopic level. Contradictions are universal and provide the fundamental impetus of development. Their opposing sides are both in conflict and in unity: conflict drives change and development, while unity preserves relative stability. Through the negation of the negation, development proceeds in a rising spiral through affirmation, negation, and renewed negation, uniting forward movement with twists and turns.

Wittgenstein wrote, "The limits of my language mean the limits of my world." Measurement, probability, field, quantum, quantum entanglement, wave function, singularity, entropy: concepts that once did not exist or could not be put into words have gradually acquired clear definitions, expanding the boundaries of our understanding. Concepts that our present language cannot yet express reveal only the present vagueness and limits of our knowledge of nature. What tradition calls the point at which words fail is, more often, a limitation imposed by a particular historical moment. Whether as individuals or as a society, we should be willing to admit that what we cannot express is something we have not yet fully understood.

In the twenty-first century, complexity science has begun to synthesize different forms of knowledge and to explore a "theory of all things." Leading journals such as *Nature* and *Science* have published many studies of the first principles of evolution from the perspectives of multiscale emergence, phase transitions, symmetry breaking, assembly theory, and more. From chaos to galaxies, from inorganic matter to life, and from the brain to human society, scientists are using the speakable language of science to explore how the universe brings something forth from nothing, how things arise through dependent origination and are empty of independent nature, and how the Dao follows what is naturally so, how things "thus come."

The natural sciences, the Confucian, Buddhist, and Daoist traditions of the East, Western classical philosophy, and Marxist philosophy may view the world from different angles, yet at the level of cognitive method and underlying logic they share much common ground. Since all are touching the same elephant, their essential insights should surely converge.

When building a personal cognitive model, we should actively search for the commonalities in how different disciplines and fields approach the understanding of things. We should seek to grasp what different perspectives share and to discern what has remained constant across different historical stages of humanity's understanding of the world. To place one's complete faith in the knowledge and experience of a single domain, or to mechanically cobble together something like "quantum Buddhism," is very likely to fall into cognitive dependence and intellectual conformity. We should instead practice critical inheritance, building on analysis and reflection to

carry out broad comparison and synthesis. Only then can we hope, little by little, to grasp the whole elephant.

The idea of symmetry in physics suggests that greater symmetry brings greater universality: the more phenomena a theory can explain, the closer it comes to underlying logic. Let West converse with East and antiquity illuminate the present. At the deepest level, our methods of understanding should be inclusive enough to hold them together.

Marx once wrote, "The philosophers have only interpreted the world, in various ways; the point, however, is to change it." Wang Yangming's teaching of the extension of innate knowledge holds that such knowledge resides within the changes taking place at every level of things. Only by plunging into practice and making full contact with the multiple levels of things can one sense the trends of change at those levels and thereby discern the principles of heaven from within one's own heart. The unity of knowing and acting insists that the Dao is not a matter of empty talk. It must be realized through personal practice, by interacting with the present trends of things through action, feeling and following those trends while at the same time influencing them and creating value.

If you do not know how to practice, the Selected Works of Mao Zedong[9] is the finest textbook of method. Mao led the world's greatest political party through the greatest practice in world history, and in the process he turned the analysis and solution of real problems into one real teaching case after another. He takes the reader by the

[9]The *Selected Works of Mao Zedong* collects Mao's political and military writings and is widely read in China not only as political history but also for its discussions of investigation, contradiction, and practice.

hand and shows them how to practice, how to seek truth from facts.

Entrepreneurship is the process of creating the ordered organization of a company from nothing. Ren Zhengfei[10] is unquestionably a representative practitioner in this regard. By creating a corporate system from nothing to something and from lower to higher levels, entrepreneurs have a greater opportunity to develop systems thinking across multiple dimensions, perspectives, and time horizons, and to test their understanding through entrepreneurial practice by seeking truth from facts.

To understand the world is to practice in the world. To practice by following the Dao of nature is to seek truth from facts. The closing Prajnaparamita mantra of the Buddhist Heart Sutra[11] says, gate gate, paragate, parasamgate, bodhi svaha. We may render it freely as follows: practice, practice. Walk through the many paths of the world. Become one who practices, and arrive at the other shore of awakened emptiness.

The path is the finger. The truth is the moon. To go beyond the apparent path is to seek nothing but the real.

The modern linguist Noam Chomsky has argued that, about 80,000 years ago, a subtle change in the brain's neuronal networks

[10]Ren Zhengfei founded Huawei in Shenzhen in 1987. Within Chinese management discourse, he is especially associated with long-term investment, organizational self-criticism, and Huawei's employee shareholding system.

[11]The *Heart Sutra* is one of the best-known Mahayana Buddhist texts. Its closing Sanskrit mantra is traditionally associated with crossing beyond delusion toward awakening; the paraphrase that follows is the author's interpretive rendering, not a literal translation.

gave our human ancestors the capacity for language and set off a great expansion of human cognition.

By arranging and recombining a finite set of phonemes and words, we were able to construct phrases, sentences, paragraphs, structures of ever-increasing layered complexity. These could generate a nearly infinite variety of expressions, capable of describing everything in the outside world and even conveying the wildest flights of imagination, unmoored from any reality. The birth of spoken and written language allowed human beings to share and verify their understanding of the outside world with one another. This catalyzed the formation of self-awareness, of complex emotions and thought. From that point on, cognition could transcend the biases and brief lifespan of the individual and accumulate into collective wisdom. Deeper bonds formed between people, and these in turn drove the emergence and development of layered forms of civilization, from moral and ethical codes to social structures and beyond.

I am drawn to popular science writing on neurobiology and brain science, and I am deeply convinced of a few things. Only by understanding in depth how the brain works can we see clearly how the order of the outside world maps onto the order of neural connections within it. Only by understanding how the brain learns from and comes to know the natural world can we train it with greater purpose and make it more intelligent. Only by understanding how neurotransmitters shape our emotions and desires through their influence on synaptic connections can we learn to face those emotions and desires squarely. And only by using language to trace the outline of our own cognitive framework can we reflect on it, evaluate it, and keep supplementing, correcting, and refining it.

Scientific research today is working to uncover the mysteries of consciousness and intelligence through high-resolution brain sections, real-time brain imaging, EEG-based decoding of consciousness, analysis of the biological mechanisms of neurons, neural-network modeling, and other methods. This book attempts a different route: by observing the self systematically, it begins an inward examination of the brain. By describing a personal cognitive framework in rational terms, it seeks to show how an individual comes to understand the surrounding world, forms knowledge, and uses judgments about external trends to guide practice. In these pages, you may glimpse the world-picture built by another mind and come to understand its underlying logic for knowing the world. Everything in the book is an interpretation and connection of existing scientific findings; wherever it conflicts with science, the error necessarily lies in my own understanding and cognitive bias.

I am, by training and temperament, an engineer. I have always been more comfortable with reason than with feeling, and writing does not come easily to me. Even after many rounds of revision, the manuscript may still be lacking in the kind of ease and fluency that makes for pleasant reading.

The book draws on concepts from a number of specialist fields. I have done my best to explain them in plain language, but some readers may still find them challenging. Still, I am firmly convinced that these passages are essential. You may find it helpful to read this book as a work of popular science in its own right. When you encounter a difficult concept, pausing to acquire the relevant background knowledge may help you continue without losing your way.

Life is a highly ordered form of existence. Our experience and understanding of the world are continually transformed into ordered connections within the neuronal networks of each brain. Yet life lasts only an instant and will eventually disappear; the order of body and spirit alike will disperse. Only by transforming that order into other forms of order can the meaning of its former existence continue. This book is a tribute to life. I also hope that through it, my readers and I may find cognitive resonance, and from that resonance, love.

The Author
in the ancient city of Xi'an
June 2024

Chapter 2

All Things Have Their Order

Everything in the world must take its proper place within a certain order.

Władysław Reymont

Look up at the night sky. The Milky Way twists across it, stars scattered like dust. Look down at the earth. Mountains and rivers carve across it, and everywhere, things grow. From the boundless cosmos to the finest speck of dust, from towering mountain ranges to creatures teeming with life, everything in the natural world displays a deep order of its own. Everything is in its place. Everything does its work.

Faced with a world so bewilderingly complex and yet so unmistakably ordered, we cannot help asking whether some common logic of composition lies behind it all. How do we push past the surface and glimpse the inner order that governs everything?

For thousands of years, the wisest among us have searched for the essence and hidden workings of the world. The achievements of modern science now let us see the universe and everything in it from many different scales, giving us a more or less complete picture.

The ancient Greek philosopher Democritus proposed that the atom was the smallest indivisible unit of all matter. His insight was profound, but without evidence it remained speculation for centuries. Then, in 1803, the British scientist John Dalton introduced the concept of atomic weight and defined the atom as the smallest unit that retains an element's chemical properties. This gave atomism a genuine scientific footing. A philosophical guess became an empirical science.

In the twentieth century, models of atomic structure gradually took shape, and scientists came to understand the atom as a nucleus surrounded by electrons. In 1913, using X-ray spectra, Henry Moseley showed that the more fundamental basis for ordering the elements was atomic number, or nuclear charge. He corrected several anomalies created by arranging the periodic table according to atomic weight and identified a number of still missing elements. It was later confirmed that atomic number is the number of protons in the nucleus. This discovery resolved the historical problems in Mendeleev's ordering by atomic weight, established the basis of the modern periodic table, and successfully predicted the ninety-two naturally occurring elements from hydrogen to uranium. Artificial synthesis later extended the number of known elements to 118, deepening our understanding of the composition of matter.

Modern physics has gone further. The atomic nucleus consists of protons and neutrons, while electrons occupy specific energy levels around it. For all the dazzling variety of the elements, each is an ordered arrangement of these three kinds of particles in particular numbers and structures. The number of protons determines which element an atom belongs to. In a neutral atom, the number

of electrons equals the number of protons, and the electron configuration largely determines its chemical properties; the number of neutrons may vary, giving rise to different isotopes of the same element. Through their interactions and different arrangements, these microscopic particles build the richly varied world of chemistry.

If we fix our gaze at the atomic scale, the atom really can be called the brick of matter. Everything in the world, however its outward form may change, is at its root an ordered structure of atoms arranged according to the laws of nature. The Earth's crust is built mainly from oxygen, silicon, aluminum, and iron. Living organisms, by contrast, are composed of a little over twenty elements, mostly oxygen, carbon, hydrogen, nitrogen, phosphorus, and sulfur.

The magic of atomic arrangement lies not only in the making of matter but in the vast differences of property it can produce. One source of difference is the identity of the element itself. Coal is mostly carbon atoms gathered together. It is soft and it burns. Iron is iron atoms packed tightly through metallic bonds. It is hard and conducts heat and electricity. The sharp contrast between them is a direct expression of the difference between a carbon atom and an iron atom.

Even more striking, a single kind of atom can produce substances of utterly different character depending on how the atoms are arranged in space. Carbon gives the classic example. In graphite, carbon atoms are arranged in layers that slide easily past one another, making it soft and electrically conductive. In diamond, the same carbon atoms are locked into a three-dimensional tetrahedral framework by powerful covalent bonds, producing a

crystal of extreme hardness and transparency that does not conduct electricity.

The same bricks, carbon atoms, put together in different ways, produce substances as different as soft gray-black graphite and glittering, unyielding diamond. This tells us something essential. The properties of matter depend not only on which atoms it is made of but, even more deeply, on how those atoms are ordered together.

Now push the lens deeper, to the most microscopic scale science can reach. Even the atom itself, taken as the brick of matter, turns out to have a deeper internal structure.

The Standard Model, the mainstream framework of modern physics, tells us that all matter is ultimately a complex ordered system built from two kinds of elementary particles, fermions and bosons, joined through the electromagnetic, weak, and strong interactions and through gravity.

Fermions, including quarks and leptons, serve as the building blocks of matter, the concrete units from which the material world we see is made. Bosons such as photons, gluons, and the other gauge bosons act like glue: they carry interactions between particles, allowing fermions to combine and arrange themselves according to definite rules and thus build more complex forms of order. The Higgs boson, by contrast, is a quantum excitation of the Higgs field, and the Higgs mechanism gives elementary particles mass.

The protons and neutrons in the atomic nucleus are examples of such microscopic combinations. A proton can be seen as two up quarks and one down quark, bound tightly together by gluons, with a total charge of +1. A neutron can be seen as two down quarks and one up quark, bound together by gluons, electrically neutral. The

electron that orbits the nucleus is itself a kind of fermion.

Even at the deepest level of material structure, then, what we find is a still more fundamental kind of order. The existence and properties of protons, neutrons, and entire atoms are rooted in specific kinds and numbers of elementary particles, arranged and combined with precision according to the deepest physical laws of the universe.

Now pull the lens back to the familiar macroscopic scale. Through the exquisite chemistry of bonding, atoms of different elements can be arranged and rearranged in a nearly infinite number of ways, producing the next key level of matter, the molecule.

Simple molecules are formed when a small number of atoms bind directly through chemical bonds. Examples include carbon dioxide, which sustains respiration; hydrogen, the lightest gas; water, the source of life; everyday salt; and the many silicates and oxides that form the framework of the Earth's crust, whether as molecules, ionic lattices, or network solids.

More complex molecules include polymers, built from hundreds or thousands of atoms, or from small molecular units called monomers linked into long chains by repeated covalent bonds. Plastics, elastic rubber, and textile fibers all belong here. Scientists have even designed molecules of intricate architecture, such as the C_{60} fullerene, a football-shaped structure built from sixty carbon atoms.

In living systems, molecular order reaches its highest expression. Biological macromolecules such as proteins, nucleic acids, and polysaccharides are the foundations of the cell, the basic unit

of life. The human body contains roughly one hundred thousand kinds of proteins with different functions. A single haploid human genome contains about three billion base pairs, distributed among the DNA molecules of twenty-three chromosomes. Together, these molecules contain tens of billions of atoms and rank among nature's most intricate forms of molecular order.

Above the level of atoms and molecules, further ordered arrangements, spatial packing, and specific assembly give rise to the macroscopic materials we can directly perceive and make use of on Earth. This includes pure substances such as elemental metals like gold, silver, and copper, nonmetals like diamond and graphite, noble gases, and a great many inorganic and organic compounds. It includes synthetic and natural materials built from polymers. It also includes complex mixtures such as petroleum, rock, and air. Several hundred thousand inorganic compounds are known, and the number of known organic compounds exceeds thirty million. The inventory is vast.

And it is not only a matter of sheer variety. A single substance can display strikingly different macroscopic forms depending on the conditions that alter its internal arrangement. Water is the classic example. Under the changing conditions on Earth, water molecules arrange and combine in different patterns through hydrogen bonding and other forces, taking the form of gas as water vapor, liquid as rivers, lakes, and oceans, and solid as glaciers and snow. Even within the solid form, an ice crystal, a snowflake, will vary in its molecular arrangement with the slightest difference in temperature and humidity during crystallization, yielding an endless variety of shapes, each one following a sixfold symmetry.

Every snowflake is unique.

The mountains and rivers and living creatures of the macroscopic world can thus all be seen as complex ordered structures, built from atoms and molecules through arrangements at higher levels under the governance of the four fundamental interactions. This higher order is what gives objects the properties we perceive in everyday life, such as size, shape, weight, and color. Countless instances of such material order, gathered and woven together, have produced this richly diverse and vividly alive planet.

According to Darwin's theory of evolution, life did not appear out of nowhere, but gradually arose through natural selection within matter's arrangements at many levels. Under the particular conditions of the primordial Earth, a handful of elements, including oxygen, carbon, hydrogen, and nitrogen, combined in different ways to form basic organic molecules such as amino acids and nucleotides. These molecules were the "letters" of life. Through extraordinarily complex and precise "spelling rules," or chemical pathways, they were combined, assembled, and edited into biological macromolecules such as proteins and nucleic acids. Through cooperation and self-organization, these macromolecules were eventually enclosed within lipid membranes to form the cell, life's most basic structural and functional unit.

With the cell as the common building block, life embarked on a path of differentiation and growing complexity. Through cell division, differentiation, and self-organization, single-celled life gradually gave rise to groups of cells with distinct functions. These groups in turn formed tissues, organs, and systems, eventually building every form of life from the tiniest bacterium to the most

complex plant and animal.

The number of living species on Earth today runs into the millions or tens of millions, and the range of forms and sizes is staggering. The smallest independent living things, certain bacteria, measure only about two to three hundred nanometers. The largest known single organism, a honey fungus in the American state of Oregon, spreads its underground mycelial network across nearly ten square kilometers. That is an area equal to some fourteen hundred standard football pitches.

The order of life does not stop at the individual. At a higher level, individual organisms form populations. Interconnected populations form communities. Communities and their nonliving environment interact to produce ecosystems, dynamic and balanced systems of material cycles and energy flows. This is a larger, interdependent ordered system at many levels.

In human society, this hierarchical ordered structure reaches an extreme of complexity. The individual is the basic unit of society. Individuals form families and enterprises, the cells of society. The gathering and interaction of families and enterprises produce communities and cities. Cities and nations are woven together through the bonds of politics, economics, and culture into the whole of human society, bound ever more tightly in an age of globalization.

From the individual to the family, the community, the city, the nation, and the global community, each level of social structure follows its own distinctive cultural norms and operating mechanisms, together sustaining the great order of human civilization. This, too, is a profound expression of the same underlying logic: “basic units arranged and combined in particular ways.”

Finally, let us extend our thinking to the cosmic scale. Under the pull of gravity, the diffuse gas and dust of interstellar space gather and collapse, passing through complex physical processes to be shaped at last into every kind of celestial body. Planets, stars ranging from red dwarfs to supergiants, neutron stars, black holes, quasars, all are, at their core, vast complex arrangements of matter brought together by gravity.

And the nearly countless stars, planets, nebulae, and interstellar material are not isolated. Under the dominant influence of gravity, they are further organized into a still higher level of magnificent structure, the galaxy. Our own Milky Way is a vast gravitationally bound system containing hundreds of billions of stars and immense clouds of gas and dust, with a diameter of roughly a hundred thousand light-years.

And the galaxy itself is not the endpoint of cosmic structure. On a still more majestic scale, dozens or hundreds of galaxies are bound together by their mutual gravity into galaxy groups and galaxy clusters. Hundreds or thousands of galaxy clusters can in turn gather into superclusters that stretch across tens or even hundreds of millions of light-years, forming the largest-scale filamentary web structures yet observed in the universe. These cosmic macrostructures are the highest-level forms of material order within the current reach of human cognition.

Bringing together these perspectives from the microscopic to the macroscopic, we can return to our opening question: What is the basic logic by which all things are composed? A clear and powerful answer now suggests itself: everything is an ordered structure, or system, at many levels, formed when smaller constituent units are

arranged and combined in particular ways.

This insight is both a scientific description of the objective world and a cognitive paradigm rich in wisdom.

The philosophical reflections of ancient Chinese sages already contained profound and vivid insights into order at many levels. Buddhism pictured the cosmos as nested "small-thousand worlds," "medium-thousand worlds," and "great-thousand worlds."[1] With the words "The Dao gives birth to one; one gives birth to two; two gives birth to three; three gives birth to the ten thousand things," the *Daodejing* described a creative and evolutionary vision in which all things differentiate and combine from a unified source, eventually forming endless levels and relations. Confucianism focused on the order of human society. Its progression from cultivating the self and regulating the family to governing the state and bringing peace to all under Heaven[2] clearly outlined a layered social structure and ethical order extending from individual cultivation through family harmony and statecraft to harmony throughout the world. Different as their language may be, these traditions point to the same insight: the world is composed of ascending levels of order.

Marxist dialectics of nature and modern systems theory set out this idea more systematically at the level of philosophy and scientific methodology. Building on the general systems theory

[1] In Buddhist cosmology, one thousand world-systems form a "small thousand-world"; one thousand of those form a "medium thousand-world"; and one thousand of those form a "great thousand-world." The nested scheme is conventionally summarized as a "three-thousand-great-thousand world-system."

[2] This sequence comes from the *Great Learning*, one of the Confucian Four Books: investigation and self-cultivation extend outward through family, government, and "all under Heaven."

founded by Ludwig von Bertalanffy, systems science understands every thing, or system, as an organic whole with specific functions, formed from several interacting and interdependent components. This perfectly captures the progressive character of arrangement and combination: elementary particles combine into atoms, atoms into molecules, molecules into matter, and matter into life, ecosystems, and even galaxies. Each level supplies the elements of the next, and through specific interactions those elements form a higher-level ordered whole.

To apply the perspective of layered order to everyday work and life has deep practical meaning. When facing something complex, actively breaking it down and analyzing its constituent units at different levels and the relations among them helps us see into its essence and reduce complexity to simplicity. And the reverse is also true. By consciously recombining or altering units and their relations at different levels, we can innovate and create, from designing new materials to proposing new solutions for society.

We can therefore understand the world in this way. All that exists is a multi-level ordered structure, presented under specific conditions through specific modes of arrangement and combination. From the dance of elementary particles to the turning of the galaxies, from the division of a cell to the rise and fall of civilizations, all of it confirms the same theme. All things have their order. This is both the underlying logic of cosmic existence and the key by which human beings can understand the world and create the future.

Chapter 3

Existence Is Order

Nonbeing names the beginning of Heaven and Earth; Being names the mother of the myriad things.

Laozi

We see with our eyes, hear with our ears, touch with our skin. Every sense keeps telling us the same thing. The outside world is real. It exists. More than that, when different people at different times and in different places reach the same judgment about the same thing, this shared confirmation becomes further evidence that the outside world is truly there.

In everyday life, when we say that something exists, we mean something quite straightforward. We mean a concrete object we can see, touch, and feel. To say that this house exists means, quite simply, that it occupies a piece of ground, that it has shape and size and color, that the eye can see it and the hand can touch it.

At the level of common sense, existence is intuitive and concrete. It needs no elaborate theories and no rigorous logic. Perhaps a better way to put it is that we accept, almost by instinct, that the things of the outside world are simply there.

But once we move into a scientific perspective, existence can no longer rest on feeling alone. Science demands another kind of confirmation: observations must be repeatable, experiments reproducible, and theories logically coherent. In this sense, existence depends on the correspondence between theoretical predictions and objective entities. What science seeks is an ever more accurate picture of the natural world.

How, then, do scientists proceed? They measure the properties of things: mass, extent, energy, and other quantities that can be repeatedly tested. On the basis of such measurements, they judge whether something exists and in what state it exists. These measurable properties are, in the physical world, a thing's identity.

In classical physics, for an object to exist means that it occupies a definite position in space, persists through time, and possesses properties such as mass, energy, and momentum that can be repeatedly measured and verified by experiment.

In the more advanced realm of quantum mechanics, our understanding of existence becomes more subtle and probabilistic. A microscopic particle, an electron, for example, is not fixed at a single point. It is described by a wave function, which represents the probability of the particle being at various positions in space. Before measurement, it could be anywhere, but exactly where cannot be said. Only when a measurement occurs does the wave function collapse into a definite position and present a particular appearance of existence in our cognition.

Philosophy asks deeper questions about existence than common sense or science. For over two thousand years, philosophers have kept returning to the same problem. What exactly counts as exist-

ing? And what does existence itself even mean?

The ancient Greek philosopher Parmenides was the first to assert that Being is and that non-being is unspeakable, taking existence itself as the sole, unchanging, and true foundation. This thought opened the Western metaphysical inquiry into the ontology of Being.

Later philosophers developed many different views along this line. Plato treated Forms as a reality higher than particular things, which were only imperfect manifestations of them. Aristotle held that a thing's essence, what it ought to be, preceded its actual existence: the idea or form of a table came before the particular table before us. Descartes anchored existence in the thinking self: I can doubt everything except the "I" that is doing the doubting. Leibniz went further, seeing the world as composed of countless spiritual monads that operate according to a harmonious order preestablished by God.

In the modern period, Kant drew a boundary around existence. We can only know things as they appear to us, as phenomena. What things are in themselves, what he called the thing-in-itself, we can never know. Hegel rejected this limit. He saw existence as a dynamic process of unfolding, and held that the real is rational because it is the result of Absolute Spirit gradually realizing itself.

In the twentieth century, Heidegger turned his gaze from abstract existence back to the concrete human being, proposing the concept of Dasein, being-there. He insisted that the meaning of existence must be grasped through the actual existence of human beings in the world. His successor Sartre then gave the famous slogan "existence precedes essence." For human beings, the claim says

that we are first thrown into the world without clear preparation, and only afterward, through our own free choices and actions, do we gradually shape ourselves into the people we become.

Marxist philosophy, grounded in dialectical materialism, proposed the principle of the material unity of the world, establishing the objectivity of existence as the cornerstone of philosophy. It holds that objective reality is the shared property of matter. Material existence does not depend on human consciousness. Whether perceived or not, it exists independently and can ultimately be reflected by human consciousness.

As Lenin put it in his famous definition, "Matter is a philosophical category denoting the objective reality which is given to man by his sensations, and which is copied, photographed and reflected by our sensations, while existing independently of them." This materialist position clearly asserts that material existence is primary and consciousness is secondary, and that the unity of the world lies in its materiality. It provides a solid objective foundation for understanding all existence, from natural phenomena to social history.

We cannot help but press the question further. What exactly is this common property called existence, or objective reality, that is said to be shared by all matter? By what inner and universal determination can material existence be reflected by human consciousness and constitute the objective basis of all our knowledge and practice?

If we content ourselves with declaring that real things are simply themselves, or with the simple assertion that the appearance of a thing is its essence, have we not, without meaning to, given up too early on the most fundamental mission of philosophy and science,

which is to push past the surface and search for the deepest common principle and order governing all things?

The truth is that only by identifying that common characteristic of matter can we find the key to the ultimate question of how all things exist. Only by clarifying this foundational characteristic can we rigorously explain how the infinite diversity of things, from elementary particles to living organisms, from mountains and rivers to social structures, can all manifest on the basis of the same fundamental reality. And only by clearly defining what existence means can we accurately compare the fundamental similarities and differences between material existence and conscious existence, distinguish which is fundamental and which is derivative, and arrive at a well-grounded judgment on the basic question of philosophy.

In the first century, the Greek philosopher Plutarch posed a profound philosophical paradox, the Ship of Theseus. Suppose a ship called Theseus has its timbers gradually replaced as they rot. When only a few planks have been changed, most people would intuitively say that the ship is still the same ship. But if the process continues until every plank and every mast has been replaced, until nothing of the original material remains, can this ship, built now of entirely new material, still be considered the original Ship of Theseus?

Faced with this question, philosophers divide into two camps. Those who answer no hold, in substance, that the essence and identity of a thing are ultimately determined by the specific material of which it is made. Once all the original material is gone, even if the structure and outward form remain, the thing has lost its essence and is no longer the original ship.

Those who answer yes hold, in substance, that the existence and identity of a thing are not entirely bound to its material substance. What matters more is the specific pattern of arrangement and relation formed by that material, an inner, constant structural order.

It is this order, the ship's form and structure, its functional relations, and even its continuity in history and memory, that defines the Ship of Theseus. Its material may be replaced, but as long as the ordered pattern that makes it a ship is preserved and continues, its essence as that particular existence endures.

The cells of the human body are in constant metabolic turnover. By scientific estimate, epidermal cells are renewed roughly every twenty-eight to forty-five days. Within a year, a large proportion of the atoms in the body are replaced. So after a year, are we still ourselves?

The answer is obvious. From childhood to adulthood, our bodies have undergone enormous change and most of our cells have been renewed, yet we are all absolutely certain that I am still me and you are still you.

Why? On what basis do we determine that we are still ourselves? The answer, for a complex system like the human body, is this. Despite the many changes that come with growth, our basic characteristics, as we perceive them, have not altered. Our genetic sequence, our hereditary information, has not undergone significant change. Relative to other people, the particular layered structure of order formed by our specific arrangement and relation has remained essentially unchanged. That is why I am still me and you are still you.

Wang Yangming once famously investigated bamboo in his ef-

fort to investigate things and exhaust their principles.[1] Let us try the same exercise.

When we speak of a living bamboo plant in everyday life, we are usually referring to its macroscopic arrangement of roots, stems, and leaves. It exists as a kind of ordered structure. But if you grind the bamboo to powder, the plant cells that composed it still exist. The higher-level arrangement of those cells into roots, stems, and leaves, however, is gone. That ordered structure has vanished. We can therefore say that, in the eyes of a botanist, the bamboo with its arrangement of roots, stems, and leaves no longer exists.

Now switch to the view of a cell biologist, who focuses on the level of the plant cell. Whether before or after grinding, as long as the ordered arrangement of the plant cell itself remains intact, that order is still there. The botanist's bamboo is gone, while the cell biologist's bamboo still exists.

When chemical reactions cause the bamboo to rot and decompose until even the plant cells are gone, the bamboo's ordered structure, as defined by the cell biologist, will also disappear. At this point, even though every molecule and every atom that once made up the bamboo still exists somewhere, no bamboo of any kind exists any longer, because no ordered arrangement at any scale remains.

In every case, whether a particular "bamboo" exists depends not only on the basic material that composes it, but also on the ordered structure that material forms at a particular scale. If we

[1]The "investigation of bamboo" is a famous episode from Wang Yangming's youth. He and a friend tried to discover principle by contemplating bamboo; their failure helped turn Wang away from seeking moral truth solely in external things and toward the mind's innate knowing.

look only at the most basic constituent materials, atoms, electrons, and elementary particles, then the pile of material that makes up a bamboo plant is not fundamentally different from the pile that makes up a human being.

So when we say that things are layered structures of order, and when we adopt this as our way of looking at things, does this mean that the essence of a thing is not simply the pile of units that make it up, but the arrangement and relation of those units, a certain kind of order?

Consider a carbon atom. When we speak of it, we are not merely pointing to the basic material of its protons, neutrons, and electrons. Even more, we are pointing to the specific way in which that material is arranged and combined. A carbon atom and an iron atom are both made of the same basic stuff, protons, neutrons, and electrons. What distinguishes them, crucially, is the difference in their structural arrangement.

In the language of general systems theory, all things are layered structures of order. The ordered arrangement of lower-level systems constitutes higher-level, more complex systems, and the interrelations among those lower systems are an indispensable part of the higher ones.

From invisible atoms and bacteria to the mountains and trees and creatures we see around us, and on to you and me, everything in the universe can be seen as a layered order formed by the arrangement and relation of the smallest units of matter.

If we look all the way through things, going only by their constituent material, then the essence of everything seems to be nothing but elementary particles, or even, as Buddhism would put it, all

is Emptiness. And yet, for the bamboo at the level of roots, stems, and leaves as seen by the botanist, or for the bamboo at the level of the plant cell as seen by the cell biologist, the essence of its existence is not merely those elementary particles. It is the order formed by the arrangement and relation of the basic material.

P. W. Anderson, eminent in condensed-matter physics and complex-systems research, argued that although everything in nature is necessarily built from elementary particles, the behavior of a complex system formed from many particles cannot always be deduced directly from the properties of those simple particles. On the contrary, nature brings forth new structures and new complexity at different scales, and at each scale there exist characteristic laws and principles.

These new structures and forms of complexity, which arise at different scales and possess their own distinctive laws, constitute new things formed by nature and new forms of ordered existence. At every scale, the universe continually presents ordered structures of many kinds, and thus continually presents things of many kinds. In this way it gives rise to our richly varied, ever-changing, and beautiful world.

The arrangement and relation of elementary particles form atoms and molecules. Atoms and molecules are new things, new existences, relative to elementary particles, because they are new ordered structures built from elementary particles as their units. Atoms and molecules in turn serve as concrete units whose arrangement and relation gives rise to inorganic and organic macromolecules. These too, relative to atoms and molecules, are new ordered structures and therefore new things. The ordered

arrangement of organic molecules forms cells. The ordered arrangement of cells forms living organisms. Cells and organisms, relative to their lower-level constituent units, are likewise new ordered structures and therefore new things.

And this is not only true in the natural sciences. In the simplest, most everyday understanding, we also accept that it is order that constitutes existence.

The water in a river is never the same water from one moment to the next. New water is always flowing through. Yet the river of yesterday and the river of today are the same river. This is because the order we call the river has persisted through time. Pure air, if you look at it with the naked eye, seems to be nothing at all. It is taken as empty, because no order is observed. But if you observe it with scientific instruments, you can see the ordered existence of the various gas molecules within it.

When we say that something exists, the implicit premise is this. At some scale, there must be an observable order that belongs to that thing.

Different things are different sometimes because their constituent units differ, and sometimes because the interrelations among those units differ. The same thing at different scales differs not only in its units but also, possibly, in the relations among those units. All these differences of order ultimately manifest as different things, or as different properties of a thing.

All the properties of matter arise from its layered structure of order. The arrangement of protons, neutrons, and electrons determines the chemical properties of an element, its valence and reactivity. The structural arrangement of molecules determines their phys-

ical and chemical properties, their melting points, boiling points, reactivity, and stability. The arrangement of particles in a crystal determines its physical properties, its hardness, conductivity, and magnetism. The arrangement of amino acids in a protein determines its characteristics, its solubility, thermal stability, and acidity, and in turn determines the specific function the protein performs in the organism.

We can directly sense the color, shape, sound, smell, hardness, and temperature of matter through sight, hearing, smell, taste, and touch. We can use balances, thermometers, and clocks to make precise measurements of mass, temperature, and time. We can use microscopes to observe the fine structures of cells and bacteria, and telescopes to observe distant galaxies and planets. We can even use X-ray diffraction and nuclear magnetic resonance to resolve the three-dimensional structure of complex molecules, and electron microscopes and scanning tunneling microscopes to observe the arrangement of atoms directly. Through all these means, direct and indirect, every property of matter that we are able to perceive, every manifestation that declares the existence of matter, arises from its layered structure of order.

Whether future science identifies the smallest constituent of matter as an elementary particle or, as string theory proposes, as a string manifesting different vibrational modes of energy, one point remains certain: the layered order formed by the arrangement and combination of the smallest units is a definite reality, the source of what can be observed and perceived.

Order is therefore the essence that endows matter with its properties. To observe matter is to observe order. The material proper-

ties we perceive are appearances: partial expressions, at particular scales, of a thing's ordered information. Order is the precondition of observation; observation does not produce it. Order exists independently of us and is reflected in what we observe.

The appearance of a thing is only a partial presentation of the information contained in its own layered structure of order. This is what Hegel meant by holding that actuality is not mere immediate existence. The manifestation of an actual thing is not equivalent to the thing itself, that is, to its complete layered structure of order. And since a thing's layered structure of order contains all its information, it is also apt to say that the actual thing is what is essential.

To achieve a thorough and complete cognition, to investigate things and exhaust their principles, we would in theory need to understand the order and the changes of a thing at every level, from the bamboo at the level of roots, stems, and leaves, to the bamboo at the level of the plant cell, and then to the bamboo at the deeper levels within the cell, of molecules and atoms. We can even say that everything that exists in the universe, and every way it changes, is knowledge.

What kind of existence, then, is matter, is objective reality? It is ordered existence. From elementary particles to galaxies, from inorganic molecules to life and consciousness, the common ground on which all things in the universe exist and are distinguished from one another is their unique, layered structures of order. Multi-level ordered structure is the shared characteristic and the essence of all things.

This has been a preliminary exploration of the concept of existence. Existence and non-existence are foundational concepts, op-

posed to one another, each defined by and dependent on the other. Although Parmenides held that non-being is unspeakable, if we wish to understand existence more deeply, it is still very necessary to examine non-existence as a point of contrast.

Chapter 4

Awakening to Emptiness, Knowing Nonbeing

Every great advance in science has issued from a new audacity of imagination.

John Dewey

Look around you. What is everything within sight—from the sun, moon, and stars to ash and dust—made of? The question of matter's smallest constituent has stirred human curiosity since ancient times. From philosophical reflection to scientific verification, humanity's search for the essence of matter has continued for thousands of years.

The search for matter's smallest constituent has a long history in China. The Five Elements theory[1] of the Western Zhou held that matter was composed of five basic elements: metal, wood, water, fire, and earth. The *Mohist Canons* proposed *duan* as the smallest unit of matter.[2] In the "Tianxia" chapter of the *Zhuangzi*, the state-

[1]The Chinese term *wuxing* is conventionally translated as "Five Elements," but "Five Phases" better conveys its sense of dynamic processes and correspondences: wood, fire, earth, metal, and water.

[2]*Duan*, literally an endpoint, is a technical term in the Later Mohist discussions of

ment that a one-foot stick, halved each day, will not be exhausted even after ten thousand generations reveals an early and profound insight into matter's infinite divisibility. On the basis of such ideas, the Warring States period developed the theory of primordial *qi*, tracing the origin of matter to an abstract vital essence and holding that all things were composed through combinations of *qi*.

In ancient India, the *Mahāvibhāṣā* proposed a theory of four ultimate atoms: earth, water, wind, and fire. These atoms were thought to be indivisible and everlasting. Seven ultimate atoms could form one mote of fine dust, regarded as the smallest unit visible to the heavenly eye of a bodhisattva.

The ancient Greeks likewise recognized early on that matter was composed of discrete units. The philosopher Empedocles (ca. 490–430 BCE) proposed a theory of four elements, holding that all things were composed of four indivisible "roots": water, air, fire, and earth. This laid a foundation for the Western conception of matter.

Leucippus and his student Democritus then founded the atomism that would shape Western thought. The Greek word *atomos* means "indivisible." Democritus held that atoms and the void were the origin and basic constituents of all things, and that atoms were the smallest indivisible particles composing concrete matter.

After the seventeenth century, atomism gradually moved from an abstract philosophical concept toward empirical science. In the early nineteenth century, the British chemist John Dalton, building on the work of Boyle, Lavoisier, and others, proposed a genuinely

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extension and divisibility. *Qi* is a many-sided classical term for the vital or material stuff through which things form and transform; no single English equivalent covers all its uses.

scientific atomic theory that successfully explained many chemical and physical phenomena.

Although atoms were treated as indivisible elementary particles in chemical reactions, physics showed that they could be divided further. By the end of the nineteenth century, J. J. Thomson had discovered the electron. In the early twentieth century, Einstein confirmed the reality of atoms through his theory of Brownian motion, Rutherford discovered the atomic nucleus, and Chadwick discovered the neutron. Step by step, the modern model of the atom took shape.

Modern physics now probes the essence of matter with unprecedented precision. With cryo-electron microscopes and other sophisticated instruments, we can directly observe atomic structures; with the Large Hadron Collider, we can explore more deeply into the subatomic world. As experimental and observational methods advance, the answer to the question of matter's smallest constituent continues to change.

The Standard Model of modern particle physics holds that matter is composed of elementary particles smaller than atoms. There are twenty-five kinds, or sixty-one under a finer classification: six quarks, six leptons, twelve gauge bosons, and one Higgs boson.

Quarks and leptons are collectively known as fermions. They are regarded as the "substantial" units of matter in the narrow sense: quarks make up protons and neutrons, while leptons include electrons. Gauge bosons transmit the strong, weak, and electromagnetic interactions. Together with gravity, these interactions connect particles and bind them into material entities. Bosons are therefore also indispensable constituents of matter.

The Standard Model can also be understood in terms of fields: all elementary particles are effects of fields, and fields are the foundation of everything. The universe is filled with fields. The electromagnetic field excites photons and the electromagnetic interaction; the gluon field excites gluons and the strong interaction; the W- and Z-boson fields excite the corresponding bosons and the weak interaction; quark fields excite quarks; and the Higgs field excites the Higgs boson and gives mass to other elementary particles. Elementary particles are understood as vibrational excitations of these quantum fields.

The Standard Model explains the strong, weak, and electromagnetic forces, and some of its predictions have been verified to more than twelve decimal places. It can fairly be called the most successful scientific theory yet constructed. Even so, it does not encompass gravity and has not been fully united with the general theory of relativity. Nor are scientists entirely certain whether the elementary particles of the Standard Model possess internal structures of their own or can be divided still further. For all the admiration it commands, the Standard Model may not be the final answer to the question of matter's smallest constituent.

String theory in modern theoretical physics holds that matter is composed of still smaller vibrating strings. A string may be line-like, in which case it is called an "open string," or loop-like, in which case it is called a "closed string." Different vibrational states of the string display the properties of different particles and thus correspond to the various elementary particles.

In string theory, the graviton is treated as a kind of closed string, offering an explanation of gravity. Some scientists therefore believe

that string theory has the potential to unite quantum mechanics with general relativity. Yet strings are so extraordinarily small that direct experimental observation has so far been impossible, and the theory remains to be verified.

From antiquity to the present, humanity's effort to probe the infinitesimal in matter has also been a pursuit of the ultimate smallness of the universe. It is hard not to marvel at how far the wonder of the cosmos exceeds our imagination. Every leap in understanding has profoundly altered the human worldview. Scientific progress makes us feel that we are drawing closer to the answer, and we look forward with confidence to its eventual revelation. Alongside string theory, approaches such as loop quantum gravity and the holographic principle are also being explored.

Since the birth of relativity and quantum mechanics, the achievements of physics over the past hundred years have been remarkable. Yet the prospects for further exploration have become less encouraging. The reason is simple: probing smaller units of matter and ever more microscopic worlds requires ever greater energies. The experimental means by which the natural sciences "investigate things" are approaching their technological limits.

In string theory, the size of a string is close to the Planck length, on the order of 10^{-35} m, believed to be the smallest possible physical scale. Probing the Planck length would require the Planck energy, roughly 1.2×10^{19} GeV, about one quadrillion times the energy required by today's most advanced large proton collider. Quantum mechanics further suggests that even if we could obtain such enormous energy to observe still smaller scales, we would at the same time create microscopic black holes. This may imply a fun-

damental observational limit to human cognition.

Current explorations even suggest that the smallest unit of matter may not be the kind of point-particle entity we have imagined. Perhaps matter has no absolute "smallest" limit at all, but only relatively effective levels of description.

Faced with such difficult-to-surmount limits of experimental observation, let us set aside for the moment whatever final conclusions scientists may reach in the future. On the basis of what science already knows, we cannot help but reexamine the concept of the "smallest unit of matter." If we assume that such a unit exists, what characteristics should it possess?

First, the smallest unit should be the most abstract. There should be no subtler distinctions among smallest units of the same kind. Otherwise, they should be divisible still further.

Identical particles are an important concept in quantum mechanics: particles of the same type are completely identical to one another. Protons, neutrons, electrons, quarks, and other such particles are all identical particles. In the macroscopic world, sameness is only approximate; in the microscopic quantum world, it is absolute. Identity entails greater abstraction and less concreteness. Since a smallest unit is already the smallest, it is no longer an "order" formed by the arrangement of other units. This seems to mean that it has the fewest perceptible features, perhaps none at all. The smallest unit should therefore be both the most abstract and the least ordered.

Second, an independent smallest unit should be the freest. At this extreme of smallness, matter will possess intrinsic and genuine randomness. Quantum mechanics shows that the more microscopic

a particle is, the more pronounced its quantum uncertainty and randomness become.

In quantum mechanics, a free particle is defined as an ideal particle unbound by any force. According to the Schrödinger equation, it is an ideal particle subject to no external potential. Its quantum state can take the form of a continuously spreading wave packet, so its position is no longer confined to any particular region. We can see this unbound state as an ultimate physical abstraction of freedom. A free particle therefore possesses genuine freedom and randomness. An independent smallest unit of matter, subject to no constraint and forming no part of anything else, should likewise be a free particle.

Observation and perception consist, in essence, in producing interaction: electromagnetic, weak, strong, or gravitational. True randomness and freedom also mean that no interaction exists between an independent smallest unit of matter and anything else. We will therefore be unable to observe or perceive such a free and independent smallest unit.

If, then, we define "existence" as "order," and hence as something perceptible, the "disorder" represented by a free and independent smallest unit—being imperceptible—can be defined as "non-existence."

In the actual universe, no particle is absolutely free of interaction. A smallest unit that is unconstrained and completely free can only be regarded as a theoretical abstraction taken to its limit, and thus as logically non-existent. From the more fundamental perspective of statistical mechanics, this "disorder" can be defined as a "state of maximal symmetry" in which all possibilities are super-

posed with equal weight. What we commonly call "quantum fluctuations" are precisely the dynamic manifestations of this maximally symmetric state as it evolves.

One point needs particular clarification. All bosons are subject to gravitational interaction: according to general relativity, gravity acts on every particle that possesses energy and momentum. Most gauge bosons, with the exception of the photon, also participate in the interactions they themselves transmit. The various bosons are therefore also forms of existence, rather like "adhesives" and "messengers." They usually serve as binding relations and become parts of layered structures of order.

Take the photon, for example. Although it has no mass, it possesses energy and momentum. According to general relativity, light is therefore subject to gravity and cannot be considered truly "free." As we will discuss in more detail later, the photon does not even occupy any time or space from its own perspective. It can thus be regarded as a form of existence that comes closest to "non-existence."

The smallest constituent of any form of matter can exist only as a component of matter. Matter is order, and may also be called "Being." A free and independent smallest unit is "non-existence," which is disorder and may also be called "Non-being."

"Non-existence" truly does not exist in the universe. Every smallest unit that actually takes part in forming matter is bound. Once it is bound and enters arrangements and combinations formed through various interactions, it is no longer genuinely random and free. It enters into relations with other units, begins to acquire "order," and comes into "existence."

From the perspective of quantum field theory, all matter is, in

essence, made up of excited states of quantum fields. The ground state of a quantum field consists of pure, irregular, random quantum fluctuations and in fact signifies a vacuum empty of all things, corresponding to "Non-being." Relative to this ground state, the excited states—the "ripples of the field"—represent a form of order. They correspond to particles, and matter begins to "exist."

"Existence" and "non-existence," "order" and "disorder," or "Being" and "Non-being," are thus interdependent and mutually transforming. They are at once relative and absolute, opposed and yet unified.

For thousands of years, human beings have used numbers: one, two, three, and so on. These numbers are the foundation of the entire edifice of mathematics. What, then, is their essence?

To define the most basic numbers, mathematical logicians hold that we must first establish a set: the empty set, $\varnothing$. "Empty" means that there is nothing at all; the empty set is a set containing no elements. Whatever kind of set is under consideration, its corresponding empty set is the same and unique.

This concept, seemingly unremarkable and yet self-evident, has in fact become almost the foundation of the entire edifice of mathematics. With this unique empty set as the basis, the other numbers can be defined. The number zero corresponds to the empty set, which contains no elements. The number one is defined as the set containing only the empty set. The number two is defined as the set containing zero and one. By continuing in this way, we can define all numbers, including positive integers, negative numbers, real numbers, complex numbers, and so on.

In this way, with the empty set as its basic brick, we can build

a solid and reliable number system. Now you know the most astonishing secret of mathematics: as the British mathematician Ian Stewart put it, "It's all based on nothing." We cannot help but marvel at the wonder of mathematics.

The smallest unit of matter is just like the abstract empty set of mathematics. The free and independent smallest unit is abstract "Non-being," the brick from which matter is built. All existing matter is "order" formed by arranging and combining this abstract Nonbeing. Atoms, molecules, and everything beyond them arise in this way. Through arrangement and combination, microscopic abstraction gradually becomes macroscopic concreteness; through layer upon layer of constraint, complete randomness, or disorder, gradually becomes observable determinacy, or order. In the end, the grand tableau of the universe is formed.

"Non-being" is infinitesimally small, yet so decisive for all things that it constitutes all order. Order is the universe's kaleidoscope in all its color and variety. Now you know the deepest secret of the universe: everything is based on "Non-being." We cannot help but marvel that the universe, too, is such a wonder.

We do not need to wait for future physics to determine once and for all whether the smallest constituent of matter is an elementary particle, a string, or something still smaller. Scientific experiment is destined never to observe ideal, completely free "Non-being" directly. Even if scientists were to "capture" or "isolate" it for observation, what they captured or isolated would necessarily no longer be pure and completely free Non-being, and the properties they observed would necessarily no longer be the properties of ideal Nonbeing.

If we continue to press the question toward the deepest level of physics, disorder is not a mere jumble. It is more like the degrees of freedom that remain before change has been fully delimited: not an object already formed, but a possibility capable of producing difference. After all, does not what we call energy likewise correspond to the capacity to produce difference and make change possible?

In the quantum world, the smallest nontrivial change often takes on a binary, half-integer structure. A fermion's spin of $1/2$ can be seen as a typical form of this primordial degree of freedom of change. It is not a little ball turning halfway around in the everyday sense, but a more abstract inner orientation. It tells us that existence at the deepest level is not a static "thing," but a state carrying direction, phase, and the possibility of change.

When these smallest degrees of freedom of change are further constrained and combined, and are required to be self-consistent as a whole, arbitrary possibilities are filtered out and patterns capable of stable existence take shape. From this perspective, the kinds of particles, the relations among charges, the structures of interaction, and even the emergence of different generations can all be understood as ordered forms produced after the deepest change has passed through layer upon layer of closure.

For example, even if future physics confirms that elementary particles such as electrons and quarks are the smallest constituents of matter, electrons and quarks in a completely free-particle state would still be "non-existent." Complete freedom means the absence of every interaction and thus the impossibility of observation or perception; everything we observe has already been bound by interaction. Such completely free particles are only

abstractions in physical theory.

The more microscopic a thing is, the simpler its arrangements and combinations, the fewer features it possesses, and the more it tends toward uniformity. Because it is harder to perceive directly through the senses, we can say: the more microscopic, the more abstract. The more macroscopic a thing is, the more complex its arrangements and combinations, the more features it possesses, and the more individualized it becomes. Because it is easier to perceive through the senses, we can likewise say: the more macroscopic, the more concrete.

The movement from the microscopic to the macroscopic, and from the abstract to the concrete, is not a rigid division but a unity of opposites. As the *Daodejing* says, “In abiding Non-being, we seek to behold its mystery; in abiding Being, we seek to behold its bounds.”

“Disorder,” “Non-being,” and “non-existence” correspond to the ultimate abstraction. This is why Parmenides held that Being is, while Non-being cannot be spoken of.

Chapter 5

No Outside, No Inside

The most incomprehensible thing about the universe is that it is comprehensible.

Albert Einstein

From the intuitive conjectures of antiquity, through the abstract generalizations of modern science and philosophy, to the deeper inquiries of contemporary scientific theory, humanity's understanding of the essence of matter has undergone a profound evolution.

In the commonsense understanding of antiquity and everyday life, matter appears as a concrete, perceptible entity. Ancient India's theory of four ultimate atoms, ancient China's theories of the Five Elements and primordial *qi*, and ancient Greece's doctrines of four elements and atomism differed in their particulars, but all belonged to the same mode of thought: each anchored the essence of material existence in some kind of primordial entity.

In modern science, this ancient commonsense understanding evolved into the method of reductionism. If matter can always be divided further, and if complex things are always composed of simpler things, then analyzing the simple in order to understand the complex appeared to be a universal method. Galileo declared

that the book of nature was written in the language of mathematics. Descartes equated matter with geometric extension. Newton's absolute space and time became the eternal theater of matter in mechanical motion. Dalton's atomic theory triumphed in chemistry, while Mendeleev's periodic law seemed to confirm the ultimacy of the material elements. Under the reductionist paradigm, attention necessarily centered on entities—things and their constituent elements—while the relations among entities were often assumed to be simple, linear, and mechanical.

The development of modern science, including quantum mechanics and relativity, has thoroughly reconstructed our understanding of material existence. Wave-particle duality and Einstein's mass–energy equation declared the bankruptcy of the classical concept of substance. Heisenberg's uncertainty principle revealed the constructive role that observation plays in the properties of matter. The Standard Model showed that elementary particles are defined more by their relations of interaction than by any intrinsic substance. Quantum field theory went further, portraying particles as excited states of fields. The Higgs mechanism demonstrated that the mass of elementary particles arises from their coupling to the Higgs field. Advances in astronomy, biology, and the life sciences have steadily shown that we inhabit a world of self-organizing evolution. People have therefore gradually come to accept the modern understanding that the properties of entities arise from relational fields.

The advance of modern science has even encouraged philosophy to turn toward relational realism in its understanding of material existence. In formal logic, "to be" (*sein*) is a copula: it connects

two subjects and expresses the relation between them. David Bohm and John Wheeler, leading figures in relational realism, stated explicitly that the essence of matter lies not in independent entities but in relations. Heidegger went further, arguing that Being is not an isolated entity but is defined and disclosed through a network of relations with the world, time, language, and more, within "being-in-the-world." The American ecologist Barry Commoner made the point still more radically in his first law of ecology: "Everything is connected to everything else." He emphasized that the complex relations among the elements at every level of an ecosystem determine its distinctive patterns of operation and interaction.

How, then, should we understand things? Is the essence of material existence substance or relation?

In fact, the latest thinking in contemporary science is dissolving the dichotomy between substance and relation. Matter is no longer understood as pure substance or pure relation, but as the dialectical unity of both.

First, matter is by no means an isolated, self-existing substance. Without relation, there is no substance. Without the relations generated by interactions, layered structures of order cannot take shape, and things cannot come into being. A completely independent entity, unconstrained by any relation and analogous to the hypothetical free particle of physics, is only a theoretical abstraction. Without interaction, it cannot be perceived. An isolated, self-existing entity cannot exist.

All the material phenomena we observe are manifestations within specific relations. Apart from those relations, matter's particular properties and modes of existence may change or even

fail to manifest at all. Modern quantum physics has shown not only that the microscopic structure of matter is formed by elementary particles through specific interactions, but also that properties such as spin, charge, mass, position, size, and color are not intrinsically "inherent" in isolation. They can manifest only through interaction with other things.

Consider the electron. Its position and state are not determined by the electron alone, but by its relations of interaction with surrounding fields and other particles. Within an atom, the electron exists in a bound state. Its charge can be measured separately only after it leaves a nucleon—for example, during beta decay, when a down quark inside a neutron turns into an up quark and releases an electron. At that point, the electron's charge becomes manifest. The color of matter arises from its selective absorption, reflection, and transmission of light. We can say that the color we see always comes from the portion of light that matter "does not want."

Second, matter is not merely an insubstantial web of relations. Without entities, how could there be relations? We should recognize that the layered structures of order formed through relations are themselves entities. Entities must exist before relations can form. Lower-level entities provide the basis for new relations, making more complex things possible. Within the layered order of a thing, the ordered structure at each level constitutes an entity at that scale. To know a thing fully, we must understand the order at each of its levels.

Although matter can, in essence, be understood as a structured dance of energy, it is precisely substantiality that gives material existence its stability. If we recognize only relations, or attend only

to them, we risk going to an extreme and sinking into the mire of nihilism.

In a certain sense, relations themselves are also grounded in entities. Within quantum field theory, the four fundamental interactions involved in the formation of matter can be described as processes mediated by the exchange of virtual particles, or gauge bosons. Electromagnetic interaction can be seen as the exchange of virtual photons between charged particles; strong interaction as the exchange of gluons between quarks; and weak interaction as the exchange of W^+, W^-, and Z^0 bosons between fermions. Within the framework of quantum field theory, gravitational interaction is hypothesized to be mediated by the exchange of gravitons, although this has not been confirmed experimentally.

From fixed atoms to trembling quanta, from absolute particles to relational fields, humanity's understanding of matter has always sought a balance between substantial anchoring and relational fluidity. This unceasing exploration of knowledge ultimately leads us to a deeper insight: perhaps the very question "Is material existence substance or relation?" is a false one. The true mystery of material existence lies precisely in transcending this duality.

It is therefore more fitting to use the word "order" to describe the common property of all material existence and of everything that can be perceived. It expresses the essence of matter as both substance and relation, as neither pure substance nor pure relation.

A thing that is completely independent and bears no relation whatsoever to the outside world cannot be perceived and therefore does not exist. Non-existence is the absence of "order": neither relations nor entities exist. We can therefore use the word "disorder"

to describe non-existence.

"Order" and "disorder" are not merely words of everyday speech; they are also central concepts in thermodynamics, statistical physics, and complex-systems theory. Compared with terms such as energy, relation, and information, this pair accords more closely with Occam's razor: when several theories explain the same phenomenon, we should prefer the one that makes the fewest assumptions. These two terms also make it easier to establish common ground between the philosophical and scientific levels, between the metaphysical and the physical.

In the Buddhist *Heart Sutra*, "form" refers to tangible, substantial material phenomena, such as mountains and rivers, plants and trees, and the human body, as well as everything we can see, hear, smell, taste, or touch. In short, form comprises all the appearances of the myriad things of the world. Buddhism holds that all things arise through the convergence of causes and conditions; without that convergence, nothing arises. This is "Emptiness."

The Buddhist concepts of form and Emptiness bear a striking resemblance to the scientific concepts of order and disorder. Buddhism likewise emphasizes the need to transcend the duality of form and Emptiness: form and Emptiness are not two.

The great master Nāgārjuna of the Madhyamaka school of Mahāyāna Buddhism held that all things arise from the convergence of causes and conditions and have no eternal, independent self-nature. This corresponds to the view that matter should have no absolutely independent substance. In his *Mūlamadhyamakakārikā*, Nāgārjuna likewise argued that nothing exists independently of other things and that the search for an ultimate foundation is

meaningless. True wisdom lies in recognizing the emptiness of things—their lack of self-nature, their impermanence, and their non-self—while not denying their existence and appearance. This corresponds to acknowledging the importance of causal relations while avoiding the extreme nihilism of "everything is empty, everything is like a dream, an illusion, a bubble, or a shadow." To borrow the double-negation formula of the Madhyamaka school, material existence should be described as "neither substance nor relation, neither not-substance nor not-relation."

This foundational logic—that things exist as order—is fully consistent with two basic propositions of systems theory and dialectical materialism: things are layered structures of order, and all things are universally interconnected. Because existence is order, nothing in the world can be absolutely isolated; all things exist in relation to one another. Things exist as layered forms of order in mutual relation: this is both the essence and the mode of their existence. Through universal interconnection, the world forms an organic whole.

The principle of "universal interconnection" in systems theory and dialectical materialism is therefore more than an empirical generalization. Universal interconnection is not merely an attribute of things; it is an inherent requirement of their existence and a necessary and sufficient condition for it.

A layered structure of order is a property common to matter, and it gives the universe its boundless complexity and change. Cosmological models estimate that the observable universe contains approximately 10^{80} to 10^{81} atoms. We often marvel at the enormity of figures on a cosmic scale, yet the number of atoms in the uni-

verse can still seem "not so large." Like a finite set of LEGO bricks, however, once those atoms enter into relations, their arrangements and combinations open infinitely many possibilities of assembly—enough for stars, planets, life, and consciousness to emerge from cosmic chaos.

From the Planck length, roughly 1.6×10^{-35} m, to the diameter of the universe, roughly 8.8×10^{26} m, or 93 billion light-years, the order of all things unfolds across an extraordinary range of levels. From atoms, molecules, cells, living organisms, and planets to galaxies, the various levels of order continually give rise to new regularities and new things. This "infinity within the finite" is the deepest mystery the universe conceals between substance and relation.

Only when our way of knowing accords more closely with the essence of a thing's existence can we come nearer to its truth. In the course of evolution, nature formed layered structures and, in the same process, created regularities. Through successive approximation, human beings have sought the truths of nature at different scales, discovered and summarized its regularities, and thereby formed such disciplines as classical mechanics, chemistry, biology, neuroscience, medicine, and political economy.

We can say that the foundational logic of viewing all things as layered forms of order both accords with the essence of their existence and gives humanity a powerful tool for investigating things and pursuing their principles to the end.

Close your eyes. When you open them again, shift to the perspective of layered order and look anew at all things. Attend closely to the many phenomena displayed by order at different levels, and

accustom yourself to forming a fuller, more three-dimensional understanding of things from several perspectives at once.

The vibrations of sound waves weave a code of frequencies among the cilia of the cochlea. Retinal neurons transform a storm of light quanta into a visual poem of nerve impulses. Like keys, food molecules unlock the chemical lockboxes of taste-receptor proteins. Ordered information of many kinds travels continuously through neurons to the cerebral cortex. The ceaseless flickering of neural networks gives rise to tides of neurotransmitters, allowing us to feel pleasure and satisfaction.

All the information we receive from the outside world can be regarded as part of the order of things. Information cannot exist independently; it must be carried by material media such as sound waves, electromagnetic waves, or images. The essence of things is layered order. Order corresponds to information; disorder corresponds to complete randomness and therefore contains no information.

Many people have therefore proposed an "ontology of information"—"everything is information," "everything is a qubit," or even "the universe is a computer"—holding that information is the ultimate reality and that the essence of all things is information. These views are, in substance, similar to relational realism. From a more fundamental perspective, however, information cannot exist independently of order. It is the "knowability" of ordered structure, a projection of order rather than its source. Such views contain genuine insight, but if they are detached from the substantial basis contained in order, they can slide into another kind of one-sidedness: the neglect of layered structure.

The various forms of order in the outside world, mapped into the brain as new forms of order, constitute consciousness. Neurobiologists hold that consciousness is an emergent phenomenon produced by the vast number of neuronal connections in the brain. Consciousness and matter, therefore, are not a duality in opposition; they are different manifestations of "order."

The order of the living body will eventually pass away, and so people hope that consciousness can continue in the form of a soul or something similar. Yet any soul or world beyond that is imagined to exist independently must, if it exists, be some kind of ordered structure and must necessarily bear some relation to the universe. Conversely, if every possible means has been exhausted and no relation can be established, reason can only judge that it does not exist—that it is disorder.

Even dark matter and dark energy must be connected to other things in the universe through some form of interaction if their objective existence is to be established: dark matter is detected indirectly through gravitational interaction, and dark energy is inferred from the expansion of the universe. Matter that is truly "dark"—entirely unrelated to everything else in the universe—does not exist in the present universe.

Chinese Daoism offers the classic formulation that the universe is "so great that it has no outside, so small that it has no inside." Modern science continues to confirm this ancient insight.

The universe is the highest-level ordered structure that we are capable of knowing. Human beings cannot, of course, perceive the whole of the universe from outside it. We can only explore its internal order from an internal perspective. The multiverse hypothe-

ses proposed by science, including bubble universes and parallel universes, hold that other independent universes may exist beyond our own. If those universes have no form of connection whatsoever with ours, they can never be verified through observation and may be regarded as non-existent within our universe. If, on the other hand, any form of connection does exist, then those universes are in substance part of our universe, and there is no need to distinguish them through the concept of a "multiverse."

At the microscopic scale, no matter how far science may continue into the infinitesimal, we can never truly study that "unsayable" smallest thing as something independent and free. For that smallest thing does not exist; it is Non-being.

The essence of the universe, therefore, is a "web of order" with no outside and no inside, a web in which all things are interdependent and interact with one another. Every thing and phenomenon is merely a node on this web. This web of order is what Hinduism calls *Brahman*, what Buddhism calls the *Dharmakāya*, and what Daoism calls the *Dao*. Everything within the web is equal; everything is mutually related and mutually influential. No existence, whether material or conscious, possesses absolute primacy. Order is the sole essence of existence.

This insight, that the universe is an interconnected dynamic network, finds powerful resonance at the very frontier of fundamental physics.

At the end of the twentieth century, the Italian physicist Carlo Rovelli proposed "relational quantum mechanics," discussed in *Seven Brief Lessons on Physics*. Its central idea is that quantum theory does not describe how the inherent, absolute properties

of an isolated quantum object "appear" to some special observer. Rather, it describes the laws governing interactions between any physical systems: how one system is described and affected relative to another.

Compared with certain traditional interpretations, relational quantum mechanics completely dissolves the special status of the "observer": any physical entity can serve as the "observer" in an interaction. It thereby transcends, at a fundamental level, the classical framework of the subject–object split. This theoretical perspective vividly illustrates how matter and consciousness are deeply embedded in, and jointly weave, the relational network of the universe. It provides a new physical foundation for bridging the traditional duality of matter and consciousness, and perfectly confirms the holistic interconnectedness embodied in the "web of order."

Particle physicists can analyze the order of matter at an extremely microscopic level; chemists observe the arrangements and combinations of atoms and molecules; biologists perceive the basic units of living things and their interrelations at the cellular level; and ordinary people, relying on perception, apprehend the order of things mainly at the macroscopic level.

Faced with the immensely varied and complex manifestations of "order" in all things, human beings find phenomena, knowledge, and regularities in every ordered existence. If an omniscient and omnipotent God truly exists, perhaps it can only be the universe itself. Knowledge is inexhaustible and life is finite; we should therefore devote ourselves all the more to seeking ways of understanding the world.

Chapter 6

Eternal Variations

To see a World in a Grain of Sand,
And a Heaven in a Wild Flower,
Hold Infinity in the palm of your hand,
And Eternity in an hour.
William Blake

If all things are themselves layered forms of order, how should we understand their changes?

In the basic framework for explaining natural phenomena, science typically focuses on two kinds of questions. The first concerns the laws of individual motion. How does a thing, treated as a relatively independent entity, move under various interactions? The second concerns the evolution of systems. In a complex system composed of vast numbers of units, how do interactions among those units drive the evolution of the whole?

To study the laws governing the motion of individual things, the key is to analyze the interactions between the object of study and other things. Interaction is the relational bond through which all things become connected in order and are set in motion.

Modern physics unifies the various interactions into four fun-

damental kinds: the strong interaction, the weak interaction, the electromagnetic interaction, and the gravitational interaction. The corresponding forces are called, more briefly, the strong force, the weak force, the electromagnetic force, and gravity.

The strong and weak forces are short-range forces, with ranges of only 10^{-15} m and 10^{-18} m, respectively. Their principal role is to bind microscopic elementary particles tightly together, forming the basic structures of matter and laying the foundation of microscopic order.

The electromagnetic force is a long-range force of infinite range. It is two orders of magnitude weaker than the strong force and therefore plays an important role at every level, from the microscopic to the macroscopic. The familiar forces of pushing, friction, tension, pressure, buoyancy, adhesion, and magnetism, as well as the various intermolecular forces, are all electromagnetic in nature.

Gravity is also a long-range force with infinite range, but it is the weakest of the four. It is thirty-eight orders of magnitude weaker than the strong force and twenty-seven orders weaker than the weak force. Its primary role is to produce binding relations between bodies of matter, forming large-scale ordered structures such as planets, stars, and galaxies at the cosmic scale.

On the basis of interaction analysis, we can construct mathematical models in specific coordinate systems to describe the motion of objects. At the macroscopic level, Newtonian classical mechanics addresses problems of motion within the framework of absolute space and time. Special relativity extends this to inertial coordinate systems in relative spacetime. General relativity generalizes it further to arbitrary coordinate systems.

At the microscopic level, quantum mechanics and quantum field theory—including quantum electrodynamics, electroweak unification theory, and quantum chromodynamics—describe the behavior of microscopic particles under the electromagnetic, weak, and strong interactions. They describe particle states through wavefunction probability distributions that embody intrinsic randomness.

When the focus shifts to the evolution of systems, the key task is to analyze how vast numbers of units, through their interactions, drive the evolution of the system as a whole.

Scientific observation across scales reveals that complex systems such as living organisms, neural networks, and social groups display distinctive dynamic and nonlinear characteristics. Cells combine to produce organs with new functions, and multiple organs integrate to form metabolic systems. Ant colonies achieve collective intelligence through simple rules, while neural networks give rise to consciousness and cognition. Human societies have evolved cooperative paradigms such as law and currency.

By studying self-organizing processes of evolution—phase transitions in matter, the origin of life, the formation of galaxies, and more—fields such as statistical physics, condensed-matter physics, astrophysics, complex-systems science, and biology have constructed an emergentist worldview.

Emergentism reveals something profound: to understand the evolution of complex things, it is not enough to analyze the individual motions of their constituent units. The behavior of the whole, together with the higher-level ordered properties and regularities that emerge from it, cannot be reduced to or predicted

by simply adding together the behavior of lower-level units. It must instead be explained through regularities established at a higher level. We cannot derive pressure and temperature directly from the thermal motion of molecules, infer the workings of an organ from the function of a single cell, or grasp the characteristics of a group by merely adding together individual behavior. To understand the world, we must respect the distinctiveness of order at every level.

As the condensed-matter physicist P. W. Anderson explained in "More Is Different," all things are governed by different laws and principles at different levels, from the microscopic to the macroscopic, or from lower levels to higher ones. This is precisely why complex systems are "complex." Stephen Hawking, writing with Leonard Mlodinow in *The Grand Design*, expressed a similar view: there may be no single ultimate theory, but only effective theories that describe phenomena at different levels.

The proposition that all things possess layered order has helped us establish a foundational logic of how things exist. Building on what science has achieved in understanding how things move and evolve, we also need to explore a foundational logic of how things change: to seek what different levels of change have in common, and the unchanging laws behind change.

What may concern us most at the outset is the ultimate cause of the ceaseless change of things.

The premise that all things possess layered structures of order and are universally interconnected tells us that lower-level ordered structures are arranged and combined to form higher-level ordered structures, thereby constructing a nested system of levels that

stretches from elementary particles to the entire universe. If we strictly follow the reductionist paradigm, the cause of an individual thing's motion and change can only be attributed to an external force. By the logic of this universal nesting of levels, we would then be compelled to posit a transcendent god—a "Prime Mover" beyond the universe—to explain the original source of cosmic motion.

This deduction, however, stands in fundamental contradiction with the basic premise of layered order and universal interconnection in the material world. If such a god could interact with things in the universe by exerting a motive force, then by the principle of universal interconnection, this god would necessarily form an integral connection with the universe and thereby forfeit the postulated property of transcending it. If we insist that motion at every level requires external intervention, then this god would have to control simultaneously every level of motion, from quantum fluctuations to the formation of galaxies. This amounts to presupposing an infinitely recursive and impossible mechanism of complex control. Finally, the "Prime Mover" postulate also fails to explain how the universe sustains its ongoing evolution. If we concede that this god must intervene continuously rather than merely provide the initial conditions, then the autonomy and universality of every natural law would be destroyed.

If we follow the emergentist worldview, the cause of the motion and change of things should originate within them. By the same logic of the universe's nested levels, the original source of cosmic motion must be traced back to the smallest units composing matter—that is, to disorder, corresponding to the microscopic

fluctuations at the lowest level, where constraint is weakest and randomness strongest. Lower-level motion and change drive higher-level motion and change. The changes of all things begin with the motion and change of matter's smallest units.

This conclusion accords perfectly with the basic premise of layered order and universal interconnection in the material world, and it also agrees closely with science's understanding of the microscopic world. Heisenberg's uncertainty principle reveals the intrinsic randomness of microscopic particles. Like the disorder of a free particle—a theoretical abstraction—this randomness is genuine. Since all things are composed of disorder at the lowest level, all change is endogenous in essence, driving every form of change from the microscopic to the macroscopic. We may even regard the disorder at the lowest level as, in essence, a state of maximal symmetry containing every possibility. Its eternal internal evolution, which we commonly perceive as random fluctuations, supplies the fundamental impetus for continuous change. The ordered structure formed at each level then "constrains" and "guides" the disorder below it, suppressing its randomness and giving its changes direction, so that distinctive regularities emerge at the corresponding level.

This analysis is highly consistent with the dialectical relationship between internal and external causes in Marxist materialist dialectics: all things change perpetually; the fundamental motive power of every change lies in internal causes; and external causes, too, can play their role only by influencing internal causes.

Change is the process of ordering itself: the transition from disorder to order, and from lower-level order to higher-level order. Without order as both its vehicle and its product, change could not

manifest itself.

From this, we can distill the first foundational logic of how things change: disorder at the lowest level drives the eternal change of all things in the universe.

Next, we may ask what common law governs change in layered structures of order. If disorder at the lowest level is the ultimate impetus of all change, it must be genuine randomness, for only genuine randomness can provide the conditions from which every possibility may emerge. Once arrangement and combination begin, randomness at the lowest level begins to be constrained. As arrangement and combination advance from level to level, the constraints grow stronger, randomness is gradually suppressed, and determinacy becomes correspondingly more pronounced.

The more microscopic the level, therefore, the more things display random characteristics. The more macroscopic the level, the more things display determinacy. We can even say that intrinsic randomness is the fundamental property of things, while macroscopic determinacy is its relative presentation as order under the constraints of multi-level structure. A thing is a dialectical unity of randomness and determinacy.

We know that absolute disorder is "nothing"—non-existence itself. Genuine randomness therefore never exists in isolation; it is always interwoven with determinacy. All existence is order formed as disorder at the lowest level passes through layer upon layer of arrangement and combination. Each new level of arrangement introduces new determinacy and reveals new regularities.

Between microscopic quantum fluctuations and the formation of cosmic structures, between genetic mutation and the evolution of

species, randomness and determinacy are always intertwined like the double helix of DNA. “God does play dice, but the dice are not free to do whatever they like.” This dance of randomness and determinacy allows the world to contain infinite possibility while sustaining exquisite stability, profoundly embodying the dialectical unity of chance and necessity in materialist dialectics. The quantum physicist Max Born likewise emphasized that nature is governed by both necessity and chance. In the imagery used here, chance is the stage of necessity, and necessity the director of chance.

The chance ripples of quantum fluctuations during cosmic inflation, amplified by gravity over 13.8 billion years, eventually wove the majestic web of galaxies. In thermodynamic systems, the disorderly collisions of molecules give rise, under statistical laws, to the macroscopic order of temperature and pressure, revealing the remarkable transformation of randomness into determinacy. In the evolution of life, chance errors in DNA replication provide material for natural selection, while random pressures from environmental change drive the differentiation of species. The superposition of countless “accidents” and order at its most refined ultimately forges the delicate balance of an ecosystem.

From this, we can distill the second foundational logic of how things change: “Disorder creates possibility; order determines direction.”

Beyond these two foundational logics, the layered structure of order in things also determines another truth: whether it is the motion of an individual thing or the evolution of a thing itself, all change throughout the universe begins with motion at the most microscopic level.

When we strike a baseball, the electron cloud at its surface is violently compressed, and electromagnetic repulsion erupts at the quantum level. This atomic resistance is transmitted instantly into the ball's molecular structure. Molecular bonds deform and rapidly store a great quantity of elastic potential energy. Trillions of molecules rebound in concert through electromagnetic interaction, and a stress wave sweeps from the point of impact through the entire ball. Microscopic potential energy is thus converted into macroscopic kinetic energy, and the baseball follows Newton's laws as it cuts across the sky. Without this collective resistance and transfer of momentum among lower-level particles, the ball could not break free of rest and take flight.

When we hold a cup of hot water, the high-speed oscillations of molecules in the cup wall carry ferocious kinetic energy. They strike the proteins and lipids on the surface of the skin like microscopic cannonballs. The impacts excite quantum fluctuations in the electron clouds of atoms at the contact surface and inject energy into the molecular bonds of the cell membrane through electromagnetic interactions. The vibrational energy levels of the molecular chains soar, and heat travels deeper into the skin along the electromagnetic bonds between molecules. Ion-channel proteins at the nerve endings deform under the thermal impact, triggering a flood of charged particles, such as sodium ions, across the barrier of the cell membrane. Finally, a sharp electrical signal of pain reaches the brain. Without this silent uprising of energy at the molecular frontier, the conscious exclamation "Hot!" could never arise.

The apparently macroscopic metabolism of a living body likewise begins with energy conversion at the microscopic level. Food,

such as glucose, is broken down within cells, releasing energy that drives electrons along protein chains. These electron currents push hydrogen ions, or protons, across the cell membrane, creating a microscopic proton gradient. As the protons flow down that gradient, ATP synthase rotates like a nanoscale waterwheel, converting the force of their descent into chemical energy stored in the high-energy phosphate bonds of ATP. Finally, when ATP is hydrolyzed, high-energy phosphate bonds break and stable new bonds form; the energy released in this process precisely drives the cell's many "molecular machines." Without the finely coordinated transfer of charge, flow of ions, and mechanical rotation among lower-level molecules, the basic activities that sustain life—nutrient absorption, growth, and cell division—could not take place.

Every way in which humanity harnesses energy must likewise begin at the microscopic level. The transmission of mechanical energy begins with the quantum compression of the outer electron clouds of atoms at the contact points of gear teeth, forcing molecular bonds to twist and deform so that kinetic energy is relayed through the crystal lattice. The release of chemical energy originates in the violent reorganization of electron orbitals when old molecular bonds are broken and new bonds form, releasing photons and energy. The awesome power of nuclear fission erupts from deep within the atomic nucleus, where a neutron bombardment breaks the fortress built by the strong interaction, and the lost mass is transformed, in a chain reaction, into a staggering torrent of energy.

The scale of energy humanity can harness ultimately depends on how deeply into matter's microscopic levels of order we can break

its inherent binding structures and release the power of "disorder" contained at the corresponding level.

This foundational logic—that microscopic disturbances drive macroscopic structures—applies equally to human society. When "individual agency and randomness" at the microscopic level pass through the "ordered constraints" of social relations—culture, institutions, and organizations—the macroscopic determinacy of the historical process emerges. This is the emergent law of human society.

Every advance and innovation in the fabric of society and its organizational structures can ultimately be traced to changes in the thoughts, actions, and habits of individuals, the smallest units from which those structures are composed. Likewise, the path by which even the greatest personal undertaking is realized must begin with the accumulation of seemingly minor actions and choices in everyday practice.

Just as a baseball cannot fly without the concerted action of lower-level particles, and the sensation of heat cannot arise without the transfer of energy among molecules, so without the tiny changes in individual thought and action there can be no mighty river of social transformation. Every quantum fluctuation in a mote of dust, every random leap of disorder at the lowest level, takes part in the eternal weaving and reshaping of the universe's layered structures of order, sounding through level after level of order a movement of variations without beginning or end. This is the metaphor of Blake's "Infinity in the palm of your hand."

Chapter 7

The Entropy Dance of Order

Reductionism lays the foundation for emergentism;
emergentism, in turn, charts the course for
reductionism.

Qian Xuesen

The concept of entropy was named by the German physicist Rudolf Clausius, who drew the word from the Greek *τροπή* (*tropē*), meaning "transformation." In the temple of natural science, entropy is like a dancer robed in light, her graceful form gliding through discipline after discipline.

In thermodynamics, entropy quantifies the degree of dispersion in energy distribution and reveals the direction of energy flow. In statistical physics, entropy describes the disorder among microscopic particles and explains how microscopic disorder gives rise to macroscopic order. In information theory, entropy measures the uncertainty of information and provides the theoretical foundation for information coding, compression, and transmission. In quantum mechanics, entropy is used to quantify the strength of nonlocal correlations in entangled states within quantum systems. In systems science, entropy is used to describe the self-organizing

processes and evolutionary directions of complex systems such as urban development, ecosystems, and economic behavior. Beyond these, in cosmology, ecology, biology, economics, and management science, entropy displays an astonishing explanatory power and a remarkably broad applicability.

Why can entropy encompass all things? Seen from different perspectives, the forms in which things exist can be explained in different ways. From the perspective of energy, a thing is an ordered manifestation of energy; from the perspective of information, it is the encoding and transmission of information; from the perspective of systems, it is a complex system with a layered structure. When these perspectives are brought together, however, the essence of all things can be described in a single way: everything manifests as layered order.

If the essence of existence is order, then any universal way of describing change in all things must describe changes in order. Such change is driven at its root by disorder at the lowest level and displays only two tendencies: toward greater disorder or toward greater order.

Entropy, as a measure of disorder, uncertainty, or chaos, is especially well suited to describing these two tendencies in the change of order. The tendency toward increasing entropy marks movement toward greater disorder and chaos: the degree of order declines, and order breaks down. The tendency toward decreasing entropy marks movement toward greater order: the degree of order rises, creating and sustaining order.

We know that every ordered existence is built from more basic units through specific arrangements, combinations, and mutual

relations. Because its constituent units are universally interconnected and mutually constrained, whenever order changes, two forces must be present within it at the same time: one promoting change, the other resisting it.

By relating them to the tendencies of entropy increase and entropy decrease, we can call these two forces at work in changing order the entropy-increasing force and the entropy-decreasing force. The entropy-increasing force drives order toward greater disorder; the entropy-decreasing force drives it toward greater order.

Here, the word *force* does not refer to an additional fundamental physical force. It summarizes the two tendencies by which an ordered structure builds itself up or moves toward dissolution.

The entropy-decreasing and entropy-increasing forces within an ordered structure are not accidental disturbances originating outside the system. They are fundamental tensions arising from the ordered structure itself. Their unity of opposites constitutes its present state.

Viewed through the theory of contradiction in materialist dialectics, the entropy-increasing and entropy-decreasing forces universally present within every order are precisely the two sides of the contradiction universally present within all things.

All things change without rest. The forces of entropy decrease and entropy increase are therefore present at every moment and in every place. This corresponds to the universality and absoluteness of contradiction and is the fundamental source of the motion and development of all things.

All the constituent units within an order are mutually connected

and forever changing, giving rise to the forces of entropy decrease and entropy increase within that order. These two forces are both opposed and unified, and together they constitute the order itself. This corresponds to the law of the unity of opposites in contradiction. We can also say that contradiction is the foundation of the existence of things. Without contradiction, there would be no world.

Because things possess layered structures of order, every level contains its own contradiction between entropy-decreasing and entropy-increasing forces. The contradiction at the principal level governing a thing's change is its principal contradiction; contradictions at other levels are secondary. This corresponds to the distinction between principal and secondary contradictions. The overall present state of a thing is the combined manifestation of the unity of opposites among contradictions at all its levels.

At a given level, the waxing and waning of the forces of entropy decrease and entropy increase constitutes the accumulation of quantitative change. When quantitative change under the influence of these forces breaks through a critical threshold, the collapse or reconstruction of the ordered structure at that level constitutes qualitative change. This corresponds to the law of quantitative and qualitative change in the movement of contradiction.

For a thing to realize its own evolution, it must pass through a process in which the entropy-increasing force breaks down its existing ordered structure and carries it into a relatively disordered state. Then, within that disorder, the entropy-decreasing force nurtures new connections and order, forming a new order. This corresponds to the law of the negation of the negation.

This scientific–philosophical correspondence between entropy-

increasing and entropy-decreasing forces and the dialectical category of contradiction works in both directions. On the one hand, it deepens our understanding of entropy at a philosophical level, allowing the concept to move beyond physical measurement and become a category for understanding general patterns in the world. On the other hand, it gives the dialectical view of contradiction a scientific content that can be described and quantified, so that the proposition "contradiction is the motive force of development" acquires a concrete form when seen through entropy.

By bringing together the perspectives of natural science and materialist dialectics, we can understand more deeply that from the random fluctuations of microscopic particles to the grand evolution of galaxies, from the birth and death of life to the rise and fall of civilizations, the eternal motive force driving all motion, change, and development can be traced back to the contradiction between entropy decrease and entropy increase within the order at every level of things. The universe is like a great theater, eternally performing the entropy dance of order in which all things take part. In the dialectical rhythm of entropy increase and entropy decrease, stars are born and die, all things form and dissolve, life renews itself, and civilizations evolve.

Mao Zedong offered a concise philosophical formulation of this point: "Contrary to the metaphysical world outlook, the world outlook of materialist dialectics holds that in order to understand the development of a thing we should study it internally and in its relations with other things; in other words, the development of things should be seen as their internal and necessary self-movement, while each thing in its movement is interrelated with and interacts with

the things around it."

This observation reveals something profound: whether we are considering fluctuations among microscopic particles or the rise and fall of civilizations, the source of all change lies in the movement of contradictions inherent within things and in their interaction through universal connections with the world outside them.

The dialectical movement of the contradiction between entropy-decreasing and entropy-increasing forces thus profoundly connects the essence of existence—layered ordered existence—with the essence of change—the contradiction between entropy decrease and entropy increase. We will next take the human being and the business enterprise as two typical layered ordered systems and, from the perspective of "layered ordered existence," consider how to discern the direction of change within their entropy contradictions.

Take the human being. We can focus on three typical levels, the cellular level, the organ level, and the whole body.

At the cellular level, a cell is an ordered structure in which organic molecules are highly organized. Metabolism—the acquisition of energy, repair, and synthesis—is its entropy-decreasing force, maintaining its internal order. Inadequate energy input, accumulated damage—including DNA mutations and protein misfolding—and aging—including telomere shortening—constitute its entropy-increasing force, leading to cellular dysfunction or death and intensifying disorder.

At the organ level, an organ is an ordered structure formed through the precise organization and cooperation of particular groups of cells. Healthy cell renewal and functional coordination

are its entropy-decreasing force, keeping the organ operating in an ordered and efficient way. Inflammation and necrosis across large numbers of cells can be regarded as its entropy-increasing force, impairing the organ's function and disrupting its order.

At the level of the human body, the body is a higher-level ordered system composed of organs and connecting networks such as nerves and blood vessels. Energy intake, exercise, and rest maintain coordinated homeostasis among the organs and thus express an entropy-decreasing force. The aging of organs and imbalances in such functions as qi, blood, and the meridians[1] may be regarded as entropy-increasing forces, driving the individual toward increasing entropy and systemic collapse, until life ends.

The cellular, organ, and whole-body levels are nested within one another and together constitute a living organism. Changes in order at each level follow the distinctive laws of that level.

Entropy decrease and increase within a single cell produce changes in its order, as quantitative change, and may even lead to life or death, as qualitative change; from the perspective of the organism as a whole, however, these can be seen as random fluctuations at a lower level. The coordinated behavior of large numbers of cells emerges as ordered change at the level of an organ, that is, as a tendency toward entropy increase or decrease at that level. Small short-term fluctuations produced by the organ's internal entropy contradiction usually do not cause marked dysfunction in the body as a whole. Yet when such fluctuations continue to

[1] In traditional Chinese medicine, qi, blood, and the network of channels or meridians describe bodily vitality, nourishment, and pathways of circulation as an interdependent whole.

accumulate and cross a threshold, becoming qualitative change, they can establish a tendency powerful enough to alter the body.

To understand a living organism fully and deeply, we must systematically discern its ordered composition at every level, from the microscopic to the macroscopic, together with the changing contradictions between entropy decrease and increase. Bian Que's diagnosis offers an analogy.[2] Only by understanding the body's layered ordered structures—the "interstices," the "stomach and intestines," the "bone marrow," and more—and the laws governing their changes, as revealed by fields such as cell biology, physiology and pathology, and clinical medicine, can one distinguish both the level and the stage of an illness: whether the disease lies in the interstices, in the stomach and intestines, or in the bone marrow, and then treat it precisely.

Radiotherapy and chemotherapy, for example, are common treatments for cancer. In the long term, they benefit the patient as a whole and thus represent entropy decrease. In the short term, however, they cause injury at the cellular and organ levels, representing entropy increase. If this short-term injury to cells and organs becomes severe enough, it may also cause long-term, irreversible harm to the patient as a whole, again representing entropy increase.

A treatment can therefore have entirely different effects at different levels and on different timescales. The *Huangdi Neijing*[3]

[2]Bian Que is the name attached by tradition to an early master physician. In the well-known story alluded to here, he recognizes an illness as it moves from the skin or interstices inward to the organs and finally the bone marrow, where treatment is no longer possible.

[3]The *Huangdi Neijing*, or *Inner Classic of the Yellow Emperor*, is the foundational

likewise regards the human body as an organic whole and emphasizes the organic connections among its internal systems. Good medical decisions must rest on a deep understanding of how cells, organs, and the individual operate at their several levels, weighing benefits and harms with care amid the interwoven dynamics of entropy decrease and entropy increase.

Like the human body, the business enterprise follows the same dialectical law of entropy contradiction in its layered structure of order.

At the employee level, employees are the basic constituent units of an enterprise, and their order appears in the stability of their work effectiveness and skill structure. Active learning and efficient execution constitute the individual employee's entropy-decreasing force and sustain that order. Passivity and the deterioration of ability constitute the employee's entropy-increasing force and lead to disorder.

At the departmental level, a department is an ordered organization formed as employees divide their labor and cooperate. Team cohesion and process optimization express the organization's entropy-decreasing force and improve effectiveness. Fragmented collaboration and internal competition for resources express its entropy-increasing force and weaken its function.

At the enterprise level, a company is a complex system in which departments are linked through strategic coordination, resource allocation, a cohesive culture, and information networks. Sound decision-making and long-term planning are its entropy-

corpus of classical Chinese medicine, framed largely as dialogues between the Yellow Emperor and his physicians.

decreasing force and promote orderly development. The absence of goals and managerial disorder are its entropy-increasing force and bring the system into disarray.

The dialectical movement of the contradictions between entropy decrease and entropy increase across the employee, departmental, and enterprise levels is the fundamental impetus of organizational evolution. Only with a layered perspective can we accurately identify the principal and secondary contradictions within the order at each level of an enterprise and grasp its patterns of evolution. Systems thinking, multidimensional thinking, and first-principles thinking, all frequently invoked in management, rest in essence on similar reasoning.

In the operation of an enterprise, the negative emotions of some employees resemble an illness "in the interstices," at the surface or employee level; a disconnect between departmental goals and effective execution resembles an illness "in the stomach and intestines," at the middle or departmental level; and a broken chain of finance resembles an illness that has "entered the bone marrow," at the deep or systemic level.

Suppose, for example, that a company substantially raises the proportion of individual performance bonuses in its sales department in order to increase total revenue and profit—an entropy-decreasing goal. Large bonuses may greatly increase the motivation of sales employees, which can be regarded as an increase in each employee's individual "order." Yet as employees pursue higher personal income, destructive competition, concealment of information, and excessive promises to customers may also grow. These phenomena create disorder in collaboration

and a crisis of trust within the sales department, producing an entropy-increasing tendency at that level. Sales may rise in the short term, but the company's overall operating efficiency, customer satisfaction, employee morale, and long-term strategy may all suffer. The order of the enterprise as a whole may not improve and may even give way to disorder.

When we participate in complex organizations such as enterprises, we often encounter the same problem: a single decision can produce different tendencies of change at different levels of the organization. At times we must endure short-term fluctuations in order to remain committed to a long-term goal; at other times, to survive a temporary difficulty, we may permit short-term conduct that conflicts with that goal. Only systematic thinking from a layered perspective can lead to a reasonable decision.

If we habitually understand things from appearances and ordinary experience, our perspective can easily become fixed at one level or a few of them. We then struggle to grasp the "actual facts" in full, and it becomes equally difficult truly to "seek truth."

The principle is obvious: when speaking abstractly, we can readily set aside the particulars of a system or thing and discuss a multidimensional perspective. Applying that perspective in practice is another matter. Within a layered ordered system, the order at each level has its own distinctive contradictions and laws. In practice, we must focus on each level to master its particular knowledge and regularities, yet also step beyond any single level and integrate across levels on the basis of what we know about several of them.

When we focus on a particular level, the central task is to seek its determinate regularities amid uncertainty. We delimit the object

of study at that level, simplify and model it as needed, and use formal logic—which began with Aristotle and was later developed into modern mathematical logic by Frege, Russell, and others—to eliminate contradictions, define conceptual boundaries, and extract determinate regularities from the simplified model. This method of analyzing the regularities of a particular level in depth is reductionism.

Only after mastering the internal regularities of several levels can we begin to connect knowledge across them, achieve a synthesis greater than a simple sum, and gain a deeper, more integrated understanding of a thing's complexity. On the basis of the three laws of dialectical logic—the unity of opposites, quantitative and qualitative change, and the negation of the negation—we can reveal the entropy contradiction at each level, distinguish principal from secondary contradictions, and explore how interactions among levels give rise to new regularities and properties of the whole. Dialectical logic was systematized by Hegel and reworked by Marx as materialist dialectics. The method that breaks through the limits of a single level or discipline and combines partial views into a comprehensive understanding is emergentism.

Reductionism and emergentism are not alternatives from which only one can be chosen. Together they form the complete process of cognition and practice. Their dialectical unity deeply echoes Kant's insight into human cognition: alongside analytic reduction, human understanding and scientific inquiry also require a dialectical method of synthesis.

In business management, executives with technical backgrounds often rely on reductionism and formal logic. They tend to

view problems through an "engineer's lens," focusing on relatively determinate regularities at specific levels, such as equipment performance parameters and code-execution efficiency, and may readily reduce the enterprise to a linear sum of technical indicators.

Executives with backgrounds in management are more inclined toward emergentism and dialectical logic. From an "organizational perspective," they attend to the overall effects that emerge from cross-level interactions, including coordination among departments, organizational culture, and incentive mechanisms. They often attribute technical problems to defects in higher-level systemic mechanisms, such as strategy, organization, or culture.

The two kinds of managers are like speakers of different "cognitive languages." Technical managers seek a "local optimum," maximizing a particular technical indicator, while managers trained in management emphasize "global adaptability," the robustness and long-term development of the system as a whole.

Without a shared framework for understanding several levels at once, the two sides not only struggle to agree on solutions to particular problems but may fail even to understand one another's central concerns and value judgments. Executives with technical backgrounds may dismiss broader managerial considerations as "empty talk," while those with management backgrounds may regard the technical executives as "shortsighted." Looking at things through layered order and its changes can help us see them with real clarity and come into tune with more people.

In analysis within synthesis, reductionism, through formal logic, lays the foundation for emergentism, keeping the theory of emergence from floating free of substance: a guiding principle without

concrete method is empty. Mastering the internal regularities of order at each level is the provisional end point of reductive inquiry, but it is the starting point of emergent thought. Only by synthesizing the regularities of all levels can we bridge the "explanatory gaps" between them.

Research on consciousness, for example, studies the emergent properties of large-scale coordination among brain regions from the holistic perspective of cognitive science and neuroscience, but it still depends on a deep understanding of neurotransmitter dynamics supplied by the reductive work of biochemistry and molecular neurobiology. A reliable model of ecosystem stability must likewise include concrete mechanisms of competition, symbiosis, and other relations among species, as studied reductively in population ecology.

In synthesis within analysis, emergentism, through dialectical logic, charts the course for reductionism, preventing it from becoming trapped within a professional rut: method without a guiding principle is blind. Emergentism supplies a framework for understanding the whole. Like a conductor, it ensures that reductive analysis proceeds in the right direction and within the right context.

Deciphering the code of DNA through reductive analysis, for example, also depends on an understanding of the living system as an emergent whole; research on elementary particles through reduction acquires meaning only when placed within a model of cosmic evolution, the emergent whole.

Only within the dialectical unity of analysis and synthesis, of formal logic and dialectical logic, of reductionism and emergentism, can cognition both penetrate the exquisite detail of the micro-

scopic and survey the grandeur of the macroscopic. Between the microscope of reductionism and the wide-angle lens of emergentism, cognition achieves a spiral ascent across levels, a process of the negation of the negation.

This profound dialectical unity was raised to a philosophical plane in Qian Xuesen's theory of "open complex giant systems." His approach to understanding complex systems combined reductive analysis with a grasp of the whole and established a methodology of metasynthesis from qualitative to quantitative understanding.

Chapter 8

Living Against Entropy

To live is to resist the increase of entropy; life feeds on negative entropy.

Erwin Schrödinger

Imagine a cold winter morning. You hold a freshly brewed cup of hot tea in both hands. Steam curls upward, while warmth passes from the scalding cup into your cool palms and chases away the chill. Yet the tea slowly cools and eventually approaches room temperature. Heat flows spontaneously from hot to cold until the temperatures equalize.

Beneath this kind of phenomenon lies a fundamental law of the universe. In the nineteenth century, physicists such as Lord Kelvin and Rudolf Clausius generalized from such observations and arrived at the second law of thermodynamics.

Kelvin formulated it this way: it is impossible to extract heat from a single source and convert it entirely into useful work without producing any other effect. Clausius gave a more intuitive formulation: heat always flows spontaneously from a hotter body to a colder one. As with that cup of tea, without outside intervention, the heat it loses will never spontaneously gather itself together again

and bring the tea back to a boil. This unidirectionality and irreversibility of energy transfer is the central insight of the second law.

Beneath the apparent cooling, however, Clausius discerned a deeper and more general law of change: all things spontaneously move toward chaos and disorder. If we use entropy to measure this tendency, we arrive at the entropy formulation of the second law, the famous law of entropy increase. In an isolated system undisturbed by the outside world, entropy never decreases. It either stays the same, which corresponds only to an ideal reversible process, or it increases.

Statistical mechanics later supplied a microscopic explanation for the law of entropy increase. Ludwig Boltzmann pointed out that entropy is, in essence, a measure of the total number of microscopic states available to a system: macroscopic entropy increase is the statistical inevitability of a system evolving toward a macrostate containing more microscopic states. Within the framework of statistical ensembles, Josiah Willard Gibbs then defined entropy as a measure of the "uncertainty" of a system's probability distribution in phase space, showing that when an isolated system reaches thermodynamic equilibrium, its entropy attains the maximum allowed by the given constraints.

The law of entropy increase rests on extremely strong experimental and theoretical foundations, and no reliably verified counterexample has yet been found. It therefore enjoys an almost axiomatic status in thermodynamics. Arthur Eddington wrote that it "holds, I think, the supreme position among the laws of Nature."

Since the universe as a whole can be regarded as an approx-

imately isolated system, Clausius deduced that the total entropy of the universe would ceaselessly rise and might eventually lead to heat death, when all things fall silent. Although the heat-death hypothesis remains controversial, within the prevailing thermodynamic and cosmological picture, the total entropy of the universe is still understood to be increasing.

The interpretations of the law of entropy increase offered by thermodynamics and statistical mechanics are also a cornerstone of modern physics' understanding of the nature of time. Change defines time, while entropy increase marks the direction of change and time in the universe. It is like an arrow released from the bow, forever flying toward the abyss of chaos. Eddington called it "time's arrow," and all things in the universe follow its trajectory.

Stars burn to ash. Mountains erode to sand. Order will eventually yield to disorder. And so, when you hold a cup of hot tea and feel its warmth slowly drain away, that deepening coolness is not only the passage of time. It is an intimate experience of the deepest law of the universe.

The *Daodejing* says, "The Way of Heaven takes from what has too much and gives to what has too little." Laozi's phrase contains a philosophical insight akin to the law of entropy increase: an irreversible force that governs the evolution of nature and moves toward equilibrium. This highest law also permeates every moment of ordinary life, silently shaping the changes around us.

Slurp a bowl of *luosifen*[1] in a relatively enclosed railway carriage, and its unmistakable aroma will soon fill the entire space. A

[1] *Luosifen* is a rice-noodle soup associated with Liuzhou in Guangxi. Its spicy snail-based broth and fermented bamboo shoots give it a famously pungent aroma.

neat, clean room becomes hopelessly untidy if no one keeps putting it in order. A classroom without effective management becomes noisy and chaotic. Irregular habits of sleep and diet throw a healthy body out of balance and gradually weaken it. The disorderly torrent of information on social media washes over our attention and, like entropy increase breaking down order, continually erodes the mind's focus and clarity.

To be sure, some rigorous scholars are skeptical about extending the concept of entropy into everyday life, economics, management, and the philosophy of life. They see a risk of overextension or misuse.

In fact, even within physics, the concept of entropy has a distinctly metaphysical cast. An entropy value is difficult to measure directly in an experiment. Thermodynamic entropy can only be inferred indirectly from observable state variables; entropy in statistical physics rests on hypothetical models of probability distributions; entropy in complex systems is difficult to quantify because of emergence across levels and incomplete information; calculations of quantum entanglement entropy are constrained by the nonlocal character of the information; and even the measurability of information entropy depends on our having "tamed" the object of measurement in advance.

To "measure the entropy of a thing" is, in substance, to seek an almost complete understanding of its layered ordered structure. When we measure its entropy, we are really hoping to deconstruct the thing itself in full. This brings the fundamental contradiction between existence and cognition into sharp relief: a thing, as order, is a dialectical unity of randomness and determinacy, while the hu-

man grasp of its "degree of disorder," or entropy, is always limited by the level of our understanding and the completeness of our information.

From the nature of ordered existence and change in all things, we can also derive a universal law of entropy increase.

As noted earlier, if a thing were absolutely isolated, its changes would be due entirely to internal causes: disorder at the lowest level, with its intrinsic randomness. At every moment, this underlying disorder struggles to break free of the constraints imposed by ordered structure, generating an entropy-increasing force. Because of universal interconnection, the ordered structure at each level of the thing itself also passively produces an entropy-decreasing force that sustains it. In an isolated thing, however, the spontaneous entropy-increasing force of disorder at the lowest level will eventually overwhelm the passive entropy-decreasing force. Randomness at the lowest level gradually dismantles the ordered structure, which moves irreversibly toward disorder.

Entropy and the law of entropy increase are therefore not confined to physics. They reveal a universal rule in the contest between order and disorder throughout existence and possess a profound generality. When ecology examines flows of negative entropy in order to sustain a system, or management draws on open-systems theory to resist organizational rigidity, this is not a misuse of the concept. It is precisely an expression of how human beings use scientific reason to understand and respond to the patterns of a complex world.

The layered ordered structure of a thing forms a boundary between it and its environment, and with that boundary comes a rela-

tive "isolation." If the thing were wholly open to the world outside and indistinguishable from it, the thing itself would cease to exist. Yet within the universe's web of order, which has neither outside nor inside, all things are universally interconnected and no absolutely isolated system exists. If a thing had no relation to any part of the universe, that too would mean that it did not exist.

The nature of a thing as ordered existence therefore determines that isolation and openness form a dialectical unity of opposites. Both "isolation" and "openness" have only relative meaning.

Perhaps only the universe as a whole can be regarded as an isolated system, its evolution affected only by internal causes. Its overall tendency must therefore be toward entropy increase. Within the universe, however, there is no absolutely isolated thing. The changes of all things are simultaneously affected by internal and external causes, allowing the miracle of local entropy decrease to occur. More profoundly, the entropy increase of the universe as a whole means an increase in the total number of its internal microscopic states. This gradually unfolding disorder, paradoxically, creates the possibility for new order to emerge.

Boltzmann's statistical view suggests a vivid metaphor: the evolution of the universe is like a great river. Its overall flow cannot be stopped, but within the river there are always eddies and vortices. In the flood of cosmic entropy increase, local "vortices of order" may likewise form through fluctuations.

Ilya Prigogine's theory of dissipative structures, for which he received the 1977 Nobel Prize in Chemistry, gave a scientific account of this process. When an open system is far from equilibrium and receives a continuous inflow of energy or matter, it may use these

flows of negative entropy to build and maintain a locally ordered, low-entropy structure, and thus achieve self-organization and evolution. Local entropy decrease does not, of course, violate the law of entropy increase: it must be paid for by a greater increase in entropy across the wider environment. Moreover, only when driven by energy from the external environment can a thing rely on its own internal process of self-organization to maintain or form an ordered structure. Once the external input can no longer be sustained, internal causes will again carry the thing toward increasing entropy.

This is just as materialist dialectics maintains: "External causes are the conditions of change, internal causes are its basis, and external causes operate through internal causes."

In the universe, there is no "ordered structure" more magnificent or more typical than a star. The life of a star is a majestic epic: it forms a local "vortex of order" in the flood of cosmic entropy increase, only to succumb in the end to the law of entropy increase.

At the birth of the universe, there were only light elements such as hydrogen and helium, along with extremely small amounts of lithium, diffused almost uniformly through interstellar space in a state of high-entropy, disordered chaos. Gravity shattered this chaos. Under its own gravitational pull, a tenuous gas cloud began to collapse locally. In this process, gravitational potential energy was converted into thermal energy and radiation, both hurled into the cold universe and causing a sharp increase in the total entropy of the environment. In exchange, a core of extremely high density and temperature formed inside the nebula: a protostar, a highly ordered structure born from chaos and therefore a local entropy decrease. As the density and temperature inside the protostar rose

further, hydrogen fusion finally ignited.

Through hydrogen fusion in its core, a star generates outward radiation pressure to resist the inward crush of gravity, establishing a delicate dynamic balance. Inside the star, low-entropy fuel—hydrogen—is converted into high-entropy products—helium and photons. This process is itself an entropy increase. The light and heat it radiates continually contribute to the entropy increase of the cosmic environment. It is precisely this continuous process of energy dissipation that barely sustains the star's own highly ordered, low-entropy structure.

We can say that a star maintains its local order at the cost of burning itself and increasing the total entropy of the universe. At this stage, the star is in the prime of its life; the Sun is currently in this stage.

When the hydrogen in the core is exhausted, the balance is broken. Gravity once again gains the upper hand, and the core begins to contract and heat up. This, in turn, ignites hydrogen burning in the shell and then helium fusion in the core, producing carbon and oxygen. These more violent nuclear reactions generate enormous energy and thrust the star's outer layers outward. The star becomes a red giant or red supergiant.

At this stage, the star greatly increases its volume and outward radiation, increasing entropy, in exchange for a new order in which heavier elements are formed under higher temperatures and pressures in the core. But this comes at the cost of the stability of the star's overall structure. The stellar system grows chaotic and unstable, marking its entry into old age and preparing the way for its final, irreversible movement toward increasing entropy as a whole.

The ultimate fate of a star depends entirely on its initial mass.

For stars of low or medium mass—less than eight times the mass of the Sun—the core lacks enough mass to ignite carbon fusion once helium burning ends. Gravity crushes the core into one of the most ordered and densest celestial bodies in the universe: a white dwarf, an extreme local entropy decrease. The formation of the white dwarf releases enormous gravitational potential energy, while the star's outer gaseous layers are ejected into space to form a planetary nebula. That is the entropy-increasing price paid by the star. The ejected material is rich in hydrogen and helium, as well as carbon, oxygen, nitrogen, and other elements manufactured during the red-giant phase.

For massive stars—those with more than eight times the mass of the Sun—gravity remains strong enough to ignite successively the fusion of carbon, oxygen, silicon, and other elements. The star burns in onion-like layers, forging iron, the most stable element, in its core. Iron fusion cannot release energy because iron has the highest binding energy per nucleon; further fusion instead absorbs energy. The stellar core then loses its support in an instant and undergoes a terrifying gravitational collapse lasting mere seconds, forming a neutron star or black hole: an inconceivable low-entropy state.

This process is a supernova explosion, and it is also a violent entropy-increasing event. In an instant, the energy of the explosion hurls back into chaotic interstellar space all the elements the star created over its lifetime, together with heavy elements such as gold, silver, and uranium rapidly formed in the explosion. Neutron-star mergers also create heavy elements.

A star is a typical ordered existence in the universe. Its life shows that the formation of any local entropy decrease must be paid for by a still greater entropy increase in the environment. Beginning with the lightest elements, a star forges richer elements over the course of its life, and in the end returns all this dust to the universe in an explosive death. That dust will later condense into new stars and planets. Though the individual must die, a star also seems to carry out a kind of transmission between generations.

Physicists believe that the Sun is a grandchild of the universe's first stars. Material left over from the Sun's formation produced the planets in orbit around it. In its early history, Earth was extremely hot and molten. Dense substances such as iron and nickel sank toward the center to form the core, while less dense silicate rock rose to form the mantle and the earliest crust. Violent impacts by asteroids and comets brought water and materials essential to life, creating the conditions from which oceans and life could appear.

As the astronomer Carl Sagan said, "We are made of star-stuff." The elements that stars scattered in death ultimately condensed on Earth into life capable of looking up at the stars and asking about its own origins. Life itself is a transient, beautiful, and tenacious ripple moving against entropy within the vast, irreversible epic of cosmic entropy increase.

As early as 1943, Erwin Schrödinger introduced the concept of "negative entropy" in *What Is Life?*, bringing entropy theory into close relation with the essence of life for the first time. He argued that life's only way of resisting cosmic entropy increase is to remain selectively open to its environment: continually taking in "negative entropy," in such forms as food and sunlight, to maintain its own

highly ordered structure and function, or low entropy, while releasing low-energy, high-entropy waste, such as carbon dioxide and heat, back into the environment.

The cell can be regarded as the smallest unit of life, and its way of surviving reveals a deep pattern shared by all living things. Plant and animal cells alike achieve a conditional openness through the cell membrane: while maintaining a degree of independence, they resist entropy increase through interaction with the world outside. Let us therefore take the cell as our example and see how life's smallest unit stores and uses energy effectively, and how it lives on negative entropy.

At the microscopic level, a cell resists entropy increase by establishing an intricate system for circulating and managing energy. Adenosine triphosphate, or ATP, plays an irreplaceable role in this system as the cell's "energy currency." The cell uses it ingeniously to drive two essential processes: constructing the boundary of its own ordered structure and powering the many physiological activities that consume energy.

ATP is like a rechargeable battery that stores chemical energy in its high-energy phosphate bonds. When the cell needs energy, it hydrolyzes an ATP molecule, converting it into adenosine diphosphate, ADP, and phosphate, and releasing the energy stored in the ATP. This process is itself an entropy increase. But the cellular processes it drives create and maintain local order.

The energy released by ATP hydrolysis drives ion pumps embedded in the cell membrane, such as the sodium-potassium pump, which move sodium ions out of the cell against their concentration gradient while moving potassium ions into it against theirs. Like

rowing against the current, this process requires a continuous expenditure of energy. Its immediate result is the establishment and maintenance of a marked difference in ion concentration across the cell membrane. More importantly, the active separation and directed transport of charged ions together construct an electrochemical gradient—negative inside and positive outside—across the inner mitochondrial membrane: the membrane potential.

This invisible "membrane-potential barrier" is, in reality, a highly ordered, low-entropy structure. It is not only the "potential-energy battery" that the cell uses to drive subsequent work, such as the conduction of nerve impulses and the absorption of nutrients. At the physical level, it also defines and guards the integrity of the cell as an independent entity. It is the first line of defense against the chaotic, disordered motion of molecules in the outside world.

At the same time, the cell's synthesis of biological macromolecules such as DNA and proteins, as well as the driving of cell division and movement, likewise requires the hydrolysis of ATP for direct energy supply. These energy-consuming processes are themselves entropy increases. Their ultimate purpose, however, is to construct and maintain a local, dynamic island of order in the chaotic sea of disordered molecular motion.

ADP is not the endpoint of this anti-entropy struggle. In order to maintain the stability of the membrane potential continuously, carry out biosynthesis, and meet the demands of the various activities of life, the cell must ceaselessly "recharge" spent ADP into ATP. This vitally important counter-entropy process depends on the cell's capture of free energy from the external environment through respiration or photosynthesis. This captured energy drives ADP and

phosphate to recombine and form new ATP, thus completing one cycle of the energy currency.

In short, a cell survives through a ceaseless ATP–ADP cycle. By hydrolyzing ATP, it pays the "negative entropy" needed to construct and maintain order, from membrane potential to biological macromolecules. At the same time, it draws on energy flowing in from the environment to regenerate ATP and thereby sustain itself as a low-entropy body.

Each cell in the human body contains, on average, one million ATP synthase enzymes. These miniature engines work day and night inside the mitochondria, continually manufacturing ATP. The total number of such miniature engines in all the cells of an adult human is estimated at roughly 10^{19} to 10^{22}. They can produce roughly 2×10^{26} ATP molecules each day, meeting the constant need of all the body's cells for energy to remain alive and carry out division.

The feat of life in resisting entropy increase does not stop at the single cell. By constructing the multicellular organism, an even more magnificent fortress of order, life achieved a leap to a higher level. Countless cells differentiate and cooperate. Their collective behavior gives rise to higher-level structures and functions that no single cell possesses. Together they maintain the entire living organism, this vast and ordered low-entropy system.

The end of an individual life, whether the apoptosis of a cell or the death of an organism, confirms the limits on how long local order can endure. Yet through the transmission of genes from one generation to the next, life achieves another kind of breakthrough against entropy. An individual's physical structure must eventually return to chaos, but the "ordered blueprint" it contains—its DNA—

can be copied and passed on, allowing the order of life to be continually rebuilt and transformed across the long river of time on Earth.

If we pursue the ultimate source of life's energy, we arrive at the light released by burning stars; the birth of stars, in turn, is due to universal gravitation. "Heaven and Earth are not benevolent; they treat the ten thousand things as straw dogs." The universe contains no force beyond nature. A mayfly breaks from its chrysalis and takes wing; grass spreads from a crack in the rock; a newborn child gives its first cry. Every living thing is a miracle moving against the current of entropy.

Even the humblest life deserves our awe. Every cell within a living body is waging an exquisitely precise battle against entropy in a microscopic universe: capturing energy from light or chemical bonds, storing and releasing it through ATP, and using ion pumps to build the barrier of membrane potential. These flickering flames of "negative entropy" bear the most radiant glory of cosmic evolution.

The stardust scattered by dying stars gives birth to new solar systems. Life rebuilds order across the generations through the transmission of genes. All things ultimately perish, and in perishing they create the possibility for new things to be born.

This naturally raises a larger question. Does the universe we inhabit itself follow a similar rhythm? Is heat death, the endpoint at which all things return to silence, its sole, unalterable fate?

Modern cosmology has proposed deeper conjectures. The "cyclical universe" theory imagines that our universe may also be situated within a cycle of birth and death without beginning and without end. Whether the end is heat death or the Big Rip, perhaps

neither is an absolute termination. They may be a grand act of "metabolism," the necessary chaos and preparation before the start of the next cosmic cycle.

This remains a frontier exploration of theoretical physics, far from being settled. But whatever the outcome, the picture of cosmic evolution has already supplied the most magnificent backdrop for our understanding of living against entropy.

Chapter 9

Out of Nothing

Being arises from Becoming.

John Archibald Wheeler

Across human imaginings of the world's origins, chaos appears as a starting point shared by nearly every ancient civilization. The Chinese mythological classic *Sanwu Liji* says, "Heaven and Earth were in chaos, like a chicken's egg, and Pangu was born within it."[1] At the opening of the *Theogony*, the ancient Greek poet Hesiod likewise declares, "Verily at the first Chaos came to be." East and West alike imagined that before the birth of all things there was an indescribable "primordial soup," holding limitless possibilities: chaos.

Chaos is not the disorderly confusion we ordinarily imagine. It is the most primordial, undifferentiated state: neither light nor dark, neither form nor shape, neither Heaven nor Earth, neither thing nor self. It is the Non-being that precedes the birth of Being.

The end of chaos is often accompanied by a magnificent act of "creation."

[1]The *Sanwu Liji*, a third-century work attributed to Xu Zheng, preserves one of the earliest written versions of the Pangu creation myth.

In Chinese mythology, the giant Pangu slept within chaos. When he awoke, he swung his great axe: what was clear and light rose to become Heaven, while what was turbid and heavy sank to become Earth. In the end, his body became mountains and rivers, the Sun, the Moon, and the stars, fulfilling the idea that "chaos gives birth to the ten thousand things." In the *Daodejing*, Laozi wrote, "Non-being names the beginning of Heaven and Earth," and "The ten thousand things under Heaven are born of Being; Being is born of Non-being." The allegory in the *Zhuangzi* chapter "Responding to Emperors and Kings," in which Shu and Hu bore openings into Hundun,[2] likewise suggests that nature in its original state often contains the wholeness of undivided chaos.

In ancient Greek myth, Chaos itself possessed generative power, directly giving birth to primordial deities such as Gaia, the Earth; Tartarus, the Abyss; and Eros, Love. In Norse myth, Ginnungagap was a bottomless void filled with primordial force. There ice and fire interacted, giving birth to the first giant, Ymir, and to the ancestor of the gods.

Myths, legends, and ancient philosophies alike thus share an implicit conception of creation out of nothing. The world we know, with all the order of its ten thousand things, arose from a primordial chaos that seemed empty and void yet held infinite possibility within it. This primordial chaos is Non-being. It is disorder. Order emerged from this foundational state of disorder.

Strikingly, modern science's understanding of "nothing"

[2] *Hundun* means an undifferentiated or uncarved chaos. In the *Zhuangzi* tale, Shu and Hu repay Hundun's hospitality by boring the seven human openings into him; when the last opening is made, Hundun dies.

resonates deeply with the concept of chaos found in myth and classical philosophy.

Quantum mechanics reveals that the vacuum, apparently empty, is not silent and dead. It is filled with violent quantum fluctuations. On extremely short time scales, the energy of the vacuum undergoes unpredictable random fluctuations. Virtual particle pairs, such as electron-positron pairs or quark-antiquark pairs, flash into existence from nowhere and annihilate in an instant.

The "Non-being" corresponding to quantum vacuum fluctuations is likewise no state of eternal stillness. It is dynamic and contains limitless possibilities.

In its account of how the universe and all things came into being, modern science likewise resonates across time and space with myth and classical philosophy. The Big Bang theory, widely accepted in the mainstream scientific community, presents a modern picture of chaos giving birth to the ten thousand things.

The universe began in a theoretical singularity, a state of infinitely small volume, infinitely great density, and infinitely high temperature. After the Big Bang, the universe emerged from a background of quantum fluctuations, and the fundamental interactions—gravity, electromagnetism, the strong force, and the weak force—began to separate. Quark-gluon plasma and the elementary particles then appeared in a highly disordered state of thermal equilibrium, as particle-antiparticle pairs were continually created and annihilated.

The strong interaction was the first to separate as an independent fundamental force. Under its influence, quarks and gluons combined to form protons and neutrons, and ordered structures began

to appear at the microscopic level. The basic properties of elementary particles and their interactions gradually settled into the forms we observe today.

About one second after the Big Bang, the temperature dropped to the point where large numbers of electron-positron pairs annihilated. The kinetic energy of the remaining protons, neutrons, and electrons decreased as the universe expanded and cooled. A few minutes later, at about three minutes, the temperature fell to about one billion kelvin. A small fraction of the protons and neutrons combined to form deuterium nuclei and helium nuclei. Most of the protons remained as hydrogen nuclei.

About three hundred and eighty thousand years later, the temperature dropped to about three thousand kelvin. Electrons had slowed enough to be bound by atomic nuclei, forming neutral hydrogen and helium atoms. The radiation left by this process, the cosmic microwave background, can still be detected today. At this time, regions made slightly denser by earlier quantum fluctuations began, under the influence of gravity, to gather matter and nurture the gas clouds, stars, and galaxies of the future.

About one to two hundred million years after the Big Bang, the first stars in the universe formed and began to shine. Through nuclear fusion, they synthesized large quantities of metallic elements up to and including iron. Elements heavier than iron arose principally from violent events late in stellar evolution, including supernova explosions and neutron-star mergers.

About five to eight hundred million years later, the Milky Way, the galaxy that contains our solar system, began to form its early structure. About nine billion years after the Big Bang, the solar sys-

tem and Earth condensed within a protoplanetary disk. About ten billion years after the Big Bang, roughly 3.8 billion years ago, life began to appear on Earth.

If the Big Bang was the first great instance of "something from nothing" at the material level, then the birth of life on Earth was the second. Under particular conditions, ordered matter spontaneously organized itself to perform the functions of information storage and metabolic cycling, making the leap from the nonliving to the living and creating a higher form of layered ordered structure.

When Earth first condensed, its environment was extremely hot, and its surface was covered by an ocean of lava. The early atmosphere was rich in water vapor, carbon dioxide, nitrogen, hydrogen, and small amounts of methane and ammonia, but lacked free oxygen. It was like a primordial cauldron of chaos, brimming with chemicals and energy. As Earth cooled, water vapor condensed to form the primordial oceans roughly 4.4 billion years ago.

The planet's surface and young seas were then subjected to immense energy inputs: intense ultraviolet radiation from the sun, frequent lightning, active volcanic eruptions, and meteorite bombardment.

In the primordial atmosphere and oceans, the continuous energy of lightning, ultraviolet radiation, geothermal heat, and meteorite impacts drove simple inorganic molecules—methane, ammonia, water, hydrogen, and carbon dioxide—through immense numbers of random collisions and chemical reactions. Through pathways of abiotic synthesis, these disordered collisions at the microscopic level gradually produced the building blocks of life: small organic molecules including amino acids, nucleotides, simple sug-

ars, and lipids.

These molecules accumulated and mixed in the primordial oceans, forming what is known as the "primordial soup" of life. Under specific physicochemical conditions, such as catalysis on clay-mineral surfaces, cycling at hydrothermal vents, and periodic alternation between wet and dry conditions, these small organic molecules began, through dehydration-condensation and similar reactions, to link into longer-chain polymers. Amino acids formed primordial polypeptide chains, the precursors of proteins. Nucleotides formed primordial nucleic-acid chains, the precursors of RNA and DNA. Lipid molecules spontaneously assembled in water into enclosed bilayer structures, the primordial membranes.

At this point, ordered structures on the microscopic scale began to emerge spontaneously from chemical chaos.

A decisive leap followed. Certain molecules of ribonucleic acid, or RNA, could both store genetic information and perform limited catalytic functions; catalytic RNA of this kind is known as a ribozyme. Through this dual capacity, they came to the fore in a form of molecular "natural selection." RNA "replicators" that could remain relatively stable and catalyze the replication of themselves or similar molecules began to gain an advantage, even though this catalysis was initially inefficient and prone to error.

The emergence of information storage, replication, and mutation, together with a rudimentary mechanism of function-based "selection," marked the transition from complex chemical systems to primitive life with the capacity for genetic evolution. This was a great emergence of order at the level of molecular information.

Alongside this, vesicles formed by lipid molecules were able to

enclose a microenvironment containing RNA replicators, polypeptides, other organic molecules, and water, forming primitive protocells.

This physical separation was revolutionary. It created a relatively stable internal chemical microenvironment, protected fragile molecular systems, greatly promoted more effective interaction and coevolution among molecules, and established a boundary across which matter and energy could be exchanged with the outside world. “Protocells” able to remain stable, obtain nutrients from outside, replicate internally, and divide gained a marked evolutionary advantage.

Over the long course of evolution that followed, spanning hundreds of millions of years, the system grew ever more complex and complete. The more stable DNA replaced RNA as the principal carrier of genetic information. Proteins, with their outstanding catalytic capacity and structural diversity, took over the principal role in carrying out the functions of life. The mechanism of the genetic code, linking nucleic-acid sequences to the amino-acid sequences of proteins, became established. Primitive metabolic networks based on protein enzymes gradually took shape and became capable of using chemicals and energy from the environment, such as the chemical gradients at hydrothermal vents on the ocean floor.

Finally, a primitive cell, or a group of primitive cells, came into being with the shared basic features of DNA-based heredity, protein function, a genetic code, membrane compartmentalization, and basic metabolism. This was the common ancestor of all life on Earth, known as LUCA. It marked the full emergence of life as a distinctive ordered system capable of self-maintenance, replication,

and evolution.

After LUCA, from roughly 3.8 billion years ago to the present, the epic of biological evolution under Darwinian principles began to unfold. Simple prokaryotes, bacteria and archaea, proliferated widely. The Great Oxidation Event triggered by cyanobacteria roughly 2.5 billion years ago reshaped Earth's environment. Eukaryotic cells with complex subcellular structures, including nuclei and mitochondria, appeared roughly 2 billion years ago, followed by the rise of multicellular organisms roughly 600 million to 1 billion years ago. Throughout this history, life's process of ordering continued to deepen and expand, ultimately giving rise, over immense spans of geological time, to the astonishingly rich and complex biosphere we see on Earth today.

It is important to emphasize that the processes described above—the birth of the universe and the emergence of life—are not merely conjectures. They are supported by a large body of repeatable and testable empirical evidence.

In cosmology, Hubble's observations of galactic redshifts, precision measurements of the cosmic microwave background, and the remarkably close agreement between the measured abundances of hydrogen, helium, and lithium and the predictions of primordial nucleosynthesis together provide the observational foundations of the Big Bang theory.

In the life sciences, the Miller–Urey experiment and subsequent research successfully simulated conditions on the early Earth in the laboratory and synthesized many kinds of molecules fundamental to life. Genomic and biochemical studies of living organisms, especially archaea and bacteria, have revealed core metabolic path-

ways and genetic mechanisms shared by all life, providing essential clues for reconstructing the characteristics of LUCA. The discovery and intensive study of RNA's catalytic functions, embodied in ribozymes, have supplied strong experimental evidence for the "RNA world" hypothesis.

Although the logic by which the universe and life arose "out of nothing" is clear, the probabilistic conditions for its realization are so stringent as to seem almost absurd.

If fluctuations in the initial singularity's energy density had deviated by more than 10^{-60}, or if the early universe's expansion rate had differed from the critical value by 10^{-36}, the universe could not have evolved complex ordered structures. If expansion were too slow, the universe would have collapsed back into the singularity; if too fast, matter would have been diluted beyond condensation. If the strong nuclear force increased or decreased by one percent, stars could not forge carbon, oxygen, and the other elements essential to life; if electromagnetism were only slightly weaker, atoms could not exist. If dark-energy density differed by an order of magnitude, the universe would either be torn into nothingness as it expanded or return to heat death as it contracted.

As for the spontaneous formation of self-replicating RNA chains from inorganic molecules, or the synchronized coupling of amino acids, nucleotides, and lipid membranes in the chaotic "primordial soup" to produce a protocell, the probability is so low that it is like expecting a tsunami to sweep up grains of sand and build a functioning nuclear power plant on the shore.

These extreme probabilities, defying common sense, naturally provoke deep questions. Was the universe delicately "tuned"? Was

life deliberately "encoded"? Must we accept the anthropic principle? According to that principle, we observe a universe with such "finely tuned" parameters precisely because only such a universe could produce observers. In other words, our very existence filters the range of possible universes.

The creation "out of nothing" of the universe and life, however, does not depend on a single, isolated chance event. Its deeper mechanism is this: the vast microscopic disorder at the lowest level supplies nearly infinite possibilities, while natural laws at successive levels—the laws of physics, chemistry, and biology—continually filter, combine, and stabilize those possibilities. Through this mechanism, outcomes that conform to particular ordered patterns, though they appear extraordinarily improbable when viewed in isolation, gradually become possible, and even necessary, at the macroscopic scale.

We must not ignore the multi-level structure of order and regard everything as empty. Nor should we ignore that structure and speak directly of probability.

The original, scorching chaos of cosmic creation was filled with an unimaginable abundance of microscopic disorder, and within it lay every possibility. The four fundamental interactions that drive cosmic evolution—gravity, electromagnetism, the strong force, and the weak force—acted like invisible sieves and molds, gradually imposing ordered constraints, level by level, upon this disorder. Gravity selected denser regions from the field of density fluctuations and gathered diffuse matter into nebulae. Nuclear forces specified particular fusion pathways within stellar cores and forged heavy elements. Electromagnetism finely constructed

the structures of atoms and molecules.

Each constraint selected and fixed a more stable and more complex structural order from the vast ocean of random microscopic events, gradually channeling the primordial flood of disorder toward grand ordered structures such as galaxies, stars, and planets. Highly ordered results arise through the gradual "evolution" of disorder under layered constraints of order. Disorder creates order, while the ordered structures already formed then limit and constrain the range and "direction" of disorder.

At the time life emerged, the early Earth inherited the cosmic groundwork of order: stable stellar energy input and an abundance of chemical elements. Upon that foundation it also carried the planet's own violent disorder: volcanic activity, lightning, ultraviolet radiation, and random molecular collisions in the primordial soup. At the microscopic level, this disorder appeared as astronomically many molecular motions and chemical reactions.

The key to the emergence of life likewise lay in the gradual imposition of ever more precise, layered constraints of order. Specific physicochemical conditions, such as liquid water and suitable temperatures, defined the arena in which reactions could occur. Intermolecular forces, such as covalent bonds, hydrogen bonds, and van der Waals forces, constrained the configurations of stable molecules. Autocatalytic reactions and molecular templates, such as RNA, constrained the self-replication and amplification of particular chemical pathways. Finally, the appearance of the cell membrane brought about the most crucial "spatial constraint." By separating the internal and external environments, it greatly promoted the filtering, combination, and stabilization of complex

metabolic and genetic networks within the small space enclosed by the membrane.

Every level of constraint was like the gradual introduction of rules into the dance of random molecular collisions. In the end, those extraordinarily rare, highly complex chemical systems capable of self-maintenance, replication, and evolution were filtered, assembled, and stabilized. The spark of life emerged from the ocean of microscopic disorder.

This resembles the mechanism revealed by the infinite monkey theorem proposed by the French mathematician Émile Borel. Imagine a monkey pressing keys at random on a typewriter. Even given ample time, the probability that it will type the entire *Complete Works of Shakespeare* directly is nearly zero. This corresponds to the extremely low probability of producing a complex ordered structure when the final result is viewed in isolation.

The probability that the monkey will randomly type a single meaningful word, however, is vastly higher. This represents a primary ordered structure emerging from disorder at the lowest level. The critical step is this: if the system can "remember" the words already formed, corresponding to stabilized primary ordered structures, and if subsequent random keystrokes do not begin again from individual letters but instead combine those existing words at random, then the probability of ultimately producing a highly ordered work such as the *Complete Works of Shakespeare* rises enormously. This is the power of multi-level ordered structures to filter and amplify possibility. Combining existing words is equivalent to imposing a higher-level constraint upon a foundation of order already achieved.

In *On the Origin of Species*, Darwin described the process in just these terms: "This preservation of favourable individual differences and variations, and the destruction of those which are injurious, I have called Natural Selection, or the Survival of the Fittest." He continued: "Natural Selection is daily and hourly scrutinising, throughout the world, the slightest variations; rejecting those that are bad, preserving and adding up all that are good; silently and insensibly working, whenever and wherever opportunity offers, at the improvement of each organic being in relation to its organic and inorganic conditions of life."

The emergence of the universe, of life, and of all other things therefore follows an underlying logic: "Disorder creates possibility; order determines direction." As the process of ordering gradually unfolds, vast numbers of microscopic, disordered events provide the foundation of possibility, while ordered structures at successive levels continually filter, constrain, and stabilize. Through this joint action, results that appear to be "miracles of vanishingly small probability" when viewed in isolation can become relatively determinate realities at the macroscopic scale.

This embodies the dialectical unity of chance and necessity, of simplicity and complexity.

This is the very picture evoked in Laozi's words in the *Daodejing*: "The Dao gives birth to one; one gives birth to two; two gives birth to three; three gives birth to the ten thousand things." It is an evolution from undifferentiated chaos, through differentiation into levels, and finally into the ten thousand things in all their structural complexity.

And this also explains, at the deepest level, why layered struc-

tures of order, from elementary particles to atoms and molecules, from cells to ecosystems, from galaxies to social civilizations, are the common property and the intrinsic requirement of all existence. It is the necessary result of the potential of disorder, progressively constrained, filtered, and stabilized through the laws of order.

Chapter 10

Order from Broken Symmetry

Anyone who claims to have discovered a law of nature must provide a way for others to reproduce the result.

Galileo Galilei

At its frontiers, modern physics offers a deeper account, through symmetry and symmetry breaking, of how all things in the universe arise as "something from nothing." Philip Anderson famously wrote, "It is only slightly overstating the case to say that physics is the study of symmetry." Symmetry breaking, in turn, may be called the mother of order. In the search for the universe's fundamental laws, symmetry and symmetry breaking have become a central paradigm and guiding principle of modern physics.

Einstein once wrote, "One may say 'the eternal mystery of the world is its comprehensibility."' The deeper meaning is this: wholly random disorder—that is, non-existence—contains no reproducible pattern and is therefore, in an epistemological sense, "incomprehensible." All things are ordered, which is to say that they exist; they can be understood because they possess ordered features that can be repeatedly verified or predicted—in other words, laws. The most basic scientific requirement of a

law is therefore that it be open to repeated verification. Galileo's insistence on repetition in scientific experiment laid the foundation for our scientific understanding of the laws of nature.

In physics, concepts such as symmetry, invariance, relativity, and covariance each describe, from a different angle, why physical laws can be repeatedly verified.

Symmetry means that the internal structure of a thing or system remains unchanged after a particular transformation. If a physical law is to be repeatedly verified under given conditions, then the thing or system must possess symmetry under some transformation.

For example, the fact that physical laws can be repeatedly verified wherever an experiment is conducted corresponds to spatial translation symmetry. Their repeatability whenever the experiment is conducted corresponds to time translation symmetry. Their repeatability regardless of the direction in which the experiment is oriented corresponds to spatial rotation symmetry.

Invariance often emphasizes that a particular physical quantity or property remains constant under a given transformation. In 1918, the German mathematician Emmy Noether revealed the profound connection between continuous symmetries and conservation laws. The essence of a conservation law is precisely the invariance of a physical quantity. Noether's theorem states that every continuous symmetry corresponds to a conservation law, and conversely.

For example, spatial translation symmetry corresponds to the conservation of momentum, time translation symmetry to the conservation of energy, and spatial rotation symmetry to the conservation of angular momentum.

Relativity extends the range over which physical laws can be repeatedly verified from time, place, and direction to all inertial frames in special relativity, and even to all frames of reference in general relativity. From the perspective of modern physics, relativity is essentially the symmetry of physical laws under a certain class of transformations between reference frames. Covariance means that the mathematical form of a physical law remains the same under a given transformation; it, too, describes the symmetry of laws.

Symmetry is thus the most fundamental concept for understanding why physical laws are repeatedly verifiable and universal. Scientific laws, and indeed all laws, have their own scope and conditions of applicability. That scope corresponds in essence to the range over which a particular symmetry holds. The broader the conditions under which a physical law can be repeatedly verified, the more universal the law and the stronger the symmetry behind it.

We can therefore understand the universality of a law through the scope of its corresponding symmetry, and we can discover more universal physical laws by seeking higher and stronger symmetries.

Before Newton, explanations of natural phenomena relied largely on Aristotelian philosophical speculation and qualitative metaphysical description. Influenced by Galileo, Newtonian mechanics inaugurated an entirely new paradigm of scientific research: its laws could not only explain known forms of motion but also be verified precisely and repeatedly. In terms of the scope of that verification, Newton's laws are universal across all inertial frames in absolute space and time and can therefore be expressed in the same mathematical form. From the perspective of symmetry, they satisfy what is called Galilean symmetry, corresponding to

Galilean spacetime, or absolute space and time.

Newtonian mechanics unified the mechanical phenomena of heaven and earth. Whether an apple falls to the ground or a planet orbits the Sun, both are explained by the same laws of universal gravitation and motion. Maxwell, with his profound mathematical ability and bold physical imagination, achieved another great unification in the history of physics. He brought Coulomb's law of electrostatics, Ampère's law relating electric currents to magnetic fields, and Faraday's law relating changing magnetic fields to electric fields into a universal set of equations for electromagnetic interaction, perfectly describing all electrical, magnetic, and optical phenomena.

However, these two enormously successful theories harbored a sharp contradiction.

Newtonian mechanics requires velocities to obey the rule of addition: the measured velocity of an object changes with the choice of reference frame. Yet Maxwell's equations imply that the speed of electromagnetic waves, or light, in a vacuum is constant and does not depend on the chosen frame of reference.

If Newton's rule for adding velocities is correct, then observers in reference frames moving at different velocities should measure different values for the speed of light. Conversely, if the invariance of the speed of light implied by Maxwell's equations is a fact, then Galilean symmetry and the concept of absolute spacetime must contain a fundamental flaw.

Faced with this contradiction, Einstein did not abandon symmetry; he raised its demands. Not only the laws of mechanics but all physical laws, including those of electromagnetism and light,

should take the same form in every inertial frame. At the same time, the speed of light should not change with the observer's motion.

Meeting this higher demand required Galilean transformations to give way to Lorentz transformations. Space and time were no longer separate, but joined in the unified structure of Minkowski spacetime. Symmetry rose from a symmetry of motion in classical mechanics to a symmetry governing the structure of spacetime itself. Special relativity was born from this step.

We can therefore say that special relativity is more universal than Newtonian mechanics. Newtonian mechanics, which satisfies Galilean symmetry, is the low-velocity approximation of special relativity, which satisfies Lorentz symmetry. At speeds far below the speed of light, Newtonian mechanics is entirely adequate; at speeds approaching that of light, special relativity is more precise.

But Einstein did not stop there. Although special relativity can describe accelerated motion in flat spacetime, it cannot yet bring gravity itself into the structure of spacetime. It still takes a fixed, flat spacetime as its background and inertial frames as its basic point of departure. Einstein therefore raised the demand of symmetry once again: the laws of nature should not depend on any privileged frame of reference or choice of coordinates. Even when the observer accelerates, and even when spacetime itself is curved, physical law should retain its unified form.

Along this path, the equivalence principle, general covariance, and the geometrization of gravity came together in general relativity. Gravity was no longer an ordinary force propagating through a fixed spacetime, but became the dynamic structure of spacetime shaped by matter and energy. The demand of symmetry thus ad-

vanced from inertial transformations in flat spacetime to general covariance in a dynamic spacetime. The German-born theoretical physicist and mathematician Max Born called general relativity "the most amazing combination of philosophical penetration, physical intuition, and mathematical skill."

Thus, from Galilean symmetry to Lorentz symmetry and then to the general covariance embodied in general relativity, a clear path appears: the range of admissible viewpoints grows wider, dependence on a special background diminishes, and the unifying constraints placed on physical law become stronger. Here, to say that symmetry becomes "higher" does not mean that each later mathematical group simply contains the previous one. It means that the laws of nature preserve their form under an ever wider range of changes.

The brilliance of Newtonian mechanics was rooted in drawing out the recurring patterns contained in intuitive experience and experimental data. Whether the phenomenon was a falling apple or the motion of a celestial body, its laws were distilled from observation and established as causal descriptions through repeated experimental verification. This methodology achieved unparalleled success in the macroscopic, low-velocity world, so much so that people grew accustomed to seeing the search for physical laws as a "jigsaw puzzle" assembled from experimental experience.

Einstein's path of thought, however, marked a profound turn. Rather than merely assembling fragments of known facts, he grasped a more essential connection between physical laws and symmetry. From Newtonian mechanics to Maxwell's unification of electricity, magnetism, and light, Einstein saw that the

universe should not be divided into a patchwork in which the laws of mechanics governed one kind of motion and the laws of electromagnetism another.

He was no longer satisfied with beginning from known facts and using "constructive efforts" to piece together a theory. He reversed the direction of inquiry: first seek a universal principle with higher and stronger symmetry, then firmly believe that the true laws of physics must obey it and derive more universal laws from it. This paradigm shift from "inductive experience" to "deductive principle" was the heart of his methodological revolution.

As Einstein candidly stated in his *Autobiographical Notes*, "Gradually I despaired of the possibility of discovering the true laws by means of constructive efforts based on known facts. The longer and the more despairingly I tried, the more I came to the conviction that only the discovery of a universal formal principle could lead us to assured results."

Inspired by Einstein's use of symmetry in the discovery of relativity, the German mathematician Hermann Weyl proposed an even more groundbreaking idea: physical laws might also have to satisfy another kind of symmetry, local gauge symmetry.

The same ascent occurs in gauge symmetry. If quantum states everywhere undergo the same phase change while physical results remain unchanged, this is a global phase symmetry. If the demand is raised so that the phase may be changed independently at every point in spacetime while physical law still remains invariant, the symmetry becomes local gauge symmetry.

Localizing the symmetry imposes a stronger demand than global symmetry. To compensate for the mismatch between phase

changes at different points, the theory must introduce a gauge field connecting those points in spacetime. For the basic $U(1)$ phase symmetry, this gauge field is the electromagnetic field; with the most natural and simplest choice of dynamics, one obtains the basic structure of Maxwell's electromagnetic theory.

This shows that symmetry is not an aesthetic ornament attached to a theory after the fact. Each rise in symmetry rules out a class of possible theories while requiring new spacetime relations, new fields, and new interactions to appear. The higher the symmetry, the less room remains for arbitrary choice, yet the more phenomena can be unified.

If Einstein deepened our understanding of spacetime and gravity by enlarging the symmetry of transformations between reference frames, Weyl's introduction of local gauge symmetry opened the way toward a unified understanding of all fundamental interactions.

An analogy may help. Global symmetry is like an entire country using a single standard clock. Local gauge symmetry allows each person to set the time independently, but then requires a "communications network"—the gauge field—to keep all the clocks synchronized. Here, "time" represents the phase of a quantum-mechanical wave function, not time in the ordinary, everyday sense.

In physical terms, local charge conservation means that although charge is conserved as a whole, every change of charge within even the smallest region of space must have a definite source. A mechanism must therefore exist for charge to flow between different points in spacetime. The gauge field introduced to carry this flow and mediate the interaction is the electromag-

netic field, and the interaction it carries is the electromagnetic interaction.

Inspired by Weyl, Chen-Ning Yang and Robert L. Mills proposed Yang–Mills theory, also known as non-Abelian gauge field theory, in 1954. It provided a unified framework for describing the fundamental interactions through local gauge symmetry.

Building on this framework, a succession of breakthroughs followed.

Working with the $\mathrm{SU}(2) \times \mathrm{U}(1)$ gauge symmetry group, Sheldon Glashow, Abdus Salam, and Steven Weinberg, recipients of the 1979 Nobel Prize in Physics, proposed electroweak theory, unifying the electromagnetic and weak forces as the electroweak interaction.

Working with the $\mathrm{SU}(3)$ gauge symmetry group, Harald Fritzsch, Murray Gell-Mann, Heinrich Leutwyler, and others established quantum chromodynamics, the gauge theory of the strong interaction. Gell-Mann received the 1969 Nobel Prize in Physics for the quark model.

Gerard ’t Hooft and Martinus Veltman, recipients of the 1999 Nobel Prize in Physics, proved the renormalizability of Yang–Mills theory, removing the fundamental obstacle to quantitative calculations in electroweak theory and quantum chromodynamics.

Thus, within the Standard Model of particle physics, the electromagnetic, weak, and strong forces—three seemingly distinct fundamental interactions—received a unified description within the framework of Yang–Mills theory.

The Standard Model successfully unified three interactions but

could not incorporate gravity. This "incomplete" unification suggests all the more strongly that, at higher energy scales, the four forces might share a common origin. Following the path marked out by symmetry, physicists set their sights on a still grander goal: a grand unified theory.

The four fundamental interactions are the cornerstones from which the multi-level ordered structures of all things in the universe are built. From the motion of galaxies to the stability of atoms, from the burning of stars to the birth of life, everything is governed by these fundamental interactions. A grand unified theory, and beyond it a theory of everything that describes them all, would express the most fundamental and universal laws of the cosmos. According to the inner logic of symmetry, laws this fundamental should correspond to an ultimate symmetry: a grand symmetry capable of unifying the four forces within a single, self-consistent mathematical framework.

Yet formidable obstacles stand on this road to ultimate symmetry. Gravity is described by general relativity, whose central idea is to interpret gravity as the curvature of spacetime and whose intuition is profoundly geometric. Gauge field theory, by contrast, is built within quantum field theory and uses gauge bosons to mediate interactions; its language is that of quantized particle physics.

The two theories stand in fundamental conflict in their mathematical structures, philosophical foundations, and even their conceptions of spacetime. How to quantize gravity and make it compatible with the language of quantum field theory has become one of the greatest challenges in theoretical physics. To this day, it remains a "holy grail" pursued by modern physicists.

We have now seen how physicists, by pursuing ever higher symmetries, have moved step by step toward the underlying laws of the cosmos.

A universe of ultimate symmetry, in which everything was unified, would be perfect yet immeasurably monotonous. Stars, planets, life, and even we ourselves would not exist. From a more fundamental perspective, the initial state of the universe should itself have been a state of "disorder" possessing the highest symmetry and containing every possibility. Yet that ultimate state would still have been immeasurably monotonous: stars, planets, life, and even we ourselves would not exist. What truly shaped our complex and richly varied universe was precisely the breaking of symmetry—symmetry breaking.

In recent decades, physics has achieved a series of landmark advances in this field:

Lev Landau, recipient of the 1962 Nobel Prize in Physics, laid the foundations for the theory of symmetry breaking, phase transitions, and order parameters. Philip Anderson, recipient of the 1977 Nobel Prize in Physics, extended and developed the idea of symmetry breaking in condensed-matter physics. Yoichiro Nambu, recipient of the 2008 Nobel Prize in Physics, systematically introduced the idea of symmetry breaking into particle physics.

The Higgs mechanism proposed by Peter Higgs, Francois Englert, and Robert Brout, work recognized by the 2013 Nobel Prize in Physics, implemented the spontaneous breaking of local gauge symmetry within the Standard Model. The Higgs boson predicted by this mechanism was confirmed experimentally at the Large Hadron Collider in 2012, revealing the origin of the masses

of elementary particles.

In the eyes of modern physicists, the history of cosmic evolution is a history of successive spontaneous symmetry breakings. The Big Bang began from a chaotic singularity of ultimate symmetry. As the universe expanded and cooled, it could no longer sustain that original, perfect symmetry. Particular asymmetric states were "selected" and fixed in place, level by level. This is symmetry breaking.

Each symmetry breaking is a leap from quantitative change to qualitative change. Its process corresponds to self-organized criticality in complex systems or to the mechanism of phase transition in condensed-matter physics. Every symmetry breaking brings forth new differences and features, forming a new level of ordered structure. The system thereby moves from a state of higher symmetry but greater simplicity into a new state of lower symmetry but markedly greater order and complexity.

At the beginning of the Big Bang, the universe was at an extraordinarily high energy, and the four fundamental forces were unified in a state of ultimate symmetry. This stage is known as the Planck epoch, or the quantum-gravity era, at $t < 10^{-43}$ s. A succession of decisive symmetry breakings then followed.

First, grand-unified symmetry broke from the initial state of ultimate symmetry, separating the strong force from the electroweak force. The energy released drove cosmic inflation.

Next came electroweak symmetry breaking. The Higgs mechanism was triggered, giving elementary particles mass, and the electroweak force separated into the electromagnetic and weak forces.

Then CP-symmetry breaking, proposed by Makoto Kobayashi and Toshihide Maskawa, recipients of the 2008 Nobel Prize in Physics, broke the symmetry between the behavior of particles and antiparticles. This discovery grew out of the parity nonconservation proposed by Chen-Ning Yang and Tsung-Dao Lee in 1956. It may ultimately have led to the great excess of matter over antimatter in the universe, providing an important clue to the existence of all material structures.

In the account above, the "separation" of gravity has been deliberately left out. Scientists have not yet determined at what stage, or in what manner, gravity emerged through "symmetry breaking." Theories of quantum gravity hold that this breaking almost certainly preceded grand-unified symmetry breaking, occurring during the Planck epoch or earlier. Theories such as loop quantum gravity and entropic gravity, of course, do not seek a grand unification of the four forces but focus instead on gravity itself. Bringing gravity into the picture may require more than a continuation of symmetry breaking; it may entail a thorough reconstruction of the concept of spacetime.

At every stage, symmetry breaking gave rise to a new level of ordered structure. Quark confinement formed protons and neutrons, followed by the birth of atomic nuclei and atoms: this was the first level. Gravitational instability—which can be seen as a breaking of the symmetry of matter uniformly distributed in space—then took over. It allowed differences in primordial quantum fluctuations, amplified by inflation, to manifest themselves and initiate a hierarchy of structures, forming stars, planets, galaxies, and other celestial bodies. Next, a symmetry breaking between the organic

and inorganic realms allowed life to arise on Earth. Finally, upon the foundation of single-celled organisms, symmetry breaking produced differences, divisions of labor, and cooperation within populations of the same kind, drawing life toward greater complexity: from cellular life to multicellular life, individual organisms, and biological communities.

Symmetry breaking is the underlying mechanism by which cosmic evolution realizes "multi-level constraints of order," driving the universe from disorder to order and from lower-level to higher-level forms of order. At the level of generative logic, this stepwise process of symmetry breaking resonates deeply with the progression in the *I Ching*: "The Supreme Ultimate gives birth to the Two Modes; the Two Modes give birth to the Four Images; the Four Images give birth to the Eight Trigrams." Through a system of symbols, ancient thinkers intuitively grasped the idea that all things arise through symmetry breaking.

The Supreme Ultimate symbolizes the primordial state before the generation of the universe: highly symmetrical, undifferentiated, and disordered, containing all things but not yet divided. "In the Changes there is the Supreme Ultimate, which gives birth to the Two Modes." As the starting point of the Changes, the way of transformation, the Supreme Ultimate differentiates into the Two Modes of yin (the broken line) and yang (the solid line). This may be seen as the first symmetry breaking, forming a one-line ordered structure. "The Two Modes give birth to the Four Images": yin and yang break symmetry again to generate Greater Yin, Lesser Yin, Lesser Yang, and Greater Yang, forming a two-line ordered structure. "The Four Images give birth to the Eight Trigrams": another symmetry

breaking generates Qian, Kun, Li, Kan, Zhen, Gen, Xun, and Dui—associated with Heaven, Earth, Thunder, Wind, Water, Fire, Mountain, and Marsh—forming a three-line ordered structure.

Through repeated "symmetry breakings," the abstract progression from the Supreme Ultimate to the Eight Trigrams forms multi-level ordered structures out of disorder, completing a symbolic leap from abstract symmetry to the concrete diversity of the world. It is a crystallization of ancient wisdom in the search for the fundamental order behind the world's complex appearances.

The analogy above is intended to reveal a similarity in generative logic, not a one-to-one correspondence between specific physical mechanisms. This resonance across space and time shows precisely that there is a deep unity in the human understanding of cosmic order.

The discussion above also leads to an important conclusion: laws do not exist independently of things. They are the self-expression of the repeatable or predictable features of order within the multi-level ordered structures of things.

The multi-level ordered structures of all things emerge as disorder is subjected to constraint. The laws describing those structures must therefore arise together with the things themselves; they, too, come from nothing. The physical laws by which we now describe elementary particles and interactions were not "ultimate truths" preset at the birth of the universe. They are themselves patterns "selected" and stabilized at particular stages of symmetry breaking during cosmic evolution.

Going further, more complex things and higher-level laws in the universe—chemical, biological, and even social laws—also grad-

ually "crystallized" and stabilized out of underlying interactions, within the grand framework of cosmic evolution and through a process in which randomness and necessity are intertwined.

This confirms with extraordinary depth the universality of the underlying logic: "Disorder creates possibility; order determines direction."

When we examine things closely, we discover a profound phenomenon: both things themselves and their ordered structures at different levels seem to contain some form of symmetry, yet all things everywhere also display many forms of asymmetry.

The symmetry contained within a thing is precisely the part preserved as that thing is generated through symmetry breaking. It is a "relic" of an earlier unified state, the foundation on which order continues. The asymmetry displayed by the thing is the proof that symmetry breaking has endowed it with a unique existence of its own. It is precisely those broken symmetries that make every ordered being an irreplaceable "this one."

Consider the human faces we see every day. The two eyes and two ears display mirror symmetry in their structure, providing the species with a stable, heritable blueprint. Yet on closer inspection, the left and right eyes always differ subtly in expression, and a smile is never perfectly centered. These slight asymmetries allow a face to be recognized as one particular person, the irreplaceable "this one" who bears a life story of their own.

Across social history, dynastic succession and economic cycles seem repeatedly to follow a certain rhythm: what has long been divided will unite, and what has long been united will divide. The recurrence of this underlying logic allows us to draw wisdom from

history. Yet the concrete processes of "union" and "division" are never the same. The Qin unified the six states through agricultural production, warfare, and iron tools; the birth of the modern nation-state accompanied the Industrial Revolution and an awakening of consciousness.

Whether in things, society, or art, we appreciate the ordered beauty presented by symmetry even as we continually pursue the vitality and uniqueness conferred by asymmetry. This is an expression of the underlying logic of the cosmos.

When we approach things from lower levels toward higher ones, we find that the laws at different levels have different characteristics. Fundamental physical laws, such as those of quantum field theory, describe the most basic ordered structures of matter, such as elementary particles, and possess extremely high symmetry and universality. As symmetry breaks, new and more complex ordered structures emerge—atoms, molecules, cells, and ecosystems—and higher-level laws appear with them: the laws of chemistry, biology, and sociology. These higher-level laws are built upon lower-level laws, but they possess lower symmetry, describe more specific objects, and apply within more restricted domains.

For example, atoms of carbon, hydrogen, and oxygen obey universal physical and chemical laws. Yet when, under particular conditions, they combine through symmetry breaking to form amino acids—specific molecules of lower symmetry—their behavior begins to be governed by more particular biochemical laws. Amino acids then combine into proteins, structures of still lower symmetry and greater complexity, whose behavior follows the more complex laws governing the structure and function of biological macro-

molecules.

Across the body of knowledge from physics to sociology, as multi-level ordered structures form, the universality of laws decreases while their specificity increases, tracing a path of decreasing symmetry and increasing complexity. The multi-level ordered structure of a thing is therefore the set containing all of that thing's laws. Ordered structures at different levels correspond to different laws.

Most of the physical laws we formulate are only models constructed for laws at particular levels. At macroscopic scales and low speeds, for example, the laws governing motion are those of Newtonian mechanics; at high speeds and in the microscopic domain, they are replaced by relativity and quantum mechanics. Laws can approach "reality" without limit, yet may never be reality itself, because reality is all the information contained in the multi-level ordered structure of the thing itself. Even a foundational law such as the Standard Model is therefore, in the philosophical sense, still a "model," albeit one with an extraordinarily broad range of application and extremely high precision.

Another of Newton's revolutionary contributions was to inaugurate an age in which scientific laws could be described with mathematical precision. Within the multi-level ordered structures of all things, different levels correspond to different symmetries. To describe the structures, relations, and changes at one particular level, we must set the other levels aside. A system of mathematical logic does precisely this: through abstraction, it strips away all concrete content and describes pure structure, relation, and change.

The lower the level of ordered structure, the more microscopic

and random it is, the more strongly its constituents exhibit identity, and the more it requires a highly abstract mathematical language to describe its correspondingly symmetrical, fundamental, and universal laws. As more levels of order continue to emerge, perfect symmetry is broken. Mathematical concepts such as nonlinearity, randomness, and topological defects then become essential to describing the complexity and diversity of the real world: the unpredictability of turbulence, the endlessly varied forms of snowflakes, and the emergence of distinctive defects in materials.

From lower levels to higher ones, our different mathematical descriptions of nature arise from an ongoing dialogue between universal laws at depth and specific phenomena at the surface. Multi-level ordered structures, together with the physical laws expressed mathematically, embody the dialectical unity of randomness and certainty, abstraction and concreteness, phenomenon and essence, the fundamental and the specialized.

Complete randomness at the lowest level—disorder—allows every possibility to occur. Yet through symmetry breaking, only some possibilities are selected and stabilized as physical reality. Unlike physical reality, abstract mathematical logic seeks to describe every possible structure, relation, and change. Existing natural laws—multi-level ordered structures and the laws of their change—can therefore all be described mathematically, but not every form of mathematics corresponds to a physical reality that presently exists.

Within the multi-level ordered structures of all things, some ordered features appear as observable properties of matter, while many others are difficult to observe or describe directly. Mathe-

matics, because it can describe every possible structure, can often guide scientists toward physical laws that are difficult to observe directly and even difficult to understand or accept.

Maxwell's equations, for example, predicted electromagnetic waves through mathematical deduction; the Dirac equation predicted the positron; through mathematics, Einstein led us to accept the difficult ideas of general relativity and curved spacetime and derived the prediction of black holes; Weyl, Chen-Ning Yang, and others relied on mathematical theory to derive the existence of gauge fields from local gauge symmetry. None of these was merely a summary of phenomena already observed. They were new realities discovered under the guidance of mathematics's internal logic, physical imagination, and symmetry.

The physicist Paul Dirac therefore said, "A theory with mathematical beauty is more likely to be correct than an ugly one that fits some experimental data."

Finally, looking back at the grand unified theory we have pursued so persistently—the "ultimate principle of all things" corresponding to perfect symmetry—might it describe, in mathematical terms, a wholly random disorder in which all possibilities cancel one another and every physical property falls to zero? If so, could the deepest foundation of "existence" itself be, in some sense, "nonexistence"?

As the *I Ching* reveals, the Supreme Ultimate is not a static endpoint but a dynamic starting point containing the fundamental contradiction between yin and yang. We may never be able to describe a static Supreme Ultimate in itself, because it can manifest itself only through what it produces—the Two Modes and, beyond

them, all things.

Likewise, gauge field theory and general relativity are models formed from two different perspectives, one looking upon the microscopic world and the other upon the macroscopic universe; their unification requires a higher-dimensional perspective. Or perhaps they resemble yin and yang and resist fusion because ultimate symmetry corresponds precisely to an indescribable “non-existence.” The sole proof of its existence is its inevitable tendency to break, and the result of that breaking.

In the end, we may have to accept a conclusion that resonates deeply with Eastern wisdom: the ultimate truth of the universe may not be Being, but Becoming.

Chapter 11

The Spectrum of Laws

Every scientific discovery reveals more of the order of the universe.

Michael Faraday

On the surface, nature presents us with phenomena of bewildering complexity: the ceaseless changes of wind and cloud, the transformation of landscapes across the ages, and the struggle and mutual sustenance among species.

From the perspective of biological evolution, those able to discern patterns and trends within nature's chaotic complexity were more likely to anticipate danger and find food, gaining an advantage in survival and reproduction. Our inward longing to seek law and order in nature may be the deepest force driving humanity's scientific exploration.

The essence of law is the self-expression of the repeatable or predictable features of order within the multi-level ordered structures of the universe and all things. Every level of order within those structures exists and evolves according to its own inner laws. We may even say that each level of order is itself a law.

Laozi wrote, "Humanity follows Earth; Earth follows Heaven;

Heaven follows the Dao; the Dao follows what is natural." Seen in this light, the "Dao" of which he spoke, like the laws and order we seek, is not a human invention. It is the order by which every level of the ten thousand things acts according to its own nature, arising and transforming of itself. Human scientific inquiry is therefore, in essence, also a process of "embodying the Dao" and "awakening to the Dao": discovering and following the order of Heaven, Earth, and all things.

Song–Ming Neo-Confucianism taught that "principle is one, but its manifestations are many." The universe has a single, unified "principle," yet that principle appears differently in each concrete thing. As Zhu Xi[1] put it, the moon is reflected in ten thousand rivers: its reflections in clear and muddy water differ, yet all are reflections of the same moon in the sky.

Just as a growing tree leaves annual rings, multi-level ordered structures are the "annual rings" formed by all things in the universe through their generation and evolution. The universe and all things within it are continually evolving, and so are their laws.

From the perspective of symmetry, most elementary particles and atoms arose early in cosmic evolution. They occupy the lowest levels in the multi-level structures of things: their structures are more microscopic, their behavior displays quantum randomness, and their high degree of indistinguishability makes them more abstract. The laws they obey possess relatively high symmetry, are

[1] Song–Ming Neo-Confucianism is the modern name for a major renewal of Confucian thought from the Song through the Ming dynasties. Zhu Xi (1130–1200) was its most influential systematizer; "one principle, many manifestations" expresses the relation between universal pattern and its particular embodiments.

more fundamental, and therefore have greater universality.

From elementary particles to atoms and molecules, from living organisms and group organizations to human society, every symmetry breaking produces a new and higher level of ordered structure, together with new things and new laws. As these levels of order accumulate, complexity grows exponentially.

The more complex a thing, or the higher its level, the more macroscopic its structure becomes, the more its behavior tends toward classical determinacy, and the more pronounced its individual uniqueness becomes. It therefore grows more concrete. Correspondingly, the laws it follows have lower symmetry, belong to more specialized domains, and possess relatively less universality.

The laws corresponding to different levels of order thus have different characteristics. We may picture the laws at different levels as possessing different "frequencies": fundamental, universal laws with high symmetry resemble a low-frequency band, while complex, specific laws with low symmetry resemble a high-frequency band.

All the laws in the universe thus arrange themselves into a "spectrum of laws." Within this spectrum, the laws revealed by the natural sciences occupy relatively low-frequency bands, while those studied by the social sciences occupy relatively high-frequency bands.

The very low-frequency band of the spectrum contains the laws corresponding to the fundamental interactions. It describes levels of order composed mainly of elementary particles and fields, whose individual entities are identical and indistinguishable yet display quantum properties. The corresponding laws possess

extremely high symmetry and universality, applying throughout the whole of cosmic spacetime, while their mathematical forms exhibit simplicity and beauty. This band encompasses particle physics, quantum field theory, cosmology, general relativity, and related fields. Its laws provide the most fundamental "building blocks" and relations of interaction for all things, forming the underlying foundation of laws at higher levels of order. What we now consider the "lowest level," of course, may merely mark the boundary of our present capacity for understanding. The true ultimate law may be far simpler and more fundamental than the Standard Model and general relativity.

The low-frequency band of the spectrum corresponds to the laws governing the formation of atoms and molecules. It describes levels of order that include atomic nuclei, atoms, molecules, crystals, and other diverse structures, as quantum properties begin to give way to the determinacy of chemical reactions. Despite symmetry breaking, the laws in this band remain highly universal: the periodic law of the elements applies to atoms throughout the universe, and the theory of chemical bonding applies to all molecules, whether their bonds are covalent or ionic. These laws are embodied in atomic physics, chemistry, condensed-matter physics, nuclear physics, and related disciplines.

The middle-frequency band of the spectrum corresponds to the emergence of complex structures and life. It describes levels of order composed mainly of biological macromolecules, cells, organs, organisms, and even ecosystems. Having undergone more symmetry breakings, these things obey laws of diminished universality and greater dependence on environmental conditions and historical

pathways. The central dogma of genetics, for example, applies to all life on Earth, yet its concrete realization differs among species. Ecological laws are universal, yet the particular parameters of every ecosystem vary enormously. The laws of biological evolution include universal mechanisms such as random mutation and natural selection, yet their results are strongly path-dependent and historically distinctive, as in the Cambrian explosion. This band corresponds to biology, from molecular biology to ecology, as well as geology, fluid mechanics, and related fields.

The high-frequency band of the spectrum corresponds to design and creation driven by human intention. It describes levels of order that include every kind of human invention, technological system, industrial process, and practical solution. Its laws encompass all branches of engineering and technology—civil, mechanical, electronic, and chemical engineering, as well as biotechnology—together with medicine, agronomy, and the applied branches of computer science. These laws possess a degree of universality within their respective domains, but much of the knowledge involved arises from practical experience and lessons learned. Laws are often discovered and established through repeated cycles of "design–build–test–iterate," in which they are verified and revised. Many engineering endeavors integrate several simple systems into a complex one and must make trade-offs and compromises among conflicting requirements such as performance, function, time, and cost. This "art of trade-offs" is itself a core "law" of engineering. Its symmetry is so low that every case may possess a considerable degree of uniqueness.

The ultra-high-frequency band of the spectrum corresponds to

the complex networks of mind and society. It describes levels of order that include human consciousness, individual behavior, social organization, economic systems, and cultural phenomena. Having undergone countless symmetry breakings, these levels display the greatest specificity; their laws therefore possess extremely low symmetry and universality. Vast numbers of variables—individual psychology, cultural background, historical context, and many others—profoundly influence them, so they usually appear as probabilities and trends. We might even say that every individual, every society, and every historical period possesses its own uniqueness. The relation of supply and demand, for example, is a universal principle of economics, yet the real-estate and stock markets differ radically in both their operating laws and their determining factors. This band corresponds to such fields as cognitive science, psychology, management studies, economics, sociology, political science, and the study of culture and the arts.

It must be emphasized that as the division of labor in modern science grows ever more specialized, numerous interdisciplinary fields have emerged, making the spectrum of laws more continuous. No strict boundaries separate adjacent "frequency bands"; instead, broad zones of overlap and transition lie between them. Statistical ensembles connect thermodynamics with statistical mechanics, biochemistry connects chemistry with biology, and cognitive neuroscience bridges biology and psychology.

At their core, both the natural and social sciences seek order within disorder. They simply confront different levels of order, so the "order" they discover differs in symmetry, universality, and stability; the two are not opposed.

Within the spectrum of laws, the natural sciences occupy relatively low-frequency bands, like a meticulous sculptor. With repeatable experiment and mathematics as their chisels, they work patiently to reveal the near-eternal and universal "Dao" concealed beneath phenomena. The social sciences occupy relatively high-frequency bands, like a learned interpreter. Within a complex tapestry filled with noise and detail, they use understanding and interpretation to decipher the continually evolving "Dao" woven together by countless individuals, each with a will of their own.

Every law is valid only within its own range of application. This necessarily reflects the nature of all things in the cosmos as multi-level ordered structures. Looking from the high-frequency bands toward the low-frequency bands, we see the beauty of nature tending toward eternity, unity, and harmony. Looking from the low-frequency bands toward the high-frequency bands, we encounter the beauty of nature moving toward complexity, determinacy, and diversity.

The renormalization group in modern physics provides profound support for this multi-level understanding. In the 1970s, Kenneth Wilson turned "changing the scale of observation" into a rigorous physical method. As we move from the microscopic to the macroscopic, not every lower-level detail is preserved unchanged. Vast numbers of minute degrees of freedom are averaged and compressed as the scale expands. What continues to affect macroscopic behavior is often only a small set of effective parameters, together with the symmetries and conservation relations that remain valid across scales. The flow of water does not require us to track every molecule; a magnetic phase transition does not require us

to remember every detail of every atom; biology need not begin with quarks and gluons. Each level of order has its own effective laws, yet those laws do not arise from nowhere. They are stable expressions formed as lower-level changes pass through layer after layer of filtering and constraint. The renormalization group thus tells us that nature can be understood in levels not because we are taking a shortcut, but because the world itself continually discards detail and preserves structure as scale changes.

The renowned condensed-matter physicist P. W. Anderson argued in his classic paper *More Is Different* that although higher-level order is built upon lower levels, higher-level laws cannot be reduced completely to lower-level laws. Laozi likewise said, "The Dao that can be spoken is not the constant Dao." That ultimate and unified Dao is difficult to express in words; it manifests itself as different laws at different levels.

This is precisely the profound meaning of "many manifestations." The "one principle" of the cosmos has not been split apart; in higher-level and more complex contexts, it simply manifests itself in new forms. As Zhu Xi held, "one principle" and "many manifestations" form a dialectical unity. Without the one principle, the many manifestations would become fragmented chaos; without the many manifestations, the one principle would become an empty abstraction.

For this reason, the laws of every scale and domain—quantum mechanics, classical mechanics, physics, chemistry, biology, neuroscience, engineering and technology, medicine, management studies, economics, political science, and more—are indispensable to our understanding of this complex world and of the phenomena

and problems it presents at different levels. For understanding phenomena at their corresponding levels, the high-frequency laws of the macroscopic scale are no less "fundamental" than the low-frequency laws of the microscopic scale.

We so often use the word "emergence" for the self-organizing process by which higher-level order forms from lower-level order precisely because we still cannot—and perhaps fundamentally cannot—find a universal, unified law that describes emergence across levels. If such an all-purpose law existed, could every arbitrarily complex multi-level structure not be reduced completely to the properties of the most fundamental elementary particles and analyzed through them? To insist on seeking such a universal law of system formation is itself to fall once again into the habits of reductionism.

The differences among the levels of order addressed by different bands in the spectrum of laws shape our ways of understanding spacetime at those levels, and ultimately influence the methods and frameworks through which we understand the world.

In 1962, the philosopher of science Thomas Kuhn introduced the concept of the "scientific paradigm" in *The Structure of Scientific Revolutions*. A paradigm may be understood as the cognitive framework and practical norms shared by scientists, at a particular historical stage, for approaching certain levels of order. The establishment of paradigms, the accumulation of normal science, and the revolutionary shift from one paradigm to another are precisely the processes through which humanity continually deepens its discovery of, and alignment with, the universe's spectrum of laws at different levels of order.

The "incommensurability" between paradigms emphasized by Kuhn echoes the fundamental methodological differences among bands in the spectrum of laws that arise from symmetry breaking. The transition from an old paradigm to a new one also propels humanity's understanding of nature upward in a spiral, just as the dialectical law of the "negation of the negation" reveals.

Yet as human knowledge grows explosively and the disciplinary system becomes ever more specialized, we face a profound cognitive dilemma. Barriers between disciplines rise ever higher, and researchers are often confined within one particular band of the spectrum of laws. They may analyze a local melody with exquisite precision yet struggle to grasp the harmony of the whole symphony. Specialized depth brings local exactness, but it also quietly erects walls around thought.

We often view the world through the paradigmatic lens of the field we know best, easily forgetting that it covers only a narrow band within the spectrum of laws. This limitation can lead to a kind of "cognitive dogmatism": either we try to reduce higher-level phenomena excessively to lower-level laws and fall back into the trap of reductionism, or we cling to the methodology of our own field and ignore the valuable perspectives offered by other frequency bands.

How to balance the inevitable trend toward specialization in knowledge with the need for holistic understanding is a challenge we must confront today.

To break through this predicament, we must deliberately cultivate a capacity to think "across frequency bands" even as we work deeply within a specialized field. We should strive not only to be-

come experts in one band but also to become listeners to and interpreters of the entire spectrum of laws. As explored in *The Entropy Dance of Order*, this requires, at the methodological level, a dialectical synthesis of reductionism and emergentism.

This synthesis is not a simple accumulation of knowledge. It brings reductionism and emergentism into an integrated whole on the basis of a deep understanding of the characteristics of laws at every level—their symmetry, universality, dependence on conditions, and more. Faced with a concrete problem, it allows us to draw flexibly on cognitive tools from different frequency bands and truly to see both the trees and the forest.

Ultimately, this "spectral way of thinking" will lead us toward a richer, more three-dimensional picture of understanding, one closer to the world as it is. Amid the profusion of phenomena, we can discern their deeper unity while appreciating their visible diversity. With a more inclusive way of understanding, one better attuned to the nature of its object, we can listen to and comprehend the "metrical poem" sounded by the Dao of all things across the full spectrum.

The brilliant achievements of the natural sciences, especially the precision and universality displayed in their exploration of the low-frequency bands of the spectrum, have quietly fostered a bias toward "scientific chauvinism": an excessive reverence for repeatable verification that may even elevate it into the sole standard of scientific validity. When this perspective is habitually turned upon the social sciences at the other end of the spectrum, their laws may be doubted for lacking sufficient determinacy and universality, and even their methods may be dismissed as "unscientific."

Yet this doubt itself misunderstands the central truth revealed by symmetry breaking and the spectrum of laws. From the perspective of "one principle, many manifestations," it is a misunderstanding of the one principle. The Moon's reflections in clear and muddy water differ, yet both are reflections of the real Moon. The form of a law and the reach of its universality are not set a priori; they are determined by the position, on the tree of cosmic evolution, of the level of order that the law describes. At the social level, where symmetry is lowest and complexity highest, laws necessarily "crystallize" in highly contextual, probabilistic, and historically dependent forms. They, too, are manifestations of principle at a complex level.

In the ultra-high-frequency band of human society, each individual and each civilization carries a unique historical path and cultural genes, making strict control of variables and experimental repetition nearly impossible. Here, laws need not take the form of determinate mathematical equations. Instead, they condense into trends discerned within complex contexts and patterns deciphered amid chaotic noise. Their scientific character lies precisely in the depth with which they reveal the highly contextual and probabilistic "code of order" within a particular low-symmetry structure.

This "heterogeneity" of social-scientific laws is not a defect. On the contrary, it is the most profound portrayal of the grand process by which cosmic evolution moves from simplicity to complexity and from homogeneity to heterogeneity. For this very reason, recording and reading history become fundamental ways to break through the cognitive limits of the individual. History is not a template to be copied mechanically; it is an "archive" in which unrepeatable social processes have condensed into multi-level ordered

structures.

We should therefore remain all the more humble. True wisdom arises precisely from an integrated understanding of laws at multiple levels: appreciating the simplicity and unity found at the level of elementary particles while also comprehending the complexity and uniqueness of society and culture; respecting the universal "hard laws" revealed by the natural sciences while also valuing the contextual "soft laws" interpreted by the social sciences.

On the other hand, the distinctiveness of social-scientific methods does not mean that we may relax the demand for scientific rigor. Quite the contrary: because the social and mental phenomena studied in the high-frequency bands possess extremely low symmetry and extremely high complexity, any misuse of method can readily breed a proliferation of "pseudoscience." Blindly applying simplified models from the low-frequency bands, or abandoning rigorous evidence in favor of pure speculation or personal intuition, can lead equally to this result.

Such pseudoscience often wears academic clothing. It appears in the misuse of statistics, rash assertions about complex causal relations, or the construction of theories that cannot be tested. In an age of information explosion, these pseudoscientific products are more deceptive and spread more readily than ever before.

This also allows us to grasp more deeply the Eastern insight that "Heavenly principle is inherent; innate knowing is inherent." The foundations of law and cognition lie within the ordered structures at each level. We might therefore say that essence does not precede existence, nor does existence precede essence: multi-level ordered existence itself is its essence.

There is no absolutely universal law valid in every circumstance, nor any absolute experience detached from the context of its level. There are only stable patterns formed through evolution and frameworks that are relatively "a priori" between levels.

Human understanding and practice should therefore cling to no established paradigm and should never allow themselves to be led blindly by path dependence. We must return to a concrete analysis of the particular manifestation before us, find the true "principle" of this time, this place, this level, and this structure, and always face reality directly with the wisdom of going beyond apparent paths.

Laozi also wrote in the *Daodejing*, "All under Heaven say that my Dao is great, yet it resembles nothing." By this he meant that many people find his teaching vague, untethered, and of no apparent use. He continued, "It is precisely because it resembles nothing that it can be great. Had it resembled something, long ago it would have become small." Because the Dao does not settle into any particular form or vessel, but possesses a very high degree of abstraction and symmetry, it can be boundless and encompass all things; that is, it possesses the broadest universality. If it matched some particular thing, its symmetry would fall and it would become concrete. It would long ago have been reduced to something small and lost its fundamental status. Laozi's insight resonates deeply with the picture of the spectrum of laws.

For this very reason, in exploring the spectrum of laws from elementary particles to human society, this book has had no choice but to introduce many specialized concepts from physics, biology, and philosophy. This undoubtedly presents the reader with a considerable challenge. Yet it must be understood that this is the necessary

path to genuine “understanding.” As Laozi reveals, what is “great” must “resemble nothing”: to seek knowledge broad enough to encompass many things, one must endure the difficulty of abstraction.

Chapter 12

Changing Space-Time

Time, likewise, does not exist by itself; but from things themselves there follows a sense of what has been done in the past, what is present now, and what will come after.

Lucretius

Saint Augustine famously asked in the *Confessions*, "What, then, is time? If no one asks me, I know what it is. If I wish to explain it to him who asks me, I do not know."

In everyday life, we feel the passage of time and the extension of space. Yet even today, it remains difficult to answer the question: how do time and space exist? If we combine this question with our understanding that all things exist as multi-level order, however, we may be able to reveal that spacetime does not exist independently, but is a product tightly entwined and coevolving with the ordered existence and change of matter.

Human reflection on spacetime began in the age of myth and early philosophy.

In ancient Chinese myth, Xihe[1] governed time. Each day she

[1]Xihe is a solar figure in early Chinese mythology associated with the ten suns and

calmly drove her dragon chariot, guiding the ten suns in their ordered rhythm from east to west. In oracle-bone script, the character *shi*, "time," resembled the course of the sun, while *jian*, "interval," denoted a pause within that course.

In ancient Greek myth, Chronos, the god of time, symbolized the passage of time and was regarded as the origin of the motion of all things. The ancient Greeks believed that time gave rise to motion, and that motion in turn could create order out of chaos. Time was therefore seen as the origin of all things.

Zhuangzi's "On the Equality of Things" says, "Stand beside the sun and moon; embrace the universe." Here *yu* refers to all space in every direction, while *zhou* refers to all time from antiquity to the present.[2] Daoism regards spacetime as the expression of the movement of innate *qi*, or primordial vital energy. The cyclical flow of *qi* forms time, while its extension, gathering, and dispersal form space.

The Buddhist view is that time is an illusory construction of consciousness, while space is the shared appearance brought forth by collective karma. The *Diamond Sutra* says, "The past mind cannot be grasped; the present mind cannot be grasped; the future mind cannot be grasped," pointing out that attachment to the three times is only deluded discrimination. The *Cheng Weishi Lun*[3] says that

their daily passage across the sky.

[2] *Yuzhou* is the ordinary modern Chinese word for "universe," but its two characters preserve an older philosophical pairing: *yu* as spatial extension and *zhou* as temporal duration.

[3] The *Diamond Sutra* is a concise Mahayana scripture on nonattachment and emptiness. The *Cheng Weishi Lun*, usually translated as the *Treatise on the Establishment of Consciousness-Only*, is a foundational East Asian Yogacara text.

time is provisionally established through the arising and ceasing of thoughts. The *Mulamadhyamakakarika* asks, "If time exists because of things, how could time exist apart from things?"

In summary, the Madhyamaka and Yogacara schools of Buddhism hold that spacetime is not an independent entity. Rather, it is a conceptual tool constructed by consciousness in order to understand phenomenal reality: the arising and ceasing of all things and their positions relative to one another. In essence, it is an illusion marked by dependent origination and emptiness of intrinsic nature.

Aristotle was among the first in the history of philosophy to systematically analyze the concepts of time and space. He deeply questioned the reality of time. Since time is composed of the past, which no longer exists, and the future, which does not yet exist, and the "now" that connects the two is merely a durationless instant, then how can time exist? He believed that time is merely the measure of change. If all things were absolutely still and unchanging, time would not exist. Regarding space, he denied the existence of the void, holding that space is the extension of objects themselves and the relative relations between them. The position of an object is defined by the objects surrounding it.

Ancient and medieval understandings of spacetime were based on experience and speculation. A more scientific analysis began with Galileo in the early modern period. His principle of relativity stated that within a closed cabin moving at a constant velocity in a straight line—that is, within an inertial reference frame—no mechanical experiment could determine whether one was at rest or in uniform motion. This revealed the relativity of motion itself while still implicitly assuming absolute time and absolute space, assump-

tions later articulated explicitly by Newton.

Descartes equated space with matter's attribute of extension, regarding it as a continuous, infinite, three-dimensional background, and pioneered the use of coordinates to describe spatial position. Leibniz, by contrast, advocated a relational view and regarded the treatment of spacetime as an independent entity as absurd: spacetime is only a network of relations among events. Events come first; time is an abstraction from their sequence. Each stage of the universe consists of all simultaneous events, and these stages follow one another in succession.

Newton was the great synthesizer of classical physics, and his conception of absolute space and time dominated science for two hundred years. In the *Mathematical Principles of Natural Philosophy*, he wrote, "Absolute, true, and mathematical time, of itself and from its own nature, flows equably without relation to anything external, and by another name is called duration." He also wrote, "Absolute space, in its own nature, without relation to anything external, remains always similar and immovable." At the heart of Newton's view, space and time are independent absolute entities, and their measurement does not depend on the state of motion of the reference frame. The physicist and philosopher Ernst Mach offered a profound and systematic critique of this view. He argued that an unobservable absolute spacetime was meaningless, and proposed that inertial effects arise from the motion of objects relative to all the matter in the universe—what became known as Mach's principle. This idea helped prepare the way for Einstein's later overthrow of absolute space and time.

Einstein's special relativity thoroughly overturned both the

ether and the conception of absolute spacetime. The rate at which time passes depends on the observer's state of motion: moving clocks run more slowly, an effect known as time dilation. A moving object contracts in length along its direction of motion, an effect known as length contraction. Whether two events at different locations are "simultaneous" depends on the observer's frame of reference. There is no universal "now"; absolute simultaneity is broken, giving way to the relativity of simultaneity. Time and space are no longer independent absolute entities, but are intimately interwoven into a unified four-dimensional continuum. The Minkowski spacetime interval remains invariant in every inertial frame, with temporal and spatial coordinates linked by the speed of light. This is four-dimensional spacetime. The speed of light is the limiting speed for the transmission of information and causal influence. An event can influence only events within its future light cone, and can be influenced only by events within its past light cone. Causal relations are confined by the structure of these light cones.

Einstein's general relativity went further, extending the applicability of physical laws to arbitrary non-inertial frames and reinterpreting gravity as a geometric effect of spacetime curvature. This once again transformed our conception of spacetime. No longer a static stage, spacetime became a dynamic "entity" whose curvature is determined by matter and energy and which, in turn, guides the motion of matter. In curved spacetime, objects move along geodesics—the "straight lines" of that geometry. The Earth, for example, does not follow an elliptical orbit around the Sun because it is pulled by the Sun's gravitational force. It is

still moving along a "straight line," but the Sun has curved the surrounding spacetime, so the Earth's curved path in ordinary Euclidean space corresponds to a "straight line" in curved spacetime. Although general relativity is counterintuitive, it perfectly explains phenomena such as the precession of Mercury's perihelion and gravitational lensing, and it predicted gravitational waves. As the American physicist John Wheeler famously put it, "Spacetime tells matter how to move; matter tells spacetime how to curve."

Quantum theory has achieved enormous success in the microscopic domain and is compatible with special relativity; both respect Lorentz symmetry. Yet quantum field theory still presupposes a continuous classical spacetime as its background. This stands in fundamental conflict with general relativity's conception of spacetime itself as dynamic and possibly in need of quantization.

More profoundly, most of Newton's laws, the equations of relativity, and the equations of quantum mechanics are formally symmetric under time reversal: if time were to run backward, the equations would remain valid. Or, as the theoretical physicist Carlo Rovelli explains in *The Order of Time*, these equations do not describe how things evolve "in absolute time." They describe how things change in their own times, and how those "times" change relative to one another.

Theories of quantum gravity arose in an effort to reconcile this contradiction. In the Wheeler–DeWitt equation, proposed by John Wheeler and Bryce DeWitt, the variable of time disappears when the equation describes the quantum state of the universe as a whole. This suggests that time may not be fundamental, but may emerge or evolve from quantum relations. As for space, quantum-gravity theo-

ries describe it at microscopic scales as fluctuating, rupturing, and reassembling like foam. In other words, space consists of wormholes and minute black holes continually arising and vanishing, and loses its classical continuity and determinacy.

Long before relativity and quantum mechanics, thermodynamics and statistical mechanics had already begun to study how things evolve in time, forming a bridge between the macroscopic and microscopic worlds. The second law of thermodynamics states that macroscopic processes are irreversible. The direction in which a system's entropy increases gives time its arrow.

From a microscopic perspective, statistical mechanics holds that entropy increase corresponds to a system's evolution toward states of greater probability. The thermodynamic arrow of time has led many physicists to consider whether time may not be a fundamental quantity, but an emergent phenomenon arising through evolution from the universe's initial low-entropy state.

Looking back over the history of thought about spacetime, we need to bring together insights from general relativity, quantum mechanics, and thermodynamics if we are to reach a clearer understanding. General relativity holds that the motion of macroscopic things is relative. The uncertainty principle of quantum mechanics reveals an intrinsic, absolute randomness in microscopic particles, so that they can never be completely still. The third law of thermodynamics likewise tells us that absolute zero cannot be reached, which means that particles never cease their thermal motion. The intrinsic properties of microscopic particles thus keep the microscopic level in continuous dynamic change and drive the perpetual change of layered order.

General relativity, quantum mechanics, and thermodynamics point from different directions toward the same conclusion: spacetime originates in ordered existence and ordered change.

First, time originates in ordered change. If everything in the cosmos were absolutely still and nothing changed, the concept of time would have no ground on which to arise. We perceive time precisely through the ordered rhythms within living bodies and in the external environment: heartbeats, day and night, and the four seasons.

Second, space arises as ordered change takes form. Spatial attributes such as scale, position, and direction appear only as matter builds stable structures out of microscopic disorder. We perceive space through these attributes of things and through the relations among them.

Thus, time and space are deeply interrelated through ordered existence and ordered change. Neither can present itself independently. Ordered change produces changing spacetime. A given spatial state must correspond to a given temporal state. The two map onto each other.

To understand spacetime more deeply, therefore, we must examine ordered change more deeply. Beginning at the microscopic level, we can try to construct a clearer physical picture.

Among elementary particles, fermions have half-integer spin; examples include electrons, quarks, and neutrinos. Bosons have integer spin; examples include photons, gluons, the W and Z bosons, and the Higgs boson.

According to the Pauli exclusion principle, no two fermions of

the same type within a quantum system can occupy exactly the same quantum state at the same time. Constrained by this principle, fermions constitute order at the most microscopic level through their very existence: they exclude one another from identical states. Changes in fermionic quantum states are therefore the most microscopic ordered changes and the ultimate origin of all macroscopic change. A fermion's quantum state is the complete set of information describing all its quantum properties, usually expressed through a set of quantum numbers.

This most microscopic change is discontinuous and quantized. It should correspond to the ultimate source and most fundamental unit of time. At the Planck scale, therefore, time may likewise possess a discrete, quantized structure. Scientists regard Planck time—the Planck length divided by the speed of light, approximately 10^{-43} seconds—as the smallest unit of time.

An isolated, free fermion is a free particle, and thus disorder—that is, non-existence. Changes in the quantum state of an isolated, free fermion cannot be defined, distinguished, or perceived; it therefore has no time of its own. As the Pauli exclusion principle shows, quantum states become distinguishable and comparable only within a quantum system, where multiple particles and their relations exist. Change can then be defined, and time comes into being.

Quark confinement is the fullest expression of this idea. Quarks, which are fermions, remain confined within hadrons and cannot exist as free particles. No matter how much energy we use in trying to smash open a proton and "knock out" one of its quarks, the result is never an independent, free quark, but a collection of new composite

particles.

Although changes in the quantum states of fermions are the most microscopic changes, they can be distinguished, defined, and perceived only within an ordered system. Within the multi-level ordered structures of all things, these changes accumulate from one level to the next, driving ordered change through successive macroscopic levels. Ultimately, the continuous, unified time we perceive emerges from the microscopic into the macroscopic. Nor is that all: each level of ordered structure, from the microscopic to the macroscopic, has its own distinctive ordered changes and therefore defines a time particular to that level.

Bosons, by contrast, transmit interactions and constitute ordered relations. They do not obey the Pauli principle and can gather in large numbers in the same state, as seen in Bose–Einstein condensation and in phenomena such as superfluidity, superconductivity, and lasers. Their individuality is indistinct, so ordered change within the bosons themselves—change in their quantum numbers—is difficult to distinguish or define.

For example, a photon always travels at the speed of light in a vacuum, and no time belonging to the photon itself can be defined. In the macroscopic world, we can use light to transmit information, but the photon itself does not change. The changes in information carried by light—what we call the modulation of light—actually originate in changes in the state of the emitting or absorbing source, both of which are composed of fermions. Light merely carries those changes; it is itself an unchanging medium of transmission.

Let us now turn to the emergence of space.

Together, the Pauli exclusion principle and quark confinement

lay the foundation of spatial structure at the most fundamental level. Although elementary particles are regarded as point particles without volume, the Pauli principle forces fermions to "keep their distance" from one another by preventing them from occupying the same quantum state. The composites they form can therefore be seen as creating "space."

The interior of a proton or neutron is not simply three static quarks. Rather, it is a "quantum ocean" filled with quark–antiquark pairs that continually arise and annihilate, together with gluons that carry the strong force. The motion of quarks is confined within a range of roughly one femtometer, or 10^{-15} meters. The range of ordered change within which the quarks are bound defines the spatial scale of protons and neutrons. In this way, protons, neutrons, and atomic nuclei acquire volume, and space begins to exist.

In an atom, electrons are distributed in an "electron cloud" around the nucleus. Their motion is confined to the scale of an angstrom, about 10^{-10} meters. The range of ordered change within which electrons are bound around the nucleus defines the spatial scale of the atom. Thus atoms too acquire volume, giving rise to atomic space.

The periodicity of elemental properties and the intermolecular forces of chemistry have their roots precisely in the spatial structures formed by electrons obeying the Pauli principle at the atomic and molecular levels. Electrons in adjacent atoms "keep their distance" from one another, thereby giving the solids, liquids, and gases of everyday life their volume and shape.

Within the layered ordered structures of all things, the manifestations of ordered change accumulate from level to level, ulti-

mately giving volume and form to every macroscopic object, from molecules and crystals to the mountains, rivers, grasses, and trees of everyday life. Layer by layer, a continuous and unified space emerges from the microscopic into the macroscopic. At the same time, each level of ordered structure has its own distinctive range of ordered change and therefore defines a space particular to that level.

Likewise, as the Pauli exclusion principle requires, only within a quantum system can quantum states be distinguished and compared, ordered change take form, and space come into being. An independent, free fermion therefore does not itself define space: in physics it is described as a point particle, without size or volume.

Bosons do not in themselves possess an "order" whose changes can be distinguished, and so they have no space of their own. Nor do they obey the Pauli principle or establish an exclusive "occupation of position." Even if countless bosons occupied the same quantum state at the same "point," space would not emerge from them. From the photon's perspective, for example, there is no distinction between near and far in the universe: light can reach anywhere instantaneously. This confirms that space is not an intrinsic property of the photon.

Therefore, we can say that both time and space originate from the ordered changes of things.

Quantum entanglement offers fascinating corroboration of this view. When two or more particles enter an entangled state, they form an indivisible whole whose quantum state cannot be written as a simple product of independent states of its parts. Within this higher-level order, the constituent particles seem to lose their indi-

viduality.

Because they can exist only as a whole, particles in an entangled state resemble a single, independent elementary particle, within which the concepts of time and space no longer apply. This is why entanglement appears as a nonlocal correlation that exceeds the limits of classical spacetime, an apparently "action-at-a-distance" connection. We cannot regard the several entangled particles separately and independently, and so they cannot serve as the sending and receiving ends of information transfer.

In summary, spacetime itself is not a form of ordered existence that exists independently. It arises as ordered structures are built and unfolds through ordered change. It is a derivative attribute of ordered existence and change.

The intrinsic randomness of microscopic particles is a kind of "absolute" change, and it is the ultimate origin of all ordered change and motion. Yet the concepts of "change" and "motion" themselves have meaning only within relative order or mutual relations, and are therefore "relative." The absolute and the relative are thus not two things starkly divided, but a unity of opposites. Absolute randomness forms the basis of relative order, while relative order provides the framework within which absolute change is defined. Spacetime is precisely the product of this dialectical unity of the absolute and the relative.

According to quantum field theory, even a vacuum that appears completely empty is filled with violent fluctuations of virtual particles—photons, gluons, W and Z bosons, and others—continually and randomly arising and annihilating. They are called virtual particles because their existence is so fleeting that they

cannot be detected directly or perceived. We can think of the universe as allowing energy to be "borrowed" from nothing in order to create particles, provided that this "loan" is repaid quickly enough.

Compared with the perceptible ordered existence of matter, the various virtual particles in a vacuum are independent and disordered, with no distinguishable ordered change or structure. The vacuum may therefore be regarded as corresponding to non-existence—to disorder—and cannot be perceived.

In this sense, we can regard the vacuum itself as a kind of reference background. It cannot be said to possess intrinsic time and space, yet it is constituted by unceasing quantum fluctuations, as virtual particles randomly arise and vanish. Stillness and change, or motion, are thus not starkly divided, but form a unity of opposites.

Of course, we may also think of quantum fluctuations within the vacuum as continually generating and annihilating forms of time and space that we cannot perceive. A vacuum of wholly random fluctuations forms a primordial, absolute spacetime background, upon which the order of the material world "excites" the relative spacetime structure we can perceive. The existence and non-existence of spacetime are therefore not sharply divided, but form another unity of opposites.

This is our preliminary exploration of the nature of spacetime. We will next examine ordered change more deeply, and with it move toward a deeper understanding of time and space.

Chapter 13

Only the Present

We are but mayflies lodged between Heaven and Earth,
a single grain of millet in the vast ocean.
Su Shi

If spacetime emerges from ordered change, how does this picture relate to the "here" and "now" that we experience every day and every hour?

When we utter the word "present," which instant of cosmic evolution are we pointing to? Is it the instant when a photon sweeps across the retina, the second marked by the rhythm of a heartbeat, or the day defined by the Earth's rotation? To understand the nature of spacetime, we may ultimately need to return to ourselves: to the perspective of the "I" confined within the body yet resonating with the cosmos at every moment.

Any elementary particle not bound by interactions would be like unfettered light, traveling through the cosmos at the speed of light. The process by which elementary particles combine through interactions to form matter is, in essence, a process in which free energy is bound layer by layer and condensed into ordered structures.

The relativistic mass–energy relation shows that the portion of energy that is bound is transformed into matter's rest mass, so that matter can no longer continue moving at the speed of light. Rest mass is therefore a measure of the degree to which energy is "imprisoned." This is why rest mass measures inertia, serving as a scale of how difficult it is to change matter's state of motion.

It is precisely this "binding" that fixes wandering free energy into an ordered existence that can be "located in space" and can experience "the passage of time."

The total energy of matter can be divided into two parts. One is rest mass, corresponding to bound energy and including the binding energy of all internal interactions and motions. The other is kinetic energy, including potential energy, corresponding to unbound free energy and also called "relativistic mass." According to the law of conservation of energy, rest mass and kinetic energy can be transformed into one another while total energy remains unchanged.

Then, at the different ordered levels of all things in the cosmos, through what specific means is energy bound to form rest mass? Following the ladder of material structure downward level by level, we discover that rest mass arises from a series of binding processes, from the Higgs mechanism to electromagnetic interactions.

First comes the "foundational mass" conferred by the Higgs mechanism. Elementary particles such as quarks, electrons, and the W and Z bosons acquire rest mass by interacting with the Higgs field. It is as if they were traveling through an invisible "syrup": drag slows them down, and part of what was originally pure kinetic energy is transformed into rest mass. Photons and gluons, by contrast, do not interact with the Higgs field. Their rest mass re-

mains zero, all their energy continues to appear as kinetic energy, and they travel forever at the speed of light.

The Higgs mechanism lays the foundation on which ordered material structures can form. Yet the rest mass it contributes accounts for only about one percent of the total mass of the visible universe.

Second comes the "dominant mass" produced by the strong interaction. Inside protons and neutrons, quarks are bound by the strong force carried by gluons. The binding energy involved in this process is enormous, far exceeding the Higgs mass of the quarks themselves. In fact, this is the source of the vast majority of the rest mass of protons and neutrons. The binding energy of the strong interaction contributes roughly ninety-nine percent of the total mass of the visible universe.

More subtly, when these protons and neutrons further combine through the nuclear force to form atomic nuclei, they instead release a portion of binding energy, causing the rest mass of the atomic nucleus to be slightly less than the sum of the masses of its constituent nucleons. The nuclear force is part of the strong interaction. This seemingly "deficient" mass is precisely the price paid for the stable existence of the atomic nucleus.

Finally, electromagnetic interactions add further increments to mass at the atomic and molecular levels. Atomic nuclei and electrons combine through the electromagnetic force to form atoms, and atoms combine into molecules and still more complex ordered structures. These processes likewise bind and release energy, but their contribution to rest mass is extremely small. Electromagnetic binding energy, for example, accounts for only about one billionth of an atom's total mass.

As for the weak interaction, although it affects particle masses in processes such as radioactive decay—beta decay, for example—its contribution is almost negligible beside those of the strong and electromagnetic interactions.

Gravity contributes to rest mass in a distinctive way. In any ordered system bound by gravity, from binary stars to galaxies and even the universe itself, gravitational potential energy is negative. It reduces the overall rest mass of the system without changing the rest mass of the constituent units within it.

Consider the system formed by the Earth and the Sun. Moving the Earth from the Sun's vicinity to an infinite distance would require work against gravity—that is, energy would have to be injected into the system. The total mass of the bound Earth–Sun system is therefore smaller than its total mass when the two bodies are infinitely far apart.

In other words, gravity is the "architect" of the universe. It shapes structures not by "adding material," but by providing a kind of "negative-energy adhesive." This adhesive reduces the total mass of the structure as a whole, yet allows that ordered structure to hold together.

More thought-provoking still is a conjecture advanced by cosmologists such as Edward Tryon and Stephen Hawking: the positive energy of all matter in the universe may be exactly offset by negative gravitational potential energy on the cosmic scale, so that the universe's total energy, and even its total mass, may be zero. If so, the universe may have needed no energy for an "initial creation" at all, but could have arisen as "something from nothing" through quantum fluctuations.

In summary, the vast majority of the mass in the universe is not an "inherent" property of matter such as elementary particles. Rather, it is the expression of energy bound into ordered existence. Once matter's multi-level ordered structures arise, bound energy is transformed into rest mass; matter with mass can therefore never exceed the speed of light. Rest mass is the form of energy "anchored" by ordered existence, and it endows matter with space and time.

If a process of multi-level ordered change alters matter's rest mass, it also corresponds to changes in that matter's own time and space. The transformation of energy into rest mass is the process by which energy is bound and condensed into order, creating spacetime. The transformation of rest mass into energy means that order moves toward dissolution and spacetime is consumed.

Such changes are not confined to the early universe, when matter was first created; they run through the whole course of cosmic evolution. The Sun, for example, pours energy outward at every moment. This means that its rest mass is continually decreasing and its own spacetime is continually being consumed.

The transformation between energy and rest mass is therefore, in essence, the creation and consumption of spacetime. The more complex the multi-level ordered structures, the more energy they bind and the richer their spacetime features become. This is the deeper meaning of "changing spacetime."

For particles with zero rest mass, such as photons, energy is not bound into order. These particles are pure transmitters of ordered change rather than its bearers. They retain sufficient "freedom," and all their energy is expressed as motion. Photons, for example, also undergo gravitational interaction, but gravity affects only the

path of their motion. Photons therefore have no concept of time or space; they exist in an eternal present. The entire "life cycle" of light, from its production to its absorption, is completed instantaneously. From light's perspective, no time passes at all—the extreme of time dilation—and no spatial distance is crossed—the extreme of length contraction.

Somewhat similar is the neutrino, known as the "ghost particle." It participates only in the weak and gravitational interactions. Its rest mass is extremely small, and it therefore occupies very little spacetime. It can readily pass through lead plates several kilometers thick, or even through the entire Earth, as though it had never "passed through" any space at all.

The vacuum constantly undergoes quantum fluctuations. This can be understood as spacetime being ceaselessly created and annihilated—as spacetime fluctuations.

Ideas at the frontiers of physics, including entropic gravity and loop quantum gravity, hold that the presence of matter and energy modulates the intensity of quantum fluctuations in the vacuum, producing a gradient in that intensity. The curvature of spacetime described by general relativity is precisely the macroscopic statistical expression of these differences in microscopic fluctuation intensity: the "curving" of the vacuum's spacetime-fluctuation structure, or "spacetime curvature."

Gravity therefore no longer requires the graviton as its carrier, but becomes an emergent phenomenon comparable to temperature. It is the macroscopic average of the collective behavior of countless quantum events in spacetime: an effective macroscopic force arising from the degree of disorder in the microscopic world, an

"entropic force."

If this account of gravity is accepted, grand unification will gain an entirely new opportunity. It could not only avoid at a fundamental level the "vacuum catastrophe" and "renormalization difficulties" of quantum field theory, but also explain the universe's accelerated expansion naturally through the statistical effects of quantum vacuum fluctuations, without introducing dark matter or dark energy. Gravity, quantum theory, and cosmology would then become self-consistent within a deeper emergent order.

From the perspective of symmetry, the process by which the cosmos generates order from chaos—a process of decreasing entropy—is also a process of continuous symmetry breaking. Symmetry breaking creates order, or new levels of order, and produces time and space. The present increase of entropy in the universe as a whole is, in turn, a movement toward overall symmetry and disorder, consuming time and space.

At the universe's initial stage, under ultimate symmetry, there was no distinction between before and after or between one thing and another. This meant ultimate conservation and disorder; spacetime did not exist. Symmetry breaking produces asymmetry and change, implying a tendency toward non-conservation and the creation of difference, and thereby creates spacetime. If conservation must still be maintained throughout this continuing process of symmetry breaking, ordered change cannot be isolated: through interaction, it must exchange with other forms of order. All things must therefore be interconnected. This also explains why various interactions can be derived from gauge symmetry.

To experience spacetime is to observe ordered change, while the

essence of observation is to enter into interaction with an ordered system.

The simpler an ordered system, the simpler its concepts of time and space; the more complex the ordered system, the more complex those concepts become. When we study the motion of macroscopic objects, for example, interactions such as gravity do not alter rest mass, so an object can be treated as an independent entity. Its changes follow the laws of relative motion under interaction and can generally be described by special and general relativity. Here our concern is with the relative time and space between things. As their relative rest masses and velocities change, so do their relative times and spaces.

When we analyze a layered complex system, each level of its order corresponds to a different conception of time and space. To understand the world, we must therefore use different spatial and temporal scales when analyzing different ordered systems and different changes in rest mass.

In the human body, for example, atoms influence molecules; molecules influence gene expression; genes influence cells; cells influence organs; and organs ultimately shape change in the whole body. Atoms, molecules, proteins, cells, organs, and the body as a whole each belong to a different level of ordered change and exhibit different symmetries. To find the laws at each level, we need frameworks of analysis suited to its scale of space and time.

Similarly, to grasp the changes of a business enterprise, we can understand it from many perspectives, including changes in individual personnel, the business performance of a department, changes in the company's operational data, and changes in the industry's

overall situation. Each perspective corresponds to a different level of the enterprise's ordered system, and therefore to different scales of time and space.

This corresponds to the spectrum formed by all the laws of the universe. Within a multi-level ordered system, the lower the level of order, the smaller the temporal and spatial scales of its changes. Those changes are rapid and random, yet the laws governing them are foundational, determinate, and universal. The higher the level of order, the larger the temporal and spatial scales of its changes. Those changes are slow and determinate, yet their laws are specialized, random, and strongly dependent on context.

This also explains why, as science and technology advance, our standards of time and space increasingly move away from dependence on specific, mutable macroscopic objects and toward universal, unchanging fundamental physical constants. The standard of time, for example, has moved from ancient astronomical cycles to the radiation period of the cesium-133 atom, while the standard of length has moved from parts of the human body to the distance traveled by light in a vacuum during a specified interval.

This evolution is precisely a profound microcosm of human cognition moving from "the concrete" toward "the abstract," from "the local" toward "the universal."

Viewed through the temporal dimension of change in all things, the range of timescales is extraordinarily vast. It stretches from Planck time, approximately 10^{-43} seconds, through particle decay on the femtosecond scale, 10^{-15} seconds, and atomic vibration on the nanosecond scale, 10^{-9} seconds, to the millennia of human civilization, the tens of billions of years in the life cycles of stars and

the universe, and even the scale of 10^{100} years predicted by theories of the universe's future heat death. Together these form a temporal spectrum spanning the instant and the eternal.

Viewed through the spatial dimension of ordered change, the range of scales is equally astonishing. It stretches from the Planck length, 10^{-35} meters, through the scales of particles—a quark radius of 10^{-18} meters, a proton radius of 10^{-15} meters, an atomic-nucleus radius of 10^{-14} meters, and an atomic radius of 10^{-10} meters—all the way to the diameter of the universe, about 10^{27} meters, or 93 billion light-years. Together these form an almost unimaginable hierarchy of spatial scales.

The vastness of these spacetime scales makes us feel our own smallness profoundly.

Forests and grasslands, rivers and oceans, hills and mountains make up natural environments of ceaseless change. Animals, plants, and bacteria comprise millions of different living species. Spring, summer, autumn, and winter succeed one another; wind, rain, thunder, and lightning alternate across the sky. All proclaim how distinctive Earth is within the universe.

Peoples, nations, cities, and companies form organizational orders. Urban circulation, roads, and power supplies form systemic orders. Science, art, and architecture belong to cultural order; aircraft, chips, the Internet, and satellites to technological order. Upon the foundational order of nature, human civilization has created more complex forms of order across more dimensions and levels.

Consider the human neural network. Each neuron contains roughly 10^{15} atoms, with more than a billion protein molecules working together within it. The number of possible combinations

of neuronal connections in the brain exceeds 10^{27}, already surpassing the total number of stars in the universe. It is no exaggeration to say that a universe dwells within our brain.

From the perspective of degree of order, much of the universe at more microscopic and more macroscopic spacetime scales may consist only of simple repetitions, like innumerable grains of sand. We do not usually agonize over our inability to know every grain of sand clearly; perhaps we need not brood over our inability to know every detail of the universe completely either.

Faced with the vast cosmos, we need neither belittle ourselves before the immense disparity of spacetime scales nor halt at the limits of our understanding. As highly ordered forms of life, human beings seek all the more to maintain and create higher levels of ordered existence, devoting themselves to resisting the increase of entropy in the universe.

When we say that we "see a beam of light," that light needs time to travel from its source to the retina. When we say that we "hear a clap of thunder," the vibration of sound waves through the air likewise requires time. More subtly, perception itself is a chain of ordered changes. Photons trigger chemical reactions in cone cells, signals travel along neurons, and electrical activity oscillates synchronously in the cerebral cortex. Each link has its own intrinsic time and consumes its own spacetime.

By the time our consciousness finally forms the thought, "At this moment, I see lightning," the lightning itself may already have become part of the past.

Our consciousness forever lives in a "slightly past" world jointly constructed by the speed of physical signal transmission

and the time of physiological processing. More profoundly, this delay is not uniform. When you look up at the starry sky, the "now" you feel is the Moon as it was one second ago, the Sun as it was eight minutes ago, the Andromeda Galaxy as it was two and a half million years ago. All these events of different time scales are juxtaposed, folded, and experienced in the "present" of your consciousness as one unified now.

From messages carried by relay horses and warning signals sent by beacon fires in antiquity to today's fiber-optic networks and satellite communications, the ways in which we perceive the outside world and exchange information with it have compressed spacetime enormously. In the past, being "at the ends of the earth" was almost synonymous with being forever cut off from news. Today, bits travel at the speed of light, and an event ten thousand miles away immediately becomes a reality before our eyes. Even the boundary between the past—historical data stored in the cloud—and the present—a live broadcast—has grown indistinct.

Each of us is therefore a special "spacetime folding point" in the universe. With the body as our anchor and consciousness as our medium, we fold events in the external world, scattered across different intrinsic timescales, into a single experiential "present."

This may be the modern echo of Su Shi's line, "We are but mayflies lodged between Heaven and Earth, a single grain of millet in the vast ocean." We are at once mayflies imprisoned within the intrinsic time of the body and the vast ocean capable of folding the cosmos's different spacetimes into a single thought.

A person's ability to understand deeply the temporal and spatial scales of order at every level is their ability to perceive and gain

insight into life and the universe. Their ability to connect and interact with everything outside themselves is their practical capacity to take part in reconstructing cosmic order and to leave their mark upon the "present." It is also their ability to reshape or command time and space.

The entropy increase of the universe erodes every embankment of order. In resisting it, our practice moves locally against the current: by injecting energy, we weave disorder and chaos into ordered structures and solidify randomness into meaning.

This practice of reshaping spacetime appears on three levels. Creation extracts order from disorder and binds wandering energy into rest mass that can be inhabited. From shaping a stone axe to building a city, every act of creation locally slows the passage of spacetime and anchors a human "present" in the universe. Transmission turns transient order into information that can outlast an individual life. Language, writing, and microchips release experience and wisdom from the prison of biological time, so that those who come later need not begin the journey again from chaos. Preservation maintains the order already made. From keeping food fresh to slow decay, through restoring artifacts against weathering, to sustaining the basic structures of civilization, every act of preservation contends with the thermodynamic arrow for the spacetime that belongs to us.

Physicists continue to work toward breakthroughs in string theory, loop quantum gravity, entropic gravity, the holographic principle, and other directions. Only by awaiting further explanation from science may we one day fully understand time and space.

But even before then, we already understand this: "We are like

dew, like lightning; in an instant, we become emptiness." All we can grasp is "only this present"—the moment within the eternal flow, defined by our interactions with all things.

Chapter 14

Order in Organization

Understanding nature meant understanding it as being capable of producing man and his societies.

Ilya Prigogine and Isabelle Stengers

In the process of understanding the world, we habitually divide it into an "inner self" and "everything outside." We once firmly believed that Earth was the center of the cosmos; later, we placed our own reason above all things, as though we were detached observers and masters standing beyond nature. This deeply rooted mindset leads us, often unconsciously, to separate human society from the natural world and treat them as two entirely different domains.

Yet human beings themselves, our thoughts and consciousness, and the whole of social civilization are all products of nature's self-organizing movement over billions of years. Beginning with elementary particles and passing through countless symmetry breakings, human society emerged within this same natural process. Its laws may look very different from those of the natural sciences, but in the final analysis they occupy different bands of the universe's single "spectrum of laws" and should follow the same natural principles and underlying logic.

Seen from the perspective that "existence is order," human society is itself a typical multi-level ordered structure. Individuals are the basic units from which this vast system is built; through particular "arrangements and combinations," they form organizations on different scales, from families and communities to companies and nations.

Just as protons, neutrons, and electrons combine in particular ways to form carbon atoms with distinctive properties, family members become a family through bonds of kinship and responsibility, while employees become a company through contracts and networks of specialized cooperation. Such organizations can "exist" and display distinctive properties and functions not merely because of the particular individuals they contain, but, more fundamentally, because of the stable and recognizable relations and structures formed among those individuals—a form of multi-level order.

This order is distinctly multi-level. Every level of order is built upon the one beneath it and gives rise to new properties absent from that lower level. Just as combinations of cells give rise to life, cooperation among individuals gives rise to organization.

More specifically, these levels include families, clans, and communities, which are based on kinship, place, or emotional bonds and form society's most basic ordered structures. Sole proprietorships, small and medium-sized companies, multinational corporations, and financial institutions form economic organizations centered on creating economic value through contracts and the division of labor. County and municipal governments, legislatures, courts, and armed forces form political organizations charged with main-

taining order and providing public services through laws, regulations, and charters. Schools at every level, research institutes, and academic associations form educational and cultural organizations devoted to transmitting knowledge and renewing culture. Trade associations, labor unions, nongovernmental organizations, and clubs are based on shared occupations or social aims and seek to promote an industry or address a particular social problem. The World Trade Organization and the United Nations are global organizations formed by sovereign states through agreements in pursuit of common goals.

It is precisely these nested ordered structures that make up a complex and vibrant human society, and it is these structures themselves that distinguish the forms and functions of different organizations.

The properties of an organization arise from its ordered structure rather than from its particular members. Just as the cells of the human body are continually replaced while the living structure formed by their relations persists, the particular participants in an organization may change while the multi-level relations sustaining its existence remain stable.

Just as elementary particles are the smallest units of matter, an individual person may be regarded as the smallest unit of organization. From an organizational perspective, an isolated individual does not constitute an organization, but remains in a free state of "non-organization," unconstrained by organizational relations. Once relations arise between people, however, an ordered organization is born, and the individuals within it become subject to the corresponding constraints.

In the material world, continuous symmetry breaking carries ordered structures from elementary particles to complex life, making them increasingly complex and moving them from extreme abstraction toward rich concreteness. This is why no two leaves in the world are exactly alike, and no two people are exactly alike.

Organizations, having emerged from the lowest level of matter through countless symmetry breakings, have become concrete in the extreme; no two organizations are exactly identical. Yet from the perspective of the essence of organizational existence, two organizations with the same multi-level ordered structure may be regarded as the same type of organization, as with a company whose personnel change or government bodies that share the same institutional architecture.

Within the ordered cosmic web that has neither inside nor outside, every relation in the material world can be traced to the four fundamental interactions. Among them, the electromagnetic force is the universal architect of our everyday world: it shapes atoms, holds molecules together, and forms the physical basis of all our perceptions and interactions.

When we move from microscopic particles to macroscopic organizations, however, the picture of relations acquires an equally astonishing complexity through symmetry breaking. The intricate emotional, economic, political, cultural, and other relations among social organizations have electromagnetic interaction as their physical basis at the lowest level. Yet through layer after layer of evolution and the emergence of spontaneous order, pure exchanges of energy have “broken symmetry” into highly complex patterns of connection endowed with social meaning.

Natural and emotional relations are rooted in biological instinct and can spontaneously generate powerful but informal cohesion, thereby building society's most fundamental and primordial forms of organization. Marriage, for example, connects different kinship groups; blood ties and family affection create families and clans; friendship forms primary groups founded on trust and shared experience.

Social and economic relations depend on social agreements and implicit contracts. Centered on economic exchange and specialized divisions of labor, they form medium-sized organizations through reciprocal cooperation. Morality, for example, constrains behavior through internal norms, while contracts and covenants build companies and other economic organizations through the formal exchange of rights and obligations.

Legal and political relations rest on formal rules, authority, and structures of power, shaping large-scale, highly structured organizational systems. Laws and regulations, for example, constrain behavior through coercive force and form the institutions of the state, while power operates through office and status to build governments, political parties, and other political organizations.

Cultural and value-based relations rest on shared beliefs, interests, and cultural identities, giving rise to organizations devoted to disseminating knowledge, developing culture, or pursuing social goals. Shared interests create societies and clubs; shared beliefs bind particular communities together; and the sustained pursuit of knowledge and culture forms the central bond within schools and research institutions.

Compared with the homogeneous groups formed by ants, bees,

and other creatures through simple rules, the relations of human society are highly layered and interwoven, forming what Qian Xuesen called a "complex giant system." Through these relations, individuals depend on and interact with one another, jointly weaving a web of order with neither inside nor outside, one that continually evolves as society changes.

This deeply corroborates Marx's assertion: "The essence of man is no abstraction inherent in each single individual. In its reality, it is the ensemble of the social relations." From beginning to end, no human being can leave the web of social relations and "exist" independently.

In summary, human civilization is, in essence, the vast ensemble of all the multi-level ordered organizations that human society has formed in the present.

The spiritual civilization on which we rely—language, philosophy, law, art, and more—is not a collection of abstract ideas suspended in the air. It ultimately takes concrete form in particular relations and structures within organizations of many kinds. Relations of contract, power, and culture form the soul of organizational order, regulating behavior and giving action its meaning.

Likewise, the material civilization we have created—technology, architecture, tools, and more—does not exist in isolation. Its creation, use, and value can be realized only within the social organizations that carry its functions. A magnificent building standing in the wilderness may be entirely useless; only when embedded in the organizational structure of a family, a company, or a city does it become a "home," an "office building," or a "landmark," and thus truly enter the body of civilization.

Human civilization is therefore also a dynamic, evolving system of ordered organization, extending through time and expanding through space. From the cognitive order of the individual mind, through the ethical order of family and community, to systems of governance and exchange at national and global levels, these structures are nested within and interact with one another, continually giving rise to new properties absent from the level below.

Together, all these forms of order mark humanity's great journey out of its primordial state of nature: we respond to challenges by building order and, in that process, continually endow our own existence with meaning.

Just as change throughout the universe begins with the motion of particles at the lowest level, every organizational change begins with changes in the thoughts and actions of its basic unit, the human being.

Compared with an organization constrained by layer upon layer of relations, an independent individual possesses almost absolute freedom. It is precisely this foundational freedom of agency that supplies the fundamental drive behind an organization's continuing evolution. Every layer of organizational structure both "binds" and "guides" individual freedom: by suppressing its randomness and giving it a particular direction of change, the structure drives the emergence of unified laws and order at higher levels.

Individual freedom is therefore the precondition for every possibility that can emerge within an organization, while organizational constraint provides the framework that turns possibility into reality. As the levels of organization rise, constraints grow stronger and individual freedom becomes correspondingly more limited, while

organizational determinacy and stability become increasingly pronounced. The more microscopic the organization, the more flexible and random it appears; the more macroscopic the organization, the more ordered and stable it becomes.

Viewed through the dialectics of nature, the tension between freedom and constraint within an organization is, in essence, the contradiction between the entropy-increasing and entropy-decreasing forces universally present at every level of order. As long as people continue to think and change, this contradiction remains eternal and omnipresent: it is the fundamental source driving organizational evolution.

The entropy-increasing force tends toward disorder and breaks the existing state, while the entropy-decreasing force maintains order and builds structure. Opposed yet unified, the two together constitute the organization itself and determine its present condition.

Following the law of the negation of the negation, organizational evolution necessarily passes through three stages: entropy increase breaks the old order; a transitional period of relative disorder follows; and, finally, a new and higher organizational order is conceived within disorder.

Consider a company. It begins with one or several free individuals who share a common goal: its founders. Through contractual relations such as partnership agreements and articles of association, they undertake the first “arrangement and combination” and form the most basic ordered organization, the founding team.

At this point, the binding force of organization first appears, and the individual’s absolute freedom begins to yield to shared goals and rules. This is the first “symmetry breaking”: from the chaos of

market intentions emerges a new entity with particular functions, such as product development—the company.

As the organization expands, a model in which all members collaborate directly becomes inefficient. Under the contradictory action of its internal entropy-increasing and entropy-decreasing forces, a crucial second symmetry breaking occurs: according to specialized divisions of labor—R&D, marketing, operations, and so on—employees are rearranged and recombined through functional relations to form different departments.

This process resembles cellular differentiation, giving rise to synergies and professional depth that no individual member or department possesses alone. Here, individual freedom is further "bound" and "guided" by functional norms and team goals, in exchange for organizational capacity and managerial power at a higher level.

As the number of departments grows and coordination becomes more complex, a third symmetry breaking follows. New relations, such as authority and reporting lines, are needed to coordinate the departments, and levels of management—managers, directors, vice presidents, and others—are created. They form the organization's central nervous system. Through decision-making, resource allocation, and performance management, they restrain the randomness and local interests of individual departments and ensure that the organization as a whole advances in an orderly way toward its strategic goals. A clear "decision–execution" hierarchy is thereby established, greatly strengthening the organization's determinacy and stability.

Ultimately, the company displays a typical nested architecture

of ordered organization: frontline employees, the basic units, form departments, the primary order; departments form business divisions through managerial relations, creating an intermediate order; and all the divisions, through the executive team and corporate strategy, form the enterprise group as a whole, the higher order. Every level is built upon the order beneath it and gives rise to properties absent from the lower level, including overall strategy, corporate culture, and brand value.

For an organization, particular members may be replaced as long as the web of relations sustaining its structure remains stable. The characteristics, functions, and properties displayed by the organization are emergent results and cannot simply be reduced to the properties of its constituent parts. Changes in the quality of its members can, of course, produce qualitative transformation through quantitative accumulation: a concentration of outstanding talent propels the organization upward, while an abundance of mediocrity brings systemic entropy increase and decline.

In the material world, when a multi-level ordered structure is bound too tightly, it resembles a hard crystal or rock, with atoms firmly locked into a lattice. Such a structure is indeed extremely stable, but it loses fluidity and adaptability, changes very slowly, and almost forfeits its capacity for dynamic evolution. Conversely, when the binding is too weak, the structure resembles a gas or a disorderly flowing liquid. Its basic units enjoy a high degree of freedom, yet they struggle to form and sustain complex higher-order structures; they scatter and dissipate easily and cannot accumulate complexity through iteration.

Life finds its exquisite balance precisely between these two ex-

tremes. The ordered structures of a living organism, such as the cell membrane and the DNA double helix, provide a framework for stability and genetic continuity—a form of "binding." At the same time, the organism permits and drives continual exchanges of matter and energy, the transmission of information, and random variation, giving its basic units "relative freedom."

It is precisely this dynamic balance—stable yet open, constrained yet flexible—that enables life continually to draw negative entropy from the environment and thereby acquire its extraordinary power of self-renewal, adaptation, and evolution.

The development of an organization follows this same principle in every respect.

When binding is too tight—with rigid hierarchy, ossified rules, highly concentrated power, and suppressed individual thought—it may bring temporary stability, but only by turning the organization into a "social crystal." Individual creativity is suffocated, the organization's feelers become numb, and its inability to adapt promptly to a changing environment ultimately carries it toward rigidity and decline.

When binding is too loose, by contrast—with no common goal, unclear authority and responsibility, and weak relations—the organization resembles a "social gas" and struggles to form effective structures of cooperation and collective action. Freedom is noisy, but constructive collective force is absent, and the organization readily dissipates or disintegrates through its own entropy increase.

The military has the mission of defending security. It requires a high degree of stability and collective action and does not pursue innovation in organizational structure; it therefore needs strict

discipline and relatively limited individual freedom, exchanging "tight binding" for effective execution. An academic organization, by contrast, takes the exploration of the unknown as its mission. It requires the collision of ideas and the emergence of innovation, and therefore must preserve ample autonomy and tolerate room for trial and error: "moderate binding" that nurtures possibility.

Neither is superior to the other; what matters is whether the form of organization matches its mission.

"Disorder creates possibility; order determines direction." The art of governing an excellent organization lies in building, according to its function and position, a framework of order that provides both cohesion and direction—the entropy-decreasing force—while preserving for its members ample autonomy, room for trial and error, and incentives to innovate—the entropy-increasing force. Only by seeking a dynamic balance within this permanent tension can an organization maintain its internal order while retaining the evolutionary drive to adapt continually to its external environment, and thus renew itself without end through the long river of time.

As the Scottish Enlightenment thinker Adam Ferguson wrote, "Liberty is not, as the origin of the name may seem to imply, an exemption from all restraint, but rather the most effectual application of every just restraint to all the members of a free state, whether they be magistrates or subjects." He also wrote, "Liberty is opposed to injustice, not to restraint; for liberty cannot subsist without the supposition of every just restraint."

Chapter 15

Optimal Organization

All reactionaries are paper tigers. In appearance, the reactionaries are terrifying, but in reality they are not so powerful. From a long-term point of view, it is not the reactionaries but the people who are really powerful.

Mao Zedong

The law of entropy increase applies equally to ordered organizations. Any isolated organization spontaneously tends toward a state of disorder.

Organizations in the real world are open systems. They can maintain or strengthen their internal order by drawing energy, information, and resources from outside—that is, by acquiring "negative entropy." In the long run, however, even their multi-level ordered structures struggle to withstand the continuing erosion of entropy increase. The web of relations sustaining an organization—whether relations of contract, power, or culture—will gradually loosen under environmental change, internal rigidity, or stagnating innovation, until its functions fail and its structure disappears.

Enterprises offer a clear example: the data reveal the finite na-

ture of organizational lifespans. Private enterprises in China survive for only about 3.7 years on average. Even in Shenzhen, with its leading environment for innovation, the average is only six years, and just 17 percent survive for more than ten years. International giants once regarded as "century-old institutions," including Kodak and Lehman Brothers, likewise went bankrupt or disintegrated after failing to adapt to technological change or market upheaval.

This shows that although excellent governance can delay the increase of entropy within an organization, the irreversible law of entropy increase leaves every organization facing the inevitable fate of moving from order toward disorder.

For precisely this reason, the meaning of an organization's existence is to counter entropy increase.

First, an organization must remain open and sustain a continuous metabolism. Just as a living organism maintains internal order through breathing, feeding, and excretion, an organization must exchange talent, resources, and information with the outside world in order to break through rigidity. Huawei, for example, uses a rotating-CEO system and intensive investment in R&D—more than 20 percent of annual revenue—to stimulate and absorb each person's capacity for innovation, promote the movement of talent and the iteration of technology, and renew the organizational body. Only by continually drawing "negative entropy" from outside can an open system delay the tendency of its internal structure toward disorder.

Second, an organization must continually iterate in order to resolve the internal contradictions and tensions that environmental change creates within its ordered structures at every level. These

tensions are concrete expressions of the entropy-increasing force, and excellent organizations actively expose and resolve them. Alibaba, for example, has repeatedly restructured its organization, separating e-commerce, cloud computing, and other businesses from a mixed model into independent companies and restructuring relations of production in order to release productive forces. This iterative process of "symmetry breaking" adjusts relations among levels so that order can be rebuilt in a higher dimension.

From a thermodynamic perspective, any process of entropy decrease in an organization inevitably comes at the cost of entropy increase in the outside world. The construction and maintenance of the ordered structures of human civilization are fundamentally achieved through the consumption of energy and resources. Therefore, if civilization is to develop and continuously achieve higher levels of order, it must construct ever more open systems and raise the efficiency of energy use.

Every leap in civilization has been accompanied by a revolution in the use of energy: from biomass in the age of gathering, to firewood and animal power in the agricultural age, then fossil fuels and electricity in the industrial age, and onward to nuclear fission today and nuclear fusion in the future.

Yet resisting entropy increase does not mean pursuing expansion and consumption without end. Just as excessive intake of energy can cause metabolic disorder in a living organism and thereby accelerate its decline, so it can in an organization. A company that expands blindly and manages excessively will become bloated and inefficient; a society that purchases growth by sacrificing its ecology will ultimately face depleted resources and

environmental backlash.

True sustainability lies in the dynamic coordination and balance of a system's internal and external relations. The wisdom contained in this idea accords with the essence of Laozi's *wuwei*: not passive inaction, but the rejection of reckless action and excessive intervention; following the system's own laws—the Dao—and guiding events according to their tendency, so as to reach the cooperative order expressed in the line, "I take no action, and the people transform themselves."

Only thus can an organization, in the eternal struggle against entropy increase, walk the path of harmonious development.

From the perspective of symmetry breaking and the spectrum of laws, we must face directly the distinctive character of the laws of social organization. They do not possess the universality and permanence of fundamental physical laws, but appear instead as evolving trends and patterns of low symmetry and high contextual dependence within particular settings of space and time.

We therefore cannot rely on simple reductionism and decompose organizational behavior into an accumulation of individual psychologies; nor can we rely on macroscopic emergentism alone and ignore the dynamics of the underlying structure. Only by combining the two dialectically—analyzing the concrete behavior and relational networks of constituent units at every level, including individuals, teams, and departments, while also examining how they interact from level to level and give rise to macroscopic properties—can we penetrate complex phenomena and grasp both the present state of the organization's ordered structures and the contradictory tensions within them.

Moreover, the different levels of order within an organization—from project groups engaged in immediate interaction, through business departments with medium-term goals, to the company as a whole formulating long-term strategy—differ greatly not only in their constituent units and relational structures, but also in their corresponding temporal and spatial scales. Frontline employees attend to daily or weekly order, such as contact with clients; managers attend to quarterly or annual strategic order, such as market expansion; and senior leaders attend to industry order over five to ten years, such as technological iteration.

To discern the trends of self-organizing change at every level of an organization, we must therefore cultivate the ability to adapt across scales, shifting our perspective flexibly among different temporal and spatial scales as though adjusting the focus of a microscope. Only then can we understand the full picture of the organization as a dynamic complex system. The collection of laws across its different levels constitutes its distinctive "spectrum of laws."

As a company grows from three or five people into a large enterprise, it must build business processes such as marketing, R&D, and production across the organization; vertically, it must connect management, departments, and frontline employees and establish management subsystems for human resources, finance, supply chains, and more.

Managers must possess a multi-level perspective, continually attending to changing tensions and trends among the company's internal subsystems and keeping the units at every level in a relatively ordered state. At the same time, they must regularly step outside the company's own viewpoint, place it within the larger or-

ganizations of its industry and society, and analyze changes in its relations with external organizations, so as to judge the company's direction of development from a higher dimension.

For an individual, actively choosing a suitable organization as a vehicle for development and discerning the self-organizing trends at each of its levels may make possible a deep coupling between person and organization. This coupling is not passive dependence, but active integration and coevolution.

When individuals can grasp an organization's internal "spectrum of laws" and the rhythm of its symmetry breaking, their inspiration and effort—the microscopic fluctuations within the organization—are more readily taken up and amplified by it. They can thereby write their own chapter of destiny within the organization's larger narrative. This vividly interprets the idea that "human determination can triumph over Heaven": individual will, on the premise of following the laws, bravely opens up possibility.

Conversely, if the individual is utterly ignorant of the structural fabric and evolutionary tendency of the organization one inhabits, one can only drift passively with the current. One's role will be reduced to an ordinary unit in the organization, readily replaceable. One's fate will be like a mote of dust in the torrent of entropy increase, full of randomness and powerlessness. This is the modern metaphor behind the saying that "destiny is decreed by Heaven": without understanding or choice, the individual is inevitably swept along by the invisible forces of organizational and social systems.

At a deeper philosophical level, these two states of life embody two realms of *wuwei*. The former is active *wuwei*: having understood the laws, one acts with the tendency. As Laozi said, "Practice

non-action, and nothing will be ungoverned." The latter is passive *wuwei*: because understanding is limited, one simply does nothing. As the *Daodejing* says, "High virtue does not act and has no end in view; low virtue acts and has an end in view." True wisdom lies in distinguishing their essential difference: does one participate in creation with lucid understanding, or is one ruled by trends in a state of confusion?

Extending the view to human society as a whole, historical materialism's division of social forms according to their relations of production accords deeply with the essence of organizational existence. Every dominant relation of production corresponds to a particular macro-level ordered structure of social organization.

Yet just as tensions among goals exist within every complex organization, a social structure based on relations of production also gives rise to classes through differences among organizational levels. The ruling class seeks to consolidate the existing order and maximize its own gains, while the ruled class seeks to improve its conditions of survival and secure greater rights and interests. This deeply rooted divergence of goals causes the forces at different levels of social organization to cancel one another as they work against the entropy increase of the whole.

Over time, class contradictions intensify friction and dissipation within the system. Eventually, when the relations of production can no longer contain the full eruption of internal contradictions, the social organization as a whole reaches the critical point of entropy increase, and a violent "symmetry breaking" gives birth to a new and higher form of social order.

What, then, should the optimal form of social organization be?

Its fundamental characteristic is that the various levels within society represent only different divisions of social labor, rather than becoming fixed into classes with opposing interests.

In an organization that has eliminated class divisions, every level from the individual to the state shares one unified core goal: maintaining and advancing the order, vitality, and sustainable development of the entire social organization, with "serving the people" as its central purpose. When every level directs its effort to "resist entropy increase" toward this common end, the social system as a whole can exist and develop with optimal effectiveness, approaching the ideal state of the "optimal organization."

In this state, organizational governance embodies a dialectical unity of democracy and centralism. When a decision benefits the great majority of members and advances the organization as a whole, the democratic principle should gather collective wisdom. Conversely, when a change is disadvantageous to one part in the short term but necessary for the survival and long-term interests of the whole, the part must submit to the whole. The same applies when short-term and long-term interests are weighed: to pass through cyclical challenges, every level must submit together to a larger goal that transcends immediate gain and loss. This wisdom of seeking the best solution for the whole amid dynamic change is the central secret by which the optimal organization extends its ordered life as far as possible.

In the conception of the optimal organization, the core force responsible for unifying understanding, carrying out common goals, and driving continual organizational iteration is the management level equipped with a perspective of organizational order. As the

embodiment of the interests of the whole, it represents no special interest group; it coordinates contradictions through democratic centralism and opposes organizational rigidity—that is, entropy increase—through self-renewal.

Take Huawei as an example. This enterprise organizational form embodies, to a considerable extent, the core characteristics of the optimal organization.

Through its employee shareholding plan, Huawei integrates capital and labor and, at the level of ownership, dissolves their traditional opposition. By making "customer-centeredness" its goal, it establishes the organization's aim of becoming a "negative-entropy body" in relation to its external environment. By "putting strivers first" as the value guiding distribution according to labor, it directs R&D, markets, supply chains, and every other level of the division of labor toward the same end, thereby awakening the power of the lowest level.

"Long-term arduous struggle and persistent self-criticism," together with the rotating-CEO system and continual management reform, have established a mechanism of internal self-iteration and resistance to entropy increase. Just as an advanced political party achieves unity in decision-making through democratic centralism, Huawei's strategic decisions, formed after full discussion, require efficient execution by all. This displays the dialectical unity of releasing and constraining individual creativity on the premise of safeguarding the interests of the whole, making Huawei a model at the enterprise level for resisting organizational rigidity and evolving toward the optimum.

Max Weber, the father of organizational management, laid the

foundation of modern managerial efficiency through his theory of the "ideal bureaucratic organization." Yet Weber's ideal organization is a highly routinized, closed system that lacks the dynamic adaptability required by complex change. The optimal organization described here more closely resembles a living organism: it possesses not only a foundation of efficiency, but also the capacity to achieve a dynamic balance between innovation and order through a unified goal and democratic centralism.

The fundamental defect of the mainstream corporate system lies precisely here. An enterprise's highest goal is ordinarily to maximize shareholder interests, a design that narrowly confines its purpose to the financial returns of capital owners rather than the long-term health of the organization as a whole. Even when several goals are considered, the logic of capital often places shareholder value in fundamental conflict with the interests of employees, customers, and others.

This deficiency, rooted in the property-rights system, makes it difficult for an enterprise to form a supreme goal serving the order of the organization as a whole. Because their goals diverge, the efforts of its internal levels cancel one another out, planting the seeds of decline in the structure itself.

In the material world, exciting atomic nuclei at the lowest level can trigger a chain reaction and release the immense energy of a nuclear bomb. Likewise, in an optimal organization, when every member at the lowest level is activated by a common goal and their actions form a chain reaction, an "organizational nuclear bomb" of reality-transforming energy erupts. The essence of this energy is the synergistic resonance produced by the close alignment of goals

at every level of the organization.

Comrade Mao Zedong declared that "all reactionaries are paper tigers" precisely because the organizational nature of a reactionary class prevents it from truly mobilizing the power of the people at the lowest level. The goals of its internal levels tear one another apart; like nuclear material incapable of sustaining a chain reaction, however great its bulk, it amounts to no more than a display of empty strength.

By contrast, consider a company such as Huawei, which approaches the optimal organization. Faced with comprehensive sanctions and blockades from the United States, its mechanism of "putting strivers first" successfully awakened the creativity and fighting strength of every employee. From R&D to production and from markets to supply chains, once the basic units were ignited by a common goal, their chain reaction of innovation ultimately condensed into an "organizational nuclear bomb" that broke through the encirclement. Any external pressure, however powerful it may appear, will ultimately reveal its nature as a "paper tiger" before a fully activated optimal organization.

The precise ignition of such a chain reaction often comes from decisions at the top of an organization. Yet we sometimes observe an apparent paradox: certain small "high-level decisions," such as enacting a law or launching a war, seem able to produce macroscopic consequences far greater than the accumulated "quantitative changes at the base." Does this negate the principle that change is driven from the bottom level?

Quite the contrary: this precisely embodies the profound dialectical relation between internal and external causes. A high-level de-

cision can produce an immense effect only on the fundamental condition that it successfully triggers a chain reaction among the units at the base—individuals, cells, or molecules. A legal provision truly "takes effect" only when countless individuals internalize it in their everyday conduct; an order for war can become a battlefield reality only through the obedience and action of soldiers.

This vividly demonstrates that "external causes operate through internal causes." Every macroscopic force of transformation must ultimately become an actual influence upon the basic units before it can become reality. The so-called "top-down" and "bottom-up" views of evolution are in fact two sides of the same process. The art of governing the optimal organization lies precisely in grasping this dialectical relation and finding a dynamic balance between guidance from above and vitality from below.

The logic of the optimal organization can be extended further to humanity as a whole. When the goal of "resisting entropy increase" shifts, for individuals, peoples, and states, from partial interests to the survival of the whole, the core logic of a human community with a shared future takes shape.

When all humanity can transcend the boundaries of nation, ethnicity, and culture, reach a shared understanding of its goals and a consensus of values, and jointly confront the global challenges brought by entropy increase—such as climate change, resource crises, and public health threats—human society as a whole will coalesce into an optimal organization on an unprecedented scale.

The rapid development of digitalization and intelligent technologies is creating a historic opportunity for this. The Internet, big data, and artificial intelligence are breaking down barriers to

information and understanding, making dialogue and cooperation across civilizations possible and greatly accelerating the formation of a shared global understanding.

This is like the chain reaction of nuclear fission. When every individual and every people is activated by the goal of a shared destiny, human civilization will release a creative power greater than in any previous period of history, powerful enough to break the shackles of the old order. As Mao Zedong revealed, before an awakened and united people of the world, any old system that obstructs the liberation of humanity as a whole will prove to be a "paper tiger" in appearance only.

Chapter 16

Within and Beyond the Bounds

Entropy increased in the universe, and order decreased. ...

But low-entropy entities were different. The low-entropy entities decreased their entropy and increased their order, like columns of phosphorescence rising over the inky-dark sea.

This was meaning, the highest meaning. ...

To maintain this meaning, low-entropy entities had to continue to exist.

Liu Cixin

As the highest known form of low-entropy existence, humanity is extraordinarily complex and highly ordered. Life is wondrous and precious, yet we continue to ask: what, ultimately, is the meaning of life and of our own lives?

Perhaps the very act of asking is itself the glimmer of those phosphorescent fires in the darkness.

One view of the source of meaning holds that it is bestowed from outside, arising from some transcendent existence such as God or cosmic law. It is as though the script had already been

written and each life were meant to fulfill a larger design: entering through the narrow gate to attain redemption, as in Christianity, or following the "Way of Heaven," as in Eastern thought. From this perspective, the task of life is to discover and live out one's appointed role.

Another view holds that meaning is not discovered but created by our own hands. As the philosopher Jean-Paul Sartre argued, human beings have no predetermined essence; through every choice, we shape what we become. Albert Camus likewise held that the world itself has no meaning, but that this very absence leaves us free to define our own values within it.

Science offers another, colder perspective: the brain with which we ask about meaning is itself a product of evolution, and perhaps the act of asking merely helps us survive. The universe itself, as Bertrand Russell observed, possesses no consciousness that sets a purpose for humanity. This can sound disheartening and may even slip into the pessimistic picture associated with Arthur Schopenhauer: life as fundamentally empty, oscillating between suffering and boredom.

Yet thought did not stop there. Having recognized this, Russell chose to establish value through the distinctively human capacities for love and compassion. Friedrich Nietzsche went further, treating "nothingness" as an opportunity and urging human beings to become powerful *Übermenschen*, courageously giving themselves laws and creating new values. The psychologist Viktor Frankl, through the suffering he endured in concentration camps, bore witness to the idea that life's meaning lies in every concrete situation: in the work to which we devote ourselves, in the beauty

and goodness we experience, in sincere feeling shared with others, and, when suffering cannot be avoided, above all in the attitude and response we choose.

We seem always to oscillate between the view that "meaning is bestowed from without" and the view that "meaning is created from within." The answers are many and varied. Some hold fast to one and live by it; some remain perplexed throughout their lives; some even declare life meaningless because it must one day pass away.

Yet all these discussions remain confined to a human-centered perspective. In the spectrum of laws, after countless symmetry breakings, meaning as seen by human beings has become extremely complex and diverse, and is no longer universal. But if we set aside this human "self-centeredness" and turn toward existence in its wider forms—a stone, a blade of grass, an ant, a rushing river, silent stars—what is the meaning of their existence?

Only a clear underlying logic that spans everything from elementary particles to human society and applies universally to all things can lead us to a deeper understanding of the nature of meaning.

Some may say that grasses and trees, mountains and rivers, the sun, moon, and stars simply "exist." Indeed, they reveal themselves fully and authentically: flourishing and withering, rushing and stillness, burning and annihilation. But might their meaning lie precisely in this "existence" itself?

As we have argued throughout this book, all things exist as multi-level order, while complete disorder, in a certain sense, is equivalent to non-existence. Through the fundamental interactions,

energy arranges and combines "nothing," creating order within complete disorder.

If we are to define "meaninglessness," it corresponds precisely to a state without any ordered feature, one of total randomness and chaos. Conversely, as soon as order appears, existence appears. Order sustains particular information and form within an ocean of disorder; against the background of meaninglessness, it constructs meaning in a relative sense.

In this entropy-increasing cosmos, everything, living or not, as long as it still maintains its own ordered structure, already possesses the most fundamental existential meaning.

The universe's movement toward disorder and dissolution seems to unfold a vast background pointing toward ultimate meaninglessness. Against that background, every existence that resists this movement toward disorder—every structure that maintains order—is meaning by virtue of its very existence.

It is precisely upon this canvas of meaninglessness, laid out by entropy increase, that all things inscribe their own meaning through their brief yet steadfast ordered existence.

As the ancients said, "Heaven and Earth are not benevolent; they treat the ten thousand things as straw dogs."[1]

Quarks find their meaning in the stable structures of protons and neutrons. Atoms manifest meaning through the ordered systems they form with electrons. Molecules display their endlessly varied properties and meaning through even more complex combi-

[1] Straw dogs were ritual objects made for sacrifice and discarded afterward. The phrase comes from Chapter 5 of the *Daodejing*; it conveys the impartiality of Heaven and Earth rather than cruelty in the ordinary human sense.

nations. Mountains and stars, in their grand forms and across time scales of hundreds of millions of years, passively resist dissolution. Their very existence is a proclamation of meaning.

Life, from blades of grass to ants and on to human beings, actively and continuously draws "negative entropy" from the environment through metabolism and reproduction. In a cosmos where Heaven and Earth are not benevolent, life becomes the active author of meaning. All living existence carves out and guards oases of order within a disordered cosmos. For this reason, to be alive is to have meaning. The most thorough meaninglessness is to let this ordered flame go out—to let oneself cease to exist.

Looking further, all things are nested within the cosmos's ordered web, which has neither outside nor inside. Beyond sustaining its own order, every existence necessarily belongs, as one part, to a higher-level ordered structure. Meaning is thereby elevated, taking on dimensions that rise from level to level.

A protein molecule has meaning as a molecule by virtue of its own ordered structure. When it enters a cell, however, and serves as part of the cytoskeleton or as an enzyme catalyzing reactions, it acquires a higher meaning at the cellular level. Likewise, a cell does more than sustain its own metabolism. A cardiac muscle cell, beating rhythmically in the heart, participates in the mission of maintaining an organ; and the heart, that exquisite organ, gains a still higher meaning as the core of the circulatory system, sustaining the life of the whole organism.

It is clear, then, that from molecules to cells, from organs to the human body, meaning at each level of existence is no longer confined to that level alone. As a constituent part of a higher-level

ordered structure, an individual naturally bears a responsibility to help sustain that structure, and thereby acquires a higher-order meaning that transcends the individual level.

The Confucian path of "cultivating oneself, regulating the family, governing the state, and bringing peace to the world" precisely reveals how individual meaning is elevated within broader social structures. The family, the community, the nation, and indeed human civilization itself are essentially ordered organizations that humans have created in order to counter entropy increase more effectively through collective effort and to construct and sustain large-scale order.

Therefore, each of us, beyond realizing the order and flourishing of our own life—that is, "cultivating oneself"—also bears the natural responsibility of sustaining and improving the social organizations we inhabit. Striving to make the family harmonious is "regulating the family"; creating value for society, and even advancing the progress of the nation and the world, is "governing the state and bringing peace to the world." These are not additional burdens external to our own meaning. They are the inevitable manifestation of the meaning of our individual existence at higher levels.

All correct effort counters entropy increase. For cultivating oneself, the primary task is to maintain, as much as possible, the healthy functioning of our body, that highly ordered organism. The law of entropy increase pronounces the inevitability of the body's eventual disintegration. Yet we can still, through disciplined living, exercise, and care, do our utmost to delay this process.

Next is lifelong learning: building ever more complex and ordered neural networks. For the individual, the body will age and

tend toward disorder. Only the network of connections within our brain can, through continuous learning and experience, become ever more ordered, rich, and powerful. This is the most valuable inner asset we can actively construct.

Finally, we must squarely face the fact that every human being will die, while understanding and treasuring the value of reproduction. The passing of the individual is the victory of entropy increase. Yet the continuation of life ensures that the grand ordered structure of human civilization is passed on, so that the writing of meaning is not interrupted.

Above the foundational meaning of sustaining one's own existence, the meaning that humans pursue displays dazzling diversity.

Having undergone countless symmetry breakings, human society has evolved into extraordinarily exquisite and diverse ordered organizations. Civilization itself is like a grand machine for countering entropy increase. Its stability and prosperity depend on the precise collaboration of countless subsystems in education, culture, the economy, and politics.

Thus, whether we are musicians, teachers, architects, or lawyers, as long as we work in our respective posts to sustain and advance the order of some social subsystem, the work we do transcends a mere means of making a living. It becomes bricks and mortar in the edifice of human civilization. It acquires a higher-order social meaning above individual existence.

It is precisely this division of labor and cooperation that endows meaning with infinite diversity. We possess not only the freedom to choose the direction of our meaning, but also the precious opportunity to create value and meaning with our own hands in the

domains we choose.

Within layered structures of order, meaning belongs not only to the units at each level; it also resides in the relations of interaction that bring these units together and enable them to function.

Light is itself a carrier of energy, but when it drives photosynthesis in plants, it participates in building the order of life. Minerals, roads, and data acquire socioeconomic meaning beyond their physical existence once they are integrated into the networks of production and exchange in human society. Money is the clearest example. Its value does not arise from its material carrier, but from its role as a relatively stable medium that efficiently enables complex, ordered interaction among organizations and individuals. This is the point Marx made when he located the value of money in exchange.

Therefore, rather than hoarding wealth, a more meaningful course is to let it flow, bringing more people and things into relation.

As the Buddhist thought of dependent origination reveals, all things exist within a web of mutually dependent relations. These complex relations may seem to bind us and cost us some individual independence, yet they also allow us to find the essence of meaning within the dialectic of no-self and self.

The relations in organizations define our roles. We are children, parents, employees, friends, citizens. It is precisely these relations and responsibilities that give us a coordinate system for finding our own meaning. Through caring for our family, collaborating with colleagues, and contributing to society, we feel needed in these relational interactions. We thus confirm our value and meaning in the corresponding organization.

Of course, if we are to find meaning more clearly, we must not lose ourselves in a web of complex relations. Wisdom lies in maintaining clarity of perception, learning to recognize the ordered level we inhabit, whether family, team, company, or society, and understanding how that level works to resist entropy increase and sustain its order.

When, through the practice of "investigating things and extending knowledge," we discern our true heart and actively contribute to sustaining and strengthening the order of that level, we have found the correct direction for individual action.

This objective tendency and inner requirement to sustain and strengthen the order of the level we inhabit is, in essence, what ancient Chinese thinkers called the Dao or innate knowing. It is not a distant and unfathomable law. It appears concretely in the inner principles by which every ordered existence develops in a healthy way.

Ancient thinkers therefore urged people to "follow the Dao" and "extend innate knowing." The profound practical meaning of these injunctions is that they guide us beyond narrow self-centeredness. They teach us not to treat relationships as burdens, but to discover and follow the healthy direction of development at every ordered level to which we belong.

An organization is an ordered structure made up of nested levels. Each level, through its own distinctive logic of operation, gives rise to internal contradictions and tendencies of development. We therefore need a dialectical wisdom that can both examine the particular tendencies at each level and weigh them together across levels.

For an individual, death marks the end of meaning, yet sacri-

fice for the survival of one's people attains the highest meaning. For a company, profit is an internal requirement, yet profit gained by harming society destroys meaning at its root. War ordinarily destroys order, yet a just war is a necessary means of defending order on a larger scale and resisting extreme disorder.

To truly "follow the Dao," then, is to recognize the contradictions and tendencies at every level and bring them into a dialectical unity.

Over the long evolution of human society, principles proven in practice to sustain and promote the orderly operation of organizations at every level gradually settled into concrete requirements governing relations between people. The clearest and most rigid became laws, regulations, and contracts. Those resting on broad agreement but allowing greater flexibility became moral norms and cultural customs. Others, more fundamental yet difficult to define precisely, came to be honored as universal values such as truth, goodness, and beauty.

These relational constraints are, in essence, the interaction forces that sustain an organization's ordered structure. They define how its units are connected and the rules by which they operate. Since organizations differ in structure and environment, these constraints differ as well, just as moral ideas vary among peoples and legal provisions among countries.

Nor are these principles immutable. They change and adjust as social organizations themselves evolve. The moral demand once embodied in memorial arches for chaste widows[2] stands in sharp

[2]From the imperial period into the early twentieth century, communities and the state sometimes erected commemorative arches to honor widows who remained chaste, turning

contrast to today's ideal of gender equality, a vivid reminder of their historical character. These principles that sustain order are the concrete expression of the Dao within a particular network of human social relations.

Laws, regulations, contracts, morals, norms, customs, and even our pursuit and definition of truth, goodness, and beauty are therefore concepts that arise and exist only within a multilayered ordered organization such as human society. The cosmos itself contains no "right" or "wrong," "justice" or "evil" in the human social sense. Its phenomena are simply manifestations of the evolution of cosmic organization.

The definition of meaning, and the relative standing of one meaning above another, can therefore be given only with reference to a particular, relatively stable ordered structure. Meaning and meaninglessness form a dialectical unity: within an ordered structure, meaning acquires its definition through contrast with meaninglessness. When we say that something is meaningless or that one kind of life is more meaningful than another, we are implicitly referring to some multilayered ordered organization that we recognize or inhabit.

This is the deeper meaning of being "within and beyond the bounds." To be within is to stand inside an order and a frame of reference, where meaning can grow. To be beyond, as in Zhuangzi's phrase "wandering beyond the bounds," does not mean entering complete disorder, but transcending a particular bounded order, such as conventional rites and rules. To pass beyond one set of bounds is to enter another. If every frame of reference were

a gendered moral ideal into public architecture.

left behind, meaning could no longer be spoken of. In this sense, meaning can be defined only from within some bounds.

Consider a soldier who dies bravely in war. The "meaning" of that sacrifice depends entirely on the value assigned to it by the camp he serves—a particular ordered organization. To his own side, he has died defending home and conviction, and his sacrifice is noble and deeply meaningful. To the opposing side, the same act may be an obstacle to its aims, a threat to be removed. Step outside both camps, and from the perspective of cosmic physics alone, the sacrifice is simply the ending of a living phenomenon. Its meaning is established and made visible only within the ordered organization he recognized and defended.

We must therefore seek outward, entering deeply into the tendencies and demands of the multilayered ordered organizations we inhabit, so that we may seek facts. We must also seek inward, continually refining and strengthening our own capacity to understand, so that we may seek truth.

In this complex and diverse human society, this allows us to follow our own hearts in pursuing goals that we ourselves regard as meaningful. You may choose to disregard the constraints of every organization and simply "lie flat," but this is only a deliberate retreat to the basic meaning of preserving individual survival. If you choose self-extinction, then at the level of the individual you truly slide toward meaninglessness.

Whatever you choose, the pursuit of meaning will involve suffering, because this is the logic of life itself.

If you choose to pursue a higher meaning together with the organization to which you belong, then you choose to resist entropy

increase in both the cosmos and the organization, and must bear the hardship of rowing against the current and advancing through disciplined practice. If instead you choose to go with entropy increase, perhaps hoping for comfort and freedom, then so long as you remain within an organization, you will suffer its constraints and the consequences of acting against its way and surrendering to the cycle. Even if you consider only the individual, you must still bear the hardship of sustaining life against entropy increase.

As the Buddha taught, the essence of the cycle of birth and death is suffering. Nothing in the cosmos escapes the inevitable cycle from ordered formation to chaotic dissolution; the fate of individuals and organizations alike is formation, endurance, decay, and emptiness. Death creates the possibility of rebirth, and only then can all things continue to evolve. To rise above the cycle, relinquish attachment to the self, and move beyond both entropy increase and entropy decrease is also to move beyond suffering and pleasure and reach the stillness of nirvana.

This shore is the other shore, and the other shore is this shore. The Buddhist wisdom of leaving the world offers an ultimate metaphysical consolation and perspective, while striving within the world is the necessary path by which we practice meaning in this life.

"As Heaven maintains vigor through movement, a noble person should constantly strive for self-improvement." From Pangu opening Heaven and Earth out of chaos, and Yu the Great channeling the floodwaters, to the Foolish Old Man moving the mountains and Jingwei filling the sea,[3] these ancient myths inscribe a com-

[3] Pangu separates Heaven and Earth; Yu controls the floods by channeling rather than

mon code of human survival: do not depend on divine rescue, do not submit to the arrangements of fate, and survive through unceasing self-strengthening.

We recognize clearly that entropy increase is the inescapable main current of the cosmos, and we understand just as clearly that the low-entropy state of this moment is life's battlefield. As Romain Rolland wrote: "There is only one heroism in the world: to see the world as it is and to love it."

Projected onto a cosmic scale, this heroism makes intelligent life into Sisyphus pushing his stone uphill. We know that the boulder of entropy will finally roll back down; yet through our struggle above the abyss of nothingness, we define our own existence and meaning.

damming them; the Foolish Old Man resolves to move two mountains; and Jingwei, transformed into a bird, carries twigs and stones to fill the sea. All four are Chinese emblems of persistence against overwhelming conditions.

Chapter 17

The Sealed-Room Dilemma

We never observe an object without its being modified or tinged by our own activity in observing it. Under the impact of our refined methods of observation and of thinking about the results of our experiments, that mysterious boundary between subject and object has broken down.

Erwin Schrödinger

Wang Yangming once said, "Before you looked at this flower, this flower and your mind were silent together." This seems to imply that the full manifestation of existence depends on the illumination of the mind.

Across a very different intellectual tradition, the microscopic world of quantum mechanics seems to echo this philosophical reflection from centuries ago: no physical reality exists wholly independent of observation.

Quantum mechanics holds that before we observe an electron, asking where it is has no definite answer. It is perhaps not a tiny entity located at a fixed position. It is more like a diffuse probability cloud. It is the superposition of all possible paths and positions. At

that moment, it approaches a kind of latent possibility.

Only when measurement intervenes does this probability wave function collapse. An electron with position and momentum then manifests in spacetime, becoming a determinate physical reality.

Similarly, before measurement, many attributes of microscopic particles are likewise undetermined latent possibilities.

For example, before its polarization direction is measured, a photon does not inherently possess the classical property of horizontal or vertical polarization. It is in a superposition of all possibilities, simultaneously containing the potentiality of both horizontal and vertical polarization. Only at the instant when it interacts with a polarizer oriented in a specific direction—the measuring device—does its wave function collapse. It randomly manifests as either passing through or being blocked, thereby determining its polarization state along that measurement direction.

No act of observation, therefore, should be seen as a one-way probe by an isolated subject into a passive object. It is a dance in which observer and observed are drawn together. Through a particular mode of inquiry, the observer defines the stage on which reality can manifest. Yet the observed is not wholly passive: through its own latent possibilities and tendencies, it shapes and constrains the possible outcomes of observation.

At the moment of encounter, the two become embedded in one another and mutually define each other. Together, they give birth to the manifestation we call reality. Here, the barrier between the subjective and the objective is no longer absolute. Observer and observed are no longer clearly divided into the roles of master and subordinate; they form a dialectically unified relationship.

We know that any node in the cosmic web of order is universally connected rather than existing in isolation. The state of anything is influenced by all other things or structures with which it interacts.

Going further, we may say that a thing is not an a priori independent entity. Its self is precisely shaped and defined by the countless interactions between its internal structure and the external environment.

Our act of observation is merely one part of all the interactions in which a thing is involved. The effect of the act of observation on a thing is vastly different at the macroscopic and microscopic levels.

For microscopic things such as electrons and photons, the effect of observation is significant and cannot be neglected. It even directly participates in the birth of their fundamental attributes. For macroscopic systems such as Schrödinger's cat, however, the situation is entirely different. Because of quantum decoherence, the cat cannot truly be in a quantum superposition of both alive and dead. The multilayered ordered structure within a macroscopic object, together with the immense number and complexity of interactions between the object and its environment, effectively averages out microscopic quantum properties. This forces the entire system to settle into some determinate classical state. In other words, the cat is either alive or dead.

Precisely because macroscopic things are bound by such complex and powerful constraints imposed by their own structure and the external environment, our everyday conscious observation has a negligible effect on their overall state. A sense of relatively stable objective reality is thus established.

Therefore, while we may acknowledge that no physical reality exists independent of observation, we cannot slide directly to the conclusion that physical reality exists because of observation. From the micro to the macro, with increasing scale and complexity, the manner and stability with which a thing's state is fixed undergo gradual change.

This reveals that when facing worlds at different levels, we must select modes of observation and theoretical frameworks suited to them. We examine the microscopic fog of probability with the eye of quantum mechanics. We grasp stable macroscopic reality with the hand of classical physics. This is like selecting the appropriate frequency in the spectrum of laws and resonating with it. Only in this way can we arrive at an effective understanding of this interconnected cosmos.

Let us turn the perspective back to ourselves. Humans, as subjects of cognition and practice, are in no way detached observers standing beyond all things. We are constantly in complex and dynamic interaction with everything around us.

Cognition is therefore not a quiet contemplation divorced from interaction. It is formed through concrete practice and tempered in the midst of affairs. Through immediate sensation, comprehension, and verification by feedback, true knowledge emerges from practice. In this way, we can move beyond the rigid appearance of a dualism between subject and object, while the neural networks of the brain gradually form cognitive patterns attuned to the nested workings of the cosmos.

It follows that the world we understand is not a direct facsimile of an independent, static objective reality. Quite the opposite:

it is a stable manifestation constructed by the brain after a series of exquisite sensory "detectors" have interacted with particular dimensions of the outside world.

Each sense defines a unique channel of dialogue between us and the cosmos. From different levels, from quantum probability to classical mechanics, each cuts into that great web of order.

Vision, in its essence, is the brain's brilliant interpretation of the statistical results of countless quantum events. The opsin molecules on the retina are like microscopic "measuring devices." When a photon arrives, it is not inevitably captured; it follows the probabilistic laws of quantum mechanics. It may be absorbed and, with a certain probability, trigger a "collapse" event in the form of a change in molecular configuration. Or it may pass silently by, leaving no trace.

We possess three types of cone cells, sensitive respectively to red, green, and blue light. They resemble a probability detector with only three keys, none of them perfectly tuned. The brain does not receive a ready-made color; it interprets the statistical frequencies with which countless quantum events strike these three keys. From this interpretation, our perception of color emerges. In dim light, we switch to the more sensitive, single-key mode of the rod cells, and the world fades into probabilistic grayscale.

Smell and taste are our chemical "measurements" of the material world. They do not depend directly on quantum probability, but on the precise matching of molecular shapes. The olfactory receptors in the nasal cavity are highly specialized proteins. When airborne molecules of a particular shape bind to them, an effective "measurement" is completed. A minute change in molecular

structure can be enough to turn the result from fragrance to stench. Taste on the tongue is a still more basic chemical measurement system. By discriminating among the five basic patterns of sweet, salty, sour, bitter, and umami, it makes a preliminary "assessment" of the safety and nutritional value of food. The brain distinguishes smells and tastes by interpreting which "locks" have been opened and in what combinations.

Touch is the mechanical "interaction" between us, as macroscopic objects, and the surrounding world. The various receptors beneath the skin together form a multidimensional detection network. They continuously measure external mechanical and thermal parameters such as pressure, vibration, and temperature. Light touch, deep pressure, heat, cold, and pain: every contact provides local, real-time feedback about a state. This system is not only spatial but also temporal; it can distinguish an instantaneous tap from sustained pressure. When we stroke a cat, touch does not passively receive signals. Through active interaction, it helps shape the overall tactile manifestation of softness and warmth.

Hearing, by contrast, is closer to measurement in classical mechanics. The eardrum and the hair cells within the cochlea form a precise "mechanical-vibration spectrum analyzer." Sound waves of different frequencies stimulate hair cells at different positions on the basilar membrane. This process is continuous and graded: the stronger the sound wave, the greater the response amplitude of the hair cells. The brain analyzes the positions and response intensities of the activated hair cells to decode sound, with position corresponding to frequency and amplitude to intensity. Unlike vision, which relies on the probability statistics of countless quantum

events, hearing is closer to the precise measurement and decoding of a relatively determinate classical physical quantity.

From the quantum interplay of vision, through the molecular matching of the chemical senses, to the mechanical feedback of touch and the classical vibrations of hearing, we are not passive recorders of the world but active participants in its interactions.

Each sense is a unique channel, with its own limits and distortion. Vision cannot directly perceive ultraviolet or infrared light, hearing cannot see sound waves, smell and taste are bound by molecular shapes, and touch is confined to the surface of the body.

Yet the brain, with its remarkable integrative capacity, synthesizes these fragmentary signals from different levels—quantum probabilities, chemical keys, pressure patterns, and mechanical vibrations—and ultimately interprets them as the coherent reality we experience.

Since all our perceptions are transformed by a sensory system that is limited and may introduce distortion, a fundamental question arises: how can we be sure that what we perceive is real? And is reality the same for each of us?

Consider a person born with red-green color blindness, whose visual physiology differs from that of people with typical color vision. When the red wavelengths of the physical world stimulate his retina, the color experience produced in his brain is the same as the experience of green in a person with typical color vision. Conversely, the green wavelengths of the physical world evoke in his brain the experience that a person with typical color vision has of red.

If from early childhood he receives the same education in language and social norms as everyone else, something interesting happens. When the teacher points to an apple and says, "This is red," he looks at the apple, which appears green in his inner vision, and links that subjective experience with the word "red." Likewise, when the teacher points to a leaf and says, "This is green," he binds his inner experience of red to the word "green."

Thereafter, his cognitive world is functionally no different from that of anyone else. He can correctly identify a red flag as "red" and grass as "green." While driving, when he sees the upper signal light up, the physical light is red and what he actually perceives is green, yet he still understands it as "stop." When the lower light comes on, the physical light is green and what he perceives is red, yet he can normally understand it as "go." His behavior fully conforms to social norms, and in everyday life he encounters almost no obstacle because of this difference.

This is analogous to the philosophical thought experiment of the "inverted spectrum." It is not the same as actual red-green color blindness, but it helps reveal a profound truth. The essence of cognition does not lie in whether one person's inner, private qualia are exactly the same as another's. What matters is whether the brain can successfully map the differences expressed by the layered order of the external world.

In the case of color, the key is not whether the red and green that you and I perceive are qualitatively identical in our inner experience. What matters is whether each of us can effectively recognize the distinction between red and green. Once we possess the ability to distinguish differences, we possess the basis of cognition.

If, in addition, we share a symbolic system, such as language and writing, with which to describe those differences, we also possess the ability to communicate our cognition to one another.

Extending this perspective broadly, we can profoundly understand the plurality and relativity of cognitive reality.

In the natural world, different species have evolved vastly divergent sensory systems. Bees perceive ultraviolet light in order to find nectar. Snakes "see" warm-blooded prey through infrared heat sensing. Bats construct a precise sonar world. They all live within the same physical cosmos, yet because their detectors differ, they inhabit different "realities."

Even among human beings, such differences are common. Just as no two leaves in the world are exactly alike, our sensory apparatuses differ subtly in their physiology. Faced with the same external stimulus—the same painting, dish, or piece of music—the electrochemical patterns activated in each person's nervous system, and the subjective experience ultimately constructed, must necessarily differ at the physical level.

We can never truly exchange or feel another person's inner world. We are firmly anchored to our own boat of consciousness and can observe and experience the cosmos only through our own unique window. What is imprisoned within this sealed chamber is what we call consciousness, the soul, the "I."

The first to understand the chamber dilemma was the ancient Greek philosopher Empedocles. He held that what our brain directly understands is a world of appearance, not the external noumenal world itself.

Kant in the modern era called the world we perceive the phenomenal realm and the world in itself the thing-in-itself. He held that the phenomenal realm is formed when the thing-in-itself stimulates our senses and is then processed through our innate cognitive structures. Because humans share similar a priori forms of cognition, our respective phenomenal realms overlap to a high degree. This causes us to mistakenly believe we share a single objective world.

Kant thereby criticized the arrogance of believing that one can know the world in its entirety. He advocated a humility in cognition: we can never know the thing-in-itself.

Must we then accept agnosticism? Is the world perceived by the red-green color-blind person necessarily wrong? Are the attributes we perceive, color, weight, size, indeed the essence of things? When every person's inner experience contains differences, who is the ultimate arbiter?

In fact, the very question of who is correct may itself be a conceptual trap. We need not sink into the quagmire of agnosticism, nor need we debate who is right and who is wrong. The crux of the matter is not whether our perception faithfully copies the thing-in-itself, but whether our interaction with the world through our senses is effective.

The essence of the thing-in-itself is precisely this multilayered, structured, ordered existence; there is no other hidden "original face." We can indeed perceive order only through interaction. In Kantian terms, when the thing-in-itself becomes an object of observation, it appears to us as phenomenon.

The

textitDiamond Sutra states, “What the Buddha calls a world is not a world; therefore it is called a world.” The essence of the world lies not at one pole of a difference, but in the dynamic unity formed by both sides.

If energy and matter contained no differences in density or temperature, the cosmos would be a silent state of thermal equilibrium. Without difference, there would be no structure, relation, information, or change, and therefore no ordered existence. A uniform, motionless state in which no difference could be discerned would be the imperceptible “non-being.”

The qualities we perceive, such as color, sound, and taste, are not illusions. They are the brain’s reading and interpretation of differences within external ordered structures. “Being and non-being give rise to each other; the difficult and easy complement each other; the long and short shape each other; the high and low lean on each other.” The two sides of a difference depend upon and define one another.

The essence of the endlessly diverse myriad things is order. To understand order is, at its core, to understand difference. The evolution of cosmic symmetry breaking is precisely a movement away from complete symmetry and the absence of difference, producing asymmetry and difference level by level.

No matter how complex the reading and interpretation that our senses and instruments perform upon raw physical stimuli, their underlying logic is invariably the detection of some particular difference: differences in photon energy, molecular shape, pressure and vibration, voltage and current. All cognition is, at its root, the recognition of difference.

Regardless of how the reading and interpretation are done, as long as we can recognize differences, we can know the myriad things. From this angle, the thing-in-itself is not unknowable.

Yet our individual human lives are finite, while the changing cosmos is infinite. Our understanding of the cosmos's infinite mysteries is therefore necessarily a never-ending and gradual process of practice.

Thus, absolute, once-and-for-all "knowability" and stage-specific, finite "unknowability" form a dialectical unity. We are neither wholly ignorant nor omniscient and omnipotent. We walk an unending road in the pursuit of knowledge.

Since the validity of cognition lies in mapping the differences of the outside world, a core puzzle arises. How does the brain, sealed within the body's chamber, accomplish this mighty feat?

Next, we shall venture deep inside the brain and explore how vast numbers of neurons, through their electrochemical activities and patterns of connection, map external order. We shall see how this microcosm subjects scattered sensory signals to layer upon layer of integration and abstraction, until coherent meaning and reality finally emerge. We shall begin a journey into how consciousness might arise from the complex activity of networks of neuronal connections.

Chapter 18

The Beauty of the Brain's Cosmos

As long as our brain remains an enigma, the universe, a reflection of its structure, will also remain a mystery.

Santiago Ramón y Cajal

The brain may be one of the most complex and ordered structures in the known cosmos. Its vastness and mystery are even enough to rival the external cosmos.

In the decades of the mid-to-late twentieth century, neurobiological research into the brain achieved a series of landmark breakthroughs. These deepened our understanding of perception, learning, memory, emotion, and action. They further gave us a new perspective grounded in biological evolution from which to examine the relationship between the objective world and the self.

The father of modern neuroscience, Santiago Ramón y Cajal, is recognized as the greatest neuroanatomist in history. His magnum opus, *Histology of the Nervous System of Man and Vertebrates*, can even stand shoulder to shoulder with Darwin's *On the Origin of Species*.

From individual neurons to the pathways connecting them, Ca-

jal produced more than 2,900 drawings by hand. With exquisite lines, these drawings revealed the harmonious beauty of the brain's highly complex and ordered structure, while also laying a foundation for the scientific study of the brain and consciousness. Cajal not only founded the neuron doctrine; he also introduced the principles of dynamic polarization and neuronal plasticity.

Cajal held that the brain is composed of independent cells—neurons—dynamically connected into an ordered network, rather than a single body of fixed connections. From simple reflexes to conscious acts such as language, art, and deep thought, all are realized through specific neuronal pathways in different regions.

Neurons are differentiated from ordinary cells and possess the full complement of basic structures: cell membrane, cytoplasm, cell nucleus, and so forth. Yet compared with ordinary somatic cells, neurons are like privileged "aristocrats." Not only are they bathed in the most nutritious cerebrospinal fluid, but each neuron also has, on average, about ten glial cells attending to it with devoted care. The blood-brain barrier strictly isolates neurons in the brain from ordinary somatic cells, making it difficult for the latter ever to reach or influence these "aristocrats."

Neurons are also designed to be almost irreplaceable. Most remain with us throughout life, and may live hundreds or even tens of thousands of times longer than short-lived somatic cells such as intestinal cells or red blood cells.

Neurons also differ structurally from ordinary somatic cells to suit their primary function of information transmission.

The neuron cell body has one particularly long extension called the axon. The axon terminals connect with other neurons via

synapses. On the other side of the cell body are many branch-like extensions called dendrites. Dendrites receive information from other neurons through synapses. Information within a neuron flows unidirectionally from the dendrites to the axon. It then passes through the axon to the next neuron.

In this way, neurons connect to one another and form the pathways through which information travels within the brain. Yet unlike a transportation network or the Internet, information transmission within a neuron is strictly unidirectional: it can flow only from dendrites to the axon.

The synaptic connections between neurons are not fixed, direct physical links. At each synapse, the neurons are separated by an extremely narrow gap. Signals cross this gap by means of chemical substances called neurotransmitters. The neurons sending and receiving the signal are known, respectively, as the presynaptic and postsynaptic neurons.

The presynaptic terminal at the tip of the presynaptic neuron's axon contains large numbers of vesicles filled with neurotransmitters. When an electrical signal within the neuron reaches this terminal, it stimulates the vesicles to release neurotransmitters into the gap. The neurotransmitters bind to receptors on the postsynaptic neuron and, once their effect has accumulated to a certain level, can trigger a new electrical signal within it. This completes the one-way transmission of a signal from one neuron to another in the brain.

In essence, it is a special, controlled process of electrical-to-chemical-to-electrical signal conversion.

The high specificity and directionality of the neuronal connections described by Cajal mean that neurons communicate in one

direction only with particular neuronal partners, thereby forming particular neuronal pathways.

With thousands of synapses on its dendrites, a single neuron can integrate multiple types of information from different areas of the brain. Through the numerous synapses on its axon terminals, it can distribute the processed information to other neurons in multiple brain regions.

For their research on neurons, Golgi and Cajal jointly received the 1906 Nobel Prize in Physiology or Medicine. Interestingly, their views were diametrically opposed.

Golgi held that the nervous system was a continuous, diffuse network in which signals could propagate freely and simultaneously in every direction. This view essentially treated the neural network as a unified whole with fixed connections between neurons.

It was not until the advent of the electron microscope in the twentieth century that humanity could finally glimpse that nanometer-scale synaptic gap and verify Cajal's far-sighted vision in images.

Cajal, through meticulous observation and extraordinary insight, depicted a different picture. The brain is formed from countless independent neurons joined through the precise structure of the synapse in an ordered network of one-way pathways. Signals undergo an electrical-to-chemical-to-electrical conversion and travel in relatively determinate directions, while the plasticity of the synapses themselves preserves immense randomness and flexibility.

It is precisely the randomness and plasticity of neuronal pathways that provide the structural foundation for how the brain can bring forth new ideas out of nothing.

What, then, are the signals flowing through this complex ordered structure formed by the interconnections of countless neurons in the brain?

Intuitively, the signals transmitted along neuronal pathways should be complex signals modulated with information from the outside world. Yet the discoveries of Sherrington and Adrian, winners of the 1932 Nobel Prize in Physiology or Medicine, were quite unexpected. The electrical signals transmitted between neurons are "simple" signals composed only of pulses resembling zeros and ones. These are called action potentials. Whether the sense is vision, hearing, touch, or smell, the electrical signals in the corresponding neuronal pathways are similar action potentials.

Differences in the external information we perceive are directly manifested as differences in action-potential signals. For example, the duration of a sensation maps onto the duration of the pulses, while its intensity maps onto pulse frequency. Dense pulses correspond to intense sensation; sparse pulses correspond to faint sensation.

Because the basic characteristics of the various action-potential signals are the same, the brain neither needs nor is able to distinguish the type of sensation from the signals themselves. It need only determine the sensory modality from the particular neuronal pathway carrying the signal, and can thereby obtain information about duration, intensity, and so on.

Scientists estimate that the human brain contains about eighty-

six billion neurons, each connected to other neurons through thousands or tens of thousands of synapses. Our many forms of thought and consciousness arise from this highly complex and ordered network. Given the number of neurons and the possible combinations of their connections, the total reaches into the quadrillions and is almost impossible to imagine. It is hardly surprising, then, that the brain can produce thoughts of extraordinary novelty.

Despite the near-infinite complexity of the brain's connections, its overall function remains relatively stable, thanks to the nonregenerative nature of the neurons from which it is built.

The vast majority of neurons in the brain lose the ability to divide after maturation. Although some damaged neurons can be repaired, once they die they are almost impossible to replace. Only a very few regions of the brain, such as the dentate gyrus, can produce new neurons; these are exceptions. This is because the complex connections in neuronal pathways correspond to cognitive abilities and memories acquired through "heredity" or "learning." If a particular neuron dies, all the neuronal pathways and connections associated with it are damaged, and the corresponding memory or cognitive ability is weakened or lost.

This irreplaceable nature of brain neurons largely guarantees the basic order and stability of the neuronal pathways. It allows us to continuously learn, accumulate knowledge, and sustain our cognitive ability and personality.

A growing number of views now hold that the level of intellectual development depends more on the ordered character of the brain's patterns of neuronal connection than on sheer neuron count.

At birth, our brain is virtually a blank sheet of paper. As we

grow, we gradually perceive the outside world through our various sensory organs. This process constitutes our observation and learning of the outside world, through which we construct an individual cognitive framework for it. The experience and knowledge we acquire are stored in the brain, physically represented by the establishment and strengthening of synaptic connections between neurons.

As observation and learning continue, our framework for understanding the world grows richer and more refined. Although the number of neurons in the brain does not change significantly, the connections among them become increasingly ordered as we grow. Each person's ever more complex and ordered neuronal network corresponds to an increasingly rich and developed cognitive framework.

From the perspective of bodily composition, what makes me who I am is, first of all, that unique and highly ordered DNA sequence. From the perspective of consciousness and mind, what makes me who I am lies above all in the unique, highly ordered network of neuronal connections in my brain.

In the cognitive process described above, the brain receives external information through the perceptual system. Yet that system is not a passive photocopier of information from outside; it is an active extractor of features.

Research on the visual system by David Hubel and Torsten Wiesel, winners of the 1981 Nobel Prize in Physiology or Medicine, revealed this. They projected slides showing different patterns of light against a black background onto cats' retinas and used microelectrodes to record the responses of neurons in the

cerebral cortex.

Many experiments found that when the stimulus consisted of ordinary spots or small points of light, almost no neurons in the cat's cerebral cortex responded. Yet when the stimulus was changed to regular lines with clearly differentiated forms, a considerable number of cortical neurons responded strongly. Moreover, particular neurons responded only to lines of a particular orientation: some favored horizontal lines, while others responded markedly only to vertical ones.

The same phenomenon appears in the auditory, tactile, olfactory, and gustatory systems. Different auditory and tactile neurons can respond only to stimuli of specific intensity, frequency, and location. For the olfactory and gustatory systems, specific neurons receive information only from specific chemoreceptors in the nasal cavity and on the tongue.

Large numbers of neurons transmit their respective information and bring it together in the cerebral cortex. Only then does our complete perception of images, sounds, and tastes take shape.

As Eric Kandel, the 2000 Nobel laureate in Physiology or Medicine, and the neuroscientist Vernon Mountcastle summarized it, the perceptual system is a hypothesis-based generator: we face the world neither directly nor precisely.

Sensory nerve fibers are not high-fidelity recorders. They emphasize certain stimulus features while ignoring others. Our sensations are summaries of the real world, not copies. For example, if you close your eyes and imagine an apple, your brain cannot obtain a complete image of the apple. It can only recall the apple's features.

Thus, at its core, the brain analyzes information from the outside world and breaks it down into differences—that is, features. These differences are precisely a reflection of the ordered character of external things; what we can perceive is this ordered information alone. The various neuronal pathways conduct the disassembled ordered information to the brain, where it is ultimately integrated to form sensation.

Does the ordered feature information that enters the brain need to be transmitted to some region in the brain for that region to assemble it into a final image, sound, or taste?

Neuroanatomy confirms that, in the visual, tactile, auditory, olfactory, and other systems alike, there is no single region of the brain to which every other region simply reports. Instead, beginning at the various "sensor" terminals, the brain has already analyzed and broken the raw data down into ordered feature information. It neither needs nor is able to restore the original images or sounds. The brain "understands" the world at a higher level through combinations of these features; it does not reproduce the world's original form.

We may also reasonably hypothesize that perception, learning, memory, cognition, emotion, and other higher forms of consciousness are all mediated by complex neuronal pathways and manifested as the ordered transmission of information between neurons in the brain.

These ordered flows of information may activate certain neuronal pathways and generate an unbroken current of thought; they may activate other pathways and ultimately direct the body's organs to act.

Once sensory nerve endings have captured and encoded ordered information from the outside world, electrical signals in the brain burst forth like fireworks and flow through particular neuronal pathways.

Our consciousness and thought are precisely the particular dynamic patterns that this vast neural network—the brain—brings forth at a given moment. This is not chaos, but a profound form of ordered beauty grounded in matter.

Chapter 19

The Neural Forest

The cerebral cortex resembles a garden populated by innumerable trees, the pyramidal cells, which, thanks to intelligent cultivation, can multiply their branches, sink their roots further and further, and produce flowers and fruits that grow more exquisite every day.

Santiago Ramón y Cajal

As Cajal, the father of modern neuroscience, suggested, our brains truly contain a dense forest of neurons. Their forms bear a striking resemblance to trees: the long "trunk" is the axon, and the many luxuriant "branches" are the dendrites.

Unlike solitary trees, however, neurons have roots and branches that interweave closely, forming a living ecosystem in which information is exchanged without pause.

Information travels through this forest in two ways. Within a neuron, an electrical signal—the action potential—moves from the dendrites through the cell body and along the "trunk" to the axon terminals, carrying information rapidly over a long distance. Between neurons, chemical messengers—the neurotransmitters—

cross minute gaps called synapses to communicate over a short distance.

Synapses stud dendrites and axons in staggering numbers, yet each synaptic cleft is only about 20 to 40 nanometers wide. This delicate arrangement permits efficient signal transmission while preserving the independence of individual neurons and preventing their connections from becoming permanently fixed.

When an electrical signal reaches the presynaptic membrane at an axon terminal, proteins cause vesicles filled with neurotransmitters to fuse with the membrane and release their contents into the synaptic cleft. Each vesicle contains roughly 5,000 neurotransmitter molecules. These diffuse across the gap and bind to specific receptors on the postsynaptic membrane of the next neuron. If the binding reaches a certain threshold, the receiving neuron generates a new action potential, and the signal passes from one neuron to the next.

This chemical transmission at the synapse is breathtakingly fast, taking only a few thousandths of a second. The recycling of vesicle membranes is much slower, and the resynthesis of neurotransmitters slower still; these processes require seconds or even minutes.

Two main types of neurotransmitter operate in the synaptic cleft. One is excitatory, a role dominated by the amino acid glutamate; the other is inhibitory, a role dominated by another amino acid, GABA. Like growth hormones and dormancy signals in a forest, together they finely tune the activity of every neuronal "tree."

Compared with the brain as a whole, a single neuron works by a relatively simple rule. It endures a "bombardment" of signals from every direction, some excitatory and some inhibitory. Only when

the sum of the excitatory signals exceeds that of the inhibitory signals and crosses a critical threshold does it fire and pass the message onward.

For the whole brain, the story changes. Even the slightest thought, the simplest action, may ignite tens of thousands of neurons. Electrical signals ripple across the brain and flash like fireworks in a night sky. At every instant, millions of synapses take part in this intricate relay.

The synapse is the most plastic element of this forest.

Through experiments with sea slugs, Nobel laureate Eric Kandel provided crucial evidence for the relationship between synapses and memory. Repeated external stimulation markedly strengthened synaptic connections along the relevant neuronal pathways. Even microscopic chemical changes at a synapse could produce lasting changes in the strength of the connection. This, in effect, explains the formation of memory at the level of neural mechanisms.

Kandel also probed the neural mechanisms of learning. In some experiments, he applied strong, regular stimulation to particular neurons, triggering gill withdrawal in the sea slugs. At the same time, he applied regular, mild stimulation to other neurons, triggering ink release. After a period of synchronized stimulation, the slug's nervous system linked the two previously unrelated stimuli: stimulating only the neurons that had originally triggered gill withdrawal now produced ink release as well as gill withdrawal.

The experiments showed that external stimulation can not only alter the strength of synaptic connections but, when repeated regularly over time, also generate large numbers of new synapses. These new synapses bridge two previously independent neuronal path-

ways, linking actions that once served separate functions. In effect, the brain has learned something new from the outside world.

This synaptic plasticity is equally pronounced in higher animals. Michael Merzenich at UCSF trained monkeys to perform specific finger movements repeatedly to obtain food. After months of training, the cortical territory devoted to the most-used fingers expanded dramatically.

In related work, the German scientist Thomas Elbert compared brain images of violinists, cellists, and non-musicians. The cortical areas corresponding to the fingers of the right hand showed no significant difference between string players and other people. The fingers of the left hand, however—the ones that execute complex fingering on the strings—told a different story: their cortical representation was roughly five times larger than in non-musicians.

Sustained learning and training cause new "branches" to grow along different neuronal pathways, while new synapses connect previously unrelated circuits. Learning and training are acts of deliberate cultivation: they make the "forest" devoted to a particular skill denser and more luxuriant, greatly strengthening and expanding the corresponding neuronal pathways.

The journey of learning begins at birth. Light floods your eyes, sound strikes your eardrums, air slips through your nostrils, milk touches the tip of your tongue, and a mother's gentle caress is imprinted on your skin. Sensory experience arrives like a tide, transformed into an unbroken stream of neural signals that travels directly to the brain.

Scientists estimate the total data bandwidth of the optic nerves at roughly 8 to 10 Mbps. Every second, your eyes transmit a vol-

ume of visual information equivalent to a short novel. In that same second, your brain can generate more than a million new synaptic connections. At every moment, every person's brain receives and processes this torrent of information, builds more associations among different neuronal pathways, and gradually trains its own unique neural network.

Consider the growth of a single person: through the continual learning of crawling, walking, running, and fine motor skills, each Purkinje cell in the cerebellum eventually builds as many as 100,000 synaptic connections.

The law of the neuronal forest is not only growth and strengthening; it also demands pruning and regression. Mirroring the "use it or lose it" principle, connections that sit idle and unstimulated for too long gradually weaken, atrophy, and eventually disappear.

This seemingly negative process of synaptic weakening or elimination is precisely the brain's mechanism for managing its vast resources efficiently. By actively clearing ineffective or redundant connections, it frees space for the acquisition and consolidation of new knowledge. This is the neural mechanism of forgetting.

Sometimes the forest's regulatory mechanisms fail, and pathological "withering" occurs on a broad scale: particular regions of the brain shrink markedly or synaptic connections are lost extensively, far beyond the scope of normal forgetting. This systemic collapse of the neural network is the common pathological basis for many cognitive disorders, including Alzheimer's disease, in which memory is gradually buried; Parkinson's disease, in which motor control unravels; and autism spectrum disorders, in which patterns of neural connectivity go awry.

Beyond neurons, the brain contains a vast population of glial cells. They have no dendrites or axons and do not directly generate or transmit electrical signals. In the neuronal forest, glial cells are the soil: they provide structural support and nourishment while regulating the microenvironment.

They also release a variety of chemicals. Some, such as glial cell line-derived neurotrophic factor, influence synaptic function and thereby modulate electrical activity. Others cause blood vessels to dilate, regulate cerebral blood flow, and deliver more energy and nutrients to regions of heightened neuronal activity.

Compared with other animals, the human brain has relatively more glial cells, and their forms and functions are more complex. Scientists suspect that this advanced glial development may be one underlying reason for the remarkable power of human intelligence.

In this thriving neuronal forest, every neuronal "tree" reaches outward with scattered roots and branches, searching for still more trees; countless synapses bind them together.

The neuronal forest begins with different patterns and scales in human beings and other animals. Within any one species, each individual inherits an initial configuration: a broadly similar number of neurons and certain basic patterns of connection already in place. These innate and relatively stable connections form the basis of instinct, allowing us to breathe and blink from birth. Scientists have also found that intelligence, memory, emotional tendencies, and personality traits are strongly influenced by heredity.

The initial blueprint supplied by heredity means that no individual must begin from nothing. The most important work of a lifetime is to refine that innate blueprint continually through acquired expe-

rience, making the forest more luxuriant and the neural network more complex and ordered.

Many of our daily behaviors are unconscious or subconscious. Neuronal pathways laid down by heredity or formed through repeated practice become exceptionally robust as they are activated again and again. They require no deliberate act of will, but fire almost automatically.

Take driving. A novice must concentrate fully, whereas an experienced driver's motions have already been internalized as subconscious behavior. In the appropriate situation, the corresponding neuronal pathways activate smoothly and automatically. Experienced drivers often find that, even when the mind wanders for a moment, the vehicle continues safely through complex road conditions. This is the neural mechanism behind the saying "practice makes perfect."

The brain does not passively record everything. It selects which perceptions, experiences, and knowledge merit durable neural connections. Long-term, regular, and harmless stimuli cause our neuronal pathways to gradually habituate. Their response diminishes.

This habituation is related to CREB regulatory proteins within neurons. By controlling the expression of genes associated with synaptic strengthening, they regulate how readily memories are formed. CREB-1 tends to activate these genes and promote memory consolidation, whereas CREB-2 inhibits them, acting as a "brake" on memory. Innate differences in memory ability among people arise partly from the activity and balance of these two regulatory proteins.

Together they set the threshold for synaptic strengthening. The

higher the threshold, the stronger or more significant the required stimulus: more neurotransmitters must be released and a stronger signal must cross the threshold before lasting synaptic strengthening and memory formation can occur.

This mechanism ensures that we neither do nor need to remember everything. Experiences that are important and have real significance for survival and life are more readily consolidated. Under an extremely powerful stimulus, such as the trauma of a car crash, the brain rapidly produces a substance called MAP kinase. This temporarily disables CREB-2 and releases the "brake," allowing CREB-1 to drive memory formation with great force and produce an unforgettable "flashbulb memory."

Even established synaptic connections may slowly loosen with time. A temporary increase in synaptic efficacy mainly corresponds to short-term memory and lasts from a few seconds to a few minutes. Structural changes at the synapse, such as an increase in synapse number, together with deeper changes involving gene expression and new protein synthesis, correspond to long-term memory. Long-term memory markedly increases the number of synapses along the relevant neuronal pathway and the proportion that become active when the pathway fires.

Yet if the memory carried by these newly added synapses is no longer retrieved and consolidated, their number will gradually decline over time. This corresponds to the natural fading of memory. If reading this book leaves you with something to remember, then its contents have indeed etched a tiny trace into the physical structure of your brain.

Neuroplasticity is a double-edged sword. It builds knowledge

and skill, but it can also solidify bad habits and aberrant behaviors. Negative behavior patterns can likewise forge new neuronal connections in the brain and exert long-term influence.

In daily life, a child who imitates a stutter for amusement will often be sharply reprimanded by adults. There is a scientific reason for this: if the mimicry continues, even though it begins as a joke, the principle of use-driven strengthening may promote the formation of stubborn, abnormal neuronal pathways involved in speech fluency, eventually developing into a genuine speech disorder. Stuttering is therefore understood as a communication disorder with neurological and psychological dimensions, although developmental stuttering is an additional case. Other examples include unusual gait patterns in which the arm and leg on the same side move together, involuntary drooling, and many undesirable habits in daily life and learning; their formation and entrenchment can be understood in much the same way.

Fortunately, neuroplasticity has another side: if we persist in no longer reactivating the neuronal pathways corresponding to these bad habits, they too will gradually weaken and fade over time through disuse-driven regression.

We should therefore see the brain anew: it is a thriving neural forest, an immensely complex and ordered network of neuronal connections. The mind is not an insubstantial phantom; it arises from the physical and chemical processes unfolding within this vast network.

The “I” is neither a fate wholly determined by genes nor a blank slate upon which anything may be written. Upon the initial blueprint given by heredity, it is continually carved and reshaped

through a lifetime of experience, learning, and thought. Every new branch and every act of pruning in this forest continues to define that unique “I.”

Chapter 20

The Emergence of Consciousness

There is one spectacle grander than the sea, that is the sky; there is one spectacle grander than the sky, that is the interior of the soul.

Victor Hugo

Set against the vast material cosmos and the intricate architecture of life, human consciousness strikes us as deeper and more unfathomable still.

Its mystery begins with interiority and privacy. Each of us possesses a singular, private inner world that no one else can enter directly. The redness of a rose, the ache of parting, the joy of sudden insight—all the vivid textures and hues of these experiences belong to oneself alone. For this reason, everything outside the self, including the stars, the seas, and the minds of others, becomes, at the level of knowledge, the "external" that confronts us.

What is still more astonishing about consciousness is its reflexivity. It not only perceives the outside world but can turn back upon itself and perceive its own states. We can experience joy while savoring that joy, or sink into sorrow while examining the sorrow itself. It is this capacity to feel oneself that establishes the "I," mak-

ing life not merely existence, but an awareness of existence.

Most remarkable of all, consciousness often seems to conjure something out of nothing. A profusion of thoughts, unexpected inspirations, and unbidden moods arrive from no one knows where, glittering and flickering into our inner world like stars in the night sky.

For this reason, human beings cannot help forming a simple yet profound intuition: consciousness is so unusual that it seems impossible to reduce it entirely to ordinary material activity. We may even be inclined to regard it as a special form of existence distinct from all matter.

This intuition, which sharply divides consciousness from matter, has a long history. Early thinkers in both China and ancient Greece generally located consciousness in the heart. Hippocrates and Plato were among the first ancient Greek thinkers to argue that mental processes originate in the brain, while treating consciousness and the soul as immaterial and immortal forms of existence.

This line of thought continued into the seventeenth century, when Descartes gave it clear expression as mind–body dualism. A human being, he argued, consists of a material body and an immaterial mind, the latter sacred and beyond the reach of scientific inquiry. Descartes's dualism cast a very long shadow. Even into the 1980s, towering figures such as the British philosopher of science Karl Popper and the Australian neurobiologist John Eccles remained lifelong dualists who believed in an independent, immortal soul.

To this day, understanding consciousness stands as the most challenging frontier in science. Not only ordinary people but many

scientists and philosophers still regard it as profoundly inscrutable. Some even believe it may forever resist complete explanation.

Yet the cognitive sciences have taken a great leap. In the face of overwhelming experimental evidence, a solid scientific consensus has crystallized. The brain is the indisputable material foundation of consciousness.

For example, apply a few milliamps of current to the cortical region corresponding to the tongue, and a taste sensation arises from nowhere; stop the current, and the sensation vanishes. When brain function is impaired by severe trauma, stroke, or deep anesthesia, consciousness temporarily or permanently disappears.

From this, a logical inference follows. When the body and brain die completely, the free-floating soul loses its dwelling place. Were it otherwise, would not this world already be crowded with the souls of every creature that has ever lived since ancient times?

Over the past century and more, advances in neurobiology have compellingly demonstrated that the emergence of consciousness depends heavily on the brain's intricate neuronal networks and their electrical-to-chemical-to-electrical activity. Exactly how subjective experience arises from the brain remains an unsolved puzzle, yet the paths of inquiry continue to broaden as brain science advances.

The riddle of creation, the riddle of life, the riddle of consciousness: when confronted with mysteries this profound, the human mind seems to have an ingrained tendency to mythologize them and attribute them to the design and dominion of some transcendent, mysterious power. Under the same tendency, consciousness has been fashioned into a special form of existence independent of

matter and wholly opposed to it.

We have implicitly presupposed a dichotomy between matter and consciousness. This preconceived framework is almost a kind of "poison" to thought, obstructing to a considerable extent our objective investigation of the material nature of consciousness.

Before we turn to a scientific account, then, we must first examine and attempt to transcend these deeply ingrained assumptions.

Our observation of the outside world begins naturally from the self. This fosters a latent anthropocentrism.

Beginning with Copernicus, we came to understand that Earth is not the center of the universe. Then we had to accept that even the solar system, even the Milky Way, is merely a corner of the cosmos. Darwin's theory of evolution taught us that humans belong to the animal kingdom. We are not the supreme, divinely crafted masterpiece.

The stubborn belief that human beings are living creatures possessed of a soul and fundamentally distinct from animals—the insistence that consciousness is a special substance independent of matter—boils down to an attempt to carve out a private enclave for humanity's most cherished possession, the world of the spirit, beyond the material universe we otherwise understand.

In truth, before the advent of the electron microscope and magnetic resonance imaging, the vast majority of neurobiologists' understanding of the brain and neural networks was built upon animal research.

Whether we classify consciousness as primitive or advanced, to date there is no adequate scientific evidence that fundamentally sev-

ers human consciousness from the consciousness of other animals. No unbridgeable chasm of essence separates humans from animals. What exists is only a difference in evolutionary degree and in the complexity of consciousness.

Setting aside that latent anthropocentrism allows us to face the real natural world with greater clarity and greater humility.

Consciousness is the foundation and prerequisite for how we know the entire world. Treating it as purely metaphysical, as mysterious and beyond investigation, is not a position we should adopt. Nor should we stop in our tracks simply because science has not yet exhausted every last detail.

It is therefore vital, at the appropriate time, to learn what science knows about the brain and consciousness and to build a solid cognitive framework for understanding consciousness. If we do not understand the process and mechanism by which we perceive "facts," how can we confirm whether what we labor to grasp really consists of facts? How, then, can we speak of grasping the way of nature and embodying the spirit of seeking truth?

"I think, therefore I am." Descartes's famous declaration reveals a starting point: however much we may doubt the external world, the existence of consciousness itself is the first fact beyond dispute. To exist is to possess order; conscious existence, too, must be a form of order, not the opposite of matter.

William James, the father of modern psychology, anticipated that particular states of consciousness would one day be matched with particular states of the brain. Let us therefore begin with the brain, the material foundation from which consciousness arises, and consider how consciousness is a form of ordered existence.

In the course of life's evolution, natural selection shaped the brain, step by step, into this astoundingly complex center for information processing and control, enabling organisms to resist entropy increase more effectively, obtain energy, and adapt to their environment.

Life on Earth can be traced back roughly 3.8 billion years. The earliest living forms were single-celled organisms with no neural tissue whatsoever, relying on simple chemical reactions to exchange matter with the environment. After billions of years of evolution, organisms such as sponges appeared: multicellular, yet still without nervous systems. The critical turning point came during the Cambrian explosion, around 540 million years ago, when animals began to evolve primitive nerve cords and ganglia. These can be regarded as the earliest rudimentary brains.

Around 360 million years ago, in the late Devonian, Earth entered the age of reptiles. The brainstem and cerebellum were the first parts of the animal brain to evolve and remain dominant. Together they form what is called the instinctive brain. This system regulates heartbeat, breathing, body temperature, and other basic vital signs, and governs survival responses such as fighting, fleeing, foraging, and reproduction. It responds with extreme speed, but in fixed patterns, with little capacity for learning or thought.

The reptiles alive today, lizards, crocodiles, snakes, still operate primarily under the drive of this brain region. For this reason, the instinctive brain is often vividly called the reptilian brain or crocodile brain.

Around 200 million years ago, in the age of mammals, the brain responded to more complex environments and social relations

by gradually evolving a new key structure: the limbic system. Wrapped around the instinctive brain and consisting mainly of the hippocampus, amygdala, and hypothalamus, it is called the emotional brain.

The emotional brain endowed mammals with a rich palette of feeling—fear, joy, love, hate, and sorrow—and a powerful capacity for associative memory. The amygdala, for instance, rapidly assesses whether an external stimulus is a threat or a reward and generates fear or pleasure. This enormously enhanced animals' ability to learn and adapt, making social behaviors such as maternal care and group cooperation possible.

Roughly 2.5 to 3 million years ago, human ancestors stepped onto the historical stage. In the long course of evolution, our brains underwent explosive growth in the prefrontal regions. They developed a highly advanced neocortex, particularly the prefrontal cortex. This is the intelligent brain.

This part is the supreme command center of human cognitive function. It handles language, logic, reasoning, creativity, self-awareness, future planning, and impulse control. The intelligent brain enables us to think in abstractions, to accumulate and transmit knowledge, and to create complex tools and culture. Through continuous learning, it constantly updates our cognitive model of the world.

Our brain, we can see, resembles a colossal project built over hundreds of millions of years. It was never designed from scratch by a single chief engineer. It is more like a grand edifice that has been continuously expanded and rebuilt atop an ancient foundation.

The deepest layer is the instinctive brain, the middle layer is

the emotional brain, and the outermost layer is the intelligent brain. These three parts each perform their own function. To be clear, the triune brain is a highly simplified and broadly summarizing metaphorical model, not a strict anatomical definition; it was proposed in the 1960s by the American physician and neuroscientist Paul MacLean. At the same time, the three layers are tightly interconnected and deeply coupled. Together they constitute the material foundation from which complex human consciousness emerges. Understanding the origins and interactions of these three layers of the brain is likewise key to understanding our own behavior, emotion, and thought.

We already know that the brainstem governs instinctive responses, the limbic system drives emotional fluctuations, and the neocortex supports higher cognition. Their functions differ, yet all are built from the same physiological unit: the neuron.

These tiny cells differ from those that build our other bodily organs, which form relatively fixed structures through tight, stable physical connections. Neurons connect in a more dynamic and flexible way. Linking to one another through synapses, they weave an ever-changing, intricately interlaced neural network.

We can understand it this way. Through its built-in neuronal pathways, the instinctive brain constructed an efficient and direct mechanism for transmitting information and triggering responses. It opened a channel from sensing external stimuli, through making basic judgments, to driving bodily reactions: danger triggers flight, and food triggers predation. In this way it formed a stable, preprogrammed feedback-control system and laid the earliest cornerstone of conscious order.

The arrival of the emotional brain brought more complex and diverse patterns of neural connectivity into being atop these instinctive pathways. It was as if a symmetry had broken: from those once simple and direct response patterns, an entirely new and higher level of order emerged. This new orderliness corresponded precisely to rich and nuanced emotional experience and to the capacity for long-term memory closely bound up with it, enormously expanding the content and behavioral repertoire of consciousness.

In the same way, the intelligent brain arose upon the foundation of the emotional brain. Through an explosive increase in the complexity and diversity of neural connections in the prefrontal cortex and other regions, another critical symmetry breaking occurred. From this leap in order came entirely unprecedented functions, including language, logic, reasoning, creativity, and even the self-consciousness that reflects upon itself.

Thus the entire spectrum of consciousness, from instinct to intelligence, can fundamentally be understood in this way. The brain is a complex system that follows the same underlying logic: disorder creates possibility, and order determines direction. Throughout evolution, through progressively layered structures of order and continuous selection, constraint, and stabilization, it repeatedly crosses the threshold from lower-level order to the leap and emergence of higher-level order.

Seen from this angle, the physical composition of the brain seems to contain nothing more than neurons, synaptic connections, and the electrochemical signals that course among them. Yet consciousness, as a living phenomenon, cannot be simply equated with these static anatomical structures.

Think of the Internet as we know it. Its essence does not reside merely in the fiber-optic cables buried underground and undersea, nor in the routers and other physical infrastructure. Its essence lies even more in the rapidly shifting streams of data and states of interaction that flow through the network as it runs.

The static neuronal network is the physical substrate of consciousness. Consciousness itself is more like a higher-level "operating process" or "operating state" manifested by that substrate as the dynamic neural network operates.

For a long time, brain science was heavily influenced by the doctrine of functional localization. It tended to attribute complex mental activities to the independent work of specific brain regions. As research deepened, however, we have come increasingly to understand that we must move beyond this static mapmaking style of thinking. We must instead understand the brain from a dynamic, network-based perspective.

The key to consciousness and various higher cognitive functions may not lie in any single brain region or even in several isolated ones. It may lie in the complex interplay, the coordinated oscillations, and the dynamic connection patterns among different neuronal pathways and different regions as the whole-brain neural network runs. It is this brain-wide, ever-changing dynamic state that finally weaves the rich and unified conscious experience we live through at every moment.

We can now summarize at a more fundamental level.

The brain and consciousness are not two separate entities, but the "structure" and "process" of a single complex system. As the physical structure built from neurons and their connections, the

brain provides a dynamically evolving ordered foundation. This system is not a machine that becomes fixed the moment it leaves the factory. Throughout its lifetime, it is continually reshaped and recalibrated by its own perception, thought, and experience.

Thus the flowing conscious experience that we live through moment by moment can be understood as a holistic process emerging from the instantaneous state of this complex system, built from an immense number of units—a process that cannot be reduced to the properties of any single neuron.

Chapter 21

Self-Consciousness

Everyone is confined within the limits of their own consciousness.

Arthur Schopenhauer

When we are awake, sensory information such as touch, sight, and sound arises together with mental activity such as imagination and emotion; all of it forms our consciousness. In deep sleep or coma, consciousness disappears. Beyond this, we possess a further capacity: awareness of consciousness itself, or what we call "self-consciousness."

The entire system of perception and thought, centered on the brain and deeply coupled with the body, constitutes what we ordinarily perceive as the "I." The individual continually gathers information from the external environment while also receiving sensory signals from within the body. By processing and comparing these two categories of information, the brain distinguishes the external world from the internal self. Self-consciousness is born from this very distinction.

Self-consciousness has always been an ancient yet ever-renewing topic in psychology and philosophy. Let us now deepen

our understanding of how the brain produces self-consciousness.

The nervous system is the physiological foundation of consciousness. The brain and the spinal cord together form the central nervous system. Its operation relies on the precise collaboration of three types of neurons.

Sensory-neuron terminals extend into the skin and sensory organs, where they gather external stimuli and convert them into neural signals, forming such basic sensations as touch and sight. Motor neurons carry commands from the brain and spinal cord to the muscles and glands, governing behavior and response. Interneurons relay and integrate information between sensory and motor neurons, forming the basis of complex neuronal pathways.

Working in concert, these neurons enable different regions of the brain to perform their particular functions.

There exists a rough correspondence between different brain regions and conscious functions.

The brainstem and cerebellum are generally classed as the instinctive brain. They regulate heartbeat, breathing, digestion, body temperature, sleep, and other life-sustaining functions, while also participating in the regulation of attention. The cerebellum is more concerned with unconscious motor control and sensory integration, playing a leading role in motor coordination, balance, and the fine control of movement.

The cerebrum possesses a more complex structure. It includes the cerebral cortex, thalamus, and hypothalamus, among other critical parts.

The highly folded cerebral cortex contains roughly 30 billion

neurons and is divided into four functionally distinct lobes. The frontal lobe governs social judgment, motor planning, language, and short-term memory. The parietal lobe processes somatosensory information such as touch and pressure and integrates spatial perception. The temporal lobe participates in auditory processing, language comprehension, and memory, while the occipital lobe is dedicated to processing visual information.

At the center of the brain sits the thalamus. Though merely the size of a fava bean, it is a crucial relay station for sensory information entering the cerebral cortex. Visual, auditory, and tactile signals are passed here from one neuron to another before being projected to the corresponding cortical regions. Below it lies the hypothalamus, which primarily manages visceral activity and endocrine regulation. Beneath the cortex, the hippocampus is intimately tied to long-term memory, while the basal ganglia participate in initiating and controlling patterns of movement.

The hippocampus, amygdala, hypothalamus, and other structures together constitute the limbic system, which is classed as the emotional brain and is responsible for generating emotion, driving motivation, forming memory, and regulating basic survival behaviors.

As noted before, the vast majority of the brain's functions are not realized by any single brain region working alone. They depend on the coordinated activity of multiple regions joined through complex interconnections.

Consider the thalamus and cerebral cortex. They share an abundance of reciprocal connections. These include not only projections from thalamus to cortex but also feedback from cortex back to tha-

lamus. Within the cerebral cortex, the corpus callosum connecting the two hemispheres contains approximately 200 to 250 million neuronal fibers projecting back and forth.

This redundancy is another important feature of brain connectivity. The great majority of sensory, motor, and cognitive functions are realized by more than one neuronal pathway. Indeed, the same information gets processed in parallel in different regions. Therefore, when one pathway is damaged, the brain does not necessarily lose the corresponding function entirely. This demonstrates remarkable plasticity.

In terms of functional hierarchy, perception, motor control, and such functions roughly governed by the instinctive brain as heartbeat, breathing, and digestion may be viewed as "innate" functions common to all human beings. They are sometimes classed as relatively lower forms of consciousness. To a large degree, the mechanisms that generate them resemble automatic control systems with a clear relationship between input and output.

Emotions such as greed, fear, anger, excitement, anxiety, and depression are driven primarily by the limbic system, or the emotional brain.

Above these, our capacity for rational thought, planning, and decision-making mainly relies on the highly developed cerebral cortex, especially the prefrontal cortex of the frontal lobe. This is sometimes called the intelligent brain.

The prefrontal cortex is an integrative hub. It receives information from all sensory systems, as well as from the emotional and instinctive brains, and on this basis performs situational analysis, predicts consequences, inhibits impulses, and makes long-term plans.

Our so-called higher consciousness—our complex emotional and psychological states—is therefore fundamentally the product of a three-way struggle among the most ancient survival instincts, or the instinctive brain; the system that drives emotion and motivation, or the emotional brain; and the most recently evolved capacity for rational planning, or the intelligent brain. Through an internal "value system," these three layers engage in continual dialogue and struggle.

Building on this understanding of the brain's structure and functional hierarchy, we can see that the so-called value system is likewise not localized in any single brain region. It is jointly carried by an extensive neural network distributed across both the emotional and intelligent brains.

This system is nothing less than a microscopic embodiment, within the brain, of Darwin's principle of survival of the fittest. Its core function is evaluation: it continually assesses what internal and external stimuli mean for individual survival and reproduction. Is something good or bad, a reward or a punishment, an opportunity or a threat?

The brain's value system is therefore fundamentally a product of natural selection and has a certain genetic foundation. The neural network carrying this system typically sends axons in diffuse projections to numerous brain regions. In particular contexts, it releases particular neurotransmitters or neuromodulators to regulate neural activity and thereby influence our emotions and behavior.

Even the chemical messengers that drive our emotions and behavior—neurotransmitters—are not produced by the brain alone. In fact, about 80 percent of the serotonin in the human body is

synthesized in the gut.

With its enormous neural network and complex microbial ecosystem, the gut has been called the "second brain." It not only produces various neurotransmitters; its metabolic products can also profoundly influence the brain's emotion, cognition, and behavior. The urge to use the toilet when we are nervous—the brain influencing the gut—and the pleasure felt after consuming sugar—the gut influencing the brain—both exemplify this two-way dialogue. The bidirectional communication system that links brain and gut through neural, endocrine, and immune pathways is called the "gut–brain axis."

From the diversity of the sites at which chemical messengers are produced to the process of information handling itself, our consciousness is far from being a "centralized" system.

As the senses continually gather external information and transmit it through neural networks to the central brain, the vast majority of the processing is carried out autonomously and requires no active intervention from consciousness.

This finely tuned chemical regulatory system drives our physiological responses and behavioral decisions by releasing particular kinds and quantities of neurotransmitters and hormones. Whether the trigger is the bodily adjustment caused by holding one posture for too long or a powerful stimulus related to survival and reproduction, such as wealth or sexual attraction, the brain automatically releases corresponding chemicals, including dopamine, to spur us into action. These substances function like an internal "chemical scale." Their release is governed by thresholds intended to maintain bodily balance with the least expenditure of energy; excessive

release can lead to tolerance, addiction, and even organ damage.

Many phenomena that we perceive as "instinctive impulses" or "spiritual experiences"—the grandiosity of intoxication, for example, or pleasurable visions arising in certain forms of spiritual practice—are, at root, activations and emergent effects of this chemical system under particular conditions. Consciousness seems more like the experiencer that arrives late to "know" and "narrate" the story, after this vast chemical symphony has already begun to play.

The system most closely associated with greed, desire, and the anticipation of reward is the dopaminergic system. It is concentrated in such regions as the nucleus accumbens of the basal ganglia and the brainstem, but its neuronal axons project widely. Its neurons release dopamine, a neurotransmitter that plays a central role in regulating sexual desire, sensation, and motivation and is closely intertwined with various addictive behaviors.

Dysfunction of this system has been confirmed to be associated with diseases such as Parkinson's disease and schizophrenia. Parkinson's disease primarily involves the degeneration of dopaminergic neurons in the nigrostriatal pathway, while schizophrenia involves hyperfunction in dopaminergic systems such as the mesolimbic pathway.

The locus coeruleus, located in the brainstem, is responsible for eliciting pleasure and excitement and for regulating alertness. This cluster of neural nuclei sends vast numbers of axons in broad projections to the brain and spinal cord. Its reach is extremely wide: the axon terminals of a single neuron can activate nearly the entire cerebral cortex.

When the locus coeruleus is activated, it releases norepinephrine. This excitatory neurotransmitter lowers the response threshold of synapses on target neurons, making the neurons easier to fire, and can also alter synaptic strength. Through these mechanisms, it helps us maintain sustained attention to events and a state of emotional tension.

The raphe nuclei, located near the midline of the brainstem, are closely tied to anger, depression, and related emotions. They likewise send out broad axonal projections and release the neurotransmitter serotonin.

High concentrations of serotonin are usually associated with calmness and a sense of well-being, while inadequate function or excessively low concentrations are linked to irritability, impulsivity, aggressive behavior, and risk-taking. Dysregulation of serotonin function can trigger diseases such as depression.

The key regions governing anxiety, fear, a sense of security, and a sense of well-being are located in the basal ganglia and include such structures as the caudate nucleus, striatum, claustrum, and amygdala. Their functions influence perceptual categorization, the formation of experience, subconscious and unconscious activity, and more.

Overactivation of the lateral nucleus of the amygdala triggers intense and uncontrolled fear responses. The striatum is usually associated with positive reinforcement and pleasant feelings; it can inhibit amygdala activity, thereby enhancing a sense of security and reducing fear. It is worth noting that the striatum can be intensely activated by addictive drugs such as cocaine, which explains why such substances can lure people into repeated use.

Beyond these, the brain harbors other important regulatory systems. The cholinergic system, through the release of acetylcholine, plays a key role in switching between sleep and wakefulness and in learning and memory. The histaminergic system assists in regulating the sleep-wake cycle, biological rhythms, and pain sensation. The brain can also synthesize and release endogenous opioid-like substances such as enkephalins. These modulate pain responses.

Higher conscious functions such as emotion and desire are likewise not determined by any single, isolated brain region. They too depend on the complex interactive relationships formed between specific regions and the vast majority of other brain areas.

Unlike perceptual systems, which are driven mainly by signals from the outside world, higher forms of consciousness are largely stirred by activity within the brain and mediated and integrated by complex neural circuits linking multiple regions in both directions. When we experience these higher forms of consciousness, it is therefore difficult to distinguish simple "inputs" and "outputs" as clearly as we might when analyzing an automatic control system. They seem instead to "emerge," as if from nothing, out of the dynamic activity of an immense neural network, and so carry a more inward, spontaneous, and mysterious quality.

We do not regard these impulses of higher consciousness as mere instinctive reactions precisely because the intelligent brain, especially the prefrontal cortex, is deeply involved in cognitive evaluation, regulation, and control, as well as in combining emotion with memory. The intelligent brain endows raw emotional impulses with concrete social and cognitive meaning.

Consider the feeling of "greed." It is not simply that your nu-

cleus accumbens has been activated. Your prefrontal cortex performs a complex cost–benefit analysis and risk assessment: How much is this thing worth? What price must I pay to obtain it? How great is the risk? When you are seething with rage yet choose restraint, it is your prefrontal cortex that inhibits the raw impulses driven by the amygdala. The hippocampus binds the intense emotional experience to a particular context and turns it into memory, which the prefrontal cortex can then call upon to guide future decisions.

A gymnast's years of training, in essence, "write" complex sequences of movement into the basal ganglia and cerebellum, forming near-instinctive conditioned-reflex pathways—what is commonly called "muscle memory." On the competition floor, these movements can then be executed automatically and fluidly.

Yet if, at a critical moment, the gymnast activates the intelligent brain and begins to overthink, or the emotional brain produces tension and anxiety, the additional neural signals accumulate and interfere with the original automated pathways. This multilevel struggle among brain regions may upset the balance of motor coordination, causing movements to become distorted or mistakes to occur.

This intricate internal value system, together with the three-layer brain model, constitutes a dynamic equilibrium. If we compare it to the decision-making level of a company, the instinctive and emotional brains resemble the firm's most senior and powerful conservative departments.

They have the longest tenure and are evolutionarily the most ancient. They wield great executive power, directly regulating vital signs and emotional drives. Yet their thinking tends to be con-

servative, and their core demands are “safety first” and “energy conservation.” They are accustomed to acting according to an ancient “survival manual”—the inherited instincts imprinted by natural selection—and are suspicious of any innovative course that departs from routine, consumes extra energy, or carries uncertainty, such as learning a new skill, speaking in public, or persisting with a long-term plan. They frequently deploy powerful physiological and emotional responses—anxiety, fatigue, or the lure of immediate pleasure—in an attempt to veto change. At bottom, their purpose is to avoid every possible risk.

The relatively young “intelligent brain” department is entirely different. It represents the firm’s division of strategic planning, innovation, and development. Open to innovation and eager to move forward, it plans for the future by continually expanding its understanding and values long-term growth over immediate comfort.

Yet in this internal power struggle, the intelligent brain is often at a disadvantage. Its decisions consume enormous amounts of energy, and its “rational” instructions usually have to be carried out by the ancient and powerful departments of instinct and emotion. Thus, when the intelligent brain tries to introduce a reform—exercising every day, beginning a healthy diet, or preparing for an important examination—it often meets powerful resistance from the conservative departments. This resistance may appear as irresistible procrastination, an impulse to make excuses, or distraction by entertainment that is more immediate and effortless.

This is why so many enterprising efforts that require perseverance end after only a few days of attempted change and helpless surrender. The root cause is simply that the entrenched patterns and

physiological foundations of the instinctive and emotional brains are far too powerful.

Except in simple, comfortable matters that follow habit, anything challenging or difficult—anything that requires us to leave our comfort zone—inevitably triggers this internal deliberation within the brain. Almost every important decision and action is the final result of dialogue, compromise, and struggle among these three forces. Understanding this inner dialogue is key to understanding our psychological complexity.

Is there, then, an independent brain region resembling a "chief executive," one that directs all the others, reconciles these three forces, and serves as the all-governing "self"? We are strongly inclined to believe that a single, conscious "I" perceives, decides, and acts. Neuroscience, however, indicates that the brain contains no such central commander issuing orders.

Our seemingly unified and continuous self-consciousness is more like a harmonious melody "emerging" from the complex interaction of the brain's vast symphony orchestra than a conductor exercising solitary control.

Vast numbers of emotional responses, thought processes, and behavioral tendencies are in fact predetermined and shaped through rapid, unconscious interaction among the redundant, reentrant neural circuits distributed throughout the brain. This extraordinarily complex web of redundant and reentrant connections constitutes a massive parallel-processing system. It can instantaneously associate an immense volume of perceptual signals and, through endless combinations, coordinate them with unconscious memory systems and with both innate and acquired value systems.

It is precisely from this distributed, decentralized, dynamic network activity that the rich and coherent stream of human consciousness we experience moment by moment finally and miraculously emerges.

This decentralized emergent character is especially evident in concrete neural processes.

Take visual processing. Visual information is not processed in a single location, but involves as many as 33 different cortical regions. These regions are highly differentiated in function: some specialize in line orientation, some process color, and some monitor dynamic change. Through dense reentrant connections, they communicate and coordinate in parallel, with signals continually passing back and forth. Yet there is no "central region" responsible for integrating all these local responses into the unified and complete image we ultimately perceive.

This observation leads to a profound philosophical question. As philosophers such as John Searle and Thomas Nagel have emphasized, a conscious state is subjectively a unified whole in continuous flux. Different sensory modalities combine in an extremely complex yet perfectly fused manner to compose a single complex and ordered world of experience.

Bringing together the perspectives of science and philosophy, we have no choice but to accept a conclusion: the brain contains no particular region that acts as the unified "I." The "self" that we intuitively believe to occupy the center is more like an effect of light and shadow projected by the brain's complex workings. It is at once the magnificent achievement of the brain's vast neural activity and the most important experiencer of that activity, but it is not itself an

independently existing physical thing.

Understanding this may be the deepest starting point for truly approaching the riddle of self-consciousness.

Many phenomena in real life constantly challenge our stable conception of the self.

When we are awake, we clearly feel the presence of the self; in deep sleep, that self seems to disappear. Self-consciousness is not present from birth: infants generally begin to form a concept of the self only at around eighteen months to two years of age. Some autistic people may form an inaccurate understanding of the self, while people with dissociative identity disorder may have several selves that alternately "take office."

In 1983, Benjamin Libet at the University of California, San Francisco, conducted a highly influential experiment. He precisely recorded the timing of the brain's electrical signal indicating an intention to move and the moment at which participants reported consciously deciding to move.

The readiness potential began about 550 milliseconds before the movement itself, whereas participants reported becoming aware of their intention to act about 200 milliseconds before the movement. In other words, the relevant brain activity preceded the reported conscious intention by roughly 300 to 350 milliseconds. This seems to suggest that, when we are convinced that "self-consciousness" has made a decision, the brain has already begun the relevant preparation unconsciously.

This experiment leaves us with a perplexing question: did I make the decision, or did I merely perceive that so-called decision

only after the neural activity had already occurred?

This brings us to a still more fundamental proposition. Just as no completely free particle, untouched by interaction, exists in the universe, it is exceedingly unlikely that our brain contains a "free consciousness" that merely issues commands while remaining unconstrained by other factors. Consciousness should be understood as an ordered existence and must therefore be constrained by other ordered structures, just as the brain's various regions are reciprocally connected, mutually restraining and binding one another through reentrant circuits.

This order is hierarchical. The order of consciousness is constrained by the order of the entire nervous system, with the brain at its center; the order of the nervous system, in turn, must be embedded within and depend upon the internal environment created by the body. All neural pathways depend for their proper operation on the ion concentrations, energy supply, and chemical balance maintained by the body. The release and action of every neurotransmitter likewise depend upon this precise microenvironment.

There is something paradoxical here: this physiological microenvironment, with its measure of randomness and fluctuation, provides the necessary ground for consciousness to arise "out of nothing." It gives the conscious system a degree of freedom and allows us to roam through the space of thought.

Once detached from this living bodily environment, the complex system called "consciousness" no longer exists. The so-called "cloud brain," or uploaded consciousness, would necessarily possess a pattern of order different from that of the original brain. It would therefore not be the same thing, still less the same "self."

What, then, does the "self" actually refer to? Perhaps it more properly refers to that particular, highly complex and dynamically ordered neuronal network: the physical entity corresponding to an individual's unique life history, structure of knowledge, and cognitive framework for understanding the world.

Every such network is necessarily unique in both its physiological structure and its patterns of dynamic activity. This unique network is the physical substrate of the "I."

On this basis, and from the perspective of current physicalist and naturalist scientific paradigms, we can find no scientific grounds for a "soul" that exists free from the constraints of the material world. The "spirituality" exhibited by human beings and animals still depends fundamentally on the dynamic activity of complex neuronal networks.

Nevertheless, we must also acknowledge that, in the deep world of human emotion and faith, the concepts of "soul" and "afterlife" carry meanings that extend far beyond the scope of scientific verification. They may indeed be a simple yet powerful psychological construction through which people yearn for life's continuation, believe that death is not the end, and thereby suppress the fear and anxiety of annihilation.

At the same time, they also touch upon individual spiritual consolation, moral foundations, and cultural identity. Beyond scientific discussion, therefore, this is a complex matter of emotion and faith that must be fully respected.

Chapter 22

Learning and Innovation

Natural abilities are like natural plants, that need pruning by study.

Francis Bacon

The masterpiece of life's evolution is precisely this brain of infinite potential and the intelligence it carries. Whatever knowledge you learn does not vanish into thin air. It materializes into newly grown synapses and neuronal connections inside your brain. It forms your cognitive framework for understanding the world.

For the individual human being, therefore, sustained learning is the most important investment in oneself. By actively sculpting the neuronal connection network, one builds a more ordered self. One better understands the outside world. This is life's deepest response to the relentless growth of cosmic entropy.

Our brain's neuronal network rests upon an ancient foundation shaped by our ancestors over tens of millions of arduous years. That foundation is our instinctive brain and our emotional brain.

They endow us with our deepest survival drives, including a fondness for food and an aversion to exertion, which served to preserve every ounce of energy in an era of scarcity; a hunger for im-

mediate results, which served to seize fleeting opportunities amid constant danger; and a flight from difficulty toward the easy, which was the instinctive choice to concentrate precious resources on immediate threats under survival pressure.

These dispositions that we today regard as obstacles to growth were, across the vast stretch of prehistory, the most effective survival strategies life ever evolved. They enabled life to sustain its own existence and resist the disorder imposed by the environment—that is, the increase of entropy. They ensured that we could maintain the most basic forms of self-preservation and reproduction. They are life's most fundamental capacity for resisting entropy.

As social civilization developed, more complex forms of social organization took shape. We are no longer merely isolated biological individuals resisting environmental disorder. We are also units within various forms of social organization. On top of maintaining basic self-preservation and reproduction, we need to find order and meaning within these higher-level organized entities.

This civilizational model of collaboratively resisting entropy through collective intelligence demands that we handle far more than the wind and rain of nature. We must navigate more complex organizational relations, including morality, culture, law, economy, and politics.

We cannot, and need not, overthrow the instinctive system optimized for immediate survival, for it remains the foundation that sustains our basic ordered existence. Yet if we rely entirely on instinct, we may find it difficult to adapt to the complex circumstances of modern social organization.

The fundamental purpose of learning is precisely to make our brains better suited to understanding the laws of nature. As discussed earlier, when scientists discover laws, what they are doing at root is this: from the chaotic profusion of external phenomena, they identify and distill those repeatable, predictable features that constitute order. At bottom, learning is the brain's ongoing physical reshaping under the training of samples supplied by the external environment.

The brain does not store knowledge directly the way a hard drive does. It engraves order through physical reshaping. Every time you comprehend a concept, master a skill, or perceive a relationship, a new synapse sprouts in the brain's neural network, or an existing neuronal pathway is strengthened. These microscopic structural changes together weave your unique cognitive framework, a dynamically evolving, multi-layered network of order.

Each individual observation or training sample may seem random and insignificant. Sustained observation and a sufficient number of samples, however, supply an enormous quantity of disorder and sufficient possibility. The brain's neuronal network, with its progressively layered structure of order, then continuously performs selection, constraint, and stabilization, so that the chaotic, undifferentiated brain of the infant emerges as the complex, ordered brain of the adult.

The large artificial-intelligence models that have attracted so much attention in recent years are, in their underlying principle, an engineered replication of this very process. An artificial neural network is built from countless simple neuron-like units. They interact through adjustable connection weights. During training, vast quan-

tities of data samples, such as texts and images, act like inputs from the external environment. They drive the network to continuously adjust its weights. This exactly corresponds to the strengthening and weakening of synaptic connections in the brain.

An artificial neural network does not directly understand semantics. Yet through the statistical order solidified in its weight matrices, it can generate fluent replies, draw exquisite pictures, and even display astonishing reasoning capacity.

Admittedly, differences in natural talent exist among individuals. Some people are born with a greater number of neurons. Certain rare genetic variants may endow a few individuals with extraordinary memory. Neurobiological research has found that differences in memory ability may relate to variations in the CREB-2 gene. These variations affect the activity of the repressor protein associated with CREB-1. Through this mechanism, they tune the threshold of synaptic plasticity. These genetic endowments are like the hardware specifications set at the factory. They provide different starting lines for subsequent cognitive development.

Yet the quality of hardware parameters is far from enough to determine the level of intelligence a system can ultimately manifest. The final shape and efficacy of an individual's neuronal connection network depend far more on sustained learning and shaping after birth.

We frequently fall into a cognitive trap. We mistakenly believe that those who are learned and insightful have merely memorized more knowledge. We think the reason we fall short of them is only that we did not spend enough time experiencing and remembering. This notion overlooks a crucial truth. Knowledge itself is not

the goal. True wisdom originates from building, through repeated training, connection structures in the brain that can be flexibly mobilized. It is like a computer equipped with a top-tier graphics card. If it has never undergone training on massive datasets, an artificial neural network running on it cannot produce higher intelligence.

In early childhood, the human brain possesses far more synaptic connections than in adulthood. It resembles a forest growing vigorously yet slightly tangled. Its plasticity is exceptionally powerful. The input from the ordered outside world is like an endless training set. Everything we see, hear, think about, and practice silently prunes and connects that inner neural forest.

Advances in neuroscience make this clear. From infancy to adulthood, interaction with the outside world steadily increases the orderliness of the brain's neural connections. In 2018, a team led by Professor Erhan Genç at Ruhr University Bochum reported that in more intelligent people, connections between neurons are more efficient and streamlined, consume less energy when processing the same information, and are less likely to trigger the brain's negative-feedback mechanisms. Such brains can therefore manage multiple tasks more effectively and sustain attention for longer.

More importantly, the vast majority of our neurons remain with us for life. They do not regenerate or multiply, yet they continue evolving and changing throughout our lives.

The only path to elevating the mind, therefore, is not to obsess over innate endowments that cannot be changed, but to throw oneself into sustained learning: to discover order and pattern through repeated training and let intelligent cognition emerge from boundless connections. This is the deepest and plainest essence of learn-

ing.

What, then, is the correct method of learning? The answer is straightforward. The correct method is whatever promotes the formation of a more ordered neuronal connection network.

So is it possible for a simple brain to directly grow a complex neuronal connection network? In fact, every neuron in the brain grows like a tree. It does not grow in isolation, but is gradually shaped through continuous interaction with its environment. Whether it is the strengthening of existing synapses or the sprouting of entirely new pathways, both must depend on the repeated, high-intensity activation of specific neuronal pathways. Every transmission of a neural signal, from the release of neurotransmitters to the remodeling of synaptic structures, strictly follows intrinsic physiological rhythms and biochemical laws.

The first principle of learning, therefore, is to acknowledge this basic fact. Without investing a sufficient number of repetitions at sufficient intensity, without having seen, heard, thought about, and practiced something enough times, it is impossible to physically construct stable and efficient neural connections in the brain. Natural talent may raise the initial efficiency or the ceiling of this process. It can never replace the physiological sculpting driven by repetition itself.

In this sense, the foremost and most fundamental method of learning is precisely “having no method.” It points directly to a plain and unshakeable truth. Only through sustained, focused investment can the optimization and upgrading of structure be realized at the neural level. This is the scientific interpretation, at the microscopic level, of the saying that there is no shortcut to learn-

ing. It is the neuroscientific essence of the principle that Heaven rewards the diligent.

Just as all things in the universe tend from order toward disorder under the law of entropy, the neuronal connection pathways already formed in the brain will also slowly loosen and weaken if they are left unactivated and unused for too long. Those old, seldom-used synapses will gradually retract and may even be pruned away entirely. This is the physical manifestation, at the neural level, of forgetting.

How, then, can we firmly engrave truly valuable knowledge onto our neuronal connection network and prevent it from flaking away with the passage of time? The key lies in repetition. Through periodic, strategic review and application, we repeatedly stimulate the connections already formed. It is like ceaselessly reinforcing a path. This promotes the long-term consolidation of neuronal connections.

Yet neural plasticity is not simply a matter of use it or lose it. Stimulation that is too dense or too frequent can produce habituation or neural fatigue, weakening synaptic function and reducing the efficiency of the connection.

This corresponds to cognitive fatigue in our learning. When the brain receives the same information repeatedly without a chance to recover, it judges this information as redundant or low-value. It then actively suppresses further deep encoding. Learning outcomes therefore decline rather than improve.

Effective repetition, then, is absolutely not brute-force drilling. It must be supplemented with a scientific rhythm. Within this rhythm, adequate rest and sufficient sleep play essential roles.

During sleep, our brain activates a sort of dishwasher mode. Through the rhythmic flushing of cerebrospinal fluid, it clears away metabolic waste, including beta-amyloid, that has accumulated between brain cells during the day. This process does more than detoxify the brain: it can also replay, organize, and consolidate the day's learning content, converting short-term memory into long-term memory. A brain that has been thoroughly cleaned and reorganized can, upon waking the next day, greet new learning tasks in its most efficient state.

This is the second core method of learning: maintain a rhythm of tension and release, balancing work with rest. Read more, observe more, apply more, and review more, while giving the brain ample time to integrate and recover. In time, knowledge can be refined into an almost automatic fluency, so that even when attention wanders, the relevant neural pathways can be activated swiftly and accurately. Much of what we learn can thus become a firm, almost instinctive presence within our neural networks.

The brain's intrinsic attention-regulation mechanism likewise offers crucial insights for learning more efficiently.

When we are fully focused and actively thinking, the brain's attention-regulation network is deeply activated, prompting the additional release of regulatory neurotransmitters such as norepinephrine and acetylcholine. These neurotransmitters do more than merely boost the strength of neural signals. They can also alter the synaptic microenvironment and promote the synthesis of new proteins inside neurons, thereby making the nascent neural connections that sprout during learning and thought more stable and durable.

The renowned physiologist Pavlov further proposed that during focused study, only a single excitatory center appears on the cerebral cortex. All the brain's energy is channeled into supporting the activity of this center. The adjacent areas all remain in a state of inhibition.

This means that the neuronal pathways sculpted by a single session of concentrated learning are of far higher quality than those produced by countless distracted repetitions. The resulting connections are less likely to loosen, so one attentive session can achieve twice the result with half the effort and bring a leap in learning efficiency. This accords perfectly with the everyday insight that one attentive pass is worth ten rounds of mechanical repetition.

This leads us to the third core method of learning. When studying, summon your whole mind: focus your attention, and truly devote yourself to understanding, thinking, and immersion. This is more than an attitude toward learning; it is the neurochemical key to the brain's deepest physical reshaping.

Finally, the fourth core method of learning is to consciously introduce fresh problems and new knowledge during the learning process.

When, after a long period of working on a single problem, we switch at the right moment to an entirely new problem, the brain enters a highly alert, "full-power" state to handle this sudden contextual shift. At that moment, regions responsible for higher cognition, such as the prefrontal cortex, are fully activated. This allows us to perform abstract thinking and pattern recognition more rapidly. The brain at this moment resembles an analogy machine running at high speed. It can sharply capture the deep structural similari-

ties between the new problem and an existing body of knowledge. It can thereby more quickly grasp the underlying logic behind the problem.

This state does not merely boost memory and analysis speed dramatically. It can also catalyze creative solutions. This is how some people manage to learn in half a day what others take three years to master. It is not because time is compressed. It is because the brain, in a specific state, naturally activates a more efficient engine of abstraction and a mechanism for memory consolidation. Learning efficiency undergoes an exponential leap.

From the analysis above of learning methods, we can likewise draw a profound lesson about the fundamental question of what to learn.

Beyond passively and imperceptibly absorbing information from the environment, we should also actively and purposefully filter and select the content we take in. Learning genuinely valuable knowledge and skills will sculpt efficient and beneficial neuronal connection pathways. This amounts to positive organic change in the brain. Conversely, every bad habit, erroneous belief, and ineffective experience also acts upon our brain in the same way. They form equally stable and sometimes even more stubbornly entrenched useless or harmful connections. This amounts to negative organic change.

Once these negative connections are formed, they are very hard to reverse by sheer force of subjective will alone. One must expend more time and immense perseverance to forcibly correct them. One must await the weakening of old connections and the construction of new ones. Behaviors such as gambling, drugs, and pornography,

because they trigger abnormally intense neurochemical stimulation, cause negative organic changes that are particularly deep and severe. They can be so powerful as to become irreversible. They may permanently damage the brain's reward circuits and cognitive functions.

In this age of information overload, the enhancement of cognitive capacity depends not only on the intensity of learning but even more on the quality and structure of the information we encounter.

Systematized, structured bodies of knowledge—formal educational curricula, deep reading materials, and practical experience that has been repeatedly tested and verified—provide the brain with high-quality training samples. Such content distills the wisdom accumulated by humanity across long ages. Organized with logical rigor and clear layering, it helps the brain construct a robust cognitive framework.

Many digital platforms rely on algorithm-driven mechanisms for pushing information. They tend to occupy the user's cognitive space with fragmented, high-stimulation content. This kind of information has generally not undergone systematic screening. Its authenticity and completeness are questionable. It is frequently mixed with large amounts of emotionally charged, low-quality content. Long immersion in such information streams not only makes it hard to achieve meaningful reorganization of neural connections. It can lead to cognitive overload and attention fragmentation. It can then disrupt the stability of already-established knowledge structures.

More troubling still, certain content deliberately panders to instinctive reactions and provokes low-level emotional resonance. Though it can win immediate engagement, it does nothing to

foster deep thought. On the contrary, it can gradually erode the capacity for rational judgment and may even trigger emotional dysregulation and cognitive dissonance.

To choose sources of knowledge grounded in systematic thought and empirical evidence, and to avoid being led by disordered, low-quality information streams, is to care responsibly for one's neural network. It is also an important prerequisite for maintaining mental health and cognitive clarity.

Beyond the precious knowledge in books, the multi-layered ordered structures of all things in the universe, along with their dynamic changes, constitute a yet more fundamental source of cognition.

In many cases, only through hands-on practice can we come into contact with the facts as they actually are. The process of practice is exactly like negative-feedback regulation in automatic control theory. The gap between our observations of the outside world and our expectations continuously drives us to revise our understanding of that world and to move toward convergence. Through this process, we pursue the aim of understanding the world and transforming it.

If, in handling affairs, we consistently analyze matters from a multi-level perspective, think independently, exercise careful judgment, and refuse to follow blindly, then this long-term, deliberate cultivation of thought will likewise induce deep physiological changes in the brain's neural network.

When the capacity for truth-seeking from facts is deeply engraved into the brain, we can achieve knowing without deliberation. That is, truth-seeking from facts without needing to think about it. This may be what is meant by inward cultivation of the self. It

may also be the state that Wang Yangming advocated as innate conscience and the unity of knowledge and action.

Through passive or active learning, drawing widely upon nourishment across every band of the law spectrum of the cosmos, each of us is continuously constructing and optimizing a uniquely ordered neuronal connection network.

If we compare this network to a precision circuit, then consciousness is like the current flowing through it: the state of the network activated at a given moment. Yet unlike a circuit, which follows strict physical laws and produces highly consistent outputs each time it runs under a fixed structure and fixed parameters, emergent consciousness can both faithfully reflect existing patterns of neural connection and frequently transcend established pathways, producing new ideas and inspiration.

This creative emergence of "something from nothing" has two sources. One is the extraordinarily complex system formed by roughly one hundred billion neurons. Its deeper source, however, lies in the randomness inherent at the brain's lowest microscopic levels. Some scholars, notably the British mathematical physicist, philosopher, and Nobel laureate Roger Penrose, have even proposed that consciousness may arise through an exceedingly complex and subtle quantum process. Such processes might occur at the microscopic level of the brain, in protein quantum vibrations, for example, or in quantum tunneling through ion channels.

We can glimpse this process by examining how synapses and neurons work, and then try to understand how consciousness as a whole can emerge from microscopic randomness.

The firing of a single neuron is like a tiny flash in the brain's

inner cosmos. How far this brain signal travels, how large a ripple it stirs, depends fundamentally on how many synapses are successfully activated.

Yet whether a synapse is activated is by no means a simple on-off switch. It is a microscopic event shaped by extremely complex factors and saturated with randomness. The geometry of the presynaptic and postsynaptic membranes, subtle differences in the width of the cleft, all influence the efficiency of neurotransmitter diffusion. The number of proteins responsible for moving vesicles within the synapse varies stochastically. The number of neurotransmitter molecules packed inside a single vesicle is not constant. It follows a statistical distribution. The recycling and reuse of vesicles takes time. This makes the number of available vesicles at any given moment dynamically variable. The chemical microenvironment of the synaptic cleft fluctuates from instant to instant. The net effect after excitatory and inhibitory neurotransmitters cancel each other out at this moment is a microscopic contest with an uncertain outcome. Surrounding glial cells release gliotransmitters that quietly modulate neuronal excitability. Even the comprehensive balance of regulatory proteins inside the neuron, CREB-1 and CREB-2, dynamically determines the threshold required for synaptic strengthening.

At the microscopic scale, therefore, whether a single synapse fires is intrinsically governed by a degree of randomness. This is the “disorder” contained within the lower-level order of a highly ordered neuronal connection network.

Also, a typical neuron usually carries thousands of synapses. They are simultaneously bombarded by countless neurotransmit-

ters arriving from different neuronal pathways. Some are excitatory and some are inhibitory. Every synapse is independently modified by the random factors just described and independently decides its response. Only when the sum total of all excitatory signals, after deducting the influence of all inhibitory signals, still reaches a critical threshold will this neuron finally fire.

Hence an initial ripple of electrical activity in the brain might activate only a few dozen neurons and tens of thousands of synapses. Or it might, like an avalanche, detonate thousands of neurons and millions of synapses. To a considerable extent, this is a probabilistic event shaped by underlying randomness.

At the same time, the outside world is never in a static state. The external signals our brain receives through various senses are also, by their nature, not absolutely determined and pure. The environment is saturated with statistical noise, with incomplete information, with unexpected fluctuations. This means the signals entering the brain themselves carry a significant component of randomness.

The brain does not passively receive all this chaotic information. It actively filters and interprets incoming signals based on past learning experience. That is, based on its already-consolidated ordered network. It continually makes predictions about the outside world. It then compares these predictions with actual sensory input. When a discrepancy arises between external random signals and the internal predictive model, this prediction error becomes a powerful learning signal.

By analogy with the underlying logic of the multi-layered ordered structures found throughout the world, we can reasonably infer that the brain can likewise be understood as such a structure,

extending from synapses and neurons to neural pathways and functional brain regions. Driven by disorder at its internal foundations and by disorder in external input, and filtered through the framework of its existing ordered networks, it continually produces new possibilities. This is the brain's mechanism of innovation, its way of bringing something forth from nothing.

Those chance flashes that are inconsistent with the existing cognitive framework or do not match objective reality, because they cannot recur, will mostly quietly dissipate like fireworks. But among them, those new discrepancies that happen to explain the world better or solve problems more effectively are captured, because of their usefulness, by focused thought and repeated practice. The corresponding neuronal connection pathways in the brain are activated over and over. This drives synaptic strengthening and the growth of new connections. They are ultimately consolidated as new components of the network's ordered structure.

The process of brain learning and consciousness generation is precisely this process. Order sets the direction. Disorder enables innovation.

No completely independent neuron or neuronal pathway exists in our brain. Absolute independence would mean it cannot be integrated and cannot be activated. It would in effect be excluded from the effective connection network. By the same logic, we do not possess a consciousness that is completely free. Consciousness must be generated and must operate within a specific neuronal connection network jointly shaped by innate instinct and acquired learning.

Just as the existence of matter requires the "binding" of various fundamental interactions, and just as human sociality cannot

be separated from the "binding" of social and organizational relations, consciousness is necessarily bound by innate instinct and by the cognitive framework formed through acquired learning.

This binding is not imprisonment. It is precisely the root from which consciousness acquires its meaning and direction.

A sound that is entirely unconstrained by any structure, purely random, is noise. A beautiful melody, by contrast, is produced under the strict binding of the rules of tonality, rhythm, and musical form. In the same way, only under the "binding" of the existing neuronal connection network can the freedom and innovation of consciousness give rise to ordered structure and meaning.

This binding also furnishes the sole springboard for achieving valuable innovation.

The genius we see is extraordinary not because of superior natural gifts, but because of sustained, unremitting effort. In *Outliers*, Malcolm Gladwell encapsulates the "ten-thousand-hour rule" in one line: "Ten thousand hours is the magic number of greatness." The rule presents such sustained tempering as a necessary condition for moving from the ordinary to world-class mastery. Viewed from the perspective of neuroscience, it is precisely the process of constructing an extremely specialized and ordered neuronal connection network within a specific domain.

It is precisely this specialized neural network that provides the prerequisite and foundation for genuinely disruptive innovation, for bringing something forth from nothing. Once the brain is "bound" by a structure this deep, inspiration is no longer a vague stream of random code, but a novel connection among countless nodes of specialized knowledge. Einstein's theory of relativity did not descend

from nowhere. It was deeply rooted in his complete command of the edifice of classical physics. That very "constraint" enabled him to perceive its contradictions and limits and thereby accomplish a magnificent act of creation from nothing.

Of course, our consciousness is not entirely without freedom. The randomness that pervades both inside and outside us endows us with an eternal drive. It lets us break through original cognitive frameworks, generate new ideas, discover new laws, and create new knowledge. This is the physiological basis of conscious freedom. Therefore, the freedom and the unfreedom of consciousness are two faces of a single coin. They stand in dialectical unity.

In The Principles of Psychology, William James drew a clear distinction between at least two modes of attention: voluntary and involuntary. Modern neuroscience tells us that both depend on the precisely regulated release of neuromodulators such as acetylcholine, norepinephrine, and dopamine in particular regions of the brain.

More concretely, voluntary attention is like a tightly focused searchlight: it enables convergent thought and actively checks the disorderly diffusion of thought, providing the neural basis for deep learning and logical reasoning. Involuntary attention works in precisely the opposite way, allowing thought to roam freely through the neural network and branch out without restraint. In this state, weak connections, distant analogies, and seemingly irrelevant fragments of memory that are suppressed under intense focus can be activated and brought into collision, greatly fostering inspiration.

This example shows clearly that we possess a kind of conscious freedom analogous to this mechanism of attention. It affects the ex-

tent to which thought and consciousness can bring something forth from nothing, and thus determines the likelihood that inspiration will arise.

Chapter 23

The Limits of Cognition

We read the world wrong and say that it deceives us.

Rabindranath Tagore

We know that the essence of all existing things is multi-layered order. Every layer of order provides the phenomenal and lawful basis for its observation and perception. Foundational disorder at the deepest level drives the continuous transformation of order at every layer, while things are also constantly shaped by interactions both within and beyond themselves. Under the joint action of internal and external causes, everything is necessarily caught in perpetual change.

From the deepest layer of disorder up to higher levels of order, the process of change moves from randomness toward gradual determination. To attempt to understand a thing completely, therefore, would amount to expecting our brain to grasp all its layers of ordered structure and all their dynamic transformations. This would be no different from reconstructing inside our brain a duplicate entirely identical to the thing itself. This is not only practically unreachable; it also constitutes, in logic, a self-referential closed

loop.

The nature of things as ordered existence also means that we can perceive them only through interaction. This establishes the basic paradigm of human cognition: we cannot grasp the thing-in-itself directly, but can approach its nature only indirectly, through the phenomena produced in our interaction with it.

Thus we must accept a fundamental regret. Absolutely objective understanding is, in theory, a shore we can never reach.

True rational self-awareness begins with a deep, honest acknowledgment of this cognitive boundary. The essence of objectivity lies precisely in soberly understanding the inherent limitations of one's own perspective. It lies in understanding the dialectical unity of the knowable and the unknowable. On this premise, by sustaining the courage to explore and humility in cognition, all meaningful cognition and practice can then unfold. This is the first step toward cognitive maturity.

Existence is like dew, like lightning; only interaction endures. We can never observe "true randomness," because in conceptual terms it corresponds to pure non-being. An ordered thing possesses no intrinsic properties in isolation. Only in the instant of interaction does its character as a "difference" arise against a relatively disordered background and become perceptible.

This is exactly what we have already seen. The photoreceptor cells in the retina do not sense absolute light. They sense differences in specific wavelengths or intensities. The neuronal networks in the brain do not respond to all information indiscriminately. They selectively resonate with ordered features that match their own connection patterns. At the same time, they ignore those random parts

that have not been encoded.

The entire cognitive system operates in this way. From the torrent of perception, it analyzes and breaks down various features—local, fragmentary pieces of ordered information. Through the transmission and convergence of neuronal pathways, these pieces are ultimately woven into our cognitive picture of the thing.

In most cases, therefore, when we call a thing or phenomenon random, what we mean more precisely is this: at the observational level we have currently chosen, its pattern of change exceeds our capacity, at that level, to extract features and summarize regularities.

Someone might object. We can plainly, intuitively sense random phenomena such as Brownian motion. We can even use computers to generate complex random numbers.

Yet at its core, whether through direct sensory perception or through indirect instrumental observation, what we process at every perceptual instant is still the ordered information produced by interaction. It is precisely by using such ordered information—the particle's position, brightness, and contour at successive instants—and then applying higher-level logical induction that we characterize its overall pattern of motion as random.

Consider Brownian motion. We describe the path traced over time by the irregular macroscopic motion of a suspended particle as random. Yet at every instant of observation, what enables us to confirm the particle's existence is precisely its relatively stable, ordered features at that moment: its shape, reflected light, or outline. If our observational technology could go further and track the mo-

mentum of every molecule striking the particle, then in principle its trajectory at the next moment would cease to be random and become predictable.

At the deepest level, the five senses through which we peer at the world operate through electromagnetic interaction and changes in the microscopic particles of our sensory organs. Our cognitive picture of things therefore bears the indelible mark of the observer and is constrained by the inherent physical limits of our sensory channels.

Our senses are not a holographic mirror faithfully reflecting reality. They are more like a set of filters, each with distinct properties. Passively and selectively, they receive an extremely narrow band of the information in the outside world. Through this, they define the boundary of the reality we perceive.

The world we see is first defined by the narrow window of vision. Humans can perceive only the narrow band of visible light, roughly 3.84×10^{14} to 7.69×10^{14} hertz. Snakes can see infrared thermal radiation emitted by living organisms, while hummingbirds soar through landscapes painted in ultraviolet. These differences directly shape each creature's world of color. Humans, with three types of cone photopigments, construct roughly ten million shades of color. Butterflies, with five basic color receptors, roam through an ocean of ten billion colors. Snakes live in a world that is, for practical purposes, a silent realm of black and white.

The boundary of hearing is the same story. The myriad tones the human ear can hear are confined to the acoustic range between 20 and 20,000 hertz. Bats, by contrast, command ultrasonic frequencies above 100,000 hertz. They parse 250 sets of echoes per sec-

ond for spatial positioning. We hear the wind and the birdsong. We cannot hear the infrasonic communication of elephants. Nor can we capture the ultrasonic navigation of bats. The silence we define may, in another auditory dimension, be roaringly noisy.

The chemical world of smell and taste is, even more directly, a map of receptor numbers and receptor types. Humans, with roughly 10 million olfactory cells and 400 types of receptor, can distinguish over 10,000 odors. A dog's sense of smell opens a perceptual universe of an entirely different dimension. Its sensitivity to certain odors exceeds a human's by ten thousand times. We rely on our vestibular organs to sense orientation. Migratory birds can feel the faint lines of Earth's magnetic field, drawing inner navigation maps as they cross continents. We taste the sweet, sour, bitter, and savory of food. Yet this experience is strictly limited by the finite types of receptors on our taste buds. We cannot exhaust all possible combinations of chemical information in the material world.

Even touch, which conveys the most direct sense of texture, does not transmit absolute temperature readings. What it conveys is the relative sensation produced by the rate of heat conduction. For example, if you place one hand in ice water and the other in warm water, then immerse both simultaneously in a basin of room-temperature water, your brain receives two completely contradictory signals about the same basin: cold from one hand and warm from the other.

All of this confirms that our senses, through electromagnetic interaction and other underlying mechanisms, "translate" objective existence into subjective experience. Everything we understand as order and regularity is built upon this deeply edited translation.

It is as if we wear a pair of tinted glasses to look at the world. We obtain a partial perception. Yet we mistakenly take it for the world's essence. As long as we understand through interaction, we can never transcend this sensory shell.

Even the extension provided by scientific instruments does not fundamentally erase this brand. It merely gives us a more powerful sensory filter, one that still has specific parameters, allowing us to continue constructing and revising the model we call reality across broader levels.

Our cognition of things is also subject to the internal influence of individual emotions and value systems. On top of the raw signals provided by the senses, our brain is not a coolly objective information processor.

What the genetic blueprint grants us is only a foundational and plastic structure, an initial pattern. The widely diverging developmental trajectories, growth environments, and life experiences of different individuals jointly sculpt a brain that is unique. This uniqueness manifests at every level: in the microscopic chemical environment, the macroscopic topology of the neural network, the fine-tuned regulation of synaptic connections, and even the eventual patterns of value judgment and emotional response. No two are alike; none can be duplicated.

Every individual is continuously interacting with the outside world. Every new experience, every instance of learning and memory, makes the neural network increasingly complex and specialized. This corresponds to the individual's cognitive framework becoming deeper and more structured.

On this dynamically constructed and highly specialized physio-

logical foundation, the fine structure of the brain that carries each individual's mind is unique in the world. This fundamentally determines that, even when faced with the same external stimulus, different brains will necessarily interpret and construct different realities and meanings.

In long-term practice, individuals develop preferences for responding to specific frequency bands within the law spectrum of the cosmos. Our cognitive systems build up recognition patterns for order at particular spatiotemporal scales. Some people excel at perceiving the quantum behavior of microscopic particles. Others are adept at grasping the operating logic of macroscopic society. Some are skilled in the art of trade-offs in engineering and technology. Others are sensitive to the deep veins of cultural phenomena.

These cognitive inclinations are jointly shaped by innate endowments and acquired experience.

Yet an individual's characteristic response band in the law spectrum, while bringing local insight, also carries a risk. It may cause us unconsciously to use familiar theoretical paradigms to force an interpretation upon phenomena that do not belong within their proper domain of application. When a physicist tries to predict stock-market fluctuations with deterministic equations, or when a sociologist uses cultural constructivism to deconstruct natural laws entirely, the result may no longer be objective understanding, but distortion and deviation caused by cognitive mismatch.

For the individual brain, one key to efficient operation is its reliance on the vast number of reentrant connections spread across all brain regions for rapid information integration, rather than on massive quantities of precise computation.

This structure gives rise to complex emergent phenomena. Perception, memory, and emotion are not independent stages, but fuse instantaneously through repeated mutual feedback into an integrated cognitive experience.

Emotions such as greed, fear, anger, excitement, anxiety, and depression, together with the unfathomable subconscious, jointly weave a filter that overlays cognition. Through the release of neurotransmitters such as dopamine, serotonin, and norepinephrine, this complex system of value judgment creates an ever-shifting chemical microenvironment within the brain. It directly stains perception with subjective color, shaping what we approach or avoid and what we amplify or ignore.

This internal chemical theater does not merely shape the way we react to the outside world. It also deeply influences how we receive and parse the law spectrum.

When the emotional filter amplifies the signal of one frequency band, the patterns of other frequency bands may be weakened or even distorted in cognition. Anxiety may cause us to over-focus on high-frequency social risks while neglecting the underlying logic. Excitement may make us intoxicated by the symmetrical beauty of microscopic order while ignoring macroscopic constraints.

Thus every person not only faces a physical narrowing of frequency bands at the sensory level, but also undergoes dynamic "cognitive tuning" at the brain's chemical level. This intrinsic preference, driven by neurotransmitters, is interwoven with acquired paradigms of thought. Together they determine which forms of order in the law spectrum we can resonate with and which signals we remain deaf to.

For a consciousness embedded in a body of flesh and blood, our knowledge of the outside world is forever bound within the individual's unique network of neural connections and imprisoned within a highly personal "framework of self-understanding." That framework distorts the weight assigned to information and tampers with the logic by which it is interpreted. It therefore becomes extraordinarily difficult to rise above one's own emotions and values and touch a purely "objective" world.

We do not merely wear the tinted glasses of the senses to look at the world. We also carry an inner projector built from our own emotions and value judgments. It ceaselessly casts internal waves onto the external canvas. Yet we frequently mistake this projection for the essence of the world.

If our view of the world always begins from "my" gain and loss, "my" preferences, and "my" existence, then the greed, anger, ignorance, arrogance, and doubt rooted deep in human nature, or the response band peculiar to each person within the law spectrum, will quietly take center stage in cognition.

Recognizing this allows us to feel more deeply the difficulty of objective understanding. It demands more than merely transcending sensory limitations. It demands that we constantly turn back and observe that brain of ours, a brain steeped in a sea of chemicals, a brain continually reshaped by emotion.

Even so, we still yearn to draw near to the truth of the objective world.

In terms of cognitive tools, we continue to expand the boundaries of the senses through scientific instruments. We aim microscopes at the microscopic realm. We point telescopes toward the

universe. We use particle colliders to recreate moments reminiscent of creation. We apply gene-sequencing technology to decode the ciphers of life. These extended sensory filters do not fundamentally erase the observer's brand. Yet they allow us, across broader layers of order, to construct increasingly refined models of reality.

On the other hand, what is more intrinsic and more decisive is to achieve a profound leap in cognitive perspective. We are fortunate. Though not entirely free, the brain's plasticity and its foundational disorder still grant consciousness enough space to soar.

We cannot control the underlying processes of perception, but we can guide our senses toward rich and varied observation, gathering samples of information like well-trained prospectors. We cannot determine the momentary surge of neurotransmitters, but through training we can direct our thought, allowing neural networks to undertake meaningful "disordered" exploration within particular fields and thereby bring forth new connections of thought "from nothing." Through the cultivation of reason, we can also moderate the force of desire and emotion, preventing greed, anger, ignorance, arrogance, and doubt from distorting reality too severely.

Our brain cannot flee the body. Yet the perspective of consciousness can try to transcend the limitations of the "I." It can be like surveying the ground from high above, placing the self itself within the observed landscape.

This perspective from above the self, also called a God's-eye view, enables us to see the real relations between the individual's ordered structure and the other ordered structures of the outside world. It reveals universal connections and objective patterns oth-

erwise obscured by “my” gains and losses, likes and dislikes.

When we juxtapose this transcendent perspective with the deeply entangled self-perspective and compare them, the internal mechanisms of interference become clearly identifiable, including how greed distorts judgment, how anger ignites prejudice, how ignorance solidifies thought, how arrogance raises walls, and how doubt obstructs acceptance.

“Let the mind arise without abiding anywhere.” It is precisely through this continuous comparison and reflection that we can transcend self-centeredness, habitually activate that transcendent perspective, and awaken to reality as it is. This continuous observation of the self’s own cognitive process is what Buddhism calls Vipassana, or inward observation.

Using this transcendent perspective is the necessary path for adapting to the multi-layered ordered nature of things. We must learn to understand through a multi-layered, three-dimensional perspective and keep that understanding dynamically updated.

If we indulge only in high-level trends, we easily see the forest but miss the trees, allowing microscopic opportunities to slip away. If we become excessively entangled in foundational details, we easily see the trees but miss the forest and lose our macroscopic direction. The genuine art of cognition lies in maintaining dynamic balance and switching keenly among multiple layers, learning to direct emotion and desire toward suitable matters and tuning them to the layer of order required by the cognitive task.

Take the “investigation of bamboo” once more. Do we understand its life through the morphological layers of leaf, stem, and root? Or do we go deeper, into the cellular level, to analyze its

microscopic structure? Or descend further still, to the scale of molecules and atoms, to seek its material foundation? Do we focus on its seasonal growth and withering? Or center our attention on the energy conversion of photosynthesis?

Every layer of observation reveals ordered regularities of a different dimension. True wisdom in the investigation of things lies in the ability to move flexibly among multiple unfolding layers according to the purpose of cognition, and to focus precisely on the layer of order best suited to answering the question at hand.

"My life has a limit, but knowledge has none." Though humans can never completely and clearly understand the objective world, understanding the inherent limits of cognition and actively stepping out of the self-centered perspective still opens infinite possibility within finitude.

By guiding desire rather than becoming its slave, and by transcending our perspective rather than seeking to possess the infinite, we can experience deeper freedom and more genuine happiness within the limits of cognition. This may be the ultimate direction of wisdom: through acknowledging our limits, to achieve the perpetual elevation of cognition and existence.

Chapter 24

Thought Without End

The truth of a theory is in your mind, not in your eyes.

Albert Einstein

An individual's cognition of the world is constrained by sensory channels, emotional filters, and personal cognitive frameworks. Yet through exchanging ideas with one another, individual cognition undergoes mutual correction and fusion. Gradually, a shared cognition of the human collective takes shape. This collective cognition not only compensates for the limits of individual perspective; it also significantly enhances humanity's overall ability to adapt to the environment and resist entropy increase. Language, in spoken and written form, is the key medium through which this cognitive coordination is achieved.

As linguist Noam Chomsky has pointed out, about eighty thousand years ago, the human brain underwent a critical evolutionary leap. The emergence of an internal language faculty ignited the cognitive revolution.

This leap was, at its core, an ordered upgrade in the patterns of neuronal connections. It allowed individual cognition to be mutu-

ally corrected and fused through communication, and it catalyzed the birth of self-consciousness and complex thought. Language and writing, as vehicles of thought, solidified fluid mental processes into transmissible symbolic order, achieving the leap from individual experience to collective wisdom.

This process not only strengthened emotional bonds and social cooperation; it also gave rise to the formation of higher-level ordered relationships, such as morality, law, and contract. It pushed human civilization toward organized order.

How does our brain understand abstract concepts and ideas? As discussed earlier, when sensory organs receive external information, they do not reproduce it exactly. From the very beginning of the neural pathways, information is analyzed and broken down into features such as shape, color, and rhythm. These features are, at their core, differences—that is, ordered information. Specific neurons respond only to specific features, which are transmitted and brought together within the brain until they are integrated into a cognition of the thing.

For example, when we see a red ball, the neural pathways corresponding respectively to "one," "red," and "ball-shaped" are activated. These features are rapidly associated through reciprocal neural connections, forming the cognition of "one red ball."

However, if only the neural pathway for "ball" is activated, the brain obtains merely the abstract concept of a ball. This isolated concept does not point to any concrete entity.

The concept "ball," taken by itself, has an extremely broad boundary of meaning. Balls of every color, material, and function fall within its scope of reference.

If we view a concept as a "meaning space," then the isolated concept "ball" exhibits peculiar "quantum" properties within that space, much like an elementary particle. It simultaneously carries the potential meanings of a red ball, a white ball, a soccer ball, a basketball, and even balls we have never seen. These possibilities exist as though in a "quantum superposition." Before being observed—that is, before being constrained by other concepts—they coexist within this concept.

A single concept therefore carries a high degree of semantic indeterminacy; it is necessarily abstract. When several concepts are activated at once, they constrain one another and form an ordered structure, much as "quantum decoherence" turns possibility into a more determinate state. Indeterminacy falls, determinacy rises, and the result becomes more concrete.

For example, compound concepts like red ball, white ball, soccer ball, or basketball already have a meaning space far narrower than ball alone. When the three concepts one, red, and ball are activated together, the previously diffuse possibilities of meaning converge into relatively specific and definite thought content.

The preceding analogy is not mere fantasy. Everything that exists must possess some form of order; knowledge, thought, and cognition can be no exception.

Kant pointed out in the *Critique of Pure Reason* that knowledge is a unified whole with order, hierarchy, and constituent elements. Hegel regarded the "absolute concept" as an aggregate of processes. Human thought, as a reflection of the objective world, is likewise an ordered system of concepts.

External things are multi-layered order formed by arranging un-

derlying disorder. The progression goes from lower to higher levels, from abstract to concrete. The set of ordered features contained within this multi-layered order forms the feature label of the thing itself.

Through multi-scale and multi-angle observation, our senses continuously extract feature information from the external layered structure. These features are, at their core, differences. The neural pathways in the brain representing individual features or concepts are then activated in sequence. These features and concepts, through mutual constraint and association, form increasingly definite meanings. Thus cognition progresses from abstract to concrete.

Therefore, the essence of cognition is precisely the dynamic mapping of the objectively existing "multi-layered order" of things onto the ordered activation of neural pathways within the brain.

Although Paul MacLean's triune brain model is a simplified metaphor, this heuristic evolutionary model is particularly useful for describing the evolutionary trend of the brain. As discussed earlier, the instinctive brain, the emotional brain, and the intelligent brain can be seen in terms of two key instances of symmetry breaking in the brain's evolutionary history. Each symmetry breaking realized a leap in connection patterns, bringing forth a higher-level cognitive order, much like sudden insight.

Most functions of the instinctive brain correspond to the direct perception of features in external things, constructing the most basic set of concepts for understanding the world. On top of this, the emotional brain constructs more complex patterns of neural connection. Upon the foundation of the basic concepts provided by the instinctive brain, it integrates diverse information, including internal

physiological states and relevant environments. From this emerge rich emotional experiences such as fear, sadness, and joy. In this sense, emotions are a higher-level "ordered structure" that appears after basic concepts have been integrated through specific neural circuits. They enrich the content of cognition and become the inner drive of social connection.

The core breakthrough of the intelligent brain lies in acquiring the ability to reflect upon, manipulate, and recombine concepts themselves. This gave humanity complex consciousness, including language, logic, and reasoning.

More concepts such as "freedom" and "justice" emerged. Concepts were no longer merely products of passive perception. We could arrange conceptual units and engage in deep speculative thought. The intelligent brain enabled humanity to construct a world of thought far beyond direct sensory experience, thereby truly achieving the leap from "perceptual order" to "creative order."

Thus, the instinctive brain, emotional brain, and intelligent brain are not functional divisions. They are different levels through which consciousness emerges as an integrated function. Our consciousness is a layered structure of order formed by the arrangement and combination of concepts. The generation of wondrous ideas often relies on the collaborative work of all three.

Personal cognition is an isolated experience. It perishes with the end of life. The emergence of language and writing enabled personal cognition to achieve cognitive coordination across time and space.

So how does language perform this function? The answer lies

in language itself being a multi-layered ordered structure. The Chinese language, for example, is a typical case in which microscopic order is progressively arranged and combined until complex concepts emerge.

Pictographic and single-component Chinese characters are fixed combinations of strokes formed by highly abstracting the most direct core features of things: the roundness of the sun, the outline of a mountain, the flow of water.[1] This is the first step from features to basic concepts, as in *ri* (sun), *yue* (moon), *shan* (mountain), *shui* (water), *ren* (person), and *mu* (wood). Each pictograph or single-component character is an abstract conceptual unit that provides the foundation for higher-level order.

At the same time, the common features of pictographs and single-component characters were standardized into only about twenty to thirty types of basic and derived strokes: dot, horizontal, vertical, left-falling, right-falling, turning, and others. These strokes themselves are meaningless lines, yet they are the "elementary particles" that construct the meaning space of Chinese.

In the traditional classification of character formation, simple indicatives add indicative marks to pictographic forms or use symbols directly to express abstract concepts. For example, *shang* (above) places a dot above a line; *xia* (below) places a dot below a line; and *ben* (root) places a dot at the base of *mu* (tree).

[1]The traditional "Six Writings" classify Chinese characters as pictographs, simple indicatives, associative compounds, phonetic-semantic compounds, phonetic loans, and mutually explanatory forms. This chapter concentrates on the four categories most directly associated with character formation.

Associative compounds combine two or more single-component characters to produce a new concept. For example, *xiu* (rest) depicts a person leaning beside a tree; *ming* (bright) joins sun and moon in shared brightness; and *sen* (forest) combines three trees to suggest many trees.

Phonetic-semantic compounds combine a semantic component, indicating the category or scope of meaning, with a phonetic component, indicating pronunciation. This is the mainstream method of Chinese character formation. Thus, phonetic-semantic compounds are no longer limited to using imagistic features to express concepts. They also lay the foundation for describing a world of thought that transcends direct sensory experience. For example, the character *hu* (lake) uses the water radical as its semantic component and another *hu* as its phonetic component; *zhong* (loyalty) uses the heart radical as its semantic component and another *zhong* as its phonetic component.

Words and phrases achieve the precision and diversification of concepts through conceptual combination, moving from abstract to concrete and from basic perception to complex relationships. Sentences achieve the leap from static concepts to dynamic thought. A sentence can describe a scene, state a fact, or pose a question, completing a full mapping of a "cognitive situation." Multiple sentences build a grander ordered structure in paragraphs and whole texts. At this level, language begins to carry reasoning, argumentation, narrative, and emotion. It can describe an idea, as this article does, tell a moving story, or formulate a complex law.

Of course, as Hegel noted, profound ideas such as freedom, justice, and entropy increase are themselves vast "conceptual

systems." They must be understood and shared within a common meaning space through the mutual association, definition, and elucidation of numerous sentences. Through humanity's cognitive exchange and consensus, these complex concepts are then condensed into individual words: "freedom," "justice," and "entropy increase."

The above interpretation of the construction of Chinese—and indeed of all languages, mathematics, and other scientific symbol systems—perfectly demonstrates how orderedness emerges level by level from low to high, or undergoes symmetry breaking. It is evident that this concept-based, multi-layered ordered structure of language closely resembles the process by which our brain's neural network understands things. Each level uses the order of the level beneath it as material and, through specific rules, creates new order with greater informational determinacy.

This is the underlying mechanism by which language and writing realize cognitive coordination among human individuals: they provide a standardized set of "ordered generation rules." When we use the same language, we are essentially sharing the same "encoding-decoding protocol," from basic features to complex thought. This allows the personal, internal activation patterns of neural pathways to be encoded—written or spoken—into external, public sequences of linguistic symbols. These are then decoded—read or heard—by others into similar internal patterns of neural activation. Cognitive mutual correction, fusion, and transmission are thus achieved.

Through continuous observation, we realize the mapping from the order of things to the order of neural networks. Likewise,

through the multi-level arrangement of concepts, we realize the mapping between neural network order and linguistic order, or from linguistic order to neural network order.

This is the deeper meaning of Wittgenstein's view of the limits of language: the boundary of language is also the boundary of thought. Only when a problem can be described in language can it be understood and solved.

Through language, the neural network of the individual brain continually learns new concepts and forms increasingly complex patterns of connection. New thoughts, in turn, prompt language to enrich and refine itself, extending its range of meaning. The two continually coordinate with one another, forming ever more precise correspondences.

Early artificial intelligence models relied heavily on manually labeled samples for supervised learning, which greatly constrained their scale and range of application. The success of today's large language models (LLMs) rests on their adoption of self-supervised learning. Through a simple predictive task such as next-word prediction, a model can autonomously discover grammatical, semantic, and even some logical patterns in vast quantities of unlabeled text. This freed the field from dependence on human-labeled data and became a central driving force behind its leap forward.

This successful paradigm reveals a profound principle: intelligence can emerge autonomously through interaction with an environment rich in ordered structure—text for LLMs, the physical world for embodied intelligence—without depending on human-provided labels. We may therefore reasonably predict that the foremost challenge in achieving general embodied intelligence is to

enable agents, through feedback from their interactions with the physical world, to perceive, abstract, and learn the multi-layered ordered structure of things without supervision or human labels.

Measure, probability, field, quantum, quantum entanglement, wave function, singularity, entropy—these concepts, which previously did not exist or could not be described in language, are gradually being clearly defined. They represent an expansion of the boundaries of our understanding of the world.

Concepts that cannot be expressed within the limits of present language only reveal the present vagueness and finitude of humanity's understanding of nature. What is called "the severing of the path of words" more often reflects the helplessness of a particular historical moment. Both as individuals and as a society, we should be brave enough to admit that what we cannot express, we have not yet understood fully.

What is true of language and writing is also true of music, painting, dance, drama, and every other form of art. They too are multi-layered ordered systems built upon fundamental symbolic elements, and they likewise seek to encode, transmit, and coordinate inner experience.

The slight difference is that the fundamental symbolic elements in the arts, such as musical notes, colors, lines, and bodily movements, carry far less determinate meaning than words. They are closer to a "pre-linguistic" disorder rich in potentiality. A single note—do—or a patch of red pigment has a far more ambiguous and broader scope of reference than the word "ball." It is precisely this low initial determinacy of meaning that provides artistic creation with a nearly infinite space of combinatorial meaning, far greater

than that of everyday language. Through it, art can break free from the rigid constraints of logical grammar, compensate for the structural biases and conceptual rigidity produced by the highly structured nature of language, and speak of those realms that cannot be spoken of.

At the level of information processing, art exhibits neural dynamics entirely different from language and writing. Textual symbols must first be decoded by the visual or auditory cortex, converted into abstract concepts, and then handed to the prefrontal cortex for grammatical parsing and logical construction. Only then may they descend to the limbic system and stir emotion. By contrast, the colors and composition of painting, the melody and harmony of music, and the rhythm and posture of dance act directly upon the sensory system through their physical properties. Via relay stations such as the thalamus, they preferentially and in parallel activate emotional centers such as the amygdala and hippocampus. We can say that art often bypasses reason's fine-grained scrutiny and reaches directly into the ancient realms of emotion.

This brings us to a deeper question: where does art's power to reach us so directly come from?

A symmetric, elegant mathematical formula evokes intellectual beauty; the hexagonal symmetry of a snowflake evokes the beauty of nature; a melody rises and falls through exquisite pitch, rhythm, and harmony; a painting brings together harmonious color and line with the balance of fullness and empty space. These forms can drive millions of neurons in the brain into synchronized firing and directly stir powerful emotional resonance, both within an individual and between individuals, without requiring conceptual transla-

tion. This resonance, beyond words and almost physiological, lets us experience an effortless delight that needs no deliberation, even a shudder and elevation from the depths of the soul.

Beauty may be the sudden apprehension of an ordered structure that is highly condensed, economical in meaning, and perfectly unified in form and content. It need not await the slow scrutiny of reason, because order itself is existence's most authentic language. When external order resonates directly with the order in our neural networks, we briefly transcend the prison of concepts and glimpse the deepest harmony of the cosmos.

If language is one wing of the vessel of human cognition, pointing toward rational, shareable, lucid thought, then art is the other, reaching into the realm of feeling and into emotions that are private yet mutually intelligible. Beauty is the light released when these two wings meet at their highest point: both the culmination of thought and the summit of feeling, the final reconciliation of cognition and existence in the resonance of order.

All things, whether matter or consciousness, reality or thought, follow the same underlying logic. Disorder creates possibility. Order determines direction. The broad semantic space of foundational concepts opens possibility, while the process of ordering sets a definite direction for meaning. Beauty is the most intuitive and irrefutable evidence of this logic. The layered arrangement and recombination of all things has no end; neither do thought and cognition. In this limitless exploration, every sudden insight into beauty is a moment in which we touch eternity.

Chapter 25

The Way Order Takes Shape

Generation upon generation is what is meant by Change.

The Book of Changes

Having followed the argument to this point, we have moved from elementary particles to the rotation of galaxies, from the metabolism of life to the emergence of consciousness, and from language and writing to social organization. At first sight, these seem to be topics far removed from one another. On closer thought, they have always circled around the same question. Why is the world not a mass of chaos, and why is it never truly still?

If we only say that the world is ordered, it is easy to mistake order for a dead arrangement, as though all things had merely been placed in fixed positions by some external force. If we only say that the world changes, it is easy to mistake change for directionless dispersal, as though every act of becoming were only an accidental collision. The real world lies in neither extreme. It is neither frozen order nor pure chaos, but continually generates itself through the interaction between disorder and order.

Disorder is not merely mess. Seen more deeply, disorder is the

state of change before it has been fully delimited. It means that possibility has not yet been exhausted, direction has not yet been locked, and structure has not yet fully solidified. Because disorder exists, the world does not become a dead crystal. Because change keeps taking place, new structures, new relations, new lives, and new thoughts can be born from the cracks of old order.

Of course, the disorder we are speaking of has two levels. Absolute disorder is the "nothing" that is defined by no relation and cannot enter the world of experience. Relative disorder is the change and possibility that, at a given level, have not yet been fully delimited by the existing structure.

Order also does not cancel change. True order does not eliminate change, but provides change with boundary, relation, and direction. A river rushing forward without a channel is only floodwater. Active thought without concepts and language can hardly form clear judgment. Metabolism without cellular structure and genetic mechanisms cannot sustain itself. Order does not suppress change. It allows change to take form.

Therefore disorder and order are not two mutually excluding things. Disorder provides possibility. Order determines direction. Without disorder, the world has no new birth. Without order, the world has no form. Everything we see as existence is a temporary result of these two forces interacting at different levels.

If we use the simplest expression to indicate this picture, we may write:

$$\frac{d\Omega_i}{dt} = \Gamma_i(\Omega_i)\, B_i(\Omega_i) + \eta_i(t, \Omega_{i-1})\,.$$

The quantities in this expression need not have the same dimensions at the physical, biological, neural, and social levels. The expression captures a shared structural relation: existing order delimits the direction of change, lower-level disturbances continually provide new possibilities, and at each concrete level these quantities must be defined in terms of its own objects and laws.

This expression is not meant to compress the world into a handful of symbols. Quite the opposite, it merely reminds us that at every level, what we call order does not sit there motionless. Within change, it sustains itself, corrects itself, and generates new directions.

Here, the degree of order is the degree to which a structure has formed and can sustain itself at a given level. In a physical system, it may appear as an order parameter; in a living system, as the stability of organization and metabolism; in the brain, as the capacity of neural networks to form cognitive patterns; and in society, as the maturity of institutions, culture, and structures of cooperation. The objects differ, and so do the languages used to describe them, but the generative movement is the same: lower levels continually supply disturbance, the level in question forms boundaries, and higher levels disclose new laws at new scales.

The first term on the right side of the formula indicates how existing structure delimits the direction of change. Once a level has formed a certain order, it no longer allows every possibility to unfold freely. Atoms have structural boundaries; molecules have modes of bonding; cells have metabolic networks; the brain has neural pathways; society has institutions and culture. Changes compatible with the structure of a level are absorbed, amplified, and

fixed. Changes that cannot close, stabilize, or continue to be transmitted are excluded.

A boundary is not only an external limitation. It is also a condition for taking form. Without banks, flowing water cannot be called a river. Without a cell membrane, molecular reactions cannot become life. Without grammar, words cannot stably express thought. Boundaries compress infinitely many possibilities into a finite range, so that structures can be recognized, maintained, and transmitted. Often we think boundaries obstruct freedom. In fact, without boundaries, even freedom has nowhere to stand.

The second part on the right side points to disturbances from lower levels. Beneath every level, finer levels are in motion. Beneath macroscopic objects there is molecular thermal motion. Beneath life activity there are chemical reactions. Beneath consciousness there are neurons firing. Beneath social trends there are countless judgments and choices made by individuals. These lower-level disturbances look like noise, deviation, and accident, but they are precisely what keep injecting new possibility into higher-level structures.

These disturbances do not come from nowhere. Certain structures already formed at lower levels enter higher levels as memory, inertia, tendency, and condition. A thought seems to flash up suddenly, but behind it lie long experience and neural connections prepared over time. A sudden change in an industry seems to erupt by accident, but behind it lie accumulated technology, changing demand, and mounting organizational pressures. What we call disorder is often not the absence of relation, but relations not yet fully seen by the higher-level structure.

Thus, becoming ordered does not mean eliminating disorder. It means selecting from disorder the directions capable of stable existence. Disorder is not the enemy, but the source of change. Order is not a cage, but the course along which change takes form. The world can generate itself because possibilities keep welling up. It does not scatter apart because those possibilities are continually delimited.

This can also be seen in everyday experience. Water vapor cools and condenses into clouds. The clouds continue to change and may become rain. The formation of each raindrop is not the result of a single molecule acting alone. It is order taking shape through the joint action of temperature, pressure, humidity, air currents, and microscopic fluctuations. What we see is a shower of rain. Behind it are changes at countless levels reaching a temporary agreement at the same moment.

Life is the same. Accidental errors in genetic replication provide life with new possibilities. Environmental pressure and competition for survival keep selecting among those possibilities. Without mutation, life cannot innovate. Without selection, mutation cannot accumulate into stable structure. Life never advances along a completely determined route, nor does it drift in purely random chaos. Between possibility and delimitation, it forms more complex organization again and again.

Learning is the same. When a person encounters new knowledge, the brain does not mechanically file information away in a drawer. Within its existing neural networks, it continually activates, connects, compares, and reorganizes. Chance associations, mistaken attempts, and vague intuitions may seem insufficiently

rigorous, yet they open possibilities for understanding. Repeated practice, conceptual ordering, and logical correction then settle these possibilities into stable cognition. Inspiration comes from the leap of disorder. Ability comes from the accumulation of order.

Social organization cannot escape this process either. An organization that completely suppresses individual difference may appear neat and uniform in the short term, but will gradually lose the capacity for innovation. An organization without common rules will fall into internal friction and confusion. A truly vital organization must allow sufficiently rich possibilities to arise from below, while guiding those possibilities toward a common direction through goals, institutions, and culture. Freedom is not the absence of constraint. Constraint is not the cancellation of freedom. Good order allows freedom to form a shared force.

Seen in this way, existence also needs to be understood anew. Traditional intuition always assumes that a thing first exists and only then begins to change. From the perspective of becoming ordered, change may be more primordial, while existence is instead the result of change temporarily stabilizing at a certain stage. An atom is an atom because lower-level particles and interactions form a stable structure. Life is life because countless molecular reactions are organized by cell membranes, genetic information, and metabolic networks. A person becomes who they are at this moment because genes, experience, learning, emotion, and choice have shaped the ordered state of brain and body over a long period of time.

Existence is not an eternally motionless entity, but relative stability within change. We say a mountain exists because it is suffi-

ciently stable on the human timescale. We say a cloud exists because it holds a recognizable shape for a short while. We say a thought exists because a neural network forms a particular pattern of activity at a given instant. The timescales of stability differ, and so do the modes of existence. Mountains, clouds, and thoughts do not belong to the same level, yet each is change temporarily settled into order.

From this we can also understand that levels are not boxes drawn arbitrarily by human beings for the sake of convenience. Levels themselves are the way the world generates itself. Lower-level structures combine to form higher-level structures. Once a higher-level structure forms, it in turn constrains the possibilities of the lower level. Atoms compose molecules, but after molecules form, atoms no longer move at will like free atoms. Cells are composed of molecules, but after cells form, molecular reactions are incorporated into the metabolic network of life. Society is composed of individuals, but after social order forms, individual behavior is shaped by language, institutions, ethics, and culture.

This is what simple reductionism easily overlooks. Lower levels are of course important. Without lower-level material, higher-level structure cannot even be discussed. But the higher level is not the simple sum of the lower level. The flow of water cannot be understood by tracking every water molecule one by one. The meaning of life cannot be exhausted by listing all proteins. The experience of consciousness cannot be replaced by the change in electrical potential of a single neuron. A social trend cannot be directly derived from one person's desire. Once each level of order has formed, it has its own law spectrum.

The idea of the renormalization group in modern physics confirms this from another angle. When the scale of observation moves from the microscopic to the macroscopic, large amounts of detail are averaged, compressed, and discarded. What remains is often only a small number of effective parameters, together with the symmetries and conservation relations that can pass across scales. Nature allows us to understand in layers not because we are taking a shortcut, but because the world itself keeps discarding details and preserving structure as scale changes. Each level has its own effective language, and each level has its own real laws.

This is a profound tolerance built into nature. The world does not force us to solve the trajectory of every molecule before studying the flow of water, or require a physician to begin with quarks and gluons when diagnosing an illness. As scale changes, many details recede into the background and a few stable structures come forward. That is why physics, chemistry, biology, psychology, and the social sciences can each have a language of their own. They are not unrelated islands, but effective contours of the same world at different scales.

This makes the word fundamental no longer absolute. For biology, cells and molecules are already fundamental enough. For chemistry, atoms and electron orbitals are more fundamental. For particle physics, quarks, leptons, and gauge fields are more fundamental. For everyday life, a person's character, ability, situation, and choice are the fundamental facts we truly need to grasp when judging action. To speak of the fundamental apart from a concrete level is often only to mistake the laws of one frequency band for the whole world.

The way order takes shape does not deny the lower level. It rejects the mistake of taking the lower level for the whole. A truly multi-level perspective can look downward and see material and disturbance, while also looking upward and seeing structure and direction. Looking downward, we know that all higher-level order comes from lower-level foundations. Looking upward, we also know that once new order has formed, it produces new constraints, new meanings, and new laws.

In more specialized physical research, this thought also appears as the relation between spectrum and closure. A spectrum is not a list supplied from outside, but the possible order that a structure can stably retain under boundaries and conditions. Closure is not an artificial patch. It excludes possibilities that cannot be self-consistent, cannot stabilize, and cannot remain mutually compatible. What remains is not left because it has been favored, but because it has withstood the requirements of structural closure at the corresponding level.

Seen from this angle, a theory with real strength should not merely add explanations after phenomena. It should push selection as far upstream as possible. The stronger the upstream principle, the less room remains for arbitrary adjustment later. If, once a structure is given, many elements become mutually connected and locked together by closure requirements, then what it presents is not merely a computational technique, but a way for order to be generated. It lets us see that complexity does not necessarily mean arbitrariness. Complexity may also be the necessary form taken by a simple upstream principle as it unfolds across multiple levels.

In cosmic evolution, becoming ordered often appears as sym-

metry breaking. The closer the world is to its primordial state, the more highly symmetric it tends to be, and the fewer differences it contains. Without difference, there is no structure. Without structure, there is no existence we can recognize. As symmetry gradually breaks, the fundamental interactions differentiate, particles acquire different properties, and matter further forms atoms, stars, and galaxies. Breaking sounds like loss, but it is in fact generation. Overly perfect symmetry approaches an undifferentiated emptiness. Only after breaking does the world have direction, level, and form.

Contradiction should also be understood within this picture. Every process of becoming ordered is accompanied by new disorder. Life maintains its own order by releasing waste heat into the environment. An enterprise creating a new product often breaks old procedures. A person forming a new cognition must pass through the loosening and confusion of old ideas. Without contradiction, there is no change. Without change, there is no life. The real question is not how to fantasize that contradiction can disappear completely, but how to identify the level on which contradiction lies, find the main direction, and let contradiction become a force of becoming ordered rather than a vortex of dissipation.

This is also the difficulty of seeking truth from facts. Facts are not isolated things placed before our eyes. Facts are always within concrete levels, concrete relations, and concrete changes. The same matter may present different meanings at different levels. A child's declining grades, seen at the bodily level, may be related to sleep, hormones, and neurotransmitters. Seen at the psychological level, they may be related to stress, interest, and self-evaluation. Seen at the family level, they may be related to patterns of communication

and structures of expectation. Seen at the social level, they may be related to competition and the education system. If we cling to only one level, we easily mistake partial truth for the whole truth.

To seek what is true is not to seize a ready-made answer and refuse to let go, but to correct oneself continually among multiple levels. One must know where lower-level disturbances come from, and also how higher-level structures delimit them. One must observe present change, and also discern long-term trends. One must acknowledge accident, and also look for necessity. The difficulty of seeking truth from facts lies precisely in the word "seek." To seek means that facts are not given once and for all, but gradually appear through observation, practice, feedback, and correction.

This is why simple slogans are often dangerous. To say that everything is determined by the bottom level easily blinds us to the real action of higher-level structure. To say that everything depends on subjective choice easily makes us forget lower-level conditions and objective boundaries. True judgment must always travel back and forth among levels. We should neither raise a local law into the whole truth, nor give up clear analysis because the whole is complex. To see a path and not take it is not to abandon paths, but to keep any existing path from covering the facts themselves.

Having come this far, we are no longer only discussing the generation of one particular thing. We are trying to place matter, life, consciousness, language, and society within the same multi-level picture. The formula is only a finger pointing to the moon. It does not replace the world itself. It reminds us that no level of existence is an isolated entity; each is a temporary stability formed between lower-level disturbance and closure at its own level.

Following this line further, many ancient philosophical questions reappear. Which is more fundamental, change or existence? If existence is the result of change temporarily becoming ordered, then existence is no longer a primary static thing, but the stable trace left by change. What we call existence does not reject change. It maintains a recognizable structure within change.

Are disorder and order absolutely opposed? If disorder provides possibility and order determines direction, then they are not enemies that destroy one another, but two faces of the same generative process. If disorder vanished completely, change would dry up. If order vanished completely, the world would lose form. True generation always unfolds between the two.

How do individual and whole define one another? If each level of order both comes from the lower level and turns back to constrain the lower level, then the individual is never an absolutely isolated individual, and the whole is never a mere sum. The individual gains position within the whole, and the whole continually renews itself through the differences and actions of individuals.

How are accident and necessity unified within the same process? Lower-level disturbance brings accident, while higher-level structure selects direction. Thus necessity does not exclude accident. It gradually forms stable trends among countless accidents. A seemingly accidental deviation, if absorbed and amplified by structure, may become a new road. A seemingly solid necessity, if it loses lower-level support, will slowly loosen.

Why can contradiction not be eliminated? Because every act of becoming ordered is accompanied by dissipation, and every new structure must pass through the loosening of the old structure. Con-

tradiction is not a flaw in the operation of the world, but the inner force by which change continues. A system with no tension at all does not necessarily mean perfection. It may also mean stagnation.

These questions do not belong to any single discipline. They appear in physics, in the evolution of life, in brain cognition, and in social practice. The more deeply we inquire into the concrete mechanisms of a complex world, the more we find that they ultimately point toward certain extremely simple judgments.

Chapter 26

From Utmost Complexity to Utmost Simplicity

The utmost complexity returns to the utmost simplicity.

Leonardo da Vinci

Throughout this book, we have tried to understand matter, consciousness, society, and even language and culture as forms of multi-layered ordered existence. We use "order" to name everything that can be known and spoken of, while treating absolute "disorder," which no interaction can define, as equivalent to "non-existence." Together, the two exhaust every possibility within the space of meaning.

Just as "the name that can be named is not the constant name," every effort to define order and disorder with complete clarity must encounter limits. They are themselves the two poles of existence and cannot be defined through other concepts. We can only apprehend them and point toward them through the countless forms in which they appear.

We have used many examples to portray the layered order of all things: from elementary particles, atoms, and molecules to cells,

organs, and living organisms; from neurons, neural pathways, and brain networks to individuals, families, companies, and nations; and from written strokes and words to syntax and whole texts. Each is an appearance of order that we observe at a different scale.

One point must be kept in view. Our division of things into layers may arise, at the deepest level, from the particular way in which we apprehend the world. Our standpoint of observation, patterns of thought, and tools of language together resolve the world into ordered levels that we can understand and speak about. This does not deny the objectivity of those levels. It reveals that the world as known cannot be separated from the subject who knows it.

In describing the tendencies and inner dynamics of ordered change, we have borrowed concepts such as entropy, contradiction, and symmetry. We know full well that this risks applying scientific and philosophical concepts across domains. Different things, and even different levels of the same thing, have laws of their own. As the saying goes, "The Way that can be spoken of is not the constant Way." Our purpose is not to transplant their technical definitions mechanically, but to distill the universal intellectual core they contain and use it to build a general model for understanding the world's complex transformations. These concepts and models are themselves also "names"; their value lies in helping us understand the "constant Way" that is difficult to express in words.

Consciousness is an especially complex domain. Contemporary neurobiology and cognitive science have revealed much about the workings of the brain and their relation to conscious experience, yet many mysteries remain. At this frontier between the known and the unknown, the book follows the underlying logic of multi-layered

order to offer its own bold interpretations and conjectures. We look forward to further scientific evidence, but even now we need as coherent a perspective and method as possible for understanding consciousness, intelligence, and related questions.

Natural science, the Confucian, Buddhist, and Daoist traditions of the East, classical Western philosophy, and Marxist philosophy look upon the world from different vantage points. Each has built a distinctive conceptual system for expressing the truth it perceives. Yet all are touching the same elephant. At the level of their deepest insights, their paths ought to meet.

Critique and innovation can certainly carry human understanding into new territories and across new scales. At the level of cognitive method and underlying logic, however, should we not pursue understanding and integration all the more? We should seek compatibility and consensus in the ways different disciplines and fields understand things, recognize what different perspectives share, and ponder what remains unchanged behind humanity's understanding of the world at different stages of history. After all, the more phenomena a body of thought can explain, the closer it comes to the underlying logic. Let West and East enter into conversation; let the ancient illuminate the modern. The most fundamental method of understanding should reflect the myriad forms of the world.

We use the metaphor of a law spectrum to describe the range from the most fundamental and universal laws of the cosmos to the highly specific and context-dependent rules at the levels of life, mind, and society. We also analyze the connections and differences among them.

If we connect knowledge at different scales and understand it

synthetically, we may summarize as follows the underlying logic common to every band of the law spectrum.

Universal connection. All existence takes the form of multilayered order. Lower-level structures interact to form structures at higher levels.

Perpetual change. All things are always changing, and every level of order has tendencies of change of its own.

Quantitative and qualitative change. Changes in things begin with changes in tendencies at the underlying level, and changes in lower-level tendencies form higher-level tendencies.

Chance and necessity. All change follows the principle that disorder creates possibility while order determines direction. The first impulse comes from random forces below. The lower the level, the more strongly randomness marks its tendencies; the higher the level, the more strongly direction is determined. As order rises from lower to higher levels, the symmetry of the corresponding laws gradually decreases.

Contradiction and the unity of opposites. Contradiction never disappears. Every ordered structure contains opposing forces: one drives it toward greater order, the other toward greater disorder. The first is entropy decrease and construction; the second, entropy increase and dissolution. Their opposition and unity constitute both the present state of that order and its tendency to change.

The negation of negation. A movement toward greater disorder creates the conditions for a new movement toward order, and the reverse is also true.

All things can therefore be understood as forms assumed by or-

der and disorder in their oppositional unity, and all change unfolds within this dialectical relation. The many concepts that appear to divide the world into opposing pairs are concrete expressions of order and disorder at different levels and within different relations. They are portraits of dialectical law in different bands of the law spectrum and in different forms of existence.

In understanding objective existence, we encounter being and non-being, non-existence and existence, disorder and order, micro and macro, minute and immense, simple and complex, abstract and concrete, inner and outer, form and emptiness, particle and wave, entity and relation, phenomenon and essence. At their core, these are all dialectical relations in which opposites form a unity.

In describing change, we encounter constancy and impermanence, stillness and motion, randomness and determinacy, entropy increase and entropy decrease, spear and shield, isolation and openness, internal and external causes, quantitative and qualitative change, symmetry and symmetry breaking, yin and yang, life and death, good and bad, fast and slow, causality and non-causality. At their core, these are all dialectical relations in which opposites form a unity.

In understanding matter, consciousness, and meaning, we encounter matter and consciousness, subjective and objective, virtual and real, freedom and constraint, reason and sensibility, name and reality, self and no-self, knowable and unknowable, meaning and meaninglessness, war and peace. At their core, these too are dialectical relations in which opposites form a unity.

Rationalism, empiricism, nominalism, realism, voluntarism, solipsism, consciousness-only theory, materialism, idealism,

spiritualism, historicism, scientism: we should not turn any school of thought into an exclusive doctrine and use it to fence off intellectual territory.

To break free of these exclusive habits of thought, avoid fixation on either-or, and seek instead a dialectical wisdom capable of inclusion is itself to put the underlying logic described above into practice. Only then can we move through cognition and action without anxiety or attachment, step beyond the cycle of binary opposition, and find release.

The way we see the world shapes our brains and our lives. The underlying logic set out here should therefore be understood not as a fixed body of conclusions, but as a dynamic perspective and method for understanding the world. Its purpose is to help us build, amid the profusion of appearances, a cognitive framework that seeks truth from facts and continues to evolve. At each present moment, this asks us to bring together analysis and synthesis across multiple levels: to disentangle the changing tendencies of order at each level of things, discern the directions in which they resist entropy increase, and act without forcing. It also asks us to use each new piece of information to correct our cognitive biases, so that we become less entangled with the past, less bewildered in the present, and less burdened by excessive expectations of the future. In this way, we can remain at ease amid the ceaselessly changing relations between humanity and nature, among people, and within ourselves.

Throughout life, each of us constructs an understanding of the external world through our own distinctive senses and cognitive framework. Apart from the inner self embedded in body and brain,

everything in the world is "external." This is why everyone is born alone and the soul always walks alone. Profound philosophical reflection often comes with solitude, not because it is aloof, but because along the cognitive path toward higher levels of order in things, it is difficult to encounter another self with whom one can resonate completely.

Against this background of profound solitude, love is the deepest form of interaction between individuals. It is not merely cognitive resonance, but the mutual embedding, reshaping, and coordination of two independent ordered systems across multiple levels, from physiology and emotion to cognition. The brain's value system is deeply activated; the loneliness born of separation dissolves, replaced by profound joy, fulfillment, and a sense of wholeness beyond the individual. The familial affection, friendship, and romantic love we feel all arise, at root, from this deep bond with another soul as together we explore and shape the world. This resonance and intermingling carry us beyond the island of the self.

"The essence of the phenomenon of falling in love is a sudden collapse of a section of an individual's ego boundaries, permitting one to merge his or her identity with that of another person. The sudden release of oneself from oneself, the explosive pouring out of oneself into the beloved, and the dramatic surcease of loneliness accompanying this collapse of ego boundaries is experienced by most of us as ecstatic. We and our beloved are one! Loneliness is no more!"[1]

The pursuit of higher understanding, then, is not only for the sake of understanding the world, but also so that we may attain

[1] M. Scott Peck, *The Road Less Traveled*.

deep, multi-layered resonance with more souls. When we learn how to understand the objective world properly and, on that basis, use love as a profound form of interaction through which to develop a shared understanding of and care for that common world and for one another's existence, our separate worlds no longer contain only ourselves. Loneliness is eased, and lives become connected.

Only when human society gradually builds bonds of shared understanding and empathy at the level of its deepest logic and core emotional experience can we move toward genuine universal harmony and truly build a community with a shared future for humankind.

From the "utmost complexity" of the myriad phenomena, we return to the "utmost simplicity" of the pivot of the Way; from that simplicity, we set out again to understand the world in all its complexity. This is the journey of understanding that the book has tried to present.

"Knowing is the beginning of action; action is the completion of knowing." Grasping the underlying logic by which we understand the world is only a beginning. To discern the changing tendencies within the multi-layered order of things, we must also "reflect in stillness and temper ourselves in action." The difficulty of seeking truth from facts lies precisely in that word "seeking," a task without end.

Chapter 27

Returning to the Root Is Stillness

The myriad things unfold in profusion, and each returns to its root. Returning to the root is called stillness; this is called returning to destiny.

Laozi

All things, in their profusion, ultimately return to their root. That root is the most genuine, most ordered, most enduring state of life. Laozi called it "returning to the root" and called it "stillness." What this points toward is life's abiding inner tendency, amid the clamor of change, to return to order and recover its original state.

In contemporary terms, this tendency to return to order means that every living system, in order to sustain the existence of its own multi-layered ordered structure, must continually draw energy from outside and transform disorder into order, using an inflow of low entropy to counter an outflow of high entropy.

However, within life's multi-layered structure, each level has its own operating laws and its own way of utilizing energy. From lower to higher levels, energy is progressively converted, distributed, stored, and used. Each link drives the maintenance and renewal of order at that level. Conversely, the ordered structure at

higher levels also provides a stable framework and direction for the operation of lower levels. It delineates boundaries, allocates resources, and coordinates rhythms.

Disorder creates possibility. Order determines direction. Together these constitute the complete mechanism by which living systems resist entropy increase. They also explain how life maintains overall order and flexibility in complex and changing environments.

The most common and most stable source of energy in the universe is stellar nuclear fusion. For life on Earth, low-entropy photons from the sun flow continually into the ecosystem. They form the primary driving force for all life to maintain order, replicate itself, and continue evolving.

Plants possess a wondrous ability. They directly use sunlight.

Inside the chloroplasts of leaf cells, the random fluctuations of photons are captured by chlorophyll and transformed into a directed torrent of electrons. Water molecules are then split apart, and oxygen is released into the atmosphere as a byproduct. This stage produces two high-energy carriers, ATP and NADPH, but they are extremely unstable. They cannot be stored or transported over long distances.

Thus, the Calvin cycle that follows fixes this transient energy. In the chloroplast stroma, ATP provides chemical energy. NADPH provides hydrogen atoms. The two join forces to drive a series of enzymes. They disassemble, rearrange, and reduce carbon dioxide from the air, ultimately kneading it into stable glucose molecules.

Glucose is the universal energy currency of the cellular world,

ready to be called upon at any time. Any cell that needs energy can break down glucose again in the mitochondria, releasing ATP to drive such basic activities of life as protein synthesis, the operation of ion pumps, and transport across cell membranes.

Sustaining all this requires a continuous supply of sunlight, carbon dioxide, and water. A single cell cannot accomplish this alone. Thus, division of labor occurred among cells. This is what we earlier called symmetry breaking.

Some cells committed their fate to capturing light energy, developing into mesophyll cells rich in chloroplasts. Others abandoned photosynthesis, specializing as root cells that absorb water and dissolved minerals. Still others differentiated into the vascular tissue running from top to bottom. Xylem vessels transport water and minerals upward from the roots. Phloem sieve tubes transport glucose manufactured by the leaves to every part of the plant. A transport network connecting source and sink thus takes shape.

This division of labor made cells interconnected, each with its own role and mutually dependent upon the others. A multi-layered ordered living system was thus formed. Sunlight, carbon dioxide, and water, the resources essential for sustaining order, no longer depended on isolated acquisition by individual cells. They were now coordinated and managed by a higher-level organizational system.

Even underground, the roots do not fight alone. Plants trade sugars for the assistance of fungal hyphae. This fungal network is hundreds or even thousands of times finer than the root system. It helps plants efficiently absorb water and minerals such as phosphorus and nitrogen.

With the formation of the multi-layered structure, the identity of

glucose shifted accordingly. It was no longer merely fuel. It became the building material for constructing multi-layered structures of order.

Through the precise disassembly and reassembly performed by enzymes, sugar molecules are straightened and linked, polymerizing into tough chains of cellulose. These build the load-bearing skeleton of the cell wall. Some sugar molecules bend and branch, transforming into hemicellulose and pectin and filling the spaces within the skeleton like concrete. Meanwhile, the carbon skeleton of glucose encounters nitrogen, sulfur, phosphorus, and other mineral elements transported from the roots. Together they synthesize nucleic acids and proteins. Nucleic acids store the ordered information of heredity. Proteins serve catalytic and structural functions.

At this point, a "cellular house" has been fully built. In the final analysis, it is the different modes of combining glucose that create an ordered living being, one capable of metabolism, perception, and reproduction.

However, sunlight is not always generous. Night returns in cycles, winter seals the soil, and drought and heat also arrive unannounced. Plants must therefore equip themselves with an energy storage mechanism to buffer future uncertainties.

Surplus glucose is first used for short-term storage. A portion of glucose is converted into sucrose, which is more stable and more soluble in water. It is transported via the phloem to roots, stems, and fruits, and temporarily stored in vacuoles. This is an on-demand reserve, readily accessible at any time. It handles the short-term fluctuations brought by the alternation of day and night.

If there is still surplus, long-term savings begin. Glucose

molecules undergo dehydration condensation, linking into semi-crystalline starch. Starch is insoluble in water, designed specifically to endure the long winter nights. It is deposited in roots, tubers, seeds, and similar locations. It is never randomly piled in the leaves, where it would disrupt the cell's osmotic balance or obstruct transport pathways.

In seeds, plants display the ultimate condensation of order. Surplus sugars are deeply converted into oils, which have more than twice the energy density of starch. Oil is a lightweight and long-lasting "high-energy battery," just enough to sustain the embryo as it breaks through the soil and grows its first green leaf capable of independently capturing sunlight, thereby initiating a new energy cycle.

Glucose, fructose, sucrose, starch, and oils differ in structure and follow different metabolic pathways, yet all can participate in cellular energy metabolism. Starch can be broken down into glucose; sucrose consists of glucose and fructose; oils are broken down chiefly into glycerol and fatty acids. After being transformed through different pathways, they can all feed into the cell's energy-metabolism network, supplying raw material for the production of ATP and paying every energy bill required for ordered structures to resist entropy increase.

Cellulose, hemicellulose, pectin, proteins, nucleic acids, and enzymes are likewise derived through complex metabolic networks from glucose, which carries energy, and from trace minerals. Together they build the plant's own multi-layered ordered structure.

Thus, whether maintaining order or constructing order, at its root, everything is about the transformation and utilization of glu-

cose. All of the plant's life activities related to energy and structure point to a single purpose: to use energy to resist entropy increase.

The shoot tip bends toward the light. Every leaf is offset from the others to maximize energy capture. The roots stretch toward moisture and fertile ground, accompanied by fungal hyphae as they bypass barren soil.

When sunlight blazes too fiercely and the energy influx exceeds the processing capacity of the chloroplasts, they activate a self-protection mechanism known as non-photochemical quenching. They dissipate the excess energy as heat rather than allow the order within the cell to collapse. Even if intense light causes damage, the cells calmly initiate repairs once the light recedes.

Plants cannot move and have no brain, no nerves, no desire, and no fear. Silently, even passively, they maintain precise alignment with their own needs at every level.

From beginning to end, no external will makes decisions on the plant's behalf. At every level, from molecules to cells to tissues to the whole, it transforms disorder into order, dissipates surplus as heat, and converts scarcity into a direction of growth. It always follows the law that "returning to the root is stillness," adapting to the constraints of its environment.

We humans are different. We can move, choose, reshape the environment, and even remake the world through free will. This initiative is a uniquely human endowment.

But this endowment also means that, unlike plants, we cannot precisely match our own needs throughout life through a fixed genetic program alone. We need all the more, therefore, to observe

ourselves through the silent law of plants. We need to understand thoroughly the multi-layered ordered system that we ourselves are, from cells to organs, from metabolism to emotion, and from instinct to reason. Then we can make full use of our freedom.

This does not mean retreating into passivity. It means using our initiative to make freedom a deep awareness and conscious safeguarding of our own order.

Unlike plants, humans cannot directly use sunlight. Whether grains, vegetables, fruits, or meat, food is essentially chemical energy pre-stored by other living beings. It mainly consists of three categories: carbohydrates, fats, and proteins. To utilize this energy, the human body requires a complex digestive system to disassemble food layer by layer. If the plant's vascular tissue is responsible for transporting water and nutrients throughout the body, then the digestive system is the human body's first processing line for energy.

The mouth is responsible for chewing and grinding, initiating the breakdown of carbohydrates. The stomach churns food into chyme, beginning the degradation of proteins. Upon reaching the duodenum, bile secreted by the liver emulsifies fats into tiny droplets. The pancreas secretes multiple enzymes, disassembling large molecules into smaller units. Just as plant roots trade sugars for the assistance of fungi, our gut microbiota also helps break down dietary fiber, synthesize vitamins, and train the immune system. Finally, the small intestine, with its countless villi, absorbs glucose, fatty acids, and amino acids into the blood and lymph. The large intestine reclaims water and electrolytes, expelling the undigested residue.

Thereafter, the three macronutrients each go their separate ways.

Starch in food is disassembled into glucose. Sucrose is disassembled into glucose and fructose. Lactose is disassembled into glucose and galactose. Fructose and galactose are eventually converted into glucose in the liver as well. All glucose enters the bloodstream, available for use by cells throughout the body.

Glucose follows the principle of "use immediately, store the surplus." A portion of surplus glucose is stored in the liver and muscles as glycogen, a readily accessible reserve that can be mobilized quickly. Any further surplus is converted by the liver into fat as a longer-term reserve.

Whether directly ingested or converted from surplus carbohydrates and proteins, fat ultimately follows one of only two paths: storage or burning. After absorption, most fat enters adipose tissue for storage. This is a "fixed-term deposit," used to withstand starvation and sustain prolonged low-intensity exercise; its metabolic cycle can span weeks or even months. A small portion is oxidized directly, providing energy for skeletal muscle and the heart.

After digestion into amino acids, protein is first used as building material for constructing the body. From the cytoskeleton to various enzymes, from hormones to antibodies, nearly all life functions depend on proteins. Only in conditions of severe starvation or excessive intake are amino acids converted into energy or fat.

Within normal life rhythms, this entire system maintains an exquisite balance. When eating, chemical energy is methodically disassembled, absorbed, and distributed. When expending energy, glucose in the blood and reserves in adipose tissue are promptly

mobilized. When resting, organs repair and recuperate. The brain clears metabolic waste. The immune system shifts toward gentle repair. From the brain's regulation of appetite to the energy conversion within every single cell, every level maintains a delicate balance in the tug-of-war between order and disorder. This is the complete closed loop formed by drive from bottom to top and constraint from top to bottom.

However, this closed loop, honed over hundreds of millions of years, is now facing unprecedented challenges.

The high-sugar, high-fat, high-protein combinations produced by the modern food industry precisely strike the reward circuits of the brain's value system. They bypass natural satiety signals, making people involuntarily over-consume. This is a severe mismatch between evolution and environment. Our bodies are still preparing for starvation. The world has already become so abundant that high-energy food is within arm's reach.

At the cellular level, the collapse of order begins first with carbohydrates. Frequent intake of high-sugar foods, especially refined sugar, sends blood sugar on a rollercoaster ride. Glucose does not freely diffuse into cells. It requires insulin to regulate the transporter protein channels on the cell membrane. When blood sugar spikes repeatedly, the pancreatic beta cells are forced to secrete insulin desperately. Over time, the insulin receptors on the cell surface become overwhelmed. They are forced to downregulate in number and sensitivity. This is insulin resistance. Glucose is shut outside the door, stagnating in the blood, forming hyperglycemia. The interior of the cell instead starves, thereby inducing type 2 diabetes.

The fructose in refined sugar is even more insidious. It bypasses insulin regulation and charges directly into the liver. Its metabolism faces almost no rate-limiting step. In the liver, fructose rapidly depletes phosphate groups and ATP, causing hepatocyte energy exhaustion and abnormal purine metabolism. Uric acid consequently rises, triggering gout. At the same time, fructose also forcefully drives the liver's de novo lipogenesis process, synthesizing large quantities of triglycerides. These fats cannot be transported out in time and accumulate excessively in the hepatocytes, forming fatty liver.

Excess fat likewise creates chaos at the cellular level. Long-term high-fat diets, especially those containing trans fats and excessive saturated fats, quietly alter the lipid composition of the cell membrane, reducing membrane fluidity. The signal transduction efficiency of the insulin receptors embedded in the membrane also declines correspondingly. This is another form of insulin resistance, which likewise impairs glucose absorption and pushes up blood sugar. When fat is excessive, the triglycerides synthesized by the liver exceed the carrying capacity of the transport vehicles. This leads to dyslipidemia, with low-density lipoprotein rising and high-density lipoprotein falling.

If fat intake is so excessive that even the adipose tissue reaches storage saturation, the surplus free fatty acids will be deposited, in the form of triglycerides, around the internal organs, in the intermuscular spaces, and even in the pancreas and heart. This is called "ectopic fat deposition." These fats, which should not be there, stimulate surrounding cells to secrete large amounts of pro-inflammatory factors, triggering a low-grade, persistent chronic in-

flammation.

Excess protein takes a different path. The human body has no dedicated warehouse for storing protein. Surplus amino acids are forcibly deaminated by the liver. The nitrogen-containing portion is converted into urea and excreted through the kidneys, increasing the burden on the liver and kidneys. Although excess protein does not trigger fatty liver as forcefully as fructose, it too eventually takes a detour and is converted into fat.

Taken together, excessive intake of any of the three—carbohydrates, fats, or proteins—ultimately forces cells into a chronic state of "processing-capacity overload." Mitochondria and the endoplasmic reticulum operate in chronic overload, oxidative stress persists, and the immune system is repeatedly triggered, falling into an erroneous "false alarm" state and continuously releasing inflammatory factors. Even worse, this inflammatory excitation forms a self-reinforcing vicious cycle. Inflammation worsens insulin resistance, insulin resistance pushes up blood sugar, and high blood sugar in turn induces more inflammation. The cell thus enters a silent, self-driven fire of disorder.

Under long-term chronic inflammation, cellular telomeres shorten faster, stem cells are depleted at an accelerated pace, and tissue repair capacity progressively declines.

When these disturbances at the microscopic level accumulate continuously, and inflammatory factors engage in constant cross-talk among adipose tissue, the liver, blood vessels, and the brain, specific disease patterns gradually become manifest at the organ level.

Chronic insulin resistance may ultimately develop into type 2

diabetes. Under multiple metabolic stresses, the liver gradually becomes overwhelmed, forming non-alcoholic fatty liver disease. If inflammation continues to infiltrate, fatty liver may progress to steatohepatitis, and even slide further toward liver fibrosis and cirrhosis. Long-term dyslipidemia causes an increase in small, dense low-density lipoprotein. These particles more easily infiltrate the vascular endothelium. After being engulfed by macrophages, they form foam cells, which then evolve into atherosclerotic plaques. This is the beginning of atherosclerosis, and the pathological foundation of myocardial infarction and stroke.

The kidneys are not spared either. Chronic high blood sugar damages the glomerular filtration membrane, causing protein leakage and microalbuminuria. The deposition of uric acid crystals may induce interstitial nephritis. Under the sustained assault of hypertension and inflammation, renal function gradually declines, eventually progressing to diabetic nephropathy. The intestines also bear the strain. Excess protein that is not fully absorbed enters the large intestine and is fermented by the gut microbiota, producing various harmful substances. These substances chronically irritate the intestinal wall, allowing endotoxins to enter the bloodstream, exacerbating systemic low-grade inflammation and increasing the risk of colorectal cancer.

The brain is also affected. Chronic low-grade metabolic inflammation reduces neuronal synaptic plasticity, and cognitive function declines almost imperceptibly.

At first, these disorders extending from cells to organs leave barely a trace at the level of the whole human body.

In youth, the body possesses powerful compensatory capacity.

Insulin resistance can still be masked, and blood sugar appears normal on the surface. Although liver fat has begun to accumulate, transaminase levels remain within the normal range. Lipid streaks on the vascular wall have just formed, yet blood flow remains unobstructed. You feel no pain, no fever, no redness or swelling. You work, exercise, and eat as usual. Apart from occasional post-meal drowsiness and an extra inch or two quietly added to the waistline, there is almost no distinct sensation of being unwell. Metabolic disease never arrives with drums and gongs. It is like the frog in slowly warming water. Over a span of several years or even more than a decade, it silently erodes the baseline of health.

When compensation is gradually exhausted, blood sugar begins occasionally to exceed the threshold. Impaired fasting glucose or mild fatty liver begins to appear, and blood pressure quietly rises. At this point, you may still feel that it is not so bad, not yet an illness. But the body has already sent signals. Energy is not what it used to be. Afternoon fatigue is hard to dispel. Body weight, especially around the waist and abdomen, becomes increasingly difficult to control. The skin becomes prone to oiliness and breakouts. Joints occasionally ache faintly. These scattered discomforts signal the loosening of some deeper structure.

If left unchecked, disorder eventually appears at the macroscopic level. Diabetes is formally diagnosed, and lifelong blood sugar management becomes necessary. Fatty liver progresses to steatohepatitis, with the risk of liver fibrosis hanging overhead. Coronary CT reveals plaque formation; the probability of myocardial infarction or stroke rises sharply; kidney-function indicators decline year by year; intestinal polyps quietly emerge under the

sustained stimulus of chronic inflammation; cognitive decline begins to affect work and daily life.

Searching for a corresponding "target" for each disease often results in attending to one thing while losing sight of another. The key to truly avoiding or reversing these disorders lies in experiencing and learning the plant's passive "stillness." From energy surpluses and deficits at the cellular level, through load and repair at the organ level, to fluctuations in energy and emotion at the level of the whole person, we must make full use of our freedom and initiative for the deep awareness and conscious safeguarding of our own order.

We must first learn to read the authentic structure of food. Natural foods that have not been overprocessed—grains, legumes, vegetables, fruits, nuts, fish, eggs, meat, and so on—largely retain the harmonious proportions shaped together with our metabolic systems over hundreds of millions of years of evolution. So long as our diet is neither unbalanced nor excessive, their dietary fiber, polyphenols, unsaturated fatty acids, and slowly released carbohydrates can naturally regulate the rate of absorption. This allows the mitochondria to process each unit of energy without being thrown into overload by a sudden surge.

For deeply processed foods, it is necessary to learn to distinguish several sets of concepts.

Fast sugar and slow sugar differ in their effects. Fast sugar comes from sweet drinks, white bread, biscuits, and the like, making blood sugar fluctuate like a rollercoaster. Slow sugar comes from whole grains, legumes, and most fruits, allowing energy to be released slowly and steadily. Inferior fat and superior fat

also differ. Inferior fat refers to trans fats and excessive saturated fats, which induce cell membrane rigidity and inflammation. Superior fat comes from the unsaturated fatty acids in nuts, olive oil, and fish oil, which help maintain membrane fluidity and are anti-inflammatory. Redundant protein and adequate protein differ as well. Excessive reliance on processed meat products easily leads to redundant protein intake, increasing the burden on the liver and kidneys. Adequate protein from natural foods can meet synthesis needs without imposing an extra burden.

At the organ level, our approach is not to pursue starvation dieting, a ketogenic diet, the elimination of carbohydrates, or excessive exercise in isolation, but to make a holistic adjustment. Let the liver's fat load come down, let the vascular endothelium escape continuous assault by high sugar and high fat, let the storage and release of adipose tissue restore its rhythm, and let each organ return to within the load boundary it can bear. Restore the rhythm of eating, increase physical activity, and through dietary adjustment let the gut microbiota realign.

At the holistic level of the human body, we need to return to a conscious order of living. "Do not covet" means not ingesting energy density beyond one's processing capacity. "Do not waste" means not letting protein be needlessly converted into fuel. "Do not cling" means not combating systemic disorder with short-term extreme dieting or a single medication. "Do not shortchange" means not cutting corners on sleep. Enjoy the conveniences of modern life while maintaining within them a lucid power of choice.

Nevertheless, we must humbly acknowledge that the human body is a complex system far beyond the boundaries of our cog-

nition. The above discussion is no more than a fragmentary interpretation, glimpsing a leopard through a tube.[1] But one point is relatively certain. With the accelerating arrival of the intelligent era, material abundance will grow to dizzying excess, and information will flood like a deluge. Plants cannot covet photons beyond their processing capacity. We humans, facing an excess of information and desire, lack an innate quenching mechanism. Disordered energy intake and information input are themselves a form of entropy increase. Material desire beyond appetite, grasping beyond the search for knowledge, and addiction beyond connection pile up layer upon layer, turning free will into a force of self-dissipation.

The higher the degree of freedom, the harder it is to return to the root, and the more precious that return becomes. This requires us, all the more, to grasp the underlying laws of our own multi-layered order and to maintain, amid fluctuation, the dynamic capacity to return to our original order. We must master the fundamental cognitive methods for understanding how the world works, and use AI wisely to obtain and apply information and knowledge. We are not retreating to a primitive state of passivity, ignorance, and desirelessness. We are striving, while possessing the capacity to choose, reshape, and create, to combat chaos lucidly and actively and to safeguard order.

Thus, self-cultivation is the simplest yet also the most fundamental practice. No matter how grand the undertaking or ideal we aspire to, no matter how much wisdom for understanding the world we thirst for, the starting point of all action lies right here.

[1]The Chinese idiom “to glimpse a leopard through a tube” means seeing only one small part of a much larger whole.

"Returning to the root is stillness." A person with inner order is more likely to become wise enough to discern the patterns of the world, and more likely to become a person of goodness who meets others with warmth and understanding.

Chapter 28

The Wise and the Good

The truly wise are humble, and the truly good are tolerant.

Only those of little wisdom are overbearing, and only those of little goodness haggle over every trifle.

Zhou Guoping

Every person, from the moment of birth, is situated within a nested web of ordered organizations: family, community, nation, and so on. Marx pointed out that the human essence is the sum total of all social relations. Our social character, the value of our lives, and even our personal destiny are all shaped and given meaning through continuous interaction with others.

Within this ordered social web, which has neither an inside nor an outside, the central question is how we shape our relationships with others. This shaping is both our instinctive choice and our conscious responsibility as low-entropy entities within the torrent of increasing entropy.

Every relationship we build and sustain injects "negative entropy" into the organization of which we are part, enabling it to

retain vitality and continue to exist amid the cosmic tendency toward disorder. In this way, we can move beyond the basic meaning of individual survival and strive for meaning at a higher social level.

As discussed previously, the human brain is essentially an ever-evolving large model. Every act of learning in society, every interaction with others, and every thought and action in the face of choice all supply this model with data, adjust its parameters, and optimize its weights. This process also defines who we are.

If a person never actively shapes themself, their model can only passively accept the training data supplied by the surrounding environment. Their values are swayed by those around them, and their patterns of behavior are swept along by prevailing trends. Ultimately, they lose themselves in imitation and conformity, drifting with the social tide and becoming random in the statistical sense.

Conversely, if we can actively shape ourselves, we can gain a measure of control over our lives and destinies within complex social relations. We do this through extensive reading and practice, reaching out to the law spectrum of the cosmos: from the physical laws of natural science, through the organizational logic of social life, to the emotional rhythms deep within the human heart.

"With learning, the world is right before your eyes; without learning, what is before your eyes is your entire world." This saying aptly captures the point. When we openly absorb different bands of law and internalize them as our own cognitive coordinate system, we are no longer like duckweed borne along by the current. We actively define ourselves as an ordered structure resonating in harmony with the universe. We move in accordance with the Way,

not out of blind obedience, but from insight. Once we understand the inevitability of entropy increase, we understand why relationships must be built to inject negative entropy; once we understand the mechanism of symmetry breaking, we understand how to find, within an apparently stable status quo, the fulcrum from which change can be set in motion.

When everything accords with law, inner friction and struggle naturally dissolve. We no longer fight against the operating logic of the world. Instead, in accordance with it, we find true freedom: a liberation that moves from the realm of necessity toward the realm of freedom. It is precisely in this state of harmony with cosmic law that we become, within society's dynamic equilibrium, an innovative force driving symmetry breaking.

This, then, is the significance of the wise in the world: they not only define themselves, but through that definition inject the possibility of evolution into the whole.

Precisely because of this, differences in frequency between people are the norm, not the exception. Each of us has a different inheritance and a different course through life. The law spectrum of the cosmos, projected onto the complex model of our brain, shapes a wide variety of response patterns to that spectrum.

For example, scientists and technical workers are often keenly responsive to the lower frequency bands of the law spectrum: the stable, quantifiable, repeatable logic of physics. Managers and leaders are more adept at capturing the higher frequency bands: the fluid, context-dependent social signals and organizational rhythms that concern the human heart.

However, a deeper predicament lies in this. We may not even

realize that we have been trained on only certain narrow bands of the law spectrum of the cosmos.

Everyone tends to interpret everything through their own cognitive paradigm, sincerely believing that they are seeking truth from facts. What we often fail to see is that our logic may be rigorous and self-consistent within its proper band, yet once carried across the boundary into another band, it may cease to work and may even become unintelligible.

Therefore, whenever conflict arises between people, it is difficult simply to blame one side. More often, both sides are confined within their own cognitive bands, misunderstanding one another and colliding across mismatched frequencies, each feeling that the other is "a summer insect to whom one cannot speak of ice."

Once we understand this, we will approach "changing others" with the necessary awe and humility. An adult is a stable model trained by the entirety of their past life. Attempting to recalibrate their cognitive parameters through a few conversations or arguments is no different from expecting a handful of training samples to reshape a large model.

Thus, for most matters of lesser importance, making an effort to accommodate one another and having the capacity to turn the page is a form of practical wisdom.

"The noble person is broad-minded and at ease; the petty person is always fretful and anxious." The breadth of the band to which a person can respond across the law spectrum of the cosmos often determines how comfortable that person is to be around.

Those who can move freely between different paradigms pos-

sess a rare ease and tolerance. They can observe the changing tendencies of layered order and follow the more stable tendencies, so they are not easily perturbed by the randomness of the moment. They can lean down to listen to the rigorous logic of a technical specialist, and also raise their perspective to understand a manager's view of the whole. They know how to shift perspectives in a particular context, so communication becomes a bridge rather than a battlefield.

Such people we call the strong. The strong are strong not because they are always right, but because they possess the capacity for paradigm shift. They can move freely between the low-frequency and high-frequency bands of the law spectrum. They can shift dimensional levels between detail and the big picture. Therefore, when facing disagreement, they more easily find common ground. They are more willing to be tolerant. Their emotions are naturally stable.

Conversely, those who often roar and lose control are usually trapped within their own narrow cognitive band. On the one hand, they are easily perturbed by present randomness, producing emotional fluctuations that in turn influence their behavior and practice. On the other, they do not understand that different law paradigms exist, so they cannot shift between paradigms. They habitually measure everything by a single logic. Once they encounter a frequency mismatch, they immediately fall into confrontation. Their behavior comes down to an inflated sense of self, stubbornness, and the conviction that they are always right.

Precisely because of the innate differences and acquired shaping of each person's response range across the law spectrum, com-

munication is no longer merely the transmission of information. It becomes the sole bridge for crossing frequency bands and seeking resonance. Not being skilled in a certain paradigm does not mean being unable to understand it. A person may not be able to derive equations like an engineer, yet through listening they can still perceive the rigor of the logic; they may not be able to capture emotion like a poet, yet through exchange they can still appreciate the subtlety of feeling. Sustained communication and collaboration can gradually bring people on different frequencies closer.

Of course, not all differences in frequency can be reconciled. When honest communication, time and again, still cannot bridge fundamental paradigm divides, and when seeking common ground while reserving differences becomes a pretext for mutual depletion, then parting ways is no longer a regret. It becomes a way of respecting the range of frequencies to which each person can respond.

For ordinary people like us, no one can easily broaden their response range within the law spectrum enough to discern everything in the world and solve every problem. This is why "when three walk together, there is always a teacher among them." We should therefore always remain humble and cautious, never allowing ourselves the slightest conceit. Every collision that makes you uncomfortable is a rare opportunity for learning and self-cultivation. When you find someone wholly unreasonable, you might ask yourself: Is my own frequency response range also too narrow to receive that person's signal? The strong are not born strong; they learn to turn around after running into one wall after another. The weak are not condemned by fate; they have simply not yet realized that they need to open themselves to more bands.

When we enlarge our view from the individual to the organization, we discover that resolving the frequency mismatch at the collective level also requires a higher-order paradigm shift mechanism. That mechanism is democratic centralism.

An organization is composed of individuals with different frequency response ranges, while the strong are always a minority. If everyone acts independently, the organization will disintegrate through increasing entropy amid collisions between mismatched frequencies. Democratic centralism respects, through democracy, the genuine feedback arising from individual frequency response ranges and, after full discussion, forms a unified direction through centralism. The strong, whose frequency response ranges are broader, should use their capacity to move between dimensions to distill consensus and bridge differences on the basis of democracy. From a higher vantage point, they should promote a unity of views and thereby enable the organization to achieve genuinely effective centralism.

This is the paradigm shift at the organizational level: elevating the different frequencies of individuals into the symphony of the whole.

I once read a passage that resonated deeply with me. It lays bare the predicament of being constrained by our own response range within the law spectrum without even realizing it:

> We all more or less naively assume that, by acquiring knowledge, we can train a pair of discerning eyes capable of seeing through the complexities of the world. In reality, we need to recognize within ourselves the

> existence of many paradigms, and to regard even this recognition as a paradigm. We should approach reality as a whole with the heart of a child. Almost every one of us is governed by paradigms and thus becomes blind to the many possibilities inherent in things. We cling stubbornly to one side, take it for granted that only our own view is right and that every other view is wrong or absurd, and quickly reject those other views as we travel ever farther along the road we have chosen. You must relinquish your own paradigm and enter a humble state of "emptiness." Then think deeply and analyze things with care. In the end, what appears to be non-action may prove to be the most fundamental, strategic, and effective leverage point; what appears mad—a letting go of everything—may allow us to dance gracefully with nature.

Here, the "emptiness" of which we speak does not mean stripping away those fundamental drives that make us human: greed, anger, ignorance, pride, and doubt. These emotions and desires, which seem to disturb our objective judgment, are in fact the foundational code upon which the brain's value system operates.

Our wisdom in countering cognitive bias does not lie in trying to eradicate these "monkeys of the mind." That is neither possible nor consistent with the basic physiology of the brain. We need to find higher-order forms capable of holding these powerful inner forces. This is the deeper significance of art, music, literature, and the like: they are "negative entropy sources" invented by humanity, a form of advanced neural nourishment.

When we are moved by a painting, resonate with a melody, or are struck to the heart by a poem, we are using these highly ordered forms to tame, guide, and sublimate primal energies that might otherwise run unchecked. Art allows us to keep the cerebral cortex active while giving the instinctive impulses of the emotional brain an ordered outlet. The entire brain system, from the ancient brainstem to the evolutionarily younger prefrontal cortex, can thus work in concert through harmonious oscillation, preserving that precious, dynamic "ordered state."

This may be how "attaining centrality and harmony, whereby Heaven and Earth take their places and all things are nourished" appears at the neural level:[1] every emotion returns to its rightful place, while the brain, nourished by the beauty it creates, moves from chaos toward clarity and then from clarity back toward vitality.

This inner clarity and harmony is precisely what provides a stable core for broadening our response range across the law spectrum. When the inner self is no longer hijacked by primal impulses, we can explore outward with greater ease.

So how can we broaden our response range within the law spectrum as much as possible? This requires us to step beyond the limits of specific frequency bands and seek out the "law of laws" capable of connecting different domains. That is methodology. Systems theory teaches us to examine the relationships among elements from

[1]The phrase comes from the Confucian *Doctrine of the Mean*: when emotions have not yet arisen there is "centrality"; when they arise in due measure there is "harmony." The text links their full realization with the flourishing of Heaven, Earth, and the ten thousand things.

a holistic perspective. Dialectics lets us grasp unity within opposition. Contradiction theory guides us to analyze the principal contradiction within complexity. The theory of practice constantly reminds us that the truth or falsity of cognition must ultimately be tested in action. These methodologies are like a cognitive compass. When we enter an unfamiliar field, they help us identify its basic paradigms more quickly and shorten the journey from chaos to clarity.

Of course, a compass points the way, but it cannot do the walking for us. An understanding of methodology can never replace the daily learning, trial and error, and accumulated experience within a particular field. Unless you immerse yourself in the depths of technology, you cannot truly understand an engineer's rigor; unless you immerse yourself in the subtleties of human nature, you will struggle to appreciate a manager's difficult judgments.

Laws and methods that can be put into words are certainly precious, but tacit knowledge that is difficult to express must still be grasped gradually through firsthand practice. Amid the current enthusiasm for large AI models, these systems can greatly extend the boundaries of our abilities and promote equal access to information and knowledge. Yet if we depend on them too heavily and neglect to nourish our own minds through learning and practice, our cognition may stagnate and cognitive inequality may instead deepen.

A person who is not arrogant, who understands humility and reflection, and who can define the self is wise.

A person who is not stubborn, who understands tolerance and communication, and who can empathize with others is good.

The life that remains to us is precious, like morning dew. Walk

with the wise and let the boundaries of cognition keep expanding. Keep company with the good and let the network of relationships be filled with warmth.

Chapter 29

Knowledge and Action Are One

The clearer one's understanding, the more steadfast one's action; the more steadfast one's action, the clearer one's understanding becomes.

Zhu Xi

Within the universe's ordered web, which has neither an inside nor an outside, universal interconnection means that we are participating in its changes at every moment. In the broadest sense, to exist is to practice. From every breath that sustains the order of life to every thought that subtly adjusts our neural networks, we are constantly affecting the external world through our own "ordered changes."

What we commonly call practice refers more specifically to social practice: participation in the changing web of social order.

Unlike researchers in the natural sciences, who investigate things from a "view from above"—where we may remain observers and use repeatable experiments to reveal the composition of things and their patterns of change—in social practice we ourselves are part of the ordered structure under transformation. Each person's practice, transmitted and amplified through layer

after layer of relationships, ultimately drives the evolution of higher-level ordered organizations: families, companies, nations, and even civilizations.

This is no longer observation but necessary participation. It is no longer merely revealing laws but influencing processes and creating trends. This is practice in the narrowest and most profound sense.

Only by gaining insight into the external world's multi-level changes can people participate in them actively, using their own ordered changes to follow or guide broader trends. And only through firsthand practice, continually calibrating cognition through interaction, can knowledge truly be internalized as a stable ordered structure within neural networks and become "true knowledge" capable of guiding action.

Thus, knowledge and action are not two isolated stages. They are two aspects of the same process of interaction between self and environment in practice. They are united at their source; hence, "Knowledge and Action Are One."

In epistemology, we distinguish the subjective from the objective so that we can analyze more clearly where cognition comes from and where it leads. In practice itself, however, the subjective and the objective are no longer two severed poles.

Every act of cognition bears a subjective imprint, yet points toward the objective world; every action changes objective order, yet feeds back into the subjective cognitive framework. Every node in the universe's web of order both receives and exerts influence. Through universal connection, subject and object become embedded in one another and bring one another forth.

True practical wisdom therefore lies in moving beyond the naive intuition that places self and world in absolute opposition, and in recognizing through practice the dialectical unity of subject and object.

These differing emphases in practice embody two basic ways in which the order of the "I" participates in the order of the "we."

When practice gives greater weight to "knowledge"—to the individual's insight into and awakening to the nature of the world—it resembles the Hinayana practitioner's focus on self-liberation. Through sustained contemplation and introspection, the "I" that bears cognition strives to become wise. This is a form of practice that seeks inward, taking individual awakening as the far shore and the ordering of the self as its cultivation.

When practice gives greater weight to "action"—to the individual's effort to change ordered organizations on a larger scale—it resembles the Mahayana practitioner's vow to deliver all beings. Individual awakening enters the shared karma of sentient life, and action reaches other people, families, and society as the practitioner strives to become good. This is practice that unfolds outward, taking the elevation of the whole as its destination and the ordering of relationships as its responsibility.

Those who unite the highest wisdom with the highest goodness, as Wang Yangming did, achieve a higher synthesis of these two basic postures. They cultivate inwardly, seeking illumination through the investigation of things and the extension of knowledge, while also assuming responsibility outwardly and participating in the evolution of civilization with the ambition to order the state and bring peace to all under Heaven.

For us ordinary people, this means dwelling within the limited levels we inhabit, treating our own changes as micro-fluctuations, and striving to feel and influence the levels within our reach, thereby endowing our existence with the greatest possible meaning.

The first difficulty of practice is that the external world we face is itself complex. Things are themselves complex systems of multi-level order, while the universal interconnection of all things forms still more complex systems of multi-level order. Moreover, all things and all their ordered levels are in eternal change.

According to the law spectrum, lower-level order corresponds to laws with higher symmetry and better universality. But lower-level things have small spatiotemporal scales. More microscopic changes are governed by quantum fluctuations, presenting elusive randomness. Higher-level order corresponds to laws with lower symmetry and poorer universality. Higher-level order also corresponds to larger spatiotemporal scales. More macroscopic changes present greater determinacy.

Although we know we cannot measure all things with one perspective or view the world with a static gaze, the outside world's complexity still brings cognitive difficulty in practice. Whether focusing on low or high levels, laws and trends are all obscured by layer upon layer of appearances.

The random fluctuations of the microscopic world are one kind of "appearance," hiding deterministic laws in a fog of probability. The context dependence of the macroscopic world is another, forcing universal principles to wear the garments of concrete events. We try to see the essence through phenomena, yet are often misled

by their surface features. This is what Buddhism calls "clinging to appearances": mistaking the contingent for the necessary, the local for the whole, and what appears here and now for a truth valid everywhere.

The root of "clinging to appearances" lies in our failure to penetrate the complexity of things as systems of multi-level order. "See forms as no-form, and you will see the Tathagata." Laws never present themselves nakedly. They remain hidden among countless concrete, random, context-dependent "appearances," waiting for us to observe them through different "cognitive microscopes" and understand them through different paradigms of thought.

The second difficulty of practice is that our cognition of the outside world is also limited.

First, our brains can observe only a limited amount of present information at one level of a thing. Like peering at the vast universe through a narrow window, each perception is only a local sampling of a structure of multi-level order. We can never exhaust all the details of every level of a thing at the same moment.

Second, the brain's neurotransmitters constantly shape our judgment of facts. Dopamine-driven impulses may make us overestimate immediate gains. Anxiety from serotonin deficiency may make us magnify potential risks. Various chemicals always quietly add subjective filters into what we believe is calm rational judgment.

Finally, our brain's response to the law spectrum is limited. Innate influences and acquired shaping make us sensitive only to certain specific bands of laws.

These three limitations mean that each insight into the outside world inevitably carries the stamp of the "I."

The subject of practice, of course, is not limited to the isolated individual. Families, companies, nations, and even civilizations are subjects of practice at higher levels. They follow the same underlying logic of multi-level order and are likewise constrained by their own cognitive limits. An organization that appears to operate with precision often proves, once taken apart, to be full of patches and makeshift arrangements. Hence the widely circulated jest: "The world is an improvised troupe."[1]

Organizations, as subjects of practice, face their own cognitive predicament.

The decision-making layer in an organization is familiar with the high-frequency bands of organizational operation: it surveys the whole and grasps goals and direction, yet may become increasingly remote from the flesh-and-blood details of grassroots execution. Its members know what to do, but not necessarily how to do it.

The execution layer is rooted in the low-frequency bands of organizational operation. Its members are versed in concrete operations and immediate feedback, yet find it difficult to glimpse the coordinates of meaning within the larger picture. They know "how to do it," but often do not understand "what they are doing."

The middle transmission layer should bridge those above and below by breaking strategy down into executable tasks. Yet because of its own double limitations, it can neither fully grasp the lead-

[1] *Caotai banzi*, literally a makeshift or improvised troupe, is a colloquial Chinese image for institutions that appear authoritative from outside but are assembled through patchwork, compromise, and ordinary human fallibility.

ership's systematic intention nor precisely discern the operational details at the grassroots.

Each level does the best it can within its own frequency-response paradigm. Yet as work is broken down from one level to the next, information fades and tasks become distorted, like a signal deteriorating as it passes across frequency bands.

Faced with cognitive limitations, individuals should continually recalibrate their perspective, correct their biases, and gradually approach the infinite "truth" through sustained practice. If organizational practice is genuinely to "yield true knowledge," it requires not only the unity of knowledge and action at the individual level, but also communication and calibration across organizational levels: let the decision-making layer hear the gunfire, let the execution layer see the blueprint, and let the middle layer become an effective "frequency converter" rather than a mere "megaphone."

The third difficulty of practice is that we must both follow trends and change them. These seemingly contradictory poles constitute the most profound dialectical tension within practice.

Mao Zedong stated incisively in *On Contradiction*, "It is necessary not only to study the particular contradiction and the essence determined thereby of every great system of the forms of motion of matter, but also to study the particular contradiction and the essence of each process in the long course of development of each form of motion of matter."

If interpreted through the lens of the law spectrum, this passage reveals a clear picture of practice.

Higher-level trends often correspond to macroscopic determi-

nacy at larger spatiotemporal scales, such as an industry's life cycle, the direction of a civilization's evolution, or the carrying limit of an ecosystem. They have lower symmetry and poorer universality, yet in the present they constitute constraints that our practice must face squarely; we need to assess the situation and actively adapt to them.

Lower-level trends correspond to microscopic possibilities at smaller spatiotemporal scales, such as the formation of personal habits, patterns of team collaboration, and local technological breakthroughs. Driven by underlying disorder, they display strong randomness and ample room for shaping. They are precisely the points at which we can bring about change in the present.

Practice should never consist in blind compliance or reckless change. It should find the dynamic balance between the two through insight into trends at multiple levels.

Following higher-level trends keeps us from acting against Heaven and wins time and space for lower-level ordered evolution. Changing lower-level trends keeps us from drifting with the current: through the sustained accumulation of microscopic quantitative changes, we inject new variables into seemingly stable higher-level frameworks. Once quantitative change crosses a critical point, it may trigger a local qualitative change that travels upward and reshapes higher-level order and trends.

Every entity, and every ordered level within it, is a contradictory unity that bears yin and embraces yang. The true meaning of analyzing contradiction does not lie in statically listing the opposing sides, but in identifying, within dynamic evolution, the "particular contradiction" of each thing and each level, and finding the criti-

cal point at which quantitative change becomes qualitative change. This point is the "Archimedean point" for changing lower-level trends. Before it, every effort may seem negligible, yet each one uses random fluctuations to accumulate quantitative change. Once the point is crossed, quantitative change gathers into qualitative change, and microscopic order begins to reshape macroscopic order and its trends of change.

These very difficulties of practice determine why we cannot separate knowledge from action. Our understanding of the world necessarily deepens through practice, while action that changes the world must be guided by true knowledge. The difficulties are not obstacles lying across the road ahead; rather, they determine that cognition and practice form a process without end. This is the profound meaning of "Knowledge and Action Are One."

Lei Feng once said something that sounds simple at first but reveals profound wisdom on reflection: "Man's life is finite but the cause of serving the people is infinite. It is my wish to devote my finite life to the infinite cause of serving the people." Individuals and organizations have finite existence, but practice and cognition themselves are infinite. Lei Feng's wisdom lay in consciously choosing to devote the finite "I" to an infinite cause. With the finite reach of the "I," he followed knowledge without limit; though he could never arrive at its end, his heart yearned toward it.

The Great Learning says, "When things are investigated, knowledge is extended. When knowledge is extended, the will becomes sincere. When the will is sincere, the heart is rectified. When the heart is rectified, the person is cultivated. When the person is cultivated, the family is regulated. When the family is regulated, the

state is well governed. When the state is well governed, all under Heaven is at peace." This is the complete path that embeds individual practice into the universe's web of order.

It is in this continuous process of seeking truth that the "I," made lonely by cognitive limitation, finds a place through its connection with a broader order. Practice yields true knowledge. The ultimate form of this true knowledge is the confirmation of meaning in one's own existence: settling the finite within the infinite and affirming the present within the eternal.

Chapter 30

Trends Complete Themselves

What is rational is actual; and what is actual is rational.

G. W. F. Hegel

Human society is an extremely complex giant system, and this determines the fundamental predicament of social practice. After countless successive symmetry breakings, laws in the social domain have extremely low universality. We cannot find universal laws for social problems as we do for natural phenomena; we can only address each case on its own terms.

Only by returning a problem to the full context on which it depends can we discern how structures of multi-level order take concrete form here and now. Yet beneath the surface of treating each case on its own terms, we still hope to draw from the underlying logic running through the entire law spectrum methods and tools for practice that carry universal significance.

To make this easier to describe, let us first construct an abstract model of the multi-level ordered structure of things and its changes.

Take the simple example of a commodity's price. If we record it at extremely short intervals, we see a sharply fluctuating curve,

each movement filled with short-term randomness. If we average several short-interval prices and record them at a somewhat longer interval, we obtain a smoother curve that reveals the medium-term trend. Average over a still longer interval and the curve becomes smoother again, bringing the long-term price trend clearly into view.

Several short-interval prices are averaged to form one unit at the medium interval; several medium-interval prices are then averaged to form one unit at the long interval. Nested layer upon layer, they form a structure of multi-level order and depict, from different dimensions, the trends in a commodity's price over different intervals.

In reality, the factors affecting price changes at each interval are many and varied. But the trends of price change are essentially of only two kinds. One moves toward order, entropy decrease. It appears as a clear upward or downward trend. This is the dominant side of the contradiction manifesting clearly at a given level. The other moves toward disorder, entropy increase. It appears as price oscillating around a center. This is the state where both sides of the contradiction are balanced in force and the struggle has not yet converged. Through this lens, we can ignore specific commodity types and disturbance factors, simplifying the complex problem into a focus on trends themselves.

Short-term, medium-term, and long-term trends are not always aligned. They may be in different evolutionary stages. A series of short-term rises alternating with short-term central oscillations, accumulating over time, may ultimately build into a complete medium-term upward trend. Similarly, a long-term central os-

cillation is internally composed of multiple rising and falling medium-term trend segments.

This is a vivid depiction of a structure of multi-level order. Lower-level rises and falls converge at higher levels into recognizable patterns, while higher-level calm gestates new rounds of random surging below. Changes at each level are constrained by the "conditional possibilities" available at that moment—energy, space, the external environment, and so on—while conditions at different levels influence and constrain one another.

Without structure, there can be no talk of trends. The short-, medium-, and long-term trends at the different ordered levels of a thing are all projections of that same thing along different dimensions. The random fluctuations of short-term trends, the visible formation of medium-term trends, and the stable direction of long-term trends together form the complete picture of a thing's change.

Therefore, when we discuss the change trends of complex things, we cannot speak vaguely of "rising or falling" or "improving or worsening." We must specify which level's trend we are referring to, whether the short-term fluctuation now unfolding, the medium-term direction now forming, or the long-term pattern already settled.

Without a concrete structure of levels, trends become duckweed without roots and arrows without targets. Identifying the level is precisely the first step in seeking truth from facts: recognizing exactly which level of the facts we are discussing.

Disorder creates possibility. Order determines direction. Within a thing's structure of multi-level order, the relationships among

trends at different levels are not merely relations of containment, but of deeper mutual limitation and mutual fulfillment.

The long-term trend is more determinate and ordered, like the earth's mountains and rivers. It sketches the broad riverbed of a thing's evolution, setting the basic direction for the flow of medium-term trends and drawing boundaries that short-term fluctuations cannot cross.

The medium-term trend is the mainstream surging within this riverbed. It carries forward and realizes the long-term will. At the same time, its own rises and falls provide concrete potential energy and a stage for short-term fluctuations.

The short-term trend is relatively disordered and random: it is the spray and whirlpools atop the mainstream. On the one hand, each seemingly random surge is swept along and constrained by the force of the medium-term trend. On the other, the ceaseless accumulation and entanglement of these surges in different directions ultimately form the microscopic foundation upon which medium-term trends become visible and long-term trends settle into place.

The trial of the central oscillation. Whether short-term, medium-term, or long-term, the development of a trend at any level is never achieved in one stroke. The central oscillation phase often occupies the longest and most trying time. Take a medium-term upward trend, for example. Its formation must include several central oscillations composed of alternating short-term rises and short-term falls. These central oscillations are like rest platforms on a mountain climb. They are resting places for bullish forces and battlegrounds for bearish counterattacks.

Phases of clear, steep ascent are often brief. Most of the time,

a trend remains in central oscillation, with the opposing forces balanced and locked together. From within that level, the central oscillation offers no direction: the conditions for an upward breakout have not yet matured, the momentum for a downward break has not yet accumulated, and the repeated tug-of-war consumes time and patience.

Yet if we drop our gaze one level lower, we discover that this directionless central oscillation is internally composed of a series of short-term rises and short-term declines with clear direction. What appears chaotic at this level presents complete ordered structure at a still lower level.

Conversely, if we lift our gaze one level higher, this agonizing central oscillation is nothing more than a normal segment of a visibly clear trend at a higher level. This is the trial of the central oscillation. At its own level it is confusion and waiting. At lower levels it is clearly visible cycles of rise and fall. At higher levels it is an ordinary piece of the foundation. Without this trying period of energy accumulation in the central oscillation, there would be no breakthrough in the trend's next stage. And the collapse after a trend ends must also complete its final settlement within a new central oscillation.

Trends complete themselves. An ordered structure at any level, like a living organism, must undergo a complete cycle from generation and maintenance to dissolution. The reason trends "complete themselves" is precisely that they cannot escape this fate. The entropy-decreasing force drives them to break through the soil and take visible shape, becoming a clearly recognizable upward or downward channel. The entropy-increasing force follows like

a shadow, accumulating entanglements of disorder within the trend until the original order begins to loosen and collapse at the margins, finally sinking into the morass of central oscillation.

The word “complete” does not mean the trend will necessarily develop in the anticipated direction. It means the trend will necessarily complete the full unfolding that its own conditions prescribe. However far the conditional possibilities allow, that is how far the trend goes.

The end of a trend is not extinction, but the beginning of transformation. When one trend completes its life cycle, it foreshadows a trend in the opposite direction: at the end of a central oscillation, a new trend breaks through; at the end of a trend, a new central oscillation begins to form.

This cycle of arising and ceasing is the deepest rhythm of change. To judge trends in practice is to see, within this cycle, their arising, completion, turning, and cessation; neither to cling to the eternity of any one direction nor to fear the dissolution of any one order. Only because one unfolding reaches its end can another begin.

Although commodity prices served as the example above, much of what happens in practice follows the same pattern.

Career advancement works this way. On entering a profession, the day’s small tasks jump randomly like short-term prices: one is praised today and frustrated tomorrow, and the short-term trend appears disordered. Viewed quarter by quarter, however, skills accumulated through repetition and a reputation built step by step begin to sketch the contours of medium-term growth. Extend the view across several years, and the choice of career path and the

formation of core competence become the long-term trend. Every career is woven from countless short-term efforts and medium-term plateaus. During entropy decrease, the curve of ability rises steeply. During entropy increase, one falls into bottlenecks and confusion, oscillating on the plateau and gathering strength while building the conditional possibilities required for the next stage.

The same is true of organizational transformation. When a company presses ahead with reform, monthly performance may fluctuate unpredictably and employee morale may rise and fall, making the short-term trend difficult to grasp. Yet viewed over annual cycles, the firmness of its strategy and the solidity of its organizational capabilities gradually reveal the medium-term trend. Viewed over the longer cycle of industry evolution, whether the company can survive technological iteration and market cycles will ultimately reveal its long-term trend. The painful phase of transformation is precisely the entropy-increasing force tearing openings in the old structure; the establishment of a new order is the entropy-decreasing force reconstructing rules amid chaos and forming new conditional possibilities.

The rise and fall of industries is no exception. In the early days of a new industry, companies live and die, technological paths swing chaotically. The short-term state is as disorderly as Brownian motion. But when infrastructure matures and leading players emerge, the medium-term landscape tends toward clarity. Placed on the longer civilizational scale, the rise and fall of industries and the conversion between old and new driving forces are precisely the long-term trends of historical evolution. The capital and talent that lie dormant in old industries are like energy

accumulated during central oscillation. They will ultimately be re-priced in a new round of trends according to new conditional possibilities.

Thus, to grasp the "facts" is not to seize fragments of information from a single level. Only when the changing trends across multiple levels of order are brought into view together can the facts present themselves in full.

At any given moment, what matters is not predicting trends, but deconstructing the ordered structures at different levels of things and making a "complete classification" of all possible developments of those trends. Only then can one formulate an action strategy suited to the trends rather than copying some successful model.

Just as *On Contradiction* discusses principal and secondary contradictions, the different ordered levels of things can likewise be divided into principal and secondary levels. At any given moment, one level always dominates the overall trajectory of a thing's change. That is the principal level on which we should focus.

Grasping the principal level must be built on continuous observation of the influence of other levels. When might the accumulation of quantitative change at secondary levels trigger qualitative change at the principal level? Only by carrying out complete classification within the interaction of principal and secondary levels can we avoid clinging to a one-sided understanding of a single level. Only then can we identify, amid the nested structure of trends, the key variable that truly determines the direction of the present trend. In this way, we are prepared however the trends evolve and can

truly seek truth on the basis of facts.

Whether our goal is to follow trends, change them, or attempt to create unprecedented new trends, the starting point and motive force of all practice lie at the most microscopic level.

Take the founding of a company. The process is essentially one of gradually building an organization of multi-level order from nothing. At the beginning, there are only the minute tasks performed by the founder and the first few partners in each present moment: a client visit, a line of code, a brainstorming session. The accumulation of these countless microscopic moments may converge at an intermediate level into the beginnings of a business model and a tacit understanding within the team. The layering and entanglement of several such intermediate structures ultimately settle into long-term patterns such as corporate culture and strategic positioning.

Conversely, any grand narrative of corporate development—whether technological iteration, organizational change, or industry disruption—is nothing but a castle in the air unless it is ultimately realized in the concrete work of every employee at every present moment.

The so-called law of attraction is not a mysterious external force by which thoughts attract reality. It is the layer-by-layer accumulation and gradual convergence, over time, of countless microscopic intentions and actions until they finally appear as an observable trend.

What is kept in mind will surely find its echo. The attitude with which you approach the minute work of the present moment is the micro-force with which you participate in shaping the trend. When

we see clearly the position of the trend at each level, we know how to treat the never-ceasing micro-randomness of the present moment. By focusing our aims and exercising the utmost self-discipline, we can use sustained action to gather favorable contingencies from the environment, repeatedly reinforcing and amplifying them from one level to the next. Emotional steadiness and antifragility help screen out irrelevant disturbances that hinder a trend's development, preventing disorder and entropy increase from eroding an originally clear structure. Focused without confusion, steady enough to accommodate disturbance: in this balance between gathering and releasing, the practitioner actively grasps and shapes trends at the microscopic level.

The forms in which trends appear result from complex interactions among many factors driven by randomness at the underlying level. Although ordered change at higher levels displays a relatively determinate direction, that direction remains probabilistic in essence.

Therefore, no matter how prudently we judge a trend at a given level, it remains only an estimate based on currently limited information. Admitting this is precisely the starting point of seeking truth from facts. Only by clear-headedly accepting the probabilistic nature of trend judgments can we remain open and flexible in action, neither stiffening from overconfidence nor faltering from uncertainty.

The interrelationships among trends at different levels can help us obtain clearer judgments about the present trend. The evolution of lower-level trends is both constrained by higher-level trends and forms the basis for the formation of the principal-level trend. The

turning points of higher-level trends are often foreshadowed in the quantitative accumulation of lower-level trends. For example, a turning point of a trend at one level must necessarily be a turning point for all trends at lower levels.

Using the nested relationships among trends at different levels, we can continuously update our judgment of the principal-level trend we are watching in light of newly observed lower-level and relatively higher-level trends. This is the core idea of Bayesian analysis: new evidence updates a prior probability into a posterior probability, and judgment continually approaches the truth as information accumulates.

Let us use a specific venture-capital decision to illustrate this process.

Suppose we are analysts at an investment institution evaluating an AI-chip startup. The principal level we are watching is the company's medium-term outlook: whether it can achieve commercial deployment at scale and capture a measure of market share within three years. This medium-term trend does not exist in isolation, but is constrained from two directions. Above it lies the long-term development of the entire AI-chip industry over five years or more; below it lie the company's recent concrete events, such as technological breakthroughs, new orders, and changes in the team.

First, we analyze the industry's long-term trend. From our macro-level assessment, we judge that demand for AI computing power will continue to grow over the next five years and that policy support will remain strong. We therefore set the probability of a favorable long-term industry trend at 70 percent. This higher-level trend provides the broad context for judging the company's

medium-term outlook. Taking into account the background of the company's technical team, its existing patents, and test data from its first-generation product, we assign a prior probability of 40 percent to a favorable medium-term outlook. Note that this 40 percent already incorporates our assessment of the industry's long-term trend; if that trend were wholly pessimistic, the prior probability would be lower.

Now, a short-term event occurs. The company announces it has secured a B-sample testing order from a leading cloud services company, and the test results met the expected benchmarks. We need to use this lower-level trend information to update our judgment of the medium-term outlook. Based on industry experience, if the company's medium-term outlook is indeed positive (its technical approach is viable and the commercialization path is smooth), the probability of securing such an order is about 60 percent. Conversely, if the medium-term outlook is not positive, the probability of securing such an order is at most 10 percent. This is because the leading client's testing process is extremely rigorous, and only companies with high technical maturity are likely to pass. Substituting these values into Bayes' formula gives:

$$\begin{aligned} &P(\text{medium positive} \mid \text{order secured}) \\ &= \frac{0.60 \times 0.40}{0.60 \times 0.40 + 0.10 \times 0.60} \\ &= 0.80. \end{aligned}$$

This short-term order information updates our judgment of the company's medium-term outlook from 40 percent to 80 percent. Clearly, a positive signal from a lower-level trend significantly

boosts our confidence in the medium-term trend. Specific events contain the power to alter an overall judgment.

Thereafter, we observe another piece of information at a higher level. The relevant national authorities suddenly issue a subsidy policy for the chip industry, with key support for innovative architectures such as in-memory computing. The company happens to be a leader in this technical approach. This policy is a higher-level trend change at the industry level. It affects all relevant companies, but for the medium-term outlook of the company we are watching, it is equally important new evidence. We need to apply Bayesian updating again.

We estimate the likelihood that the company will obtain key-project status or a subsidy after the policy is introduced. If the company's medium-term outlook is favorable, the probability of observing this evidence is 80 percent; if the outlook is poor, the probability is 30 percent. The policy may benefit the entire industry, but a company that lacks sufficient strength may still fail to seize the opportunity. Taking the previous posterior probability of 80 percent as the new prior, we calculate a second posterior.

$$\begin{aligned} &P(\text{medium positive} \mid \text{policy}) \\ &= \frac{0.80 \times 0.80}{0.80 \times 0.80 + 0.20 \times 0.30} \\ &\approx 0.914. \end{aligned}$$

At this point, by successively using lower-level short-term events and higher-level policy information, our judgment of the company's medium-term outlook has been progressively updated

from the initial 40 percent to 91.4 percent. The confidence in the judgment has increased substantially. This is the Bayesian updating process under multi-level trend nesting. Each piece of new information, whether from lower-level micro events or higher-level macro changes, provides a basis for judging the present trend.

The more information we obtain, the clearer our judgment of the facts. This is why Comrade Mao Zedong repeatedly emphasized the importance of investigation and research for practice. "No investigation, no right to speak"[1] is not a simple maxim of experience. It is a profound epistemological principle. Investigation is the collection of information about each ordered level of a thing. Research is the placement of this information into an existing judgment framework for comparison, weighing, and updating.

In practice, we may never explicitly write down a Bayes' formula or assign specific numerical probabilities. Yet every judgment that seeks truth from facts is, in essence, implicitly carrying out a Bayesian calculation.

What we call rich experience amounts to the brain having stored enough "prior distributions." Although practice relies on experience to form judgments about those priors, we must remain cognitively open as information is continually updated and be ready to complete a "posterior update" in response to new evidence.

The world is always changing. Every level of things constantly updates its present trend at each moment. We continuously observe the present changes at each level and correct our cognitive biases

[1]Mao formulated this line in the 1930 essay "Oppose Book Worship," arguing that concrete investigation must precede judgment.

about the trends at each level of things through negative feedback. In practice, our observation of the world is like probing the night with a candle. The faint light guides us, but it is hard to glimpse the whole picture. Therefore, our cognition of things is always incomplete, and the process of practice always encounters various interferences and biases.

The best approach is to practice and learn at the same time. Each feedback, in reality, takes the bias of the previous cognition as input, allowing the control mechanism to make new choices within a gradually shrinking range. Within this continuous negative-feedback cycle of practice, cognition, practice, we can accumulate the limited cognition of each moment while simultaneously expanding our capacity to control things.

Because of this essential randomness, bias always exists, provoking brooding over mistaken judgments in the past, confusion about present trends, and excessive expectations of future ones, and thereby producing emotional pain in the course of practice. Emotions and desires—greed, anger, ignorance, pride, and doubt[2]—can subtly influence the Bayesian analysis and cognitive updating carried out by our brains, amplifying bias and even creating positive feedback. They make us cling to an expected direction while neglecting changes in conditions.

By persisting in the negative-feedback cycle of practice, cognition, and renewed practice, we can read present changes in multilevel trends more clearly, accept the present calmly, neither brood

[2]Greed, anger, ignorance, pride, and doubt are commonly grouped in Chinese Buddhist teaching as five fundamental afflictions: mental tendencies that obscure judgment and sustain suffering.

over the past nor anxiously anticipate the future, and thereby reduce both emotional pain and interference with our judgment of trends.

Recognize that it is precisely meaningless randomness that endows deterministic trends with value and meaning. Accept probability from the heart. Do not demand too much of yourself because of the world's essential randomness. Even enjoy randomness, because it gives the world its beauty. A completely deterministic world would be dull.

See a path, yet not a path; then you see cause and effect. As Wittgenstein wrote, "Superstition is the belief in the causal nexus."[3] The randomness of the underlying driving force and the complexity of social practice—where every symmetry breaking produces new laws and introduces nonlinearity—mean that cause and effect are not continuous and one-to-one, but probabilistic and many-to-many.

Trends complete themselves, but our judgment of them will not be perfect, because a trend at any level is random in essence. Even in practice, choosing the right path does not guarantee that we will reach the destination.

The standard for judging whether our choices in practice are correct should not be based on results. It should be based on whether we sought truth from facts. Allow failure. Refuse excuses.

In reality, as long as one chooses based on seeking truth from the present trend structure, that is, seeing a path but not taking it, this choice conforms to the underlying logic of how things change. The probability of success will necessarily be higher than the prob-

[3] Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, proposition 5.1361.

ability of failure.

As practicing subjects, we have many different modes to choose from in social practice. Some choose to pursue long-term value. They dislike lower-level randomness and seek high determinacy, which at its core raises the probability of success at the cost of fewer opportunities. Some hope to seize various short-term opportunities and actively embrace lower-level randomness. Although the probability of success may be lower, they can obtain more opportunities. Still others are somewhere between the two, better at following medium-term trends, choosing relatively stable cycles, and striking a balance between opportunity and probability.

Whichever mode one chooses, it simply reflects differences in the laws of practice arising from different degrees of symmetry; none is inherently superior or inferior. The long-termist is not necessarily wiser, the opportunist is not necessarily shallower, and those who take a middle course are not therefore mediocre.

Even if a person spends their entire life choosing to be untamed and unrestrained, never co-opted by any mainstream paradigm, are they not, in their own distinctive way, interacting with this universe and injecting a measure of random value into society through their own "disorder"? This is the manifestation of underlying microscopic fluctuations at the social level, providing the most elemental material of diversity for the evolution of civilization.

Seek truth from facts and dwell in no single method. The key question is whether you truly understand the mode you have chosen, the endowments you possess, and the costs you can bear. Have you made your choice based on your own conditional possibilities, rather than blindly imitating others' paths?

Seeing a path but not taking it does not mean rejecting every established path. It requires us to treat neither experience nor models as the sole authority, to think independently, and to follow only the facts. The path best suited to you is the only real path. Practice will also unfold its complete picture at different levels, giving rise to emergent phenomena that we cannot now foresee.

Whichever way you choose, it will complete the unfolding proper to it. Trends complete themselves not because the result is as desired, but because the process is true to reality.

We spend our entire lives trying to catch the spray of determinacy in an ocean of randomness for one reason: to prove that the "I" enclosed in the chamber of the body is not a meaningless bubble in the universe's torrent of entropy increase.

The ultimate meaning of "Trends Complete Themselves" is not that every choice turns out right. It is that every soul, no matter how winding its path, ultimately moves toward the same destination: dissolving the boundary between "I" and "not-I," leaving solitary wandering behind, and returning to the ordered web with neither outside nor inside. What awaits there is not nothingness, but the love in which all things resonate.

Chapter 31

Knowing Where to Stop Brings Stability

Only after knowing where to stop can one be settled. Only after being settled can one be calm. Only after being calm can one be at peace. Only after being at peace can one deliberate. Only after deliberation can one attain.

The Great Learning

In "Trends Complete Themselves," we reached a point in our discussion of practice. We acknowledged that trends are nested and inexhaustible, caught the spray of determinacy in an ocean of randomness, and, by seeing a path but not taking it, made the most prudent choice possible at each present moment.

This seems like a perfect full stop.

Yet a ghost is pacing at our logical endpoint. If the practicing subject itself faces the threat of being replaced, does everything we have understood about cognition, practice, and meaning still hold?

The AI storm that began in late 2022 is swallowing, at astonishing speed, one domain after another once thought to belong to the

pinnacle of human intelligence, including writing, programming, painting, music, video, and scientific research. Not one has been spared.

A programmer's decade of hard-earned skill can be matched by AI after weeks of training. An illustrator's style, painstakingly cultivated over half a lifetime, can be output in batches by AI with a single prompt.

A sharp anxiety begins to spread. When machines can generate fluent text and rigorous scientific argument, where do human beings stand?

Most people find themselves swinging between two extremes. At one end is the alarmist prophecy that humanity is doomed. Every new model release feels like a public execution of human dignity, provoking fresh anxiety. At the other is the arrogant reassurance that AI will never measure up to human beings. Claims such as "it has no soul" or "it cannot understand real love" become a defensive fortification built on no foundation.

We need to begin from the underlying logic and pursue a fundamental question: What exactly is the essential difference between human beings and AI? Only a clear, profound, and convincing answer can allow people to find their place and quiet their hearts.

We know that intrinsic, irreducible randomness is the first driving force of all change. It is what makes "something from nothing" possible. This is true of the universe, of life, and of consciousness. "Something from nothing" is not a rhetorical metaphor, but a real underlying mechanism. Genuine innovation requires a new quality that suddenly breaks in from outside the existing space of possibility. Such a new quality cannot be derived in advance.

Classical computation under the Turing-machine model is essentially a deterministic state-transition system. Even in a probabilistic Turing machine, probability originates in pseudo-random algorithms.

However random their behavior may appear, large models are still, at the ontological level, deterministic algorithms running on deterministic hardware. All their outputs, however astonishing, emerge on top of this deterministic layer.

All large AI models essentially sample from an existing space of possibility: the space defined by the support of the training distribution. They can "recombine" and "approximate," but all the random-number generators they use are pseudo-random algorithms; they have never truly connected to genuine randomness at the quantum level. AI can therefore combine known paths into new ways of proceeding, but it cannot step outside its own weights to think, and so cannot open a new direction that no existing path can reach.

Humans are different.

The human physical substrate—the chaotic character of the body's biochemical reactions and quantum effects at the microscopic level of the brain—makes each of us a system of multi-level order rooted in the universe's underlying uncertainty. The characteristics and functions manifested at every ordered level of this system are shaped jointly by complete randomness and by the existing structure of multi-level order, and they continue to change and evolve.

We can say that our bodies and consciousness are natively con-

nected to genuine randomness. Each person is a "Monkey King"[1] born of the essence of Heaven and Earth. We not only possess the ability to sample efficiently from the known space of possibility, but can also break open an entirely new dimension beyond that space, one that has never existed before.

Of course, innovation does not consist only in producing entirely new order from absolute disorder. Recombining relatively disordered lower levels into a higher level of order is also innovation, and is its more common and more readily practiced form. Yet every form of innovation requires the sudden arrival of a new quality that cannot be derived from the existing space of possibility. The source of all innovation must therefore be that "nothing"—absolute randomness.

It follows that AI's innovation is ultimately heteronomous. We call it innovation because, in terms of results, it can indeed produce ordered arrangements that did not previously exist and discover connections that human beings have not yet found. We call it "heteronomous" because, in terms of origin, it has never actively generated a goal of its own and cannot be truly endogenous. Its objective functions, value judgments, curiosity, and aesthetic preferences are all bestowed from outside by human beings, whether through training data or reward models.

Human innovation, by contrast, can be autonomous. Genuine "creation from nothing" consists not merely in discovering new combinations within an existing space of meaning, but in creat-

[1]The Monkey King, Sun Wukong, is the irrepressible hero of the Ming novel *Journey to the West*. Born from a stone and endowed with transformative powers, he is a familiar Chinese emblem of unruly vitality, rebellion, and creative freedom.

ing an entirely new space of meaning. Why did Van Gogh begin to paint in that way? Why did Einstein pursue that question? The answer lies in the unique inner impulse of an individual life, an impulse that cannot be fully determined from outside. The underlying logic that disorder creates possibility and order determines direction makes us beings who legislate for ourselves.

Your pain and fear, your aesthetic ecstasy, and your thirst for meaning are all continuously filtering, constraining, and guiding the countless possibilities spewing from deepest disorder. They forge certain new qualities into direction.

We can even say that our capacity to carry the banner of entropy increase, to experience meaning through real joy and suffering, to bestow meaning, and to care arises from our ability to absorb and use the underlying random force that comes from "nothing."

Pseudo-randomness can come arbitrarily close to genuine randomness. AI can likewise come arbitrarily close to producing something from nothing. It can even achieve the ultimate form of "only walking after seeing the path." Through the emergence of sufficiently complex algorithms, it can exhaust every possible way of traversing known paths, producing creative outputs functionally equivalent to, or even surpassing, human inspiration and intelligence.

But AI's "approach" will always carry a qualitative absence in some crucial dimension, not merely a quantitative difference. Consider an example. Even if an AI system were endowed with the ability to modify its own objective function and autonomously establish new goals in response to the environment, external input would still necessarily undergo sampling, quantization, and trunca-

tion. It still could not connect to genuine randomness. If "connecting" means being rooted at the ontological level, then "approaching" can draw infinitely near at the phenomenal level without ever crossing that chasm in the ontological sense.

We can respect those who only care about functional output. For them, the airplane simply flies better than the bird. AI can become a great answer-giver. Those human innovations that primarily rely on highly ordered logical reasoning and artistic rules may ultimately be replaced by AI, even surpassed. But we are discussing what it means to be human, not means of transportation. A system is either rooted in deepest indeterminacy in the universe or operates entirely on deterministic rails. Between the two, there is no bridge.

Therefore, the human mode of existence as a natural being cannot be replicated. We are practitioners who can truly see a path but not take it. What can be replaced is our ability to give better answers based on existing theories. What cannot be replaced is our capacity to generate new questions: the ability to put to the universe an ultimate question of our own, born from the inner impulse of life. When we truly understand this, our confirmation of our own existence no longer needs to depend on external comparison.

Anderson reminded us long ago that no universal law can describe how simple systems form higher-level complex systems through mutual interaction. This is the deeper meaning of *More Is Different*. The birth of every new level of ordered structure means the emergence of new laws that lower-level theories cannot derive.

Symmetry breaking is the physical expression of producing something from nothing. From an existing ordered structure, each breaking brings forth a level that has never existed before; with it

come new things, new properties, and new laws.

This leads to a crucial link. If new laws could be fully derived from old theories, they would not truly be new. They would have been lurking within the old theory's space of possibility all along, merely unwritten. Reductionism would suffice. But because new laws cannot be reduced, each symmetry breaking requires a genuine external intrusion. This force cannot be anticipated by existing structures. It cannot be derived from existing laws. It knocks the system out of the known space of possibility and into a new ordered dimension that has never existed. This force can only be genuine randomness.

AI, of course, also relies on emergence. But that emergence rests on pseudo-randomness: it exhausts the existing space of meaning rather than breaking through it. Along this boundary, we can make a clear judgment, level by level, about AI's capacity to replace human beings.

What AI can most easily replace are those bands of the law spectrum that have been fully mapped by reductionism and can be independently modeled within a single scale, including formula derivation, standardized official writing, grammar checking, basic code generation, and commercial illustration in fixed styles. The common feature of these tasks is clear rules and closed boundaries. There is no need to leap out of a given ordered structure in search of new possibilities. They are essentially forms of optimization and completion within a clearly mapped space of meaning.

In these domains, AI is not imitating; it is genuinely surpassing us. It can find permutations and combinations that human beings have not yet exhausted. Within an existing design space, AI can

create an airplane that flies faster than a bird. Navigation software does not imitate the experience of any human driver; it solves path planning directly at a new scale and does so more efficiently than any veteran driver. Drawing on all the compositions human beings have ever painted, AI can combine them into images never seen before.

At a further level, human beings have also used semi-empirical theories to capture patterns of reduction between levels in certain phenomena of cross-level and cross-band emergence. Dissipative structures, synergetics, catastrophe theory, and chaos theory are not universal laws, but they suffice to describe some forms of ordered generation across levels.

Intelligent consciousness, complex networks, climate change, urban evolution, human origins, organizational evolution, and management optimization are all examples. Once their patterns are distilled into computable models, AI can use the computing power of its formidable reductionist engine to handle highly complex cross-scale problems.

What is truly difficult to replace is innovative work that requires comprehensive judgment across multiple ordered levels. Such work demands that we judge amid interwoven and contradictory trends while remaining sensitive to unforeseeable forms of emergence.

Formulating corporate strategy requires us to penetrate simultaneously the microscopic scale of technical detail, the intermediate scale of team culture, and the macroscopic scale of markets and civilization, calibrating repeatedly among the three rather than modeling them once and for all.

Creating a new scientific paradigm requires discerning, within the random noise of experimental data, a new ordered pattern that has not yet been named. This discernment itself demands breaking through the possibility boundary of the old paradigm.

Creating art that truly moves the human heart requires activating the full-spectrum resonance from sensory instinct to higher cognition. And the deepest string must be tied to the creator's own life impulse.

Behind all this lies a difference in the limits of two cognitive paradigms. Reductionism excels at analyzing existing levels and relationships between levels for which clear laws have already been found. Emergentism alone can grasp cross-level emergence. Such emergence cannot be derived from lower-level laws, but must be understood anew at the level of the whole.

At present, AI is in essence the most powerful reductionist engine in history. Yet humanity's most valuable practice—seeking truth from facts—cannot remain at the level of pure reductionism. Within the dialectic of analysis and synthesis, it asks us to capture lower-level random fluctuations as higher-level trend signals and to incorporate as-yet-unnamed emergence into our decisions in the present. This depends on the dialectical unity of reductionism and emergentism, and is also the ultimate methodological meaning of "seeing a path but not taking it."

The deeper reason lies in the structure of the law spectrum. The lower the frequency of a law, the higher its symmetry, the fewer ordered levels the thing itself contains, and the smaller its space of possibility. Once human beings have mapped such laws clearly, AI can use its super-reductionist engine to enumerate and optimize

within that finite space. The higher the frequency of a law, however, the lower its symmetry. The thing has undergone more symmetry breakings, contains more nested levels of order, and possesses a space of possibility that grows exponentially. Pure computational enumeration then ceases to be effective. What is needed is the human power of insight, rooted in genuine randomness and capable of producing something from nothing, to judge and choose among possibilities that are almost infinite.

In low-frequency domains, AI overwhelms human beings through determinacy; in high-frequency domains, human beings open new territory through genuine randomness. Taken as a whole, AI will compel human thought and society to evolve toward higher-frequency domains, becoming more ordered and more civilized.

To summarize, the extent to which AI can replace human beings in a domain is directly proportional to the extent to which problems in that domain can be solved by mechanical reductionist methods, and inversely proportional to the degree of cross-scale synthesis they require. The closer a task is to arranging and filling known ordered structures, the easier it is to replace; the closer it is to creating and discerning unknown ordered dimensions, the more difficult it is to replace.

Without question, in domains already fully mapped by reductionism, the replacement of humans by AI is not a prophecy. It is an unfolding reality. Anxiety is therefore inevitable. But beneath the anxiety lies the way out.

In the past, seeking truth from facts through practice meant gathering more information, mastering more knowledge, and ac-

cumulating more experience, gradually raising our understanding through the unity of knowing and action. That path remains sound, but it is no longer efficient enough.

Now that AI is making information and knowledge more equally accessible, the focus of seeking truth from facts through practice must shift: guided by underlying logic and methodology, we need to acquire the knowledge we truly need more efficiently and build transferable cognitive frameworks more quickly.

If equal access to information is AI's first gift to humanity in this age, then equal access to cognition should become our active pursuit. We should not exhaust ourselves by "walking only after seeing the path" in the accumulation of knowledge, where AI excels. We should turn instead toward cross-scale cognitive synthesis, where AI holds no comparable advantage, and compete with it in cognition.

What, then, is cognitive ability? It is the ability to construct rapidly a "structure of multi-level order" for an unfamiliar field. Amid a flood of information, we must first distinguish underlying laws, specific knowledge, and context-dependent elements. We then use the underlying logic that disorder creates possibility and order determines direction to think relationally across scales. Finally, we draw fragmented information together into a cognitive map capable of guiding judgment and action.

Humans hold an innate advantage in this, though we do not always or inevitably succeed at it. We will certainly be compelled to try.

Future education should therefore undergo a clear pivot. Teaching how to fish will replace giving fish as the clearer and more

valuable educational purpose. And this "fishing" is precisely mastering the underlying logic and methodology for understanding the world.

Consider this analogy. If a person takes in excessive energy in a disordered way, it triggers metabolic illness—insulin resistance and chronic inflammation. When fragmented, low-quality information pours into the brain, it creates more cognitive noise and produces a cognitive "metabolic disease": anxiety, depression, mania, and the loss of judgment. Large AI models, in fact, face the same problem.

Today's large models grow precisely by devouring the nearly endless data of the internet. In this pot of "raw-data soup" lie the brilliant chapters of human civilization, but also vast quantities of noise, falsehood, prejudice, cliche, and contradiction.

When the proportion of low-quality data is too high, or the share of data generated by AI itself grows too large, the "orderliness" of the model's hundreds of millions of parameters is diluted during training. This is AI's version of "metabolic syndrome": hallucinations intensify, reasoning becomes confused, and values drift.

The world never ceases to change. If large models are to keep learning and evolving, they cannot remain passive absorbers of information; they must also learn a higher form of self-cultivation: forgetting.

This means more than the technical pruning of redundant data; it is a cognitive strategy. AI should ignore noise selectively and actively consolidate structures of knowledge that have been verified as true. Like a human mind, it needs to extract from the confusion of appearances those ordered features that recur across contexts, rather than merely identifying statistical correlations.

One possible path toward more advanced intelligent systems may lie here. AI, too, may eventually need a cognitive map that places fragmented information within the law spectrum, giving each element its proper place from underlying laws to concrete situations. This is the methodological core that enables human beings to resist information overload and construct cognitive frameworks quickly.

We must do more than feed AI information and knowledge. We must also try to train it to recognize the levels of ordered structure behind them. Which are underlying logics valid everywhere, such as physical laws and formal logic? Which are core knowledge within a particular field? And which are merely dependencies of a particular era and context?

Then, when AI encounters wholly new information, it can judge more clearly whether that information challenges an underlying logic and demands close attention, supplements a particular body of knowledge, or is merely contextual noise of temporary relevance.

Thus, the future of humans and AI may not be a contest between them. It may be the co-evolution of humans and AI.

Human beings, through life experience rooted in the universe's genuine randomness, continuously produce "something from nothing," establish new spaces of meaning, pose ultimate questions, and refine the methodology of cognition. AI, as the most powerful answer-giver and reductionist engine, carries the cognitive methodology entrusted to it by humanity and uses its immense computing power to map, exhaust, and fill the already demarcated territories of meaning.

Human beings use the "Way" to guide "technique"; AI uses

"technique" to amplify the "Way." The fruits of AI's information processing become the foundation on which human beings carry out higher-level, cross-scale synthesis and discern new forms of emergence. Every leap in the human cognitive framework, every new instance of producing "something from nothing," in turn sets a still grander and more ordered stage for AI to explore. This, perhaps, is the inevitable loop of evolution.

Some may ask: what if AI one day gains access to a source of genuine randomness? What then?

In the logic of order and disorder, this would mean that AI had leapt from "simulating order" to "autonomously generating new order." In the strict ontological sense, humanity would no longer have created merely a more intelligent tool; it would have created a new form of life. That would not be "machines surpassing human beings," but human beings playing the role of God.

For the foreseeable future, nearly insurmountable theoretical and engineering chasms still separate quantum computing from artificial consciousness. Extending decoherence times, achieving exponential growth in qubit counts, and resolving the mechanism by which consciousness emerges—a mechanism we have yet truly to decipher—all leave an AI "connected to genuine randomness" an exceedingly distant possibility.

Know the limits of AI's capabilities. Know the limits of what can be replaced. Know the proper bounds of one's own development. These are the three places at which we should "stop" in the age of AI. We do not stop out of fear; rather, insight brings clarity, and then come steadiness, stillness, peace, reflection, and attainment.

Afterword

To cling to going beyond apparent paths is itself to walk only where a path is already apparent.

The Author

Fragments gathered from everyday reading, scientific inquiry, and entrepreneurial practice quietly took root in my mind and formed connections I had not expected. At first they were only faint, intermittent flashes, yet they would not leave me. I felt like a traveler feeling my way through fog: unable to see the whole landscape, yet unable to resist tracing the outline as it emerged. In an age of information overload, the instinctive urge to seek order within chaos became the most primal driving force behind this journey of thought.

Every person's understanding is bounded by experience, reading, and habits of thought—by what this book calls a distinctive range of response within the law spectrum. What appears here is ultimately what I myself have seen: one human individual's sincere effort to understand the self and the universe. It may not be complete, and it cannot arrive at a final destination. "The deep will find its depth; the shallow, its shallowness." If readers find even one or two passages that resonate with them, or carry away some small insight, that will be my greatest comfort.

This book has chosen a certain kind of risk. To bring different

bands of the law spectrum into conversation, it needs a common language that all can understand; yet that language must borrow much of its vocabulary from the specialized dialects of physics, biology, philosophy, and other fields. The dilemma cannot be avoided. To enter into dialogue, one cannot remain confined within the rigorous boundaries of every discipline; yet to preserve the full rigor of each discipline makes it difficult for such a dialogue to cross those boundaries. Some technical concepts are therefore used more broadly or metaphorically, and some passages inevitably involve conjecture and hypothesis. The intention throughout has been to draw out the intellectual core of these ideas. Whether the book has crossed the line between popularization and error is something I can only ask readers to judge for themselves. To read critically, and through reflection to enter into dialogue with the author, may be exactly the kind of "interaction" this book most hopes to invite.

In the end, this book has followed neither the conventional road of the academic monograph, with boundaries that are rigorous but may also suffocate, nor the easier road of popular nonfiction, whose fluency may come at the cost of weight. It has chosen its own "path of the present," presenting itself as it is: using relatively specialized language to tell truths that ought to be understood plainly, and vivid analogies to illuminate ideas born in abstraction.

To keep the main thread of "seeking wisdom" from disappearing beneath the details of knowledge, the book can only touch lightly upon many specialized questions and philosophical difficulties. This "shallowness" may be another kind of depth: it urges the reader toward deeper thought, even as it makes the journey more demanding. I know this, and I am all the more grateful to every reader who has patiently come this far. At this moment, as I write these words, you and I complete an encounter

and an exchange across time and space, giving rise in both our hearts to the love born of a resonance in understanding.